CHAPTER 13
ENVIRONMENTAL CONTROL SYSTEMS

CHAPTER OVERVIEW

SECTION 13-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

13-1-1 HEATING AND VENTILATION SYSTEM DESCRIPTION AND DATA.

13-1-1.1 Description.

The Heater System functions from bleed air supplied by the main engines under flight conditions or the Auxiliary Power Unit (APU) during ground operations.

A bleed air mixing valve is used to mix engine or APU bleed air and ambient air at a cockpit selected mixture temperature. Heated air is distributed to the cockpit and cabin through a system of ducts.

A mixture temperature sensor works along with the bleed air mixture valve, regulating the bleed air flow to match the mixture temperature selected at the cockpit heating control.

The Heater gives a temperature of 40°F (4.4°C) when the outside ambient temperature is -65°F (-53.8°C).

The Ventilation System provides ventilated air to the cockpit and cabin.

Air obtained from outside the helicopter by an air intake duct is distributed by the blower unit through the Heater System ducts.

Refer to Figure 13-1-1-1 and 13-1-1-2 for system component location and block diagrams.

13-1-1.2 Power Distribution.

Electrical power of 28 VDC is supplied by NO. 2 DC PRI BUS through HEAT VENT circuit breaker, pilot's circuit breaker panel, to the VENT BLOWER switch.

Electrical power of 28 VDC is supplied by the NO. 1 DC PRI BUS through the AIR SOURCE HEAT/START circuit breaker, copilot's circuit breaker panel, to the AIR SOURCE HEAT/START switch, the HEATER switch, and to the mixing valve.

Electrical power of 115 VAC is supplied by NO. 2 AC PRI BUS through the HEAT & VENT circuit breaker, pilot's circuit breaker panel, to the open contacts of the blower relay.

13-1-1.3 Ventilation System Operation.

Placing the VENT BLOWER switch on the upper console to ON completes the electrical circuit to the blower motor. The blower pulls in outside air through the external air intake and circulates it through the cabin heat ducting.

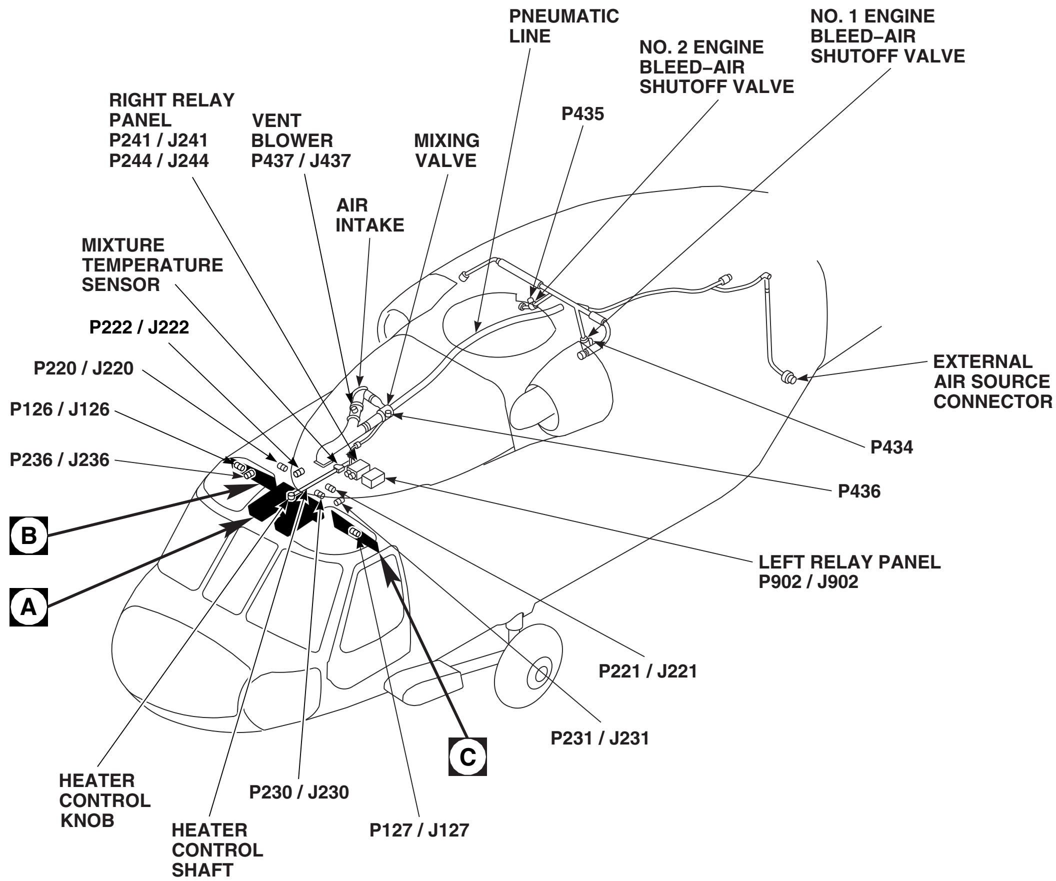
13-1-1.4 Heating System Operation.

Placing the HEATER switch ON opens the mixing valve to allow bleed air to circulate through the cabin heat ducting.

The HEATER knob on the upper console controls the temperature of the heated bleed air entering the cabin. Turning the knob from OFF to MED or HI regulates the temperature of the air entering the cabin by allowing more bleed air to pass through the mixing valve into the cabin heat ducting.

With the engines operating, placing the AIR SOURCE HEAT/START switch on the upper console to ENG opens the No. 1 and No. 2 engine bleed air shutoff valves allowing engine bleed-air to reach the mixing valve.

Placing the AIR SOURCE HEAT/START switch to either OFF or APU closes the No. 1 and No. 2 engine bleed-air shutoff valves allowing APU bleed air (or air from an external source) to reach the mixing valve.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P126 / J126	COCKPIT CEILING, BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P222 / J222	UPPER CONSOLE
P230 / J230	UPPER CONSOLE
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P241 / J241	RIGHT RELAY PANEL ASSEMBLY
P244 / J244	RIGHT RELAY PANEL ASSEMBLY
P434	NO. 1 ENGINE BLEED–AIR SHUTOFF VALVE
P435	NO. 2 ENGINE BLEED–AIR SHUTOFF VALVE
P436	MIXING VALVE
P437 / J437	VENTILATION BLOWER
P902 / J902	LEFT RELAY PANEL ASSEMBLY

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FIGURE 13-1-1-1. HEATING AND VENTILATION SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)

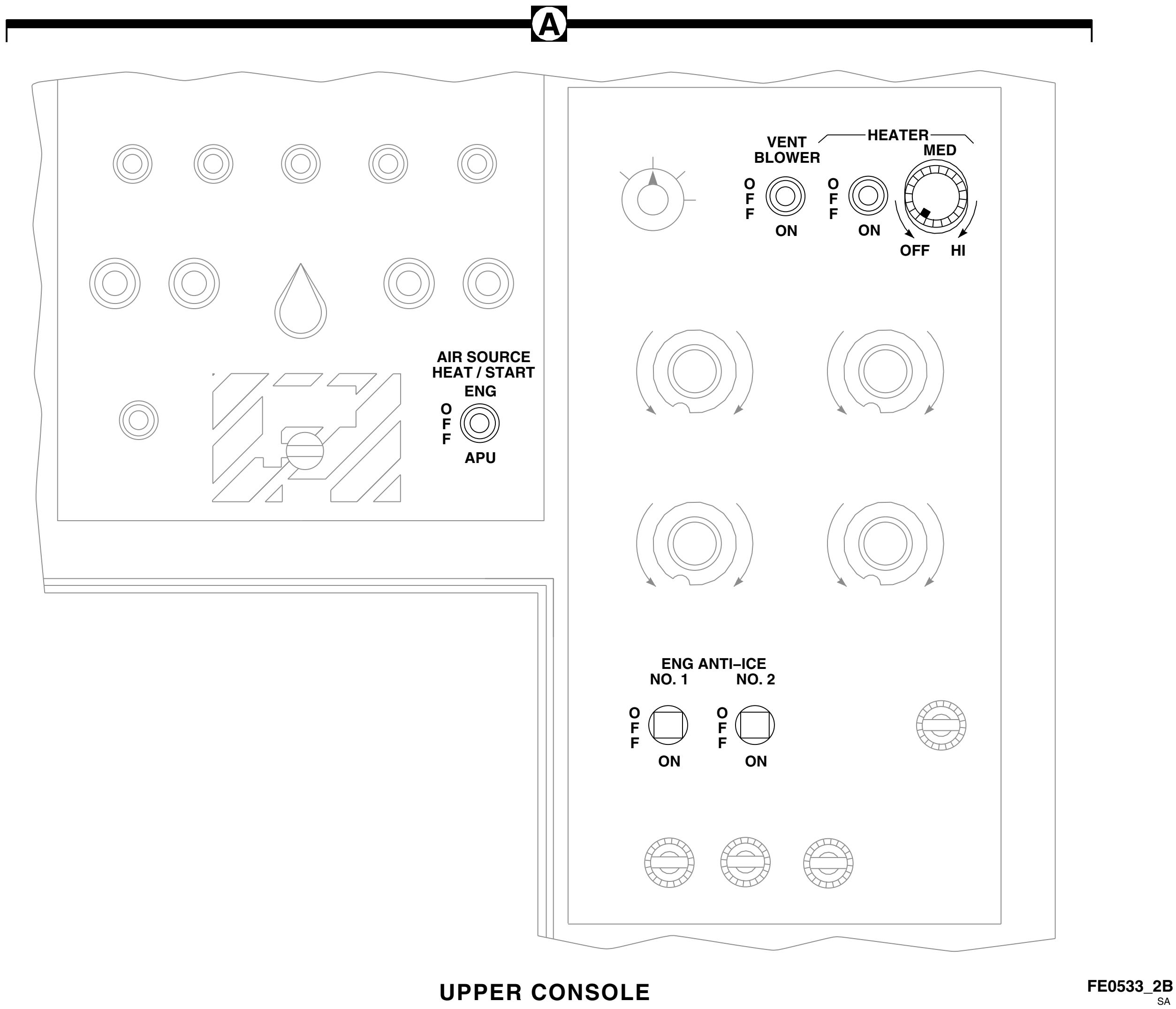


FIGURE 13-1-1-1. HEATING AND VENTILATION SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)

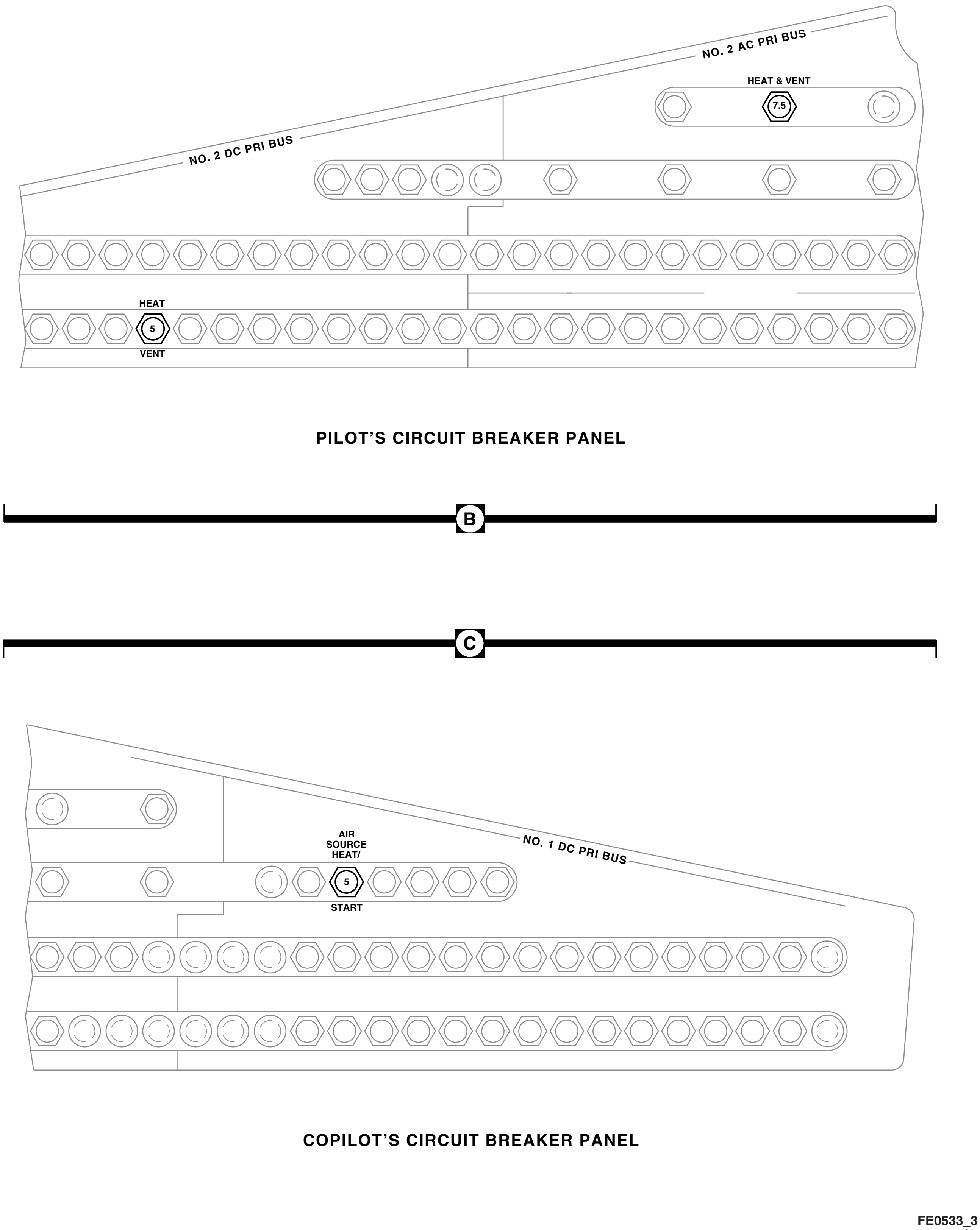


FIGURE 13-1-1-1. HEATING AND VENTILATION SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

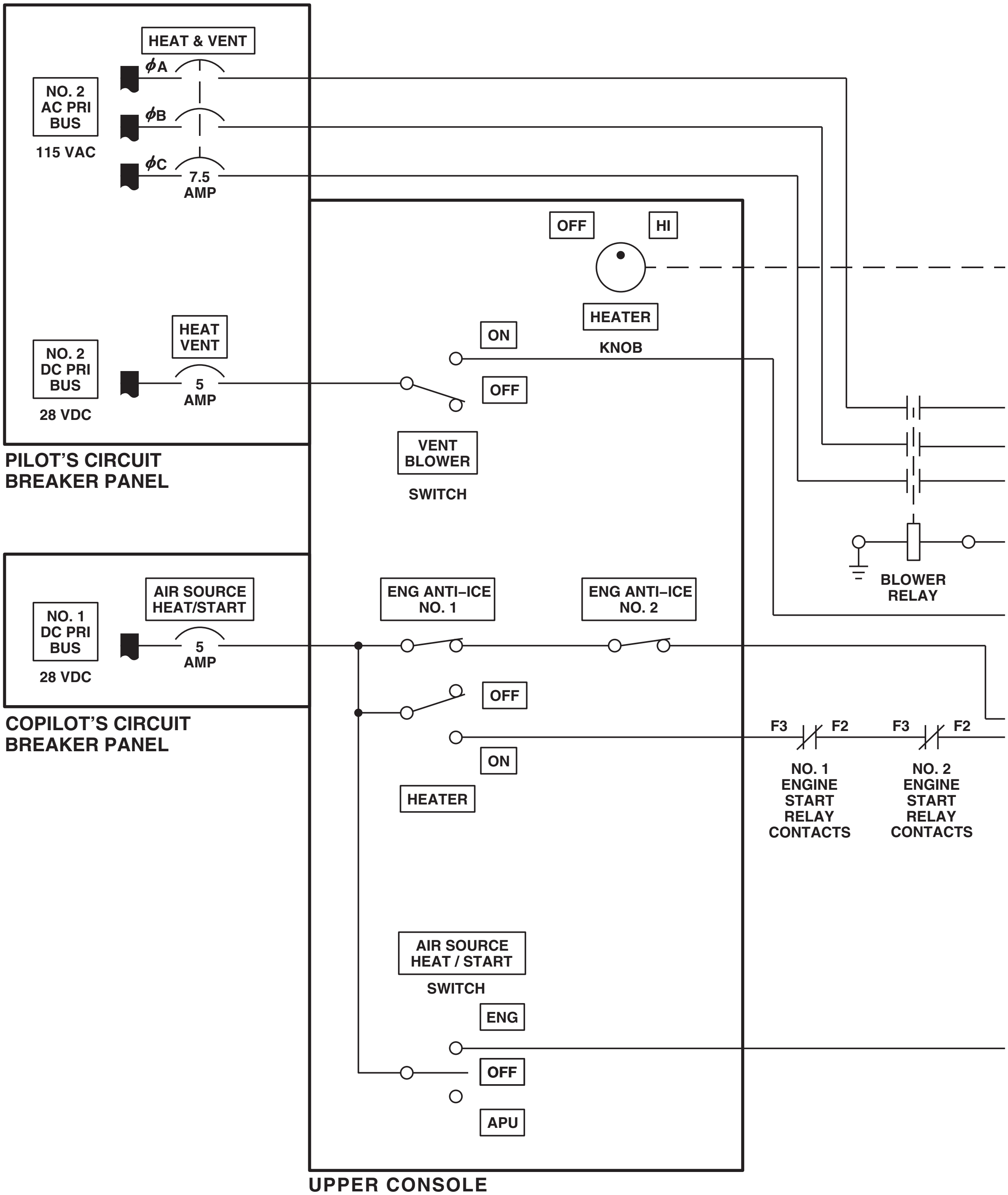


FIGURE 13-1-1-2. HEATING AND VENTILATION SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

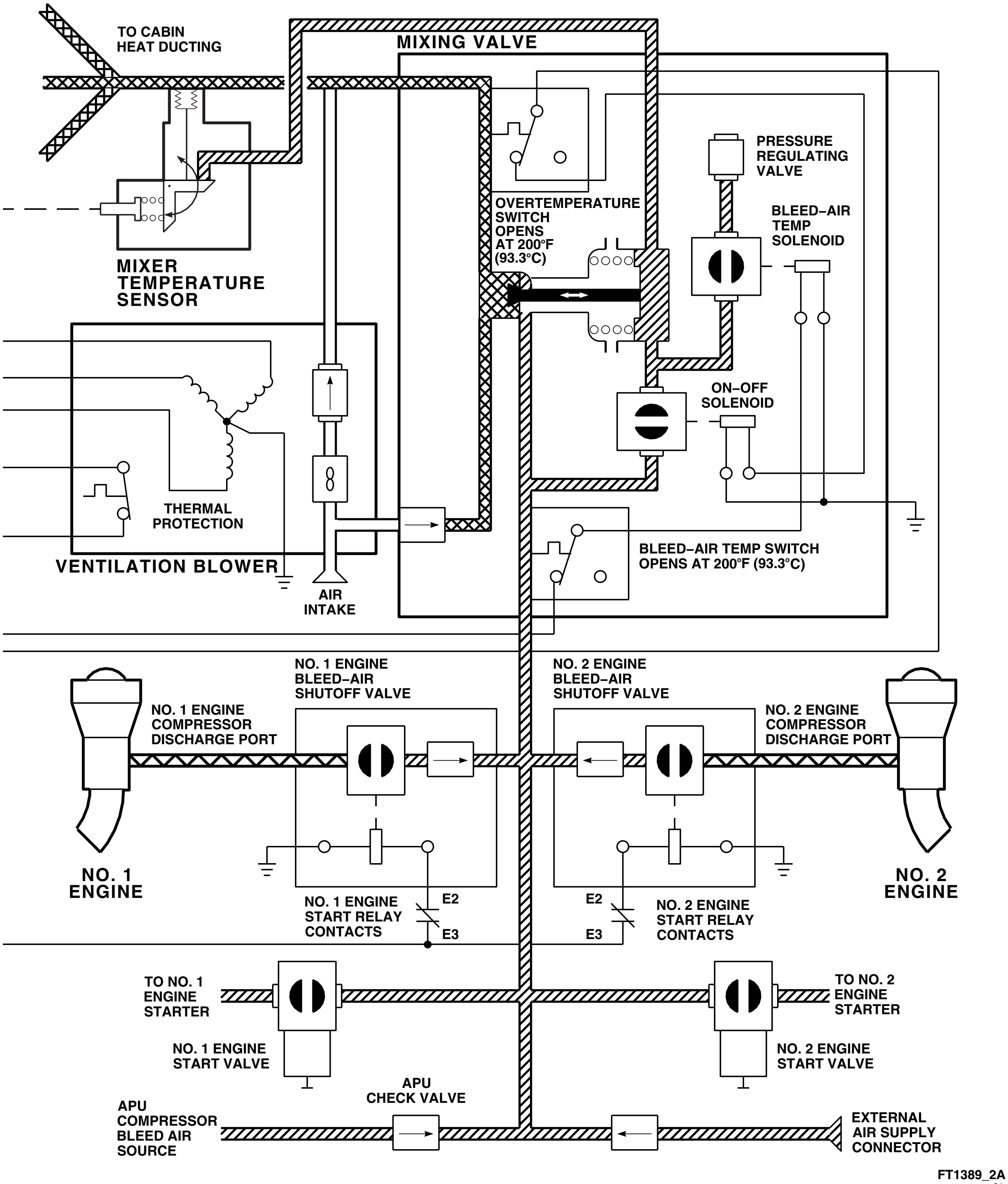


FIGURE 13-1-1-2. HEATING AND VENTILATION SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

13-1-2 ENVIRONMENTAL CONTROL SYSTEM DESCRIPTION AND DATA.

13-1-2.1 Description.

The environmental control system (ECS) consists of a heater/demister and a vapor cycle air conditioner that provides heating, cooling, ambient air circulation, and humidity control in the helicopter's cockpit and cabin in addition to the standard heating and ventilation system. Table 13-1-2-1 lists the ECS components and their locations. The system is controlled from the ECS control panel on the upper console. The ECS functionally interfaces with the caution/advisory warning system and the ac electrical system through the electrical control unit (ECU). Panel lighting for the air conditioning controls is provided by the upper console lights. See Figure 13-1-2-1 and Figure 13-1-2-2.

Table 13-1-2-1. Environmental Control System Component Locations.

Component	Location
Compressor/motor	Evaporator pallet
Condenser	Condenser pallet
Condenser fan	Condenser pallet
Filter/drier	Electrical pallet
Expansion valve	Evaporator pallet
Evaporator	Evaporator pallet
Evaporator fan	Evaporator pallet
High temperature switch	Evaporator pallet
Low temperature switch	Evaporator pallet
Hot gas bypass regulator	Evaporator pallet
Heater/demister assembly	Evaporator pallet
Temperature sensor	Evaporator pallet
Return air temperature sensor	Return air plenum
ECU	Electrical pallet
Air conditioning control panel	Upper console
Sight glass	Electrical pallet
Service valves	Electrical pallet
High pressure switch	Electrical pallet

Table 13-1-2-1. Environmental Control System Component Locations. (Cont)

Component	Location
Low pressure switch	Electrical pallet
Ambient air valve	Evaporator pallet
Thermistor sensor	Evaporator low pressure line

13-1-2.1.1 Evaporator Pallet.

The evaporator pallet is horizontally-mounted on the left side of the helicopter, rear of the cabin bulkhead. Its major components are the compressor/motor, evaporator, inlet transition duct, evaporator blower, heater/demister housing, discharge transition duct, expansion valve, filter/dryer, and hot gas bypass valve. The evaporator and heater/demister housing are insulated with neoprene foam insulation.

The expansion valve controls the rate of refrigerant evaporation allowing enough refrigerant to flow into the evaporator to keep the evaporator operating efficiently, depending on its heat load. Refrigerant enters the expansion valve in a liquid state and passes through a small orifice. It emerges as a vapor at a lower pressure. As a vapor and under low pressure, the refrigerant then enters the evaporator and begins to evaporate. The evaporator is a crossflow plate-fin type aluminum heat exchanger that absorbs heat by boiling off liquid refrigerant flowing through it. The blower draws hot cabin air through the evaporator. The heat in this air is absorbed by the evaporating refrigerant. The resultant cool air is then recirculated through the aircraft air distribution system back into the cabin. At this point, the refrigerant is a low pressure gas and is drawn back to the compressor through the suction line.

13-1-2.1.2 Compressor/Motor.

The compressor/motor is a single, hermetically sealed unit. An eight horsepower motor is mounted vertically over a rotary pump to compress and circulate the R-500 refrigerant throughout the system. The compressor's function is to draw the low pressure refrigerant gas (suction) from the evaporator and compress it to a high pressure gas for routing to the condenser (discharge). An electrical connector for the electrical power supply, discharge port, suction port, and oil level sight glass are mounted in the compressor/motor housing.

13-1-2.1.3 Inlet Transition Duct.

The fiberglass evaporator inlet transition duct connects the evaporator with the evaporator blower. The duct includes guide vanes to ensure an even air flow through the evaporator.

13-1-2.1.4 Evaporator Fan.

The evaporator fan is a six-inch vane, axial-type fan that draws return air from the cabin interior and circulates it through the evaporator.

13-1-2.1.5 Heater/Demister Housing.

The heater/demister housing is mounted between the evaporator and the evaporator outlet transition duct. It is a steel housing insulated with neoprene foam. It contains three heating elements, a heater high-temperature switch, an air conditioner low-temperature switch, a heater temperature limiting switch, and an aluminum mesh demister pad for water elimination.

13-1-2.1.6 Outlet Transition Duct.

Conditioned air is ducted to the aircraft air distribution system through the evaporator outlet transition duct. This duct is constructed of fiberglass and insulated with neoprene foam.

13-1-2.1.7 Filter/Dryer.

The filter/dryer is mounted in the high pressure (liquid) refrigerant line just before the expansion valve on the electrical pallet or evaporator pallet. Through its molded porous core, both contaminants and moisture are filtered from the refrigerant. Moisture in the air conditioning system interferes with the proper operation of the expansion valve and reacts with the refrigerant to form corrosive hydrofluoric acid.

13-1-2.1.8 Hot Gas Bypass Valve.

The hot gas bypass valve regulates the evaporator pressure, hence temperature, to provide evaporator outlet air temperature adjustment. The cooling system is designed to produce not lower than 5.6°C (42°F) conditioned air. The valve will discharge hot refrigerant gas directly into the evaporator to increase its pressure and, therefore, temperature. The valve receives refrigerant from the compressor discharge line to discharge it directly into the evaporator. The valve receives sensing input from a duct temperature sensor mounted in the helicopter return air plenum, and control voltage from the TEMP CONT rheostat through the temperature controller in the ECU.

13-1-2.1.9 Condenser Pallet.

The condenser pallet is horizontally-mounted on the right side of the helicopter aft of the cabin bulkhead. Its major components are the heat exchanger, condenser fan, transition duct, pressure relief valve, thermal protection switch, and burst disc. The condenser unit extracts the heat absorbed by the refrigerant in the evaporator, exhausts it overboard, and changes the refrigerant to a liquid state for routing back to the expansion valve. The refrigerant enters the condenser from the compressor as a compressed gas under high pressure. Ambient air is blown over the condenser by the fan. Because the refrigerant is at a higher temperature than the ambient air, the heat is transferred from the refrigerant to the air. The condenser is constructed similarly to the evaporator. The condenser is a crossflow, plate-fin type heat exchanger that uses air from the fan to extract heat from the refrigerant gas, allowing it to condense into a manifold that connects to a sub-cooler, across the condenser lower face, then out the discharge port.

13-1-2.1.10 Condenser Transition Duct.

A fiberglass condenser transition duct separates the condenser and blower. It is bolted to flanges on both components and supported by aluminum brackets.

13-1-2.1.11 Pressure Relief Valve.

A pressure relief valve protects the air conditioning system from over pressure. If system pressure reaches 475 psig, the relief valve will open, discharge refrigerant directly into the condenser transition duct and exhausting it overboard. As pressure decreases below 475 psig, the pressure relief valve will automatically reseal and air conditioner operation will resume. The valve is mounted to the high pressure (liquid) discharge refrigerant line.

13-1-2.1.12 Burst Disc.

A burst disc is included in the system to provide redundancy in high pressure protection. Mounted "in-line" with the pressure relief valve, the disc will burst at 500 psig. This would occur under a high pressure condition when both the high pressure switch and pressure relief valve failed. The burst disc must be replaced after rupture before resuming air conditioner operation.

13-1-2.1.13 Electrical Pallet.

The electrical pallet is vertically mounted aft of the condenser and evaporator pallet assemblies in the helicopter aft transition avionics compartment. The major components are: ECU, air conditioner circuit breaker panel, air conditioner fault indicator panel, high pressure switch, low pressure switch, filter/dryer, refrigerant liquid indicator (sight glass), and high/low pressure service valves.

13-1-2.1.14 Electrical Control Unit (ECU).

The ECU is a sealed unit containing practically all of the electrical components necessary for operation. Contents include the temperature controller, relays, circuit breakers, and compressor/motor capacitor.

13-1-2.1.15 Circuit Breaker Panel.

The circuit breaker panel is mounted on the ECU cover. The panel includes four ac circuit breakers and one dc circuit breaker. A description is as follows:

- COMPRESSOR MOTOR, 35 amps ac
- CONDENSER BLOWER, 25 amps ac
- EVAPORATOR BLOWER, 10 amps ac
- HEATER ELEMENTS, 10 amps ac
- DC POWER CONTROLS, 7.5 amps dc

13-1-2.1.16 Fault Indicator Panel.

A fault indicator panel, containing four one-amp circuit breakers, is mounted on the ECU cover to indicate extremes in either pressure or temperature during ECS operation. The circuit breakers are labeled HIGH PRESSURE, LOW PRESSURE, HIGH TEMPERATURE, and LOW TEMPERATURE. High pressure, low pressure, high temperature, or low temperature conditions will cause the affected circuit breaker to pop. The affected breaker should be reset before flight.

13-1-2.1.17 Service Valves.

The air conditioner contains two service ports for system maintenance; one in the high pressure line and one in the low pressure line. The service valve in the high pressure line (from compressor to condenser) allows access to the high pressure side of the system for attaching a service hose and pressure gauge. The service valve in the low pressure line (from evaporator to compressor) allows access to the low pressure side of the system. Access to both pressure lines is required for monitoring the system during maintenance operations and for servicing the system.

13-1-2.1.18 High and Low Pressure Switches.

The high and low pressure switches protect the air conditioning system from abnormally high and low pressures. Both switches are located in the servicing manifold and exposed to either the high or low side pressures. The high pressure switch will disengage the compressor/motor if system high side pressure reaches between 345 and 355 psi. The compressor/motor will reengage as pressure decreases to between 270 and 280 psi. The low pressure switch will disengage the compressor/motor at between 7 and 23 psi. At between 17 and 23 psi the compressor/motor will reengage.

13-1-2.2 Power Distribution.

Ac power is supplied by the No. 1 generator contactor K1 in the No. 1 junction box and routed through the No. 2 junction box to the ECU PWR circuit breaker in the remote control circuit breaker box. The ECU PWR circuit breaker is controlled remotely by the ECS PWR circuit breaker on the pilot’s circuit breaker panel. From the ECU PWR circuit breaker, ac power is routed to connector J1 on the ECS electrical control unit. In the ECU, the 115 vac three-phase power arms the normally-open contacts of four relays (K1, K2, K3, and K4), each protected by its own circuit breakers for ac operation of the ECS compressor, condenser, evaporator, and heater/demister. These circuit breakers are identified on the ECU air conditioner circuit breaker panel as COMPRESSOR MOTOR, CONDENSER BLOWER, EVAPORATOR BLOWER, and HEATER ELEMENTS.

Table 13-1-2-2. Air Conditioning System Power Source Priority.

Power Source	Air Conditioning System Operation
APU generator (helicopter on ground)	Air conditioning interrupted if: (1) backup pump is on or (2) windshield anti-ice is on. Windshield anti-ice interrupted when backup pump is on.
APU generator (helicopter in air)	Air conditioning interrupted while helicopter in air. Windshield anti-ice interrupted when backup pump is on.
Dual main generator (No. 1 and No. 2)	Air conditioning, backup pump, and windshield anti-ice can operate simultaneously.
Single generator (helicopter in air or on ground) or external ac power (weight on or off wheels)	Air conditioning interrupted when weight off wheels.

Dc power controls the operation of the 115 vac components through dc interlock circuitry. The 28 vdc is supplied by the No. 2 dc primary bus and routed through the ECS CONTR circuit breaker on the pilot's circuit breaker panel to the ECU on the electrical control pallet and to the ECS AIR COND COOL-OFF-FAN switch on the upper console. Air conditioning power source priorities, listed in Table 13-1-2-2 are established by circuitry in the ac electrical system and the No. 3 relay panel.

13-1-2.3 System Operation.

Control of the air conditioning system is accomplished by the ECS TEMP control, AIR COND, and HTR controls on the upper console. The temperature rheostat (TEMP COOL-WARM control) has two arrows. One arrow indicates an increase to COOL (counterclockwise), the other an increase to WARM (clockwise). The AIR COND switch is marked FAN-OFF-COOL. The TEMP control rheostat R6 is used with the AIR COND switch to set the desired cabin temperature.

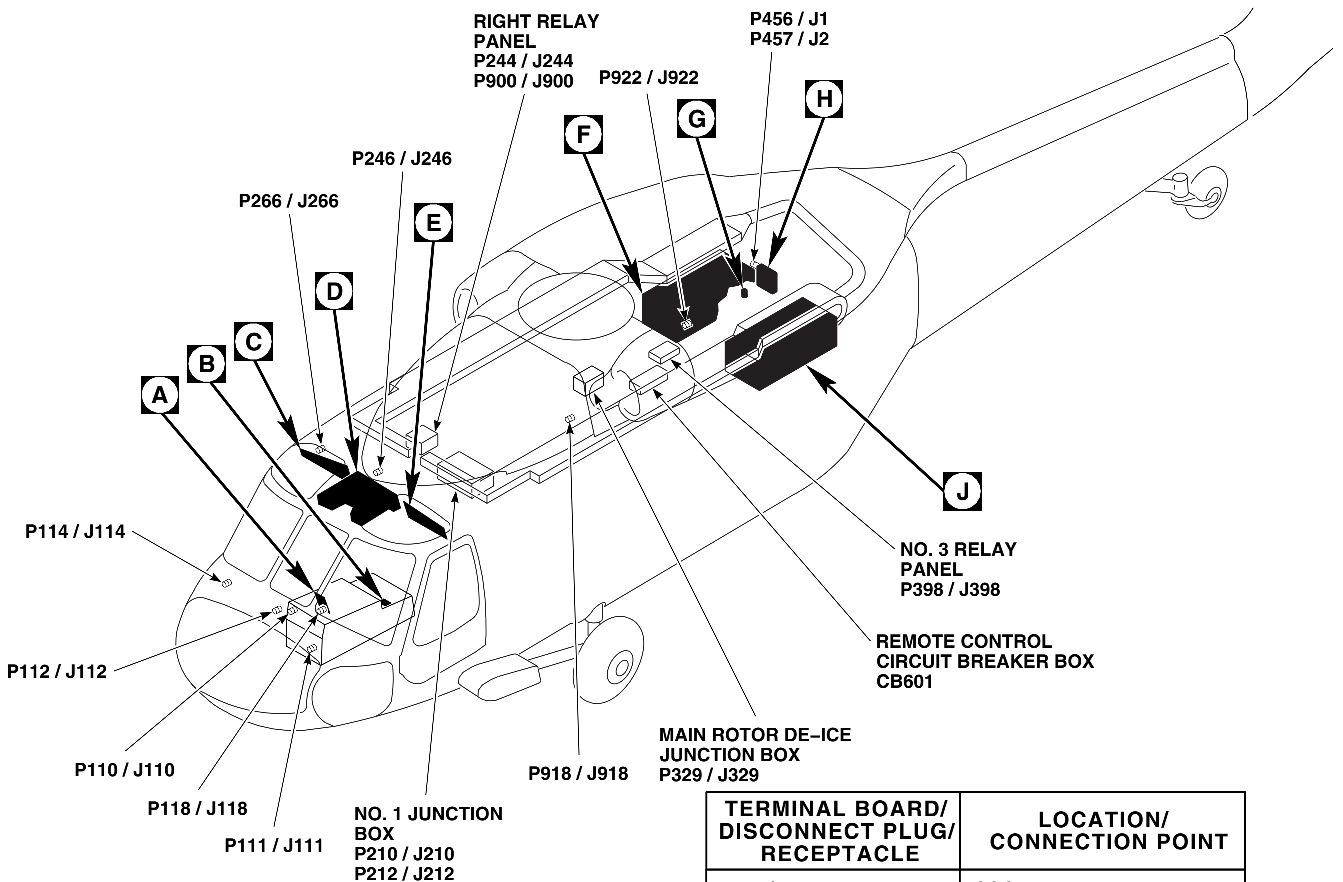
When the switch is placed to either COOL or FAN, the evaporator immediately starts, providing air flow to the cockpit and cabin. When the manually-operated ambient air valve is open, fresh air is drawn from outside the helicopter into the plenum chamber, mixed with inside cool or vent air, and circulated through the helicopter. With the ambient air valve closed, inside air will be recirculated through the helicopter by the evaporator fan. When air conditioning is desired, the switch is placed to COOL, starting a sequence of events leading to full air conditioning operation. Major electrical components are started at spaced intervals to prevent surges in 115 vac electrical power. The evaporator fan operates first, followed by the condenser fan, after a five-second delay. Finally, after an additional 10-second delay, the compressor motor operates. Temperature control is accomplished by mixing the cool refrigerant in the evaporator with warm refrigerant in the compressor. This is in response to a signal from the temperature sensor in the return air rear plenum. This signal is processed by the temperature controller and is adjusted by the TEMP CONT rheostat R6 to open the hot gas bypass valve solenoid when the desired cabin and cockpit comfort level is reached. Safety of the air conditioning system is maintained by high and low pressure and temperature switches, and by a pressure relief valve and a burst disc. The switches also latch individual fault indicators on the ECU (identified as air conditioner fault indicator panel) to provide visual indication of an air conditioning system malfunction.

When the helicopter is on the ground, the 28 vdc interlock circuitry is as follows: the AIR COND control is armed with 28 vdc through contacts of relay K82 in the No. 3 relay panel, and the ECS can then be used either with or without the APU running (through relay K95). Placing the AIR COND switch to COOL energizes evaporator fan relay K3 to switch on ac power to the evaporator fan. When the hydraulic backup pump and windshield anti-ice are off, placing the AIR COND switch to COOL also energizes the expansion valve and thermistor through relays K96 and K80A. After a five-second delay, this energizes relay K5 to in turn energize condenser fan relay K2, which connects 115 vac to start the condenser fan. After an additional 10-second delay, relay K6 energizes. Compressor motor relay K4 then energizes through the high and low pressure switches, the low temperature switch, and contacts of relay K6. Relay K4 connects ac power to start the compressor, and dc power to light the AIR COND ON capsule on the caution/advisory panel.

The high and low pressure switches and the low temperature switch are connected in series between 28 vdc from DC POWER circuit breaker on the ECU and contacts B1-B2 of relay K6. This causes the compressor motor to stop running when the pressure in the high pressure line exceeds 300 psig, the pressure in the low pressure line drops below 50 psig, or the temperature in the evaporator duct drops below 1.7°C (35°F). Either of these conditions can be monitored on the air conditioner fault indicator panel on the ECU. The circuit breakers are marked: HIGH PRESSURE, LOW PRESSURE, and LOW TEMPERATURE. When one of the switches is activated, 28 vdc is shorted to ground through the corresponding circuit breaker, which pops the breaker.

Placing the AIR COND switch to FAN arms the HTR switch on the ECS control panel when either the backup pump or windshield anti-ice is off, or when No. 1 and No. 2 generators are on. Placing the HTR switch to ON energizes relay K7 which provides a ground to energize relay K1 through contacts of relay K3 and the NC contacts of the high temperature switch in the evaporator duct. When duct temperature exceeds 86.7°C (188°F), the high temperature switch activates to short out and pop the HIGH TEMPERATURE circuit breaker on the fault indicator panel. Relay K1 switches ac power to energize the heater/demister coils and relay K7 switches dc power to light the CABIN HEAT ON capsule on the caution/advisory panel. The heater/demister is also protected by a temperature limiting switch between the high temperature switch and relay K1, which is set to disconnect power to the heater coils when evaporator duct temperature exceeds 71.1°C (160°F).

When the helicopter is airborne, the AIR COND switch can be armed only when the APU is off (through relay K95). Operation of the ECS is the same with the helicopter airborne as it is with the helicopter on the ground except that the backup pump or the windshield anti-ice will not interrupt air conditioning or heater operation because No. 1 and No. 2 generator relay K81A connects the AIR COND and HTR switches to the system components.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P1 / J1	ELECTRICAL CONTROL UNIT
P2 / J2	ELECTRICAL CONTROL UNIT
P3 / J3	ELECTRICAL CONTROL UNIT
P4 / J4	ELECTRICAL CONTROL UNIT
P5 / J5	ELECTRICAL CONTROL UNIT
P6 / J6	ELECTRICAL CONTROL UNIT
P7 / J7	ELECTRICAL CONTROL UNIT
P8 / J8	COMPRESSOR / MOTOR
P9 / J9	EVAPORATOR FAN
P10 / J10	CONDENSER FAN
P11 / J11	HIGH PRESSURE SWITCH
P12 / J12	LOW PRESSURE SWITCH
P110 / J110	COCKPIT BL 7.5 RH, STA 197

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	INSTRUMENT PANEL DISCONNECT
P114 / J114	COCKPIT BL 33, STA 197
P118 / J118	CAUTION / ADVISORY PANEL
P210 / J210	NO. 1 JUNCTION BOX
P212 / J212	NO. 1 JUNCTION BOX
P244 / J244	RIGHT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P329 / J329	MAIN ROTOR DE–ICE JUNCTION BOX
P398 / J398	NO. 3 RELAY PANEL
P456 / J1	ELECTRICAL CONTROL UNIT
P457 / J2	ELECTRICAL CONTROL UNIT
P900 / J900	RIGHT RELAY PANEL
P918 /J918	MAIN CROSSOVER
P922 / J922	CABIN CEILING BL 10.7 LH, STA 380

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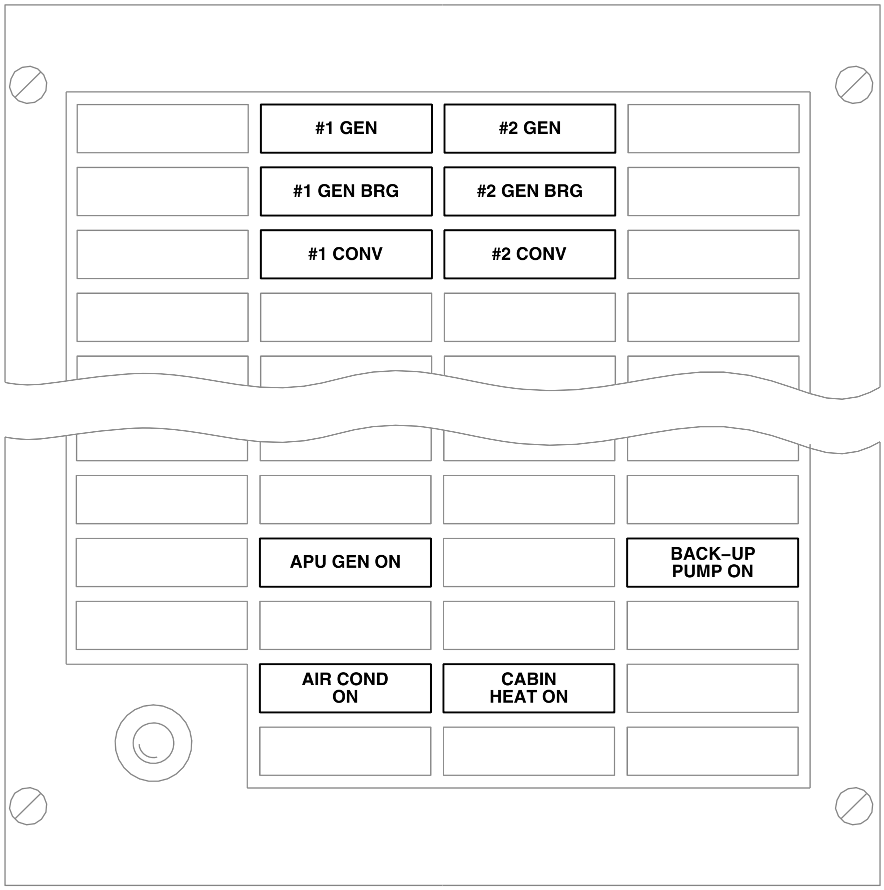
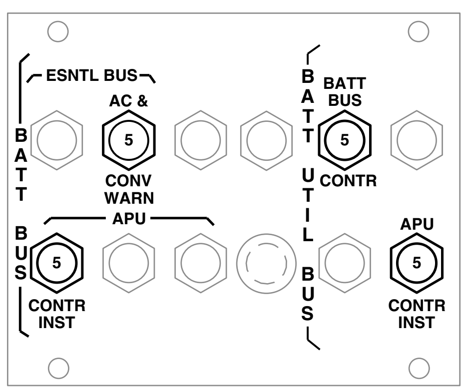
FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 1 OF 6)



CAUTION/ADVISORY PANEL

A

B



LOWER CONSOLE
CIRCUIT BREAKER PANEL

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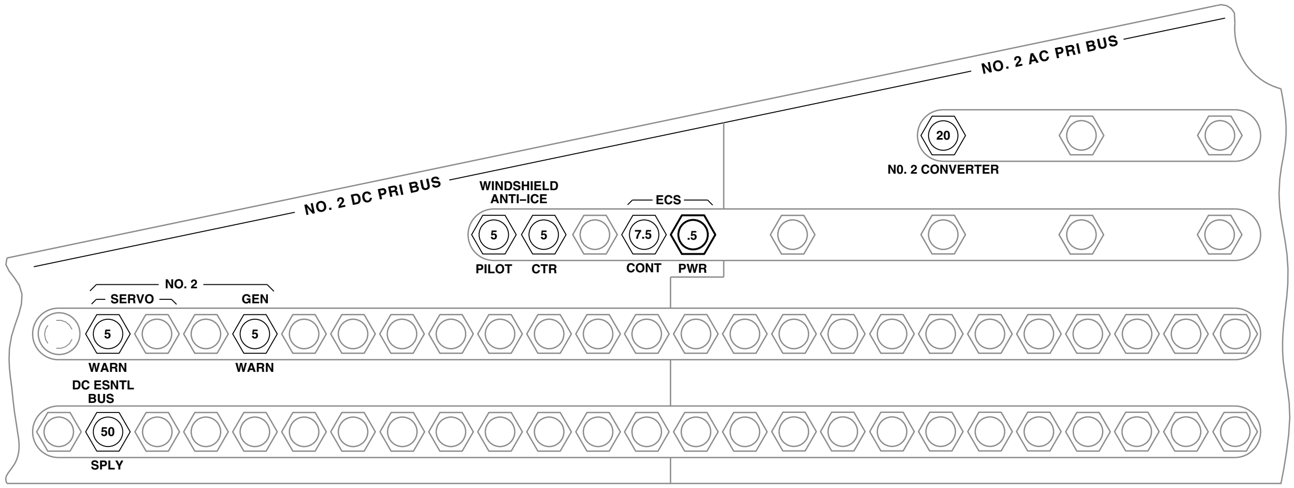
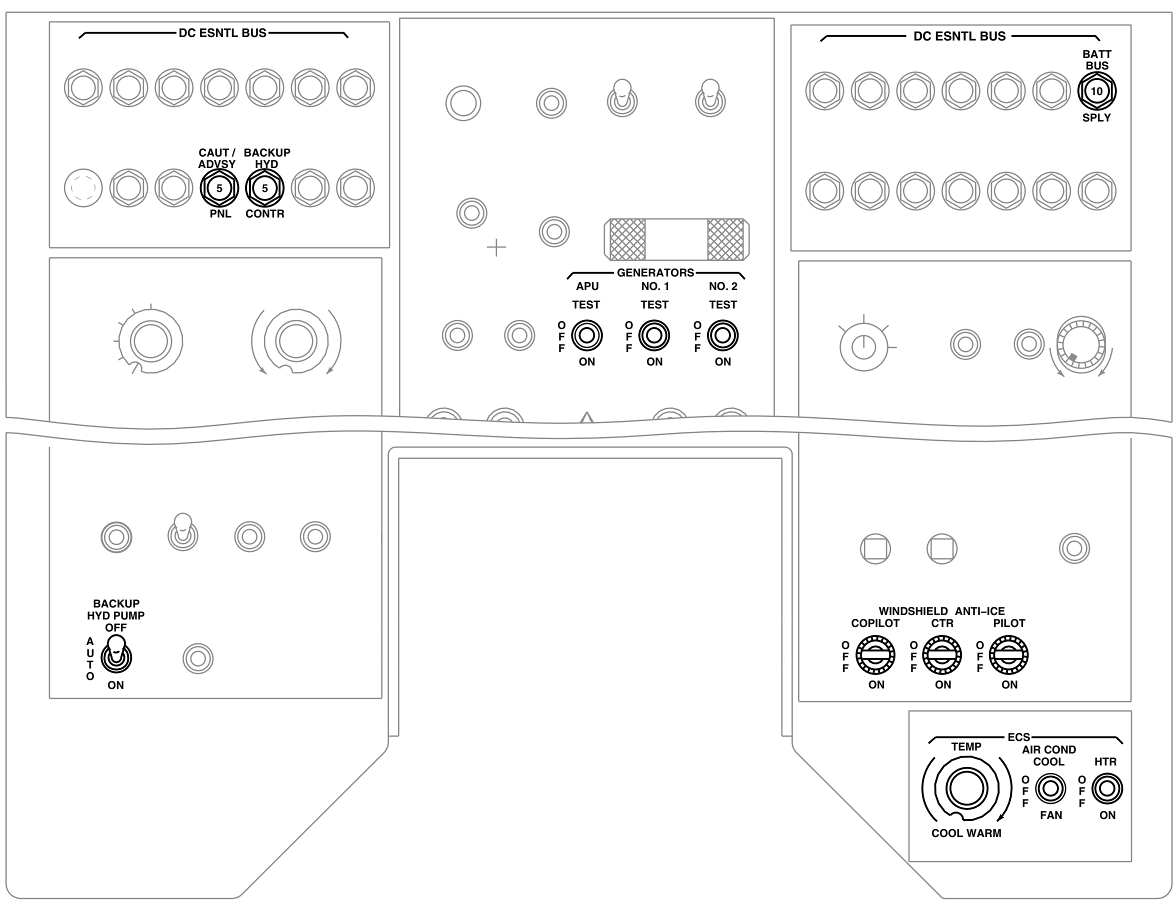
FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 2 OF 6)



PILOT'S CIRCUIT BREAKER PANEL

C

D



UPPER CONSOLE

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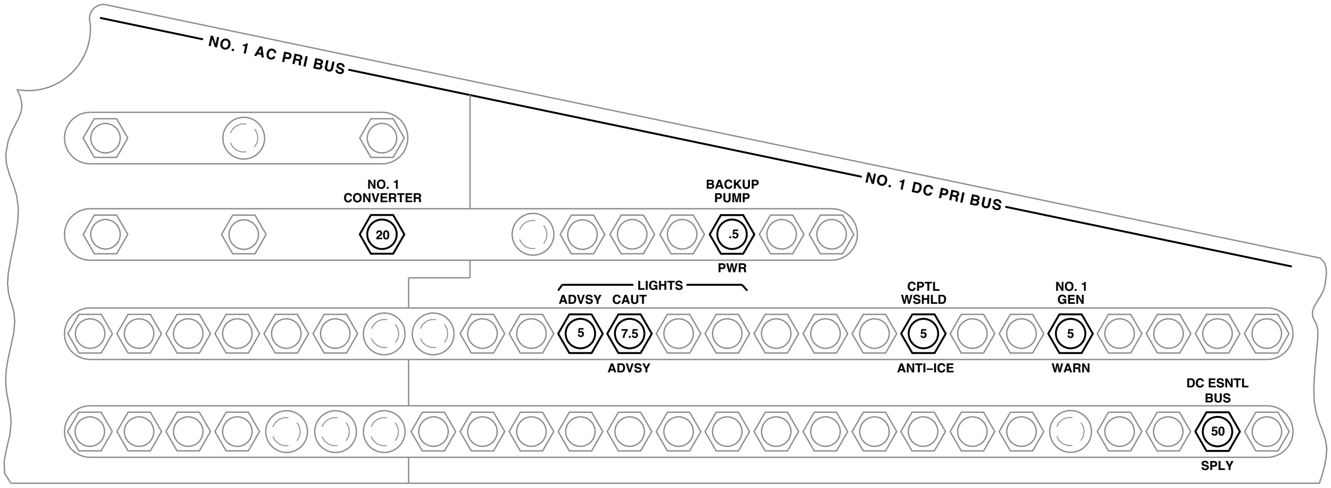
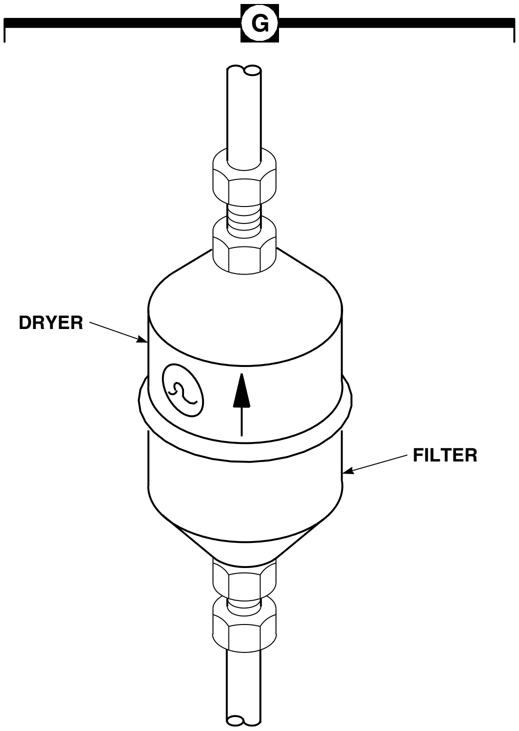
FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 3 OF 6)

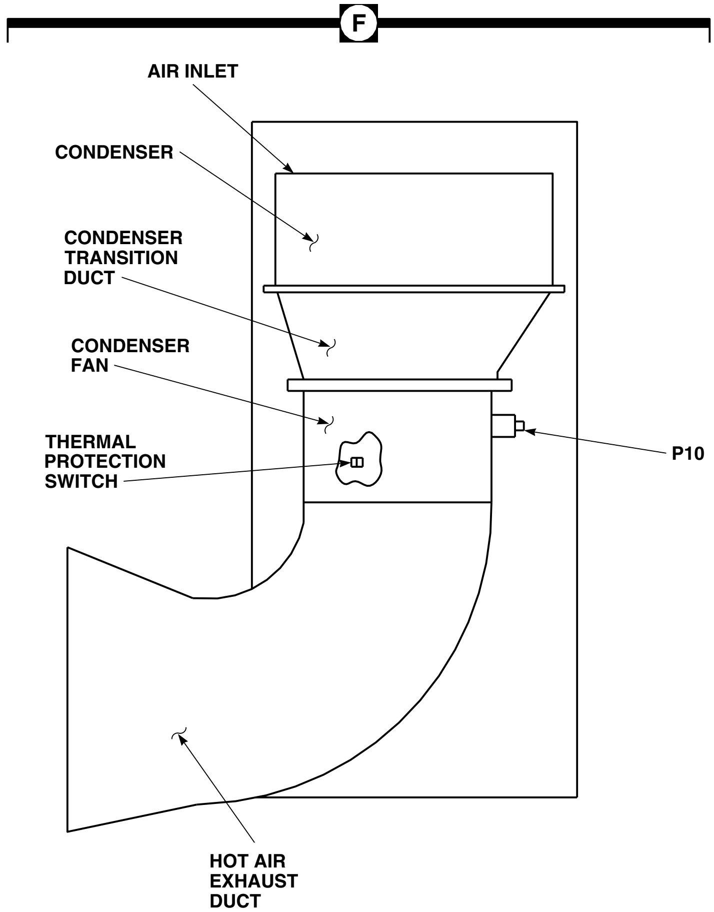


COPILOT'S CIRCUIT BREAKER PANEL

E



FILTER/DRYER



CONDENSER PALLET

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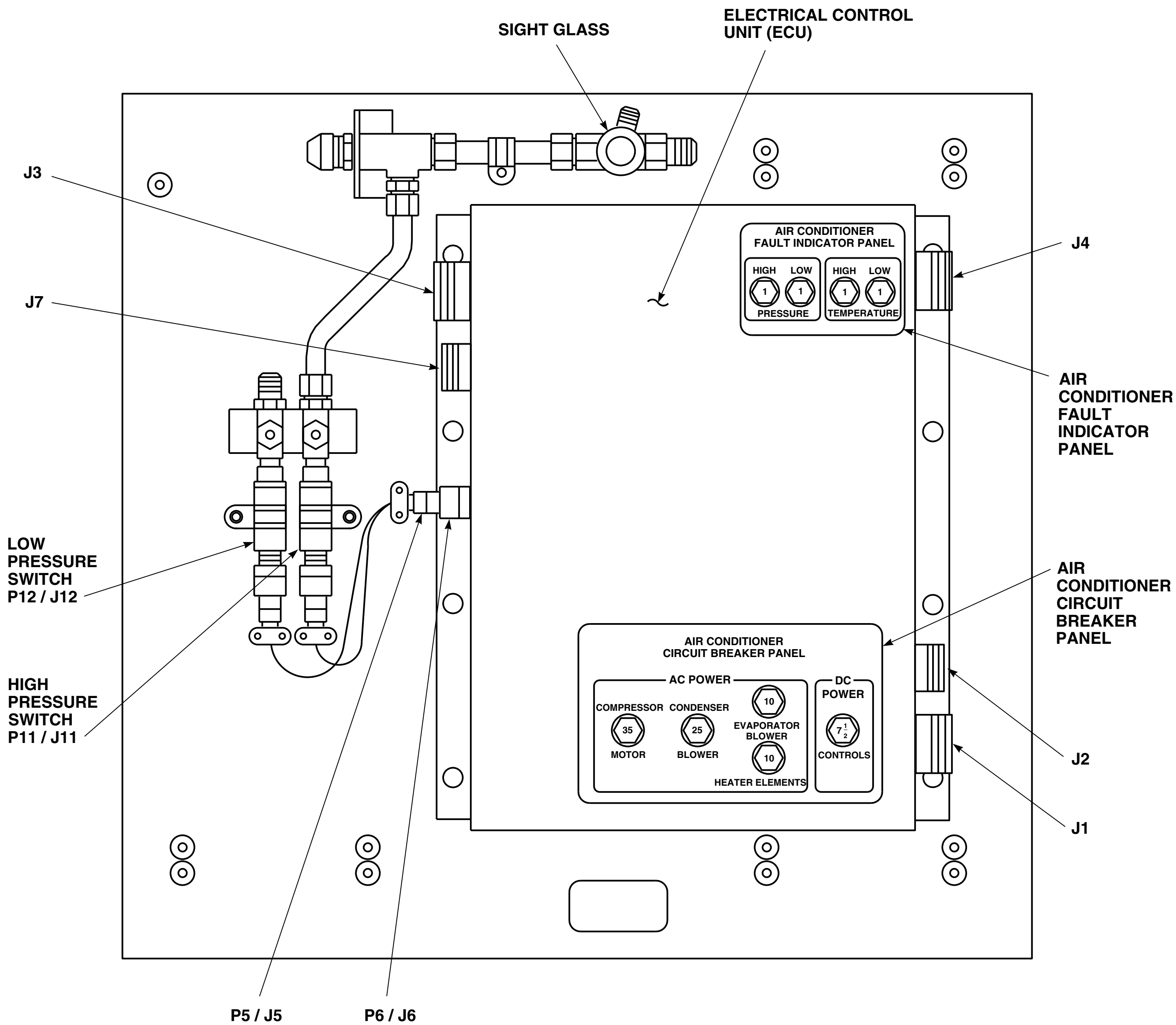
FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 4 OF 6)

H



ELECTRICAL PALLET

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FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 5 OF 6)

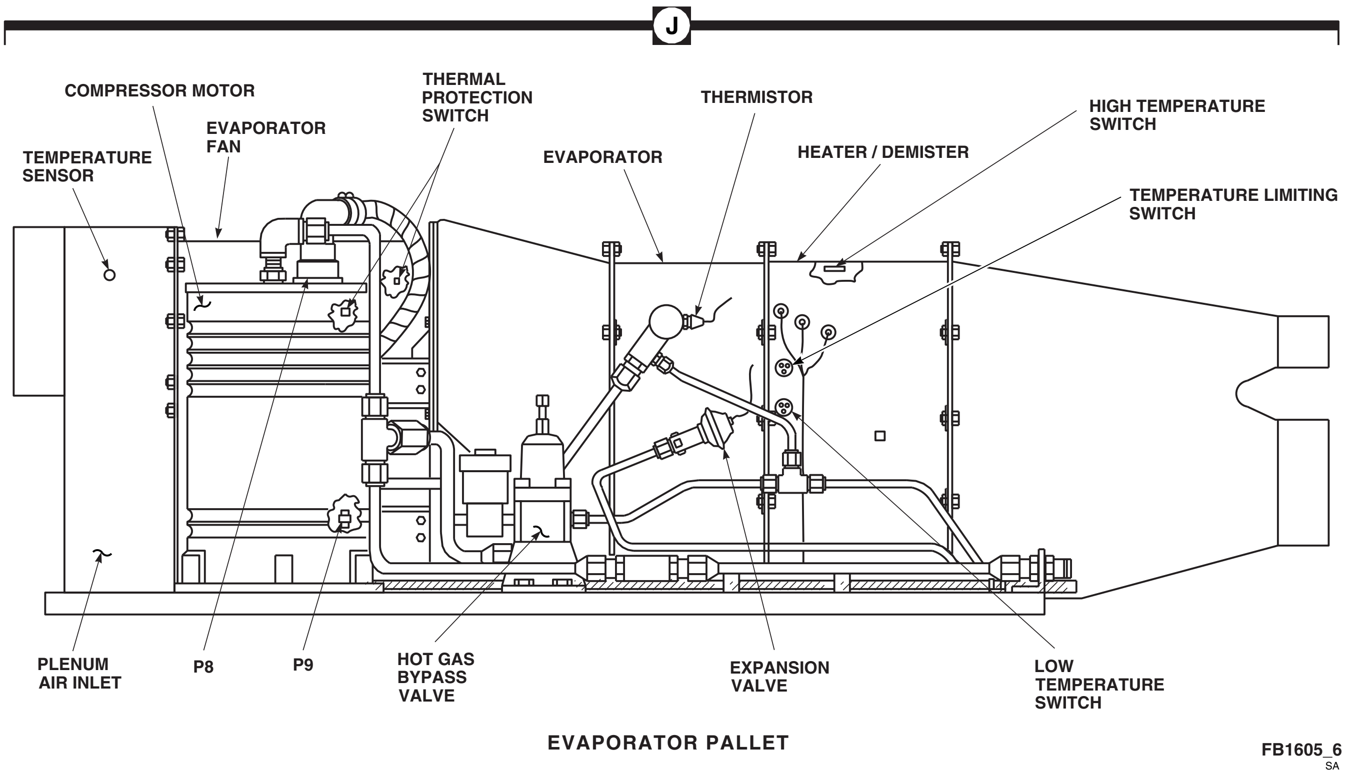

FIGURE 13-1-2-1. ENVIRONMENTAL CONTROL SYSTEM LOCATION DIAGRAM (SHEET 6 OF 6)

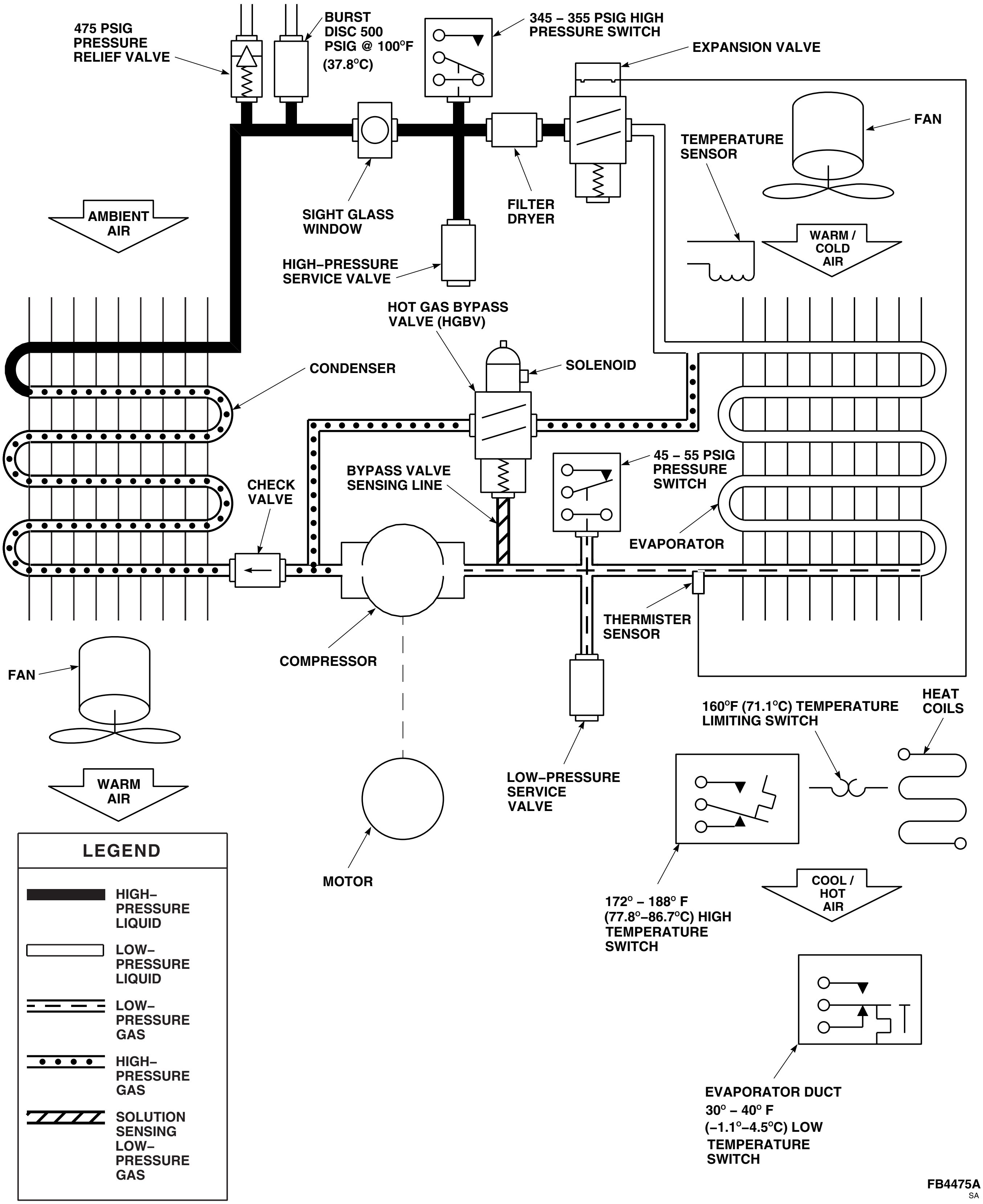

FIGURE 13-1-2-2. ENVIRONMENTAL CONTROL SYSTEM REFRIGERANT FLOW DIAGRAM

SECTION 13-2

TROUBLESHOOTING

SECTION OVERVIEW

13-2-1 HEATING AND VENTILATION SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- Appropriate Flight Manual

Equipment Conditions

- APU Operational
- No. 1 and No. 2 Engines Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

13-2-1.1 Operational/Troubleshooting Procedure.

a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

b. Start APU (Appropriate Flight Manual).

NOTE

Open air valves in cockpit and front cabin area before turning on blower.

c. Place upper console VENT BLOWER switch to ON.

RESULT:

(1) Ventilation blower shall operate.

(2) Air shall come out from each louver.

- If result (1) is not as specified, go to TABLE NO. 13-2-1-1.
- If result (2) is not as specified, repair/replace ducting (PARA 13-4-11).

d. Place upper console VENT BLOWER switch to OFF.

RESULT: Air shall stop flowing out of louvers.

- If result is not as specified, replace right relay panel (PARA 9-4-2).

NOTE

Hot air may cycle on and off during operational/troubleshooting procedure.

e. Place upper console HEATER switch to ON.

RESULT: Mixing valve shall open and warm air shall come out of louvers.

- If result is not as specified, go to TABLE NO. 13-2-1-2.

f. Slowly turn upper console HEATER knob from OFF to HI.

RESULT: Temperature of air coming out of louvers shall increase.

- If result is not as specified, go to TABLE NO. 13-2-1-3.

g. Turn upper console HEATER knob to OFF.

RESULT: Temperature of air coming out of louvers shall decrease.

- If result is not as specified, go to TABLE NO. 13-2-1-3.

h. Place upper console HEATER switch to OFF.

RESULT: Air shall not be flowing out of louvers.

- If result is not as specified, replace mixing valve (PARA 13-4-1).

i. With qualified personnel at controls, start No. 1 and No. 2 engines (Appropriate Flight Manual).

j. Shut down APU (Appropriate Flight Manual).

k. Place upper console AIR SOURCE HEAT/START switch to ENG.

NOTE

When HEATER switch is placed at ON (engine bleed air shutoff valves open), a noticeable increase in TGT (T4.5) for both engines should occur.

l. Place upper console HEATER switch to ON.

RESULT: No. 1 engine and No. 2 engine bleed air shutoff valves shall open. Mixer valve shall open and warm air shall be coming out of louvers.

- If air is coming out but No. 1 engine TGT does not increase, go to TABLE NO. 13-2-1-4.
- If air is coming out but No. 2 engine TGT does not increase, go to TABLE NO. 13-2-1-5.
- If No. 1 and No. 2 engine TGT does not increase, go to TABLE NO. 13-2-1-6.
- ESSS If fuel fumes are noticed, clean pneumatic check valve (PARA 7-4-4).

m. Turn upper console HEATER knob to HI and note temperature of air coming out of louvers.

n. Turn upper console ENG ANTI-ICE NO. 1 switch to ON.

RESULT: Temperature and flow of air coming out of louvers shall decrease.

- If result is not as specified, go to TABLE NO. 13-2-1-7.

o. Turn upper console HEATER knob to OFF.

p. Place upper console HEATER switch to OFF.

RESULT: Air shall stop coming out of louvers and a noticeable decrease in TGT of both engines shall occur.

- If result is not as specified, replace engine bleed air valve (PARA 7-4-3).

q. Shut down both engines (Appropriate Flight Manual).

TABLE NO. 13-2-1-1. Ventilation Blower Does Not Operate.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure VENT BLOWER switch is placed ON. Go to 2.

2. Check continuity between P437-5 and P437-6.

Step 1. If continuity is present, go to **3.**

TABLE NO. 13-2-1-1. Ventilation Blower Does Not Operate. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If continuity is not present, replace ventilation blower (PARA 13-4-12). Go to **14.**

3. Check for 28 VDC between J437-6 and ground.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **11.**

4. Check continuity between J437-5 and P241-j.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **14.**

5. Check continuity between J437-4 and ground.

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **14.**

6. Place an insulated jumper wire between J437-5 and J437-6. Go to 7.

7. Check for 115 VAC between:
J437-1 and J437-4
J437-2 and J437-4
J437-3 and J437-4

Step 1. If voltage is as specified, remove jumper, then replace ventilation blower (PARA 13-4-12). Go to **14.**
Step 2. If voltage is not as specified, remove jumper wire, then go to **8.**

8. Check continuity between:
J437-1 and P241-g
J437-2 and P241-h
J437-3 and P241-i

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **14.**

9. Check for 115 VAC between HEAT & VENT circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel):

TABLE NO. 13-2-1-1. Ventilation Blower Does Not Operate. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **14.**

10. Check continuity between HEAT & VENT circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel):
Terminal A1 and P241-E
Terminal B1 and P241-n
Terminal C1 and P241-N

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **14.**

11. Check for 28 VDC between VENT BLOWER switch terminal 3 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 3 and J437-6 (Appendix F). Go to **14.**
Step 2. If voltage is not as specified, go to **12.**

12. Check for 28 VDC between VENT BLOWER switch terminal 2 and ground.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **14.**
Step 2. If voltage is not as specified, go to **13.**

13. Check for 28 VDC between terminal 1 (HEAT VENT circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and VENT BLOWER switch terminal 2 (Appendix F). Go to **14.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **14.**

14. Procedure completed.

TABLE NO. 13-2-1-2. No Heated Air Coming Out Of Any Louvers.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check bleed air line from APU check valve to mixing valve for proper connections.**

   Step 1. If connections are good, go to **2.**
   Step 2. If connections are not good, repair/replace lines and/or rubber connections, as required. Go to **19.**

2. **Check for 28 VDC between P436-1 and ground.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, go to **4.**

3. **Check continuity between:**
   P436-2 and ground
   P436-3 and ground

   Step 1. If continuity is present, replace mixing valve (PARA 13-4-1). Go to **19.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

4. **Install an insulated jumper wire between P244-K and P241-G. Go to 5.**

5. **Check for 28 VDC between P436-1 and ground.**

   Step 1. If voltage is as specified, go to **6.**
   Step 2. If voltage is not as specified, go to **8.**

6. **Remove jumper wire. Go to 7.**

7. **Replace right relay panel (PARA 9-4-2). Go to 19.**

8. **Remove jumper wire. Go to 9.**

9. **Check for 28 VDC between P244-K and ground.**

   Step 1. If voltage is as specified, repair/replace wiring between P241-F and P436-1 (Appendix F). Go to **19.**
   Step 2. If voltage is not as specified, go to **10.**

10. **Install an insulated jumper wire between P902-b and P902-d. Go to 11.**

11. **Check for 28 VDC between P244-K and ground.**

TABLE NO. 13-2-1-2. No Heated Air Coming Out Of Any Louvers. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, go to **14.**

12. Remove jumper wire from P902. Go to 13.

13. Replace left relay panel (PARA 9-4-2). Go to 19.

14. Remove jumper wire from P902. Go to 15.

15. Check for 28 VDC between P902-b and ground.

Step 1. If voltage is as specified, repair/replace wiring between P902-d and P244-K (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, go to **16.**

16. Check for 28 VDC between J231-c and ground.

Step 1. If voltage is as specified, repair/replace wiring between P231-c and P902-b (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, go to **17.**

17. Check for 28 VDC between terminal 1 (AIR SOURCE HEAT/START circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.

Step 1. If voltage is as specified, go to **18.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **19.**

18. Check wiring to HEATER switch for proper connections.

Step 1. If proper connections are present, replace switch (PARA 9-4-5). Go to **19.**
Step 2. If proper connections are not present, reconnect wiring to switch (Appendix F). Go to **19.**

19. Procedure completed.

TABLE NO. 13-2-1-3. Air Temperature Cannot Be Regulated.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Unlock fastener studs securing upper console panel and hinge down panel. Turn HEATER knob and check shaft for proper turning.**

   Step 1. If shaft turns properly, go to **2.**
   Step 2. If shaft does not turn properly, secure setscrews attaching shaft to adapter or shaft to HEATER knob. Adjust system as necessary (PARA 13-4-14). Go to **11.**

2. **Disconnect control shaft at temperature regulator. Go to 3.**

3. **Turn HEATER knob and visually check if inner core of control shaft turns.**

   Step 1. If inner core of control shaft turns, go to **4.**
   Step 2. If inner core of control shaft does not turn, go to **9.**

4. **Check temperature tube for signs of damage or loose connections.**

   Step 1. If damage or loose connections are present, secure temperature tube or replace (PARA 13-4-2). Go to **11.**
   Step 2. If damage or loose connections are not present, go to **5.**

5. **Operate APU (Appropriate Flight Manual). Go to 6.**

6. **Place HEATER switch to ON. Go to 7.**

7. **Turn HEATER knob to HI. Go to 8.**

8. **Check that temperature tube is warm.**

   Step 1. If tube is warm, shut down APU (Appropriate Flight Manual), then replace mixture temperature sensor's pressure bellows (PARA 13-4-5). Go to **11.**
   Step 2. If trouble remains, shut down APU (Appropriate Flight Manual), then replace mixture temperature sensor (PARA 13-4-5). Go to **11.**
   Step 3. If tube is not warm, shut down APU (Appropriate Flight Manual), then replace mixing valve (PARA 13-4-1). Go to **11.**

9. **Disconnect control shaft at adapter. Go to 10.**

10. **Turn HEATER knob and visually check if adapter output shaft turns.**

    Step 1. If shaft turns, replace control shaft (PARA 13-4-7). Go to **11.**

TABLE NO. 13-2-1-3. Air Temperature Cannot Be Regulated. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If shaft does not turn, replace adapter (PARA 13-4-8). Go to **11.**

11. Procedure completed.

TABLE NO. 13-2-1-4. No. 1 Engine TGT Does Not Increase When HEATER Switch Is Placed ON.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure AIR SOURCE HEAT/START switch is placed to ENG. Go to 2.

2. Check for 28 VDC between P434-1 and P434-2.

Step 1. If voltage is as specified, replace No. 1 engine bleed air shutoff valve (PARA 7-4-3). Go to **10.**
Step 2. If voltage is not as specified, go to **3.**

3. Check for 28 VDC between P434-1 and ground.

Step 1. If voltage is as specified, repair/replace ground wire on P434-2 (Appendix F). Go to **10.**
Step 2. If voltage is not as specified, go to **4.**

4. Install an insulated jumper wire across P902-a and P902-c. Go to 5.

5. Check for 28 VDC between P434-1 and ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **8.**

6. Remove jumper wire from P902. Go to 7.

7. Replace left relay panel (PARA 9-4-2). Go to 10.

8. Remove jumper wire from P902. Go to 9.

9. Check for 28 VDC between P902-a and ground.

TABLE NO. 13-2-1-4. No. 1 Engine TGT Does Not Increase When HEATER Switch Is Placed ON. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between P902-c and P434-1 (Appendix F). Go to **10.**

Step 2. If voltage is not as specified, repair/replace wiring between P902-a and S18-3 (Appendix F). Go to **10.**

10. Procedure completed.

TABLE NO. 13-2-1-5. No. 2 Engine TGT Does Not Increase When HEATER Switch Is Placed ON.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure AIR SOURCE HEAT/START switch is placed to ENG. Go to 2.

2. Check for 28 VDC between P435-1 and P435-2.

Step 1. If voltage is as specified, replace No. 2 engine bleed air shutoff valve (PARA 7-4-3). Go to **10.**

Step 2. If voltage is not as specified, go to **3.**

3. Check for 28 VDC between P435-1 and ground.

Step 1. If voltage is as specified, repair/replace ground wire on P435-2 (Appendix F). Go to **10.**

Step 2. If voltage is not as specified, go to **4.**

4. Install an insulated jumper wire between P244-J and P241-F. Go to 5.

5. Check for 28 VDC between P435-1 and ground.

Step 1. If voltage is as specified, go to **6.**

Step 2. If voltage is not as specified, go to **8.**

6. Remove jumper wire. Go to 7.

7. Replace right relay panel (PARA 9-4-2). Go to 10.

8. Remove jumper wire. Go to 9.

TABLE NO. 13-2-1-5. No. 2 Engine TGT Does Not Increase When HEATER Switch Is Placed ON. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check for 28 VDC between P244-J and ground.

Step 1. If voltage is as specified, repair/replace wiring between P241-G and P435-1 (Appendix F). Go to **10.**
Step 2. If voltage is not as specified, repair/replace wiring between P244-J and P231-a (Appendix F). Go to **10.**

10. Procedure completed.

TABLE NO. 13-2-1-6. No. 1 And No. 2 Engine TGT Does Not Increase When HEATER Switch Is Placed ON.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Make sure AIR SOURCE HEAT/START switch is placed to ENG. Go to 2.

2. Check for 28 VDC between J231-a and ground.

Step 1. If voltage is as specified, repair/replace wiring between P231-a and P902-a (Appendix F). Go to **4.**
Step 2. If voltage is not as specified, go to **3.**

3. Check wiring between J231-a and switch S18 for proper connections.

Step 1. If wiring connections are good, replace switch (PARA 9-4-5). Go to **4.**
Step 2. If wiring connections are not good, reconnect wiring to switch (Appendix F). Go to **4.**

4. Procedure completed.

TABLE NO. 13-2-1-7. Air Temperature Coming Out Of Louvers Does Not Change When Engine Anti-ice Is Turned ON.

TEST OR INSPECTION
CORRECTIVE ACTION

1. With both engine anti-ice switches OFF, check for 28 VDC between J230-C and ground.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **3.**

2. Check for 28 VDC between P436-4 and P436-2.

Step 1. If voltage is as specified, replace mixing valve (PARA 13-4-1). Go to **4.**
Step 2. If voltage is not as specified, repair/replace wiring between P436-4 and P230-C (Appendix F). Go to **4.**

3. Check for broken/disconnected wires between AIR SOURCE HEAT/START switch terminal 2 and J230-C.

Step 1. If wires are broken or disconnected, repair/replace wiring between switch S18-2 and J230-C (Appendix F). Go to **4.**
Step 2. If wires are not broken or disconnected, replace switch ENG ANTI-ICE NO. 1 or ENG ANTI-ICE NO. 2 (PARA 9-4-5). Go to **4.**

4. Procedure completed.

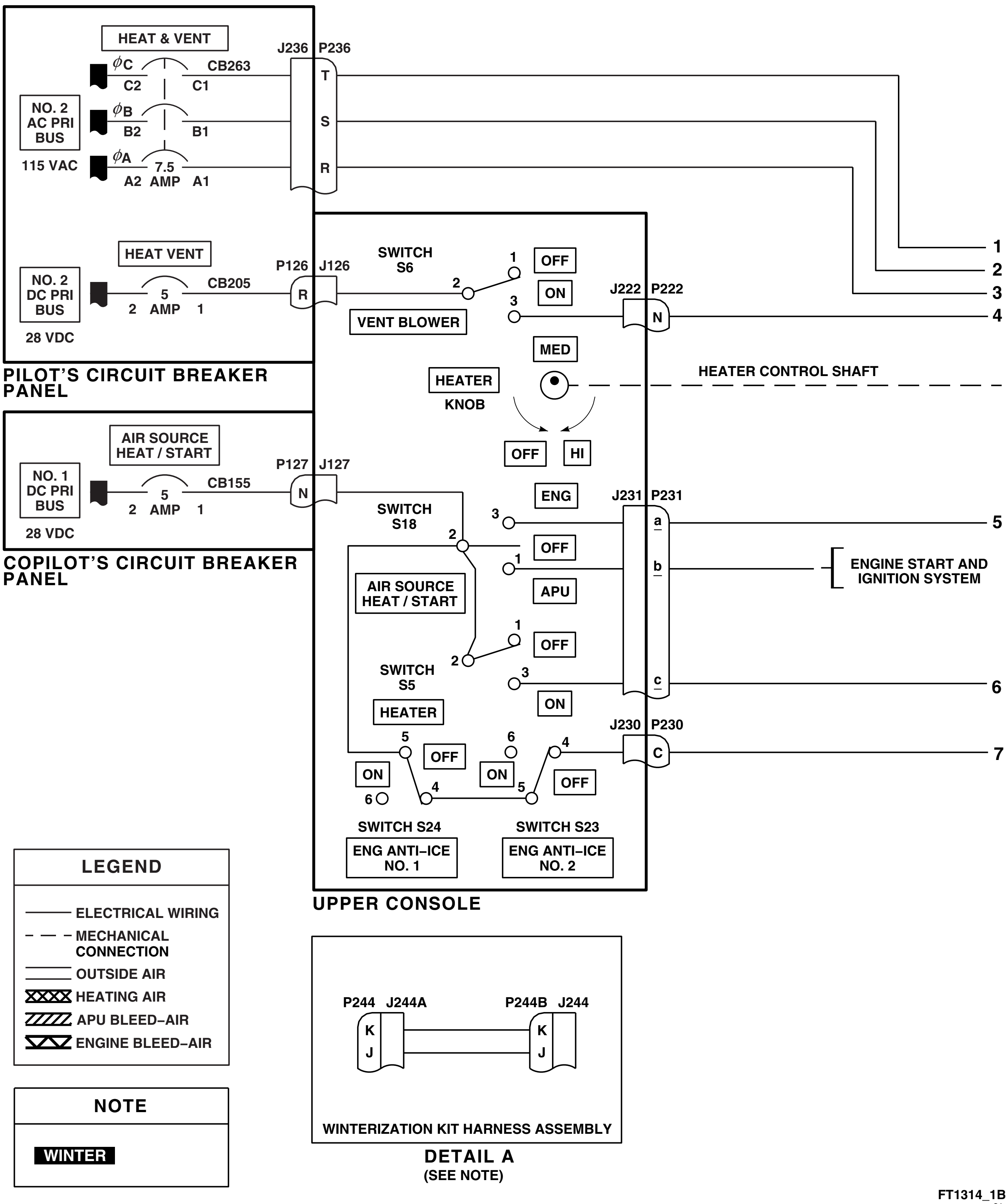

FIGURE 13-2-1-1. HEATING AND VENTILATION SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 3)

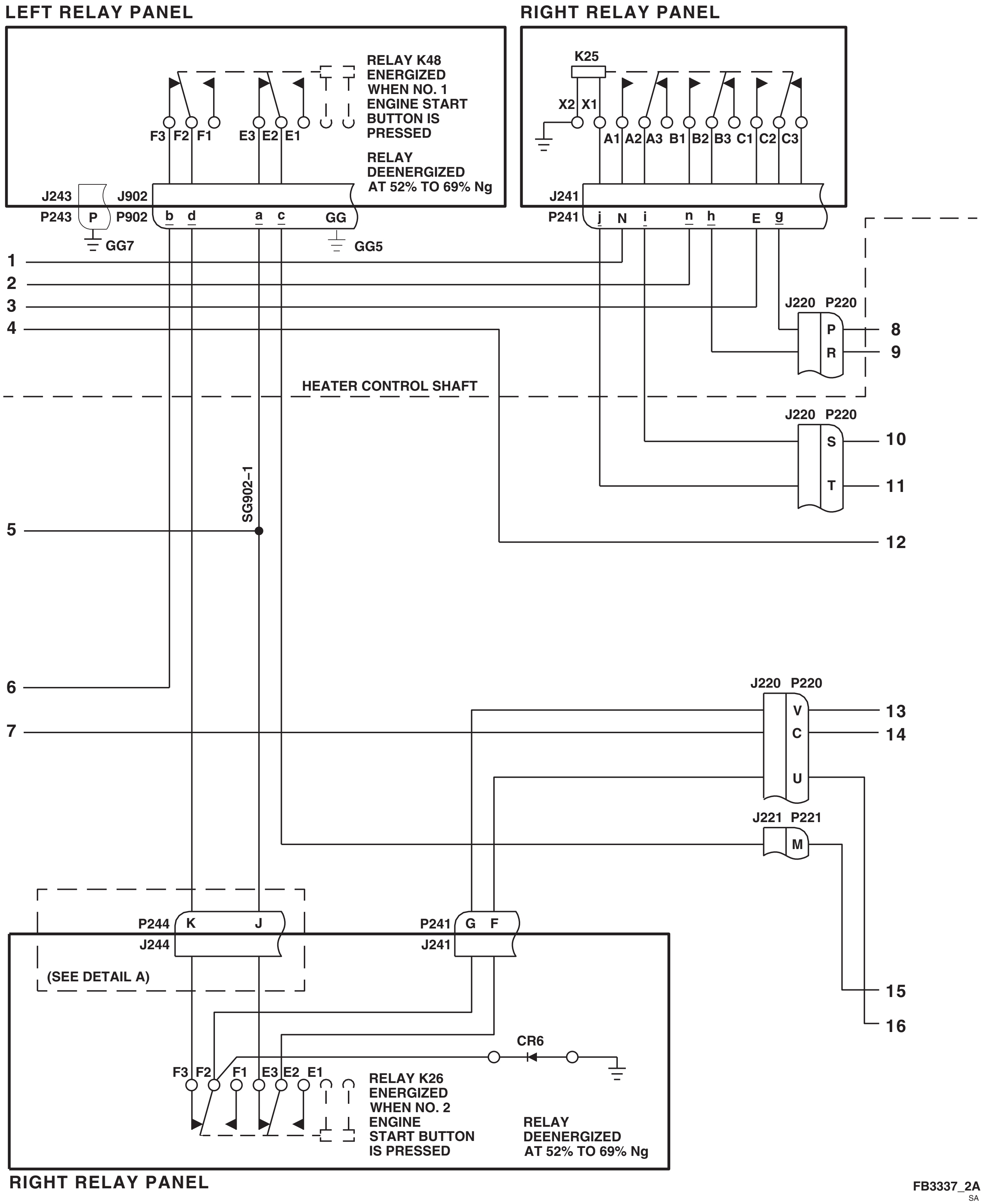

FIGURE 13-2-1-1. HEATING AND VENTILATION SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 3)

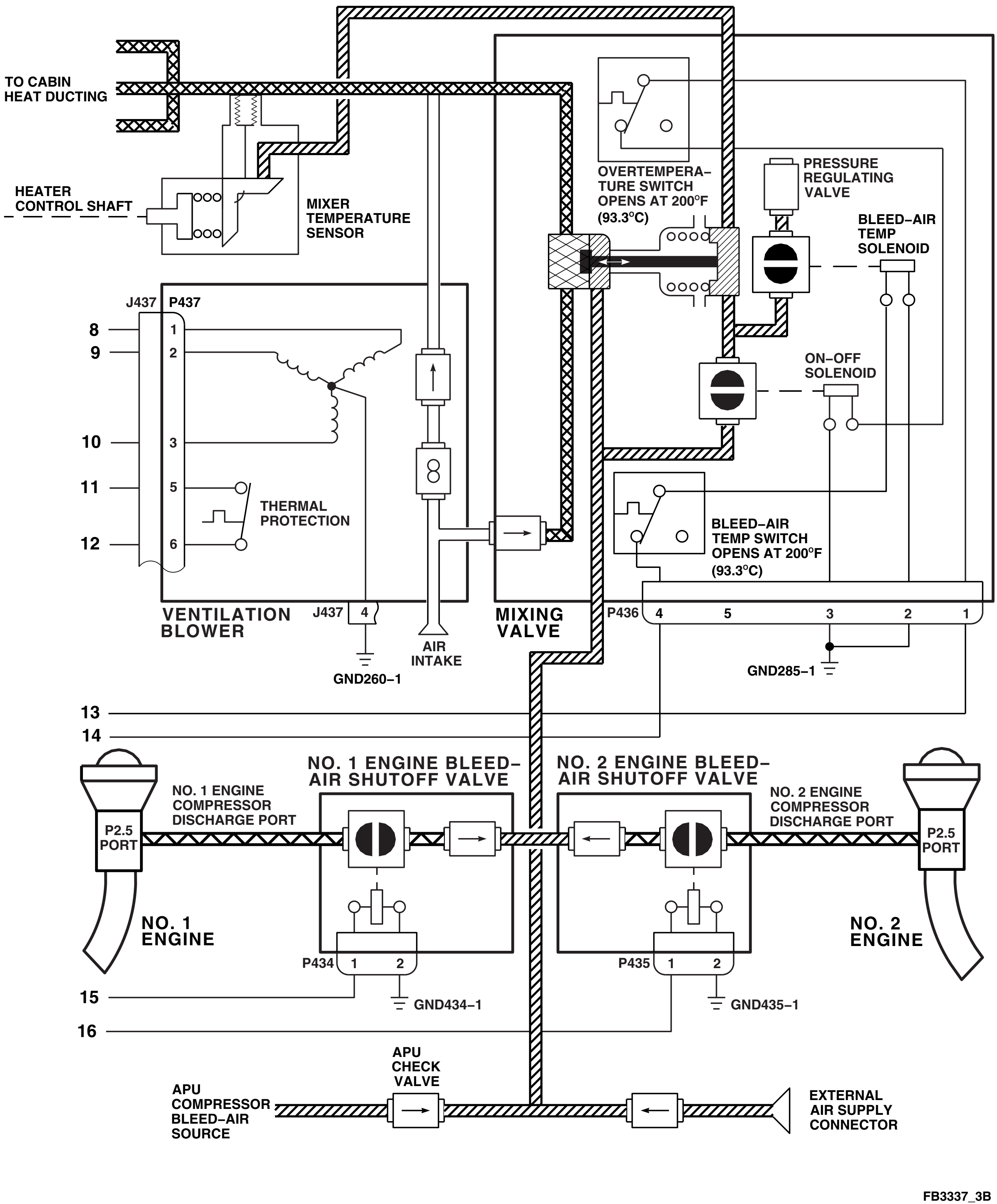

FIGURE 13-2-1-1. HEATING AND VENTILATION SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 3)

13-2-2 ENVIRONMENTAL CONTROL SYSTEM (ECS).

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Ammeter, Clamp-On, 749
- Electrical Repairer Toolkit
- Locally-Made Drag Beam Wedge, Figure H-167, Appendix H
- Pin Extractor, MS3447-20
- Refrigeration Unit Service Toolkit
- Thermometer, 389-3L

References

- Appropriate Flight Manual
- PARA 1-3-27
- PARA 1-6-16

Equipment Conditions

- ECS, Properly Serviced
- External Electrical Power Available and APU Operational
- Plenum, Not Installed
- Servicing Manifold, Connected
- Supply Ducting, Disconnected

CAUTION

Make sure cabin doors are closed and personnel are clear of exhaust duct area.

CAUTION

To prevent overheating the backup pump hydraulic system, use this chart for limits when running backup pump.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
32° (90°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
101° - 125° (39° - 52°)	16	48

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

13-2-2.1 Operational/Troubleshooting Procedure.

a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

b. Make sure upper console WINDSHIELD ANTI-ICE PILOT, CTR, and COPILOT switches, and BACKUP HYD PUMP switch are OFF.

c. Turn on electrical power (PARA 1-6-16).

NOTE

Backup hydraulic pump may run to charge APU accumulator. ECS will not operate until backup pump stops.

d. Place caution/advisory panel BRT/DIM-TEST switch to TEST.

RESULT: All capsules on caution/advisory panel shall go on.

- If result is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

e. On air conditioner circuit breaker panel, pull out COMPRESSOR MOTOR, CONDENSER BLOWER, EVAPORATOR BLOWER, and HEATER ELEMENTS circuit breakers. Make sure DC POWER CONTROLS circuit breaker is pushed in.

f. On air conditioner circuit breaker panel, push in EVAPORATOR BLOWER circuit breaker.

RESULT: Evaporator blower shall not run.

- If result is not as specified, check continuity between J246-J and J246-X.

    (a) If continuity is present, replace AIR COND switch S205 (PARA 9-4-5).

    (b) If continuity is not present, replace ECU (PARA 13-4-27).

g. Place upper console ECS AIR COND switch to FAN and then to COOL.

RESULT:

(1) Evaporator blower shall run with AIR COND switch at FAN.

(2) Evaporator blower shall run with AIR COND switch at COOL.

(3) Air shall flow from evaporator outlet duct.

- If results (1) and (2) are not as specified, go to TABLE NO. 13-2-2-1.

- If result (1) is not as specified, replace AIR COND switch S205 (PARA 9-4-5).

- If result (2) is not as specified, check continuity between terminals S205-1 and S205-4 and between terminals S205-2 and S205-6.

    (a) If continuity is present, replace switch S205 (PARA 9-4-5).

    (b) If continuity is not present, repair/replace wiring between terminals S205-1 and S205-4 or between terminals S205-2 and S205-6 (Appendix F), as required.

- If result (3) is not as specified, go to TABLE NO. 13-2-2-2.

h. Place upper console ECS AIR COND switch to OFF.

i. Clamp ammeter around wire going to P7-D.

j. On air conditioning circuit breaker panel, push in HEATER ELEMENTS circuit breaker.

k. Place upper console ECS HTR switch to ON.

RESULT: Ammeter shall indicate zero amperes.

- If result is not as specified, replace ECU (PARA 13-4-27).

l. Place upper console ECS AIR COND switch to FAN.

RESULT:

(1) CABIN HEAT ON capsule on caution/advisory panel shall go on.

(2) Ammeter shall indicate between 7 and 9 amps.

- If result (1) is not as specified, go to TABLE NO. 13-2-2-3.

- If result (2) is not as specified, go to TABLE NO. 13-2-2-4.

m. With an air temperature no higher than 90°F (32°C), allow the evaporator to run for 30 minutes.

n. Measure evaporator inlet and outlet airstream temperatures.

o. Repeat steps m. and n. for a total of three measurements.

RESULT: The average temperature differential (inlet minus outlet) shall be 10°F (-12.2°C) minimum.

- If result is not as specified, go to TABLE NO. 13-2-2-4.

p. Place upper console BACKUP HYD PUMP switch to ON.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall remain on.

- If result is not as specified, replace No. 3 relay panel (PARA 9-4-4).

q. Place upper console BACKUP HYD PUMP switch to OFF and place WINDSHIELD ANTI-ICE PILOT switch to ON.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall remain on.

- If result is not as specified, replace No. 3 relay panel (PARA 9-4-4).

r. Place upper console BACKUP HYD PUMP switch to ON.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall go off.

- If result is not as specified, replace No. 3 relay panel (PARA 9-4-4).

s. Place upper console BACKUP HYD PUMP and WINDSHIELD ANTI-ICE PILOT switches to OFF.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 13-2-2-3.

t. Place upper console ECS HTR and AIR COND switches to OFF.

u. On air conditioner circuit breaker panel, pull out HEATER ELEMENTS and EVAPORATOR BLOWER circuit breakers.

v. On air conditioner circuit breaker panel, push in CONDENSER BLOWER circuit breaker.

CAUTION

To prevent damage to the helicopter from hot condenser exhaust, make sure the right side passenger door is closed while the air conditioner is running.

w. Place upper console ECS AIR COND switch to COOL.

RESULT:

(1) After five seconds, condenser blower shall go on.

(2) Air shall come out of condenser exhaust duct.

- If result (1) is not as specified, go to TABLE NO. 13-2-2-5.
- If result (2) is not as specified, go to TABLE NO. 13-2-2-6.

x. Place upper console ECS AIR COND switch to OFF.

y. On air conditioner circuit breaker panel, push in all circuit breakers.

z. Turn upper console ECS TEMP COOL-WARM control counterclockwise to COOL, and place AIR COND switch to COOL.

RESULT:

(1) Evaporator blower shall start immediately.

(2) After five seconds, condenser blower shall start.

(3) After an additional 5 to 10 seconds, compressor motor shall start.

(4) When compressor motor starts, AIR COND ON capsule on caution/advisory panel shall go on.

(5) Cool air shall come out of evaporator outlet duct.

- If result (1) is not as specified, go to TABLE NO. 13-2-2-1.
- If result (2) is not as specified, go to TABLE NO. 13-2-2-5.
- If result (3) is not as specified, go to TABLE NO. 13-2-2-7.
- If result (4) is not as specified, go to TABLE NO. 13-2-2-8.
- If result (5) is not as specified, go to TABLE NO. 13-2-2-9.

aa. Place upper console BACKUP HYD PUMP switch to ON.

RESULT: AIR COND ON capsule on caution/advisory panel shall go off.

- If result is not as specified, check continuity between P398-j and P244-A.

(a) If continuity is present, replace No. 3 relay panel (PARA 9-4-4).

(b) If continuity is not present, repair/replace wiring (Appendix F).

1 If trouble remains, troubleshoot Hydraulic Systems (PARA 7-2-1).

2 If trouble still remains, replace right relay panel (PARA 9-4-2).

CAUTION

Allow ECS to remain off for at least 2 minutes between restarts to allow internal pressures to stabilize or damage to compressor may occur.

ab. Place upper console BACKUP HYD PUMP switch to OFF.

RESULT: AIR COND ON capsule on caution/advisory panel shall go on.

- If result is not as specified, troubleshoot Hydraulic Systems (PARA 7-2-1).

ac. Allowing at least 2 minutes between ECS restarts, alternately place upper console WINDSHIELD ANTI-ICE COPILOT, CTR, and PILOT switches to ON.

RESULT: When each switch is ON, AIR COND ON capsule on caution/advisory panel shall go off.

- If result is not as specified, go to TABLE NO. 13-2-2-10.

ad. Make sure upper console WINDSHIELD ANTI-ICE COPILOT, CTR, and PILOT switches are OFF.

RESULT: AIR COND ON capsule on caution/advisory panel shall go back on.

- If result is not as specified, troubleshoot Windshield Anti-Ice System (PARA 12-2-4).

ae. Disconnect ammeter and clamp it to TEST LOOP wire of compressor harness (located near P7).

RESULT: Ammeter shall indicate between upper and lower limits when cross referenced with ambient temperature (Figure 13-2-2-5).

- If ammeter indicates less than lower limit, go to TABLE NO. 13-2-2-11.
- If ammeter indicates more than upper limit, go to TABLE NO. 13-2-2-12.

af. Check the ECS sight glass (on electrical control pallet).

RESULT: A green dot shall be seen in the sight glass, and no bubbles or smears shall be seen (shades of green or a wisp of bubbles are normal).

- If result is not as specified, purge and service the system and add freon until sight glass clears (PARA 1-3-27).

ag. Allow system to run for 15 minutes and stabilize.

ah. Place temperature probe in evaporator outlet duct.

ai. Cover about 95% of plenum inlet face.

aj. On air conditioner fault indicator panel, check temperature probe reading when LOW TEMPERATURE fault indicator pops out.

RESULT: When LOW TEMPERATURE fault indicator pops out, temperature probe shall indicate 27°F - 43°F (-2.7°C - 6.1°C) and compressor shall shut down.

- If result is not as specified, replace low temperature switch (PARA 13-4-18).

ak. Remove cover from evaporator inlet face.

al. Reset LOW TEMPERATURE fault indicator.

RESULT: Compressor shall restart when temperature probe indicates 48°F - 63°F (8.8°C - 17.2°C).

- If result is not as specified, replace low temperature switch (PARA 13-4-18).

am. On air conditioner circuit breaker panel, pull out CONDENSER BLOWER circuit breaker.

an. On air conditioner fault indicator panel, check refrigerant gage reading when HIGH PRESSURE fault indicator pops out.

RESULT: High pressure gage shall indicate between 295 and 325 psig, and compressor shall shut down when HIGH PRESSURE fault indicator pops out.

- If result is not as specified, replace high pressure switch (PARA 13-5-14).

ao. On air conditioner circuit breaker panel, push in CONDENSER BLOWER circuit breaker.

ap. Reset HIGH PRESSURE fault indicator.

RESULT: Compressor will restart when high pressure gage on servicing manifold indicates 275 psig.

- If result is not as specified, replace high pressure switch (PARA 13-5-14).

aq. Remove ammeter from compressor harness and connect it to jumper wire at bottom of heater elements.

ar. Place upper console ECS AIR COND switch to FAN.

as. Place upper console ECS HTR switch to ON.

at. Place temperature probe in evaporator outlet duct.

au. Cover about 95% of the evaporator inlet face.

RESULT: When temperature probe indicates 152°F - 168°F (66.6°C - 75.5°C), the heater/demister shall shut down (ammeter shall indicate zero amps) and HIGH TEMPERATURE fault indicator shall pop out.

- If heater/demister shuts down at a temperature outside the specified range, turn adjustment screw of temperature limiting switch first clockwise and then counterclockwise until heater shuts down.

- If adjustment of temperature limiting switch fails to cause heater/demister shutdown, replace temperature limiting switch (PARA 13-4-17).

av. Reset HIGH TEMPERATURE fault indicator.

aw. Remove cover from evaporator inlet (the rise to 180°F (82.2°C) should occur rapidly when cover is removed).

RESULT: When temperature probe indicates 172°F - 188°F (77.7°C - 86.6°C), the HIGH TEMPERATURE fault indicator shall pop out.

- If result is not as specified, replace high temperature switch (PARA 13-4-18).

ax. Reset HIGH TEMPERATURE fault indicator.

CAUTION

If only using APU generator power, make sure 60 Hz converter is not operated at full load (30 amps) or air conditioner will exceed generator rating when ECS control panel AIR COND switch is selected to COOL.

ay. Place upper console ECS AIR COND switch to COOL.

az. Place upper console ECS HTR switch to OFF.

ba. Make sure REF and VAC handwheels on servicing manifold are closed.

bb. Connect yellow hose to vacuum side of manifold and place other end outside of work area.

NOTE

- Bleed freon very slowly to minimize the amount of lubricant discharged from the system.

- The following result may take up to an hour to occur.

bc. Slowly open VAC handwheel and allow refrigerant to vent through yellow hose.

RESULT: When high pressure gage indicates between 45 and 55 psig, the LOW PRESSURE fault indicator shall pop out and the compressor shall shut down.

- If result is not as specified, replace low pressure switch (PARA 13-5-14).

bd. Service the ECS (PARA 1-3-27).

be. Reset LOW PRESSURE fault indicator.

bf. With air conditioning operating (AIR COND switch on COOL and HTR switch OFF) and servicing manifold attached, **RECORD** low pressure gage pressure (P1).

bg. Turn upper console ECS TEMP CONT knob clockwise to full WARM.

RESULT: Low pressure gage indication shall increase from steady state.

- If result is not as specified, go to TABLE NO. 13-2-2-13.

bh. **RECORD** low pressure gage indication (P2).

bi. Calculate difference between first pressure (P1) and second pressure (P2).

RESULT: Pressure differential shall be between 9 and 11 psig.

- If result is less than specified, turn adjustment screw on hot gas bypass valve clockwise to increase pressure differential.

- If result is more than specified, turn adjustment screw on hot gas bypass valve counterclockwise to decrease pressure differential.

  (a) If adjustment fails to produce change in pressure differential, replace hot gas bypass valve (PARA 13-5-8).

  (b) If trouble remains, replace temperature sensor in plenum air inlet (PARA 13-4-26).

bj. Place upper console ECS AIR COND switch to OFF.

bk. Disconnect and remove servicing manifold.

bl. Turn off electrical power (PARA 1-6-16).

bm. Make sure upper console GENERATORS NO. 1, NO. 2, and APU switches are set to OFF/RESET.

bn. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

bo. Start APU (Appropriate Flight Manual).

bp. Place upper console APU GENERATORS switch to ON.

RESULT: On caution/advisory panel, APU GEN ON capsule shall be on, and #1 CONV and #2 CONV capsules shall be off.

- If result is not as specified, troubleshoot AC Electrical System (APU generators) (PARA 9-2-1).

NOTE

Backup hydraulic pump may run to charge APU accumulator. ECS will not operate until backup pump stops.

bq. Place upper console ECS AIR COND switch to FAN and place HTR switch to ON.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 13-2-2-3.

br. Place drag beam wedge between left drag beam switch plunger and switch plunger actuating bracket to simulate weight-off-wheels.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall go off.

- If result is not as specified, check continuity between P398-X and P398-a.

  (a) If continuity is present, troubleshoot AC Electrical System (interface panel ground power) (PARA 9-2-1).

  (b) If continuity is not present, replace No. 3 relay panel (PARA 9-4-4).

bs. Remove wedge from drag beam switch.

bt. Place upper console ECS-AIR COND and HTR switches to OFF.

bu. Place upper console GENERATORS APU switch to OFF/RESET (Appropriate Flight Manual).

NOTE

If #1 GEN BRG and #2 GEN BRG capsules on caution/advisory panel goes on, troubleshoot AC Electrical System (PARA 9-2-1).

bv. With qualified personnel at controls, start No. 1 and No. 2 engines and operate with main rotor turning at 100% (Appropriate Flight Manual).

bw. Place upper console GENERATORS NO. 1 and NO. 2 switches to ON.

RESULT: #1 GEN and #2 GEN capsules on caution/advisory panel shall be off.

- If result is not as specified, go to TABLE NO. 13-2-2-14.

bx. Place upper console ECS AIR COND switch to FAN and place HTR switch to ON.

RESULT: CABIN HEAT ON capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 13-2-2-3.

by. Place upper console WINDSHIELD ANTI-ICE PILOT and BACKUP HYD PUMP switches to ON.

RESULT:

(1) CABIN HEAT ON capsule on caution/advisory panel shall remain on.

(2) BACK-UP PUMP ON capsule on caution/advisory panel shall go on.

- If result (1) is not as specified, go to TABLE NO. 13-2-2-3.
- If result (2) is not as specified, troubleshoot Hydraulic Systems (PARA 7-2-1).

bz. Place upper console WINDSHIELD ANTI-ICE PILOT and BACKUP HYD PUMP switches to OFF.

ca. Place upper console HTR switch to OFF and place AIR COND switch to COOL.

RESULT: AIR COND ON capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 13-2-2-8.

cb. Place upper console WINDSHIELD ANTI-ICE PILOT switch to ON.

RESULT: AIR COND ON capsule on caution/advisory panel shall remain on.

- If result is not as specified, replace No. 3 relay panel (PARA 9-4-4).

cc. Place upper console WINDSHIELD ANTI-ICE PILOT switch to OFF.

cd. Place upper console ECS-AIR COND switch to OFF.

ce. Place upper console GENERATORS NO. 1 and NO. 2 switches to OFF/RESET.

cf. Shut down No. 1 and No. 2 engines (Appropriate Flight Manual).

cg. Shut down APU (Appropriate Flight Manual).

TABLE NO. 13-2-2-1. Evaporator Blower Does Not Operate With AIR COND Switch At COOL Or FAN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 115 VAC between:
P456-A and P456-D
P456-B and P456-D
P456-C and P456-D

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **8.**

TABLE NO. 13-2-2-1. Evaporator Blower Does Not Operate With AIR COND Switch At COOL Or FAN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check for 28 VDC between P457-B and P457-K.

Step 1. If voltage is as specified, replace electrical control unit (PARA 13-4-27). Go to **19.**
Step 2. If trouble remains, replace evaporator blower (PARA 13-4-15). Go to **19.**
Step 3. If trouble still remains, repair/replace wiring (Appendix F), as required between the following, then go to **19:**
P3-R and P9-A
P3-S and P9-C
P3-T and P9-B
P7-D and P9-D
P3-L and P9-E
P3-M and P9-F
Step 4. If voltage is not as specified, go to **3.**

3. Check for 28 VDC between AIR COND switch terminal 5 and ground.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, go to **5.**

4. With AIR COND switch at FAN, check for 28 VDC between AIR COND switch terminal 6 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 6 and P457-B, or between P457-K and GND457-1 (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **19.**

5. Check for 28 VDC between P398-a and ground.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P398-X and AIR COND switch terminal 5.

Step 1. If continuity is present, replace No. 3 relay panel (PARA 9-4-4). Go to **19.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

TABLE NO. 13-2-2-1. Evaporator Blower Does Not Operate With AIR COND Switch At COOL Or FAN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

7. Check for 28 VDC between terminal 1 (ECS CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P398-a (Appendix F). Go to **19.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **19.**

8. Check for 115 VAC between ECU PWR circuit breaker:
Terminal A2 and ground
Terminal B2 and ground
Terminal C2 and ground

Step 1. If voltage is as specified, repair/replace wiring (Appendix F), as required between the following, then go to **19:**
Terminal A2 and P456-A
Terminal B2 and P456-B
Terminal C2 and P456-C
GND456-1 and P456-D
Step 2. If voltage is not as specified, go to **9.**

9. Check for 115 VAC between ECU PWR circuit breaker:
Terminal A1 and ground
Terminal B1 and ground
Terminal C1 and ground

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, troubleshoot AC Electrical System (PARA 9-2-1). Go to **19.**

10. Using pin extractor, disconnect wire connected to ECU PWR circuit breaker terminal 5A. Go to 11.

11. Check continuity between ECU PWR circuit breaker terminal 5A and ground.

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **19.**

12. Using pin extractor, disconnect wire connected to ECU PWR circuit breaker terminal 3. Go to 13.

TABLE NO. 13-2-2-1. Evaporator Blower Does Not Operate With AIR COND Switch At COOL Or FAN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

13. Pull out ECS PWR circuit breaker. Go to 14.

14. Check continuity between ECU PWR circuit breaker terminal 3 wire and ground.

Step 1. If continuity is present, troubleshoot system (ECU PWR) circuit breaker (PARA 9-2-3). Go to **19.**
Step 2. If continuity is not present, go to **15.**

15. Leave wire disconnected from ECU PWR circuit breaker terminal 3. Go to 16.

16. Check continuity between ECS PWR circuit breaker (pilot's circuit breaker panel) terminal 1 and ground.

Step 1. If continuity is present, repair/replace wiring between circuit breaker terminal 1 and wire connected to ECU PWR circuit breaker terminal 3 (Appendix F). Go to **19.**
Step 2. If continuity is not present, go to **17.**

17. Pull out ECS PWR circuit breaker. Go to 18.

18. Check continuity between ECS PWR circuit breaker terminal 2 and ground.

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **19.**
Step 2. If continuity is not present, repair/replace wiring between ECS PWR circuit breaker terminal 2 and ground (Appendix F). Go to **19.**

19. Procedure completed.

TABLE NO. 13-2-2-2. Air Does Not Flow From Evaporator Outlet Duct.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for blocked distribution ducts.

Step 1. If distribution ducts are blocked, remove obstruction. Go to **4.**
Step 2. If distribution ducts are not blocked, go to **2.**

TABLE NO. 13-2-2-2. Air Does Not Flow From Evaporator Outlet Duct. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check for clogged demister.

Step 1. If demister is clogged, replace demister pad (PARA 13-5-9). Go to **4.**
Step 2. If demister is not clogged, go to **3.**

3. Check for blocked evaporator core.

Step 1. If trouble remains, replace evaporator (PARA 13-5-9). Go to **4.**

4. Procedure completed.

TABLE NO. 13-2-2-3. CABIN HEAT ON Capsule Does Not Go On With ECS HTR Switch At ON And AIR COND Switch At FAN.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 VDC between P457-C and ground.

Step 1. If voltage is as specified, go to **2.**
Step 2. If voltage is not as specified, go to **8.**

2. Check for 28 VDC between P457-J and ground.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (ECS CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and P457-J (Appendix F). Go to **12.**

3. Check helicopter configuration.

Step 1. **EMEP** Go to **4.**
Step 2. **W/O EMEP** Go to **6.**

EMEP **4. Remove and check pin filter adapter P118 (PARA 9-4-98).**

Step 1. If pin filter adapter is good, go to **5.**

TABLE NO. 13-2-2-3. CABIN HEAT ON Capsule Does Not Go On With ECS HTR Switch At ON And AIR COND Switch At FAN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to **12.**

EMEP

5. Install pin filter adapter (PARA 9-4-98). Go to 6.

6. Check for 28 VDC between P118-M and ground.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **12.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between P118-M and P457-E.

Step 1. If continuity is present, replace electrical control unit (PARA 13-4-27). Go to **12.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **12.**

8. Check for 28 VDC between P398-e and ground.

Step 1. If voltage is as specified, go to **9.**
Step 2. If voltage is not as specified, go to **11.**

9. With HTR switch at ON, check continuity between P398-d and P457-C.

Step 1. If continuity is present, replace No. 3 relay panel (PARA 9-4-4). Go to **12.**
Step 2. If continuity is not present, go to **10.**

10. With HTR switch ON, check continuity between HTR switch terminals 2 and 3.

Step 1. If continuity is present, repair/replace wiring between switch terminal 2 and P398-d or between switch terminal 3 and P457-C (Appendix F). Go to **12.**
Step 2. If continuity is not present, replace switch (PARA 9-4-5). Go to **12.**

11. With AIR COND switch at FAN, check for 28 VDC between AIR COND switch terminal 3 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 3 and P398-e (Appendix F). Go to **12.**
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **12.**

TABLE NO. 13-2-2-3. CABIN HEAT ON Capsule Does Not Go On With ECS HTR Switch At ON And AIR COND Switch At FAN. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

12. **Procedure completed.**

TABLE NO. 13-2-2-4. Ammeter Indicates Less Than 7 Amps Through Heater Demister.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check ammeter.**

   Step 1. If ammeter indicates more than 0 but less than 7 amps, replace heater/demister assembly (PARA 13-5-9). Go to **3.**
   Step 2. If ammeter indicates 0 amps, go to **2.**

2. **Check continuity between P3-D and P3-E.**

   Step 1. If continuity is present, replace electrical control unit (PARA 13-4-27). Go to **3.**
   Step 2. If trouble remains, replace heater/demister assembly (PARA 13-5-9). Go to **3.**
   Step 3. If continuity is not present, replace high temperature switch (PARA 13-4-18). Go to **3.**

3. **Procedure completed.**

TABLE NO. 13-2-2-5. Condenser Blower Does Not Go On With AIR COND Switch At COOL.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 VDC between P457-A and ground.**

   Step 1. If voltage is as specified, replace electrical control unit (PARA 13-4-27). Go to **7.**
   Step 2. If trouble remains, go to **2.**
   Step 3. If voltage is not as specified, go to **3.**

TABLE NO. 13-2-2-5. Condenser Blower Does Not Go On With AIR COND Switch At COOL. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check continuity between:
P4-A and P10-A
P4-B and P10-B
P4-C and P10-C
P4-D and P10-D
P4-E and P10-E
P4-F and P10-F

Step 1. If continuity is present, replace condenser blower assembly (PARA 13-5-3). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **7.**

3. Check for 28 VDC between P398-f and ground.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, repair/replace wiring between P398-f and AIR COND switch S205 terminal 1 (Appendix F). Go to **7.**

4. Check for 28 VDC between:
P398-V and ground
P398-W and ground
P398-U and ground

Step 1. If voltage is as specified, troubleshoot Windshield Anti-Ice System (PARA 12-2-4). Go to **7.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P398-j and ground.

Step 1. If voltage is as specified, troubleshoot Hydraulic Systems (PARA 7-2-1). Go to **7.**
Step 2. If voltage is not as specified, go to **6.**

6. Check continuity between P398-B and P457-A.

Step 1. If continuity is present, replace No. 3 relay panel (PARA 9-4-4). Go to **7.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **7.**

7. Procedure completed.

TABLE NO. 13-2-2-6. Air Is Not Coming Out Of Condenser Exhaust Duct.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for blocked condenser core.

Step 1. If condenser core is blocked, remove obstruction. Go to **3.**
Step 2. If condenser core is not blocked, go to **2.**

2. Check for blocked airduct.

Step 1. If airduct is blocked, remove obstruction. Go to **3.**
Step 2. If airduct is not blocked, check for blocked air outlet on side of aircraft and remove obstruction. Go to **3.**

3. Procedure completed.

TABLE NO. 13-2-2-7. Compressor Motor Does Not Operate With AIR COND Switch At COOL.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that low pressure gage reads at least 60 psig.

Step 1. If gage reads at least 60 psig, go to **2.**
Step 2. If gage does not read at least 60 psig, perform leak test and service system (PARA 1-3-27). Go to **11.**

2. Check for 115 VAC between:
J7-A and ground
J7-B and ground
J7-C and ground

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **4.**

TABLE NO. 13-2-2-7. Compressor Motor Does Not Operate With AIR COND Switch At COOL. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

3. Check continuity between:
P7-A and P8-A
P7-B and P8-B
P7-C and P8-C

Step 1. If continuity is present, replace compressor motor assembly (PARA 13-5-7). Go to **11.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required. Go to **11.**

4. Check continuity between P3-K and P3-J.

Step 1. If continuity is present, go to **5.**

Step 2. If continuity is not present, go to **10.**

5. Check continuity between P6-B and P6-C.

Step 1. If continuity is present, go to **6.**

Step 2. If continuity is not present, go to **9.**

6. Check continuity between P5-A and P5-B.

Step 1. If continuity is present, go to **7.**

Step 2. If continuity is not present, go to **8.**

7. Check continuity between P3-A and P3-B.

Step 1. If continuity is present, replace electrical control unit (PARA 13-4-27). Go to **11.**

Step 2. If continuity is not present, replace low temperature switch (PARA 13-4-18). Go to **11.**

8. Check continuity between J11-A and J11-B.

Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between the following, then go to **11:**
P5-A and J11-A
P5-B and J11-B

Step 2. If continuity is not present, replace high pressure switch (PARA 13-5-14). Go to **11.**

TABLE NO. 13-2-2-7. Compressor Motor Does Not Operate With AIR COND Switch At COOL. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check continuity between J12-B and J12-C.

Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between the following, then go to **11:**
P6-B and J12-B
P6-C and J12-C

Step 2. If continuity is not present, replace low pressure switch (PARA 13-5-14). Go to **11.**

10. Check continuity between J8-D and J8-E.

Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between the following, then go to **11:**
P3-K and P8-E
P3-J and P8-D

Step 2. If continuity is not present, replace compressor assembly (PARA 13-5-7). Go to **11.**

11. Procedure completed.

TABLE NO. 13-2-2-8. AIR COND ON Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP Go to **2.**

Step 2. W/O EMEP Go to **4.**

EMEP **2. Remove and check pin filter adapter P118 (PARA 9-4-98).**

Step 1. If pin filter adapter is good, go to **3.**

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to **6.**

EMEP **3. Install pin filter adapter (PARA 9-4-98). Go to 4.**

TABLE NO. 13-2-2-8. AIR COND ON Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. Check for 28 VDC between P118-G and ground.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4 or PARA). Go to **6.**
Step 2. If voltage is not as specified, go to **5.**

5. Check continuity between P118-G and P457-D.

Step 1. If continuity is present, replace electrical control unit (PARA 13-4-27). Go to **6.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **6.**

6. Procedure completed.

TABLE NO. 13-2-2-9. Air From Evaporation Outlet Duct Does Not Turn Cool.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Over a period of five minutes, observe that low pressure gage decreases and increases.

Step 1. If gage increases and decreases, replace evaporator pallet (PARA 13-5-6). Go to **2.**
Step 2. If gage does not increase or decrease, replace condenser pallet (PARA 13-5-3). Go to **2.**

2. Procedure completed.

TABLE NO. 13-2-2-10. Windshield Anti-Ice System Does Not Disable ECS.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is ECS disabled with WINDSHIELD ANTI-ICE COPILOT switch ON?

Step 1. If ECS is disabled, go to **2.**
Step 2. If ECS is not disabled, go to **7.**

2. Is ECS disabled with WINDSHIELD ANTI-ICE PILOT switch ON?

Step 1. If ECS is disabled, go to **3.**
Step 2. If ECS is not disabled, go to **5.**

3. With WINDSHIELD ANTI-ICE CTR switch ON, check for 28 VDC between P398-U and ground.

Step 1. If voltage is as specified, replace No. 3 relay panel (PARA 9-4-4). Go to **9.**
Step 2. If voltage is not as specified, go to **4.**

4. Check continuity between P398-U and P244-G.

Step 1. If continuity is present, troubleshoot Windshield Anti-Ice System (PARA 12-2-4). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

5. With WINDSHIELD ANTI-ICE PILOT switch ON, check for 28 VDC between P398-W and ground.

Step 1. If voltage is as specified, replace No. 3 relay panel (PARA 9-4-4). Go to **9.**
Step 2. If voltage is not as specified, go to **6.**

6. Check continuity between P398-W and P900-F.

Step 1. If continuity is present, troubleshoot Windshield Anti-Ice System (PARA 12-2-4). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

7. With WINDSHIELD ANTI-ICE COPILOT switch ON, check for 28 VDC between P398-V and ground.

Step 1. If voltage is as specified, replace No. 3 relay panel (PARA 9-4-4). Go to **9.**
Step 2. If voltage is not as specified, go to **8.**

TABLE NO. 13-2-2-10. Windshield Anti-Ice System Does Not Disable ECS. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Check continuity between P398-V and P900-D.

Step 1. If continuity is present, troubleshoot Windshield Anti-Ice System (PARA 12-2-4). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

9. Procedure completed.

TABLE NO. 13-2-2-11. Compressor Draws Less Than Minimum Allowable Current.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that low pressure gage indicates less than 60 psig.

Step 1. If gage indicates less than 60 psig, go to **TABLE NO. 13-2-2-7.**
Step 2. If gage does not indicate less than 60 psig, go to **2.**

2. Check for leak in system.

Step 1. If leak is present, repair leak and service Air Conditioning System (PARA 1-3-27). Go to **3.**
Step 2. If leak is not present, service Air Conditioning System (PARA 1-3-27). Go to **3.**

3. Procedure completed.

TABLE NO. 13-2-2-12. Compressor Draws More Than Maximum Allowable Current.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check that high pressure gage indicates more than 305 psig.

Step 1. If gage indicates more than 305 psig, go to **2.**

TABLE NO. 13-2-2-12. Compressor Draws More Than Maximum Allowable Current. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If gage indicates less than 305 psig, replace compressor (PARA 13-5-7). Go to **3.**

2. Remove freon from system to reduce compressor current to below maximum allowable current.

Step 1. If trouble remains, replace compressor (PARA 13-5-7). Go to **3.**

3. Procedure completed.

TABLE NO. 13-2-2-13. Low Pressure Does Not Increase.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between J246-s and J246-t.

Step 1. If continuity is present, go to **2.**
Step 2. If continuity is not present, replace TEMP rheostat (PARA 9-4-5). Go to **3.**

2. Check continuity between P457-F and P457-G.

Step 1. If continuity is present, replace ECU (PARA 13-4-27). Go to **3.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F), as required between the following, then go to **3:**
P457-F and P246-s
P457-G and P246-t

3. Procedure completed.

TABLE NO. 13-2-2-14. ECS Does Not Operate With Both Main Generators Running.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 VDC between P398-g and ground.**

   Step 1. If voltage is as specified, replace No. 3 relay panel (PARA 9-4-4). Go to **3.**
   Step 2. If voltage is not as specified, go to **2.**

2. **Check for 28 VDC between K81A-X1 and ground.**

   Step 1. If voltage is as specified, repair/replace wiring between P398-g and P329-P (Appendix F). Go to **3.**
   Step 2. If voltage is not as specified, troubleshoot AC Electrical System (dual generators power) (PARA 9-2-1). Go to **3.**

3. **Procedure completed.**

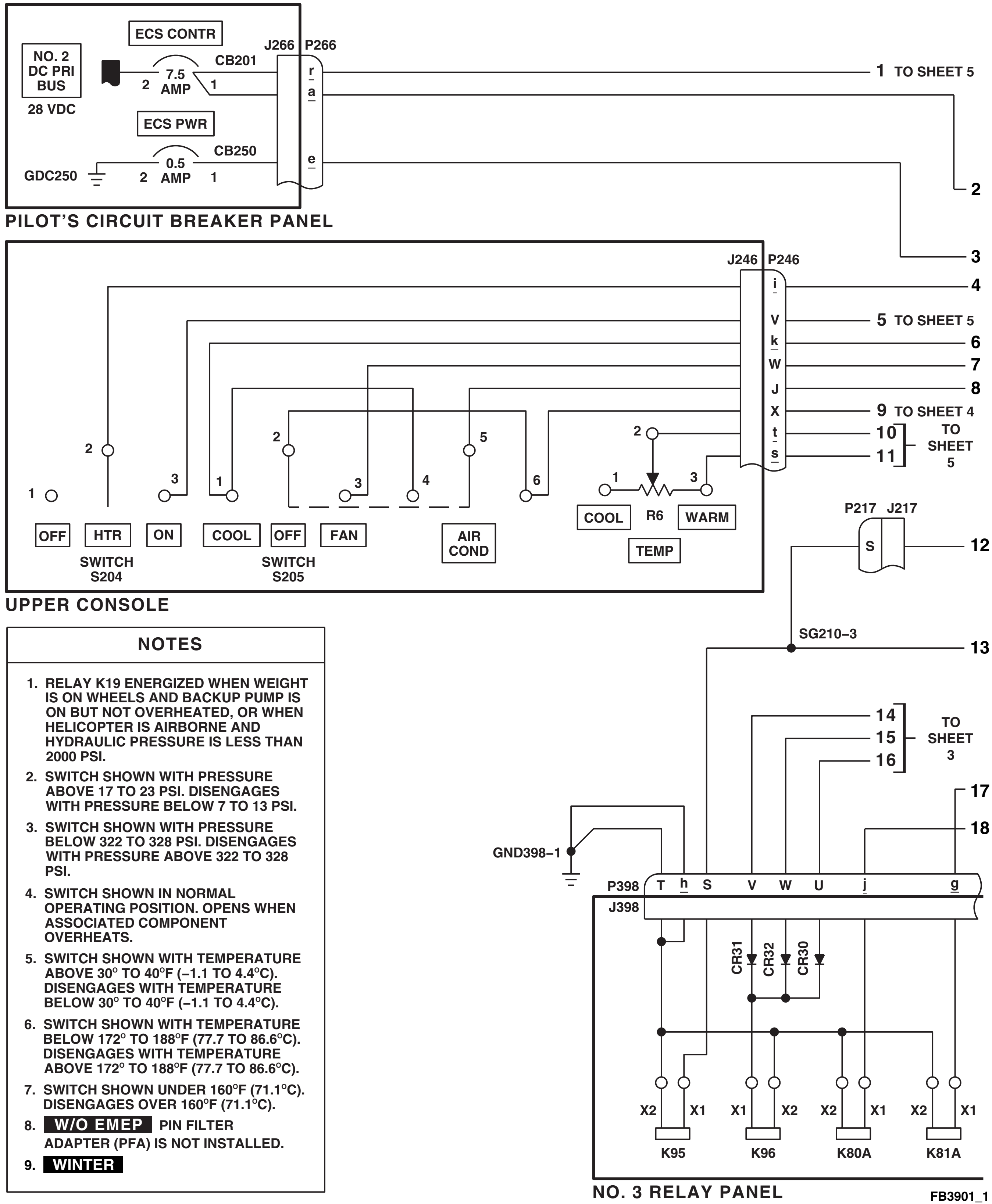

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 9)

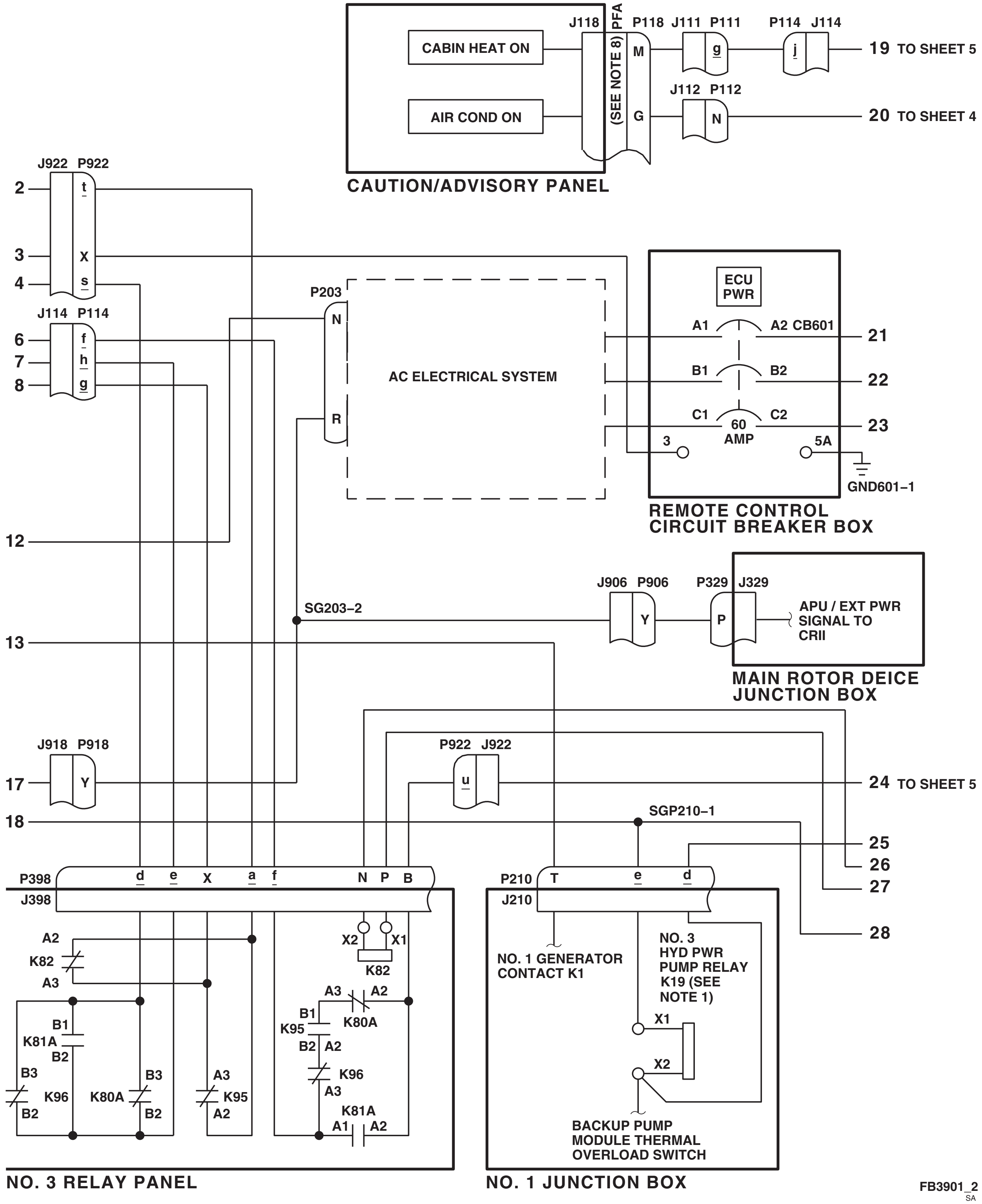

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM
(SHEET 2 OF 9)

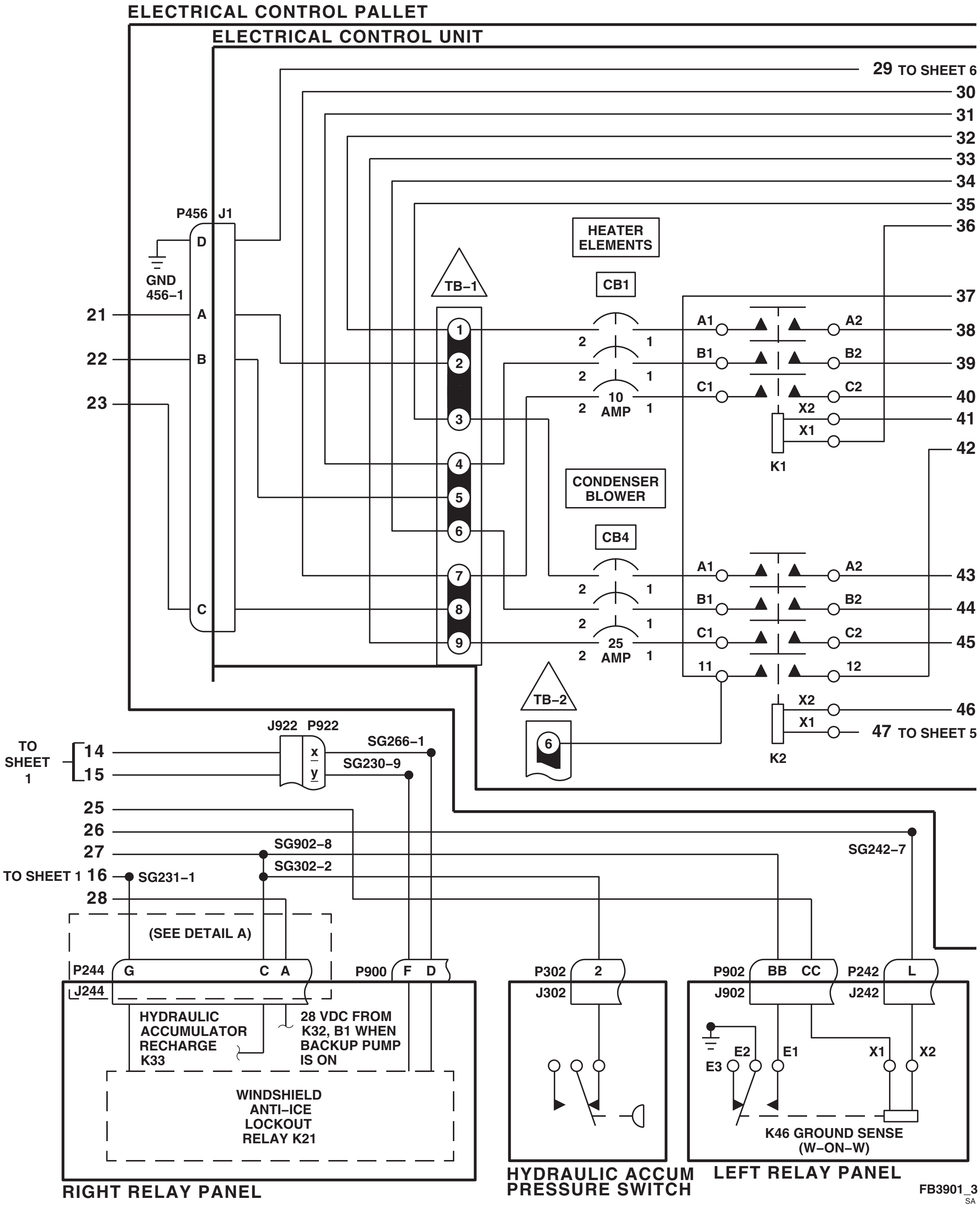

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 9)

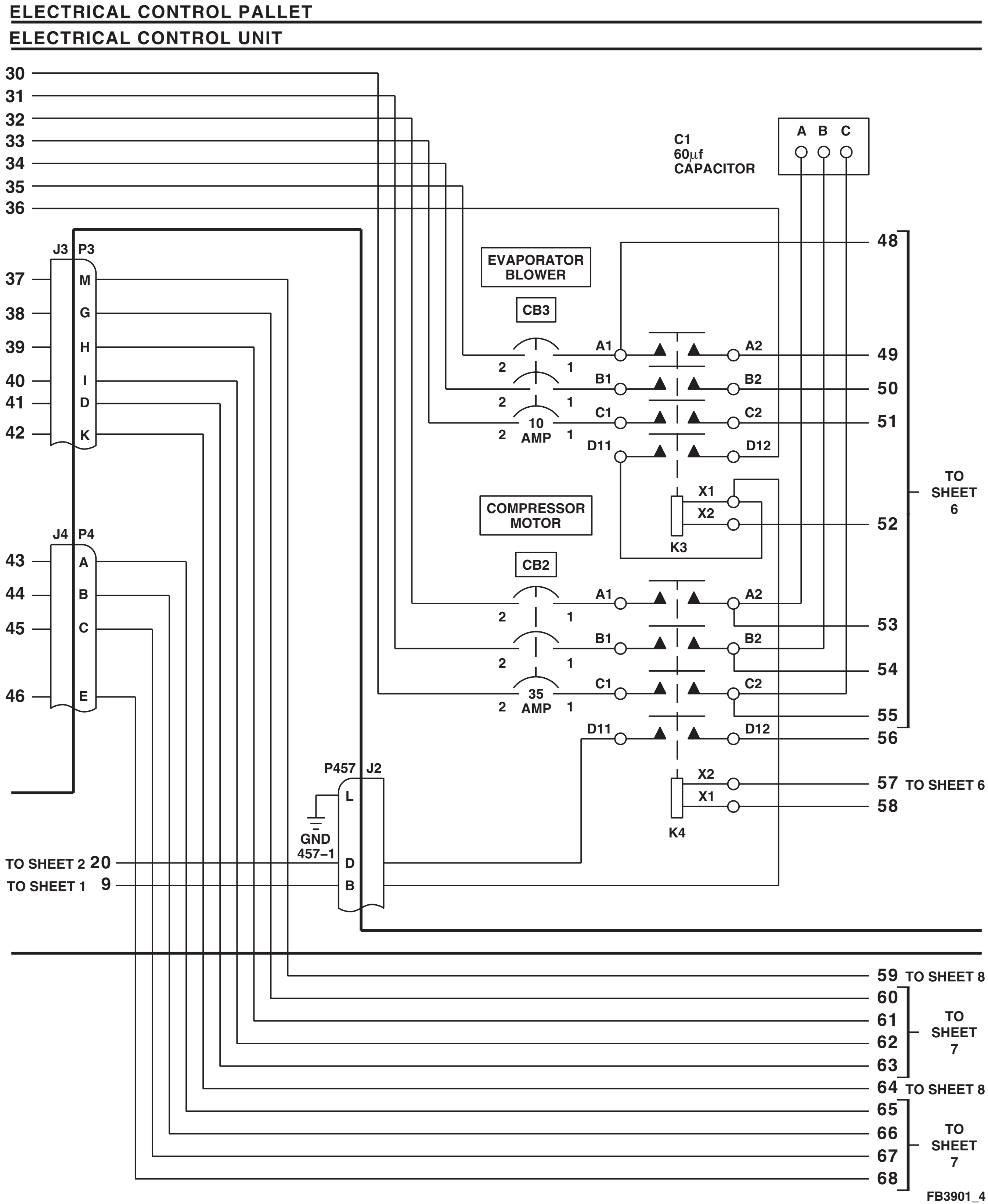

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM
(SHEET 4 OF 9)

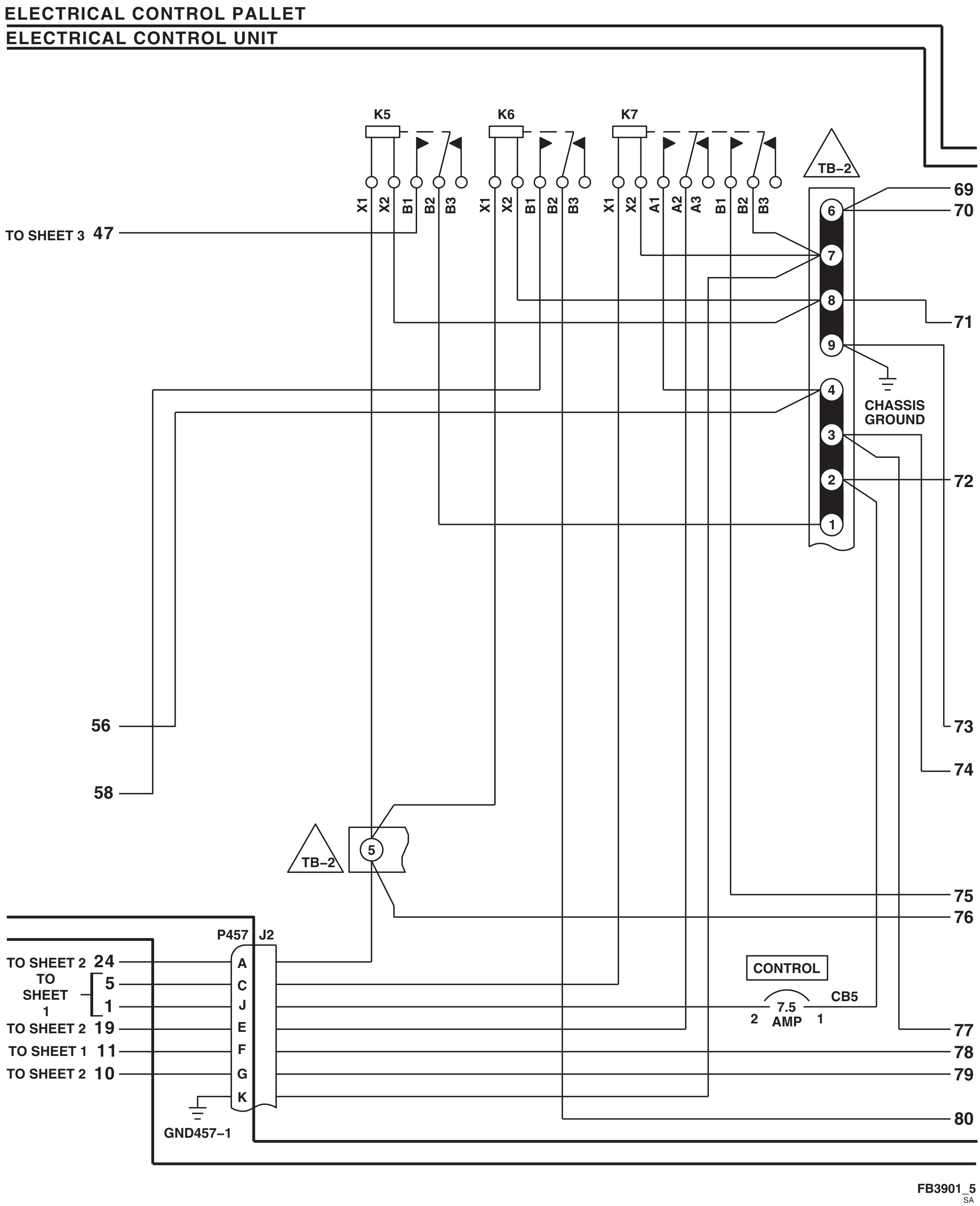

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 9)

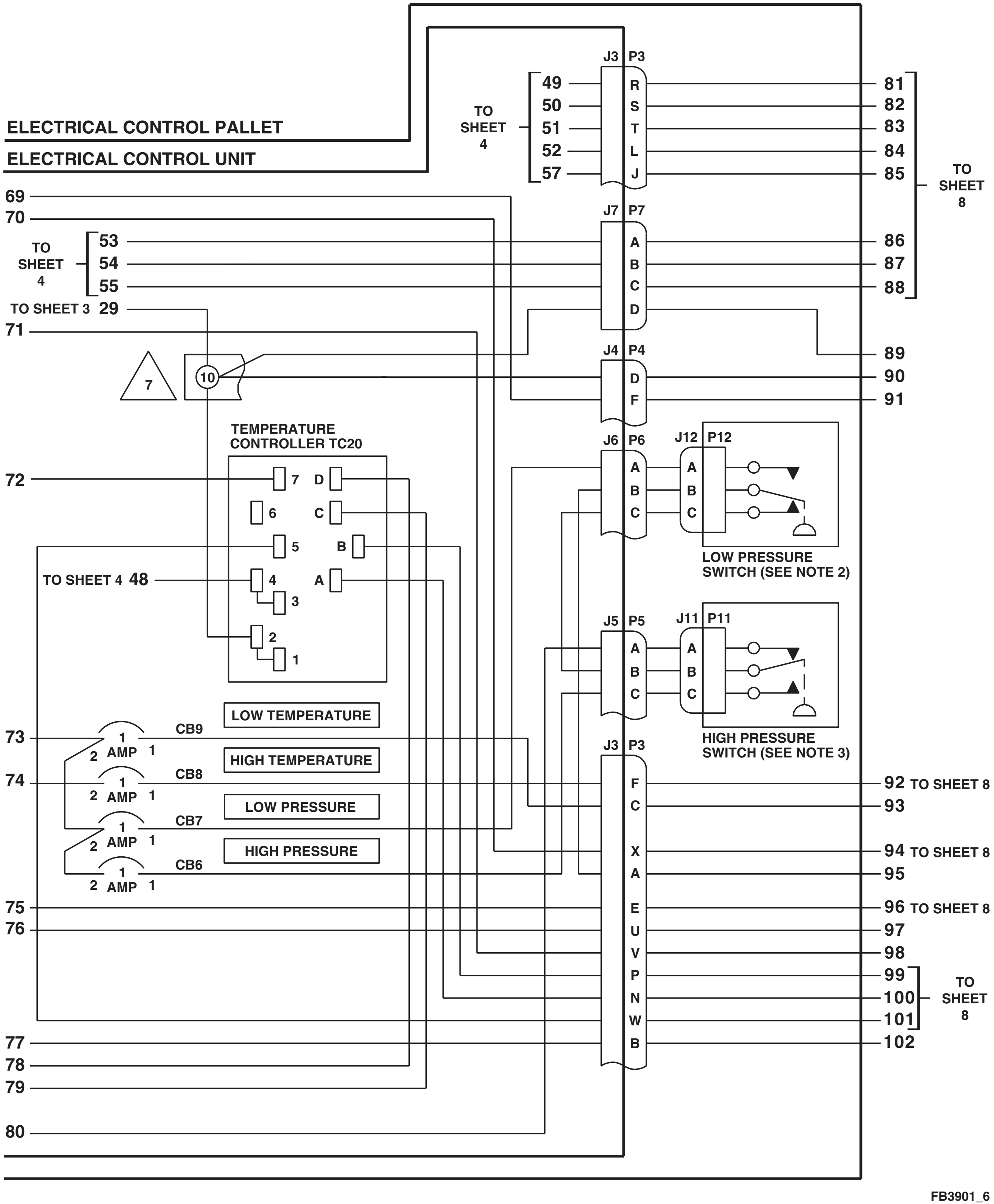

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 9)

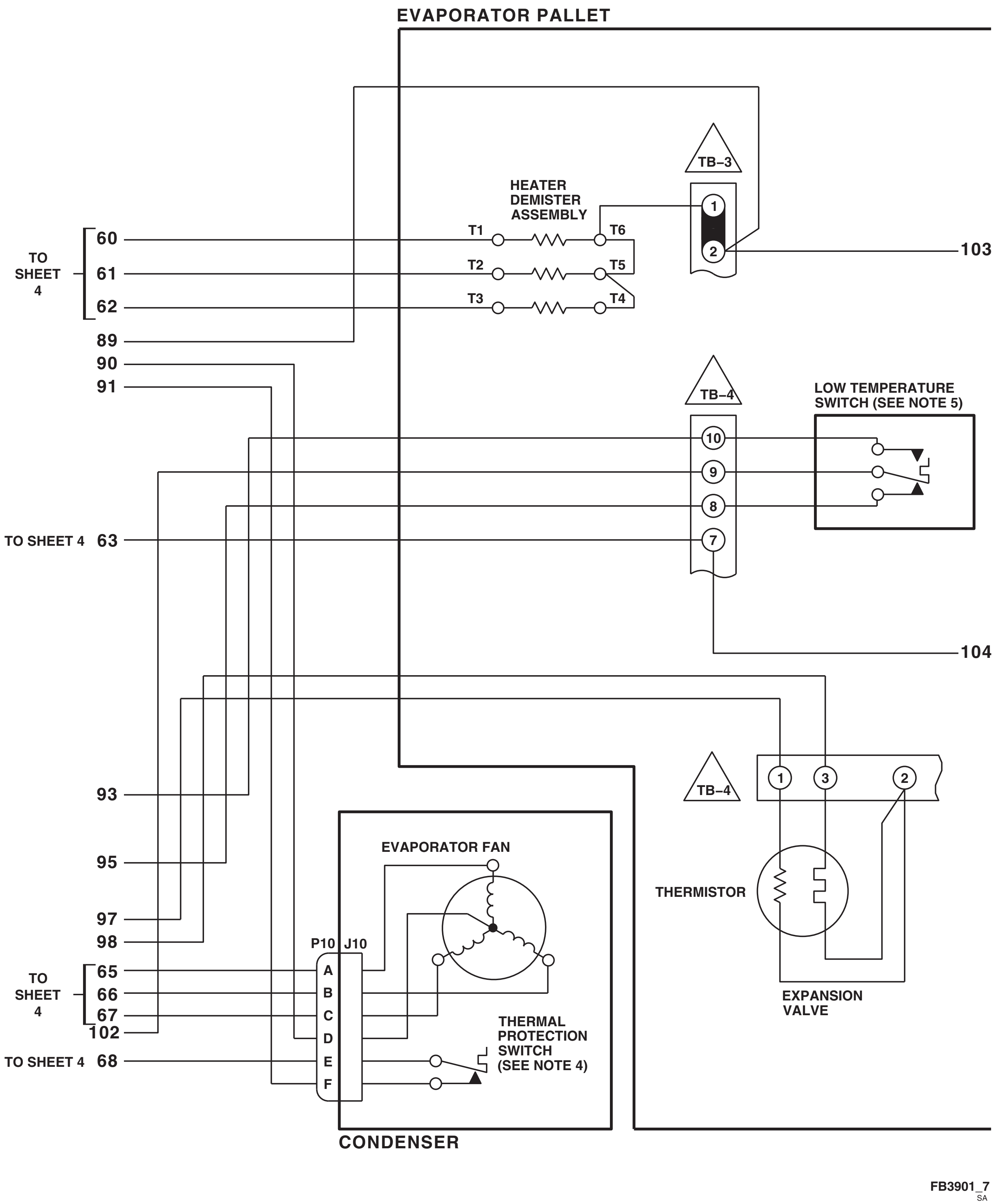

FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 9)

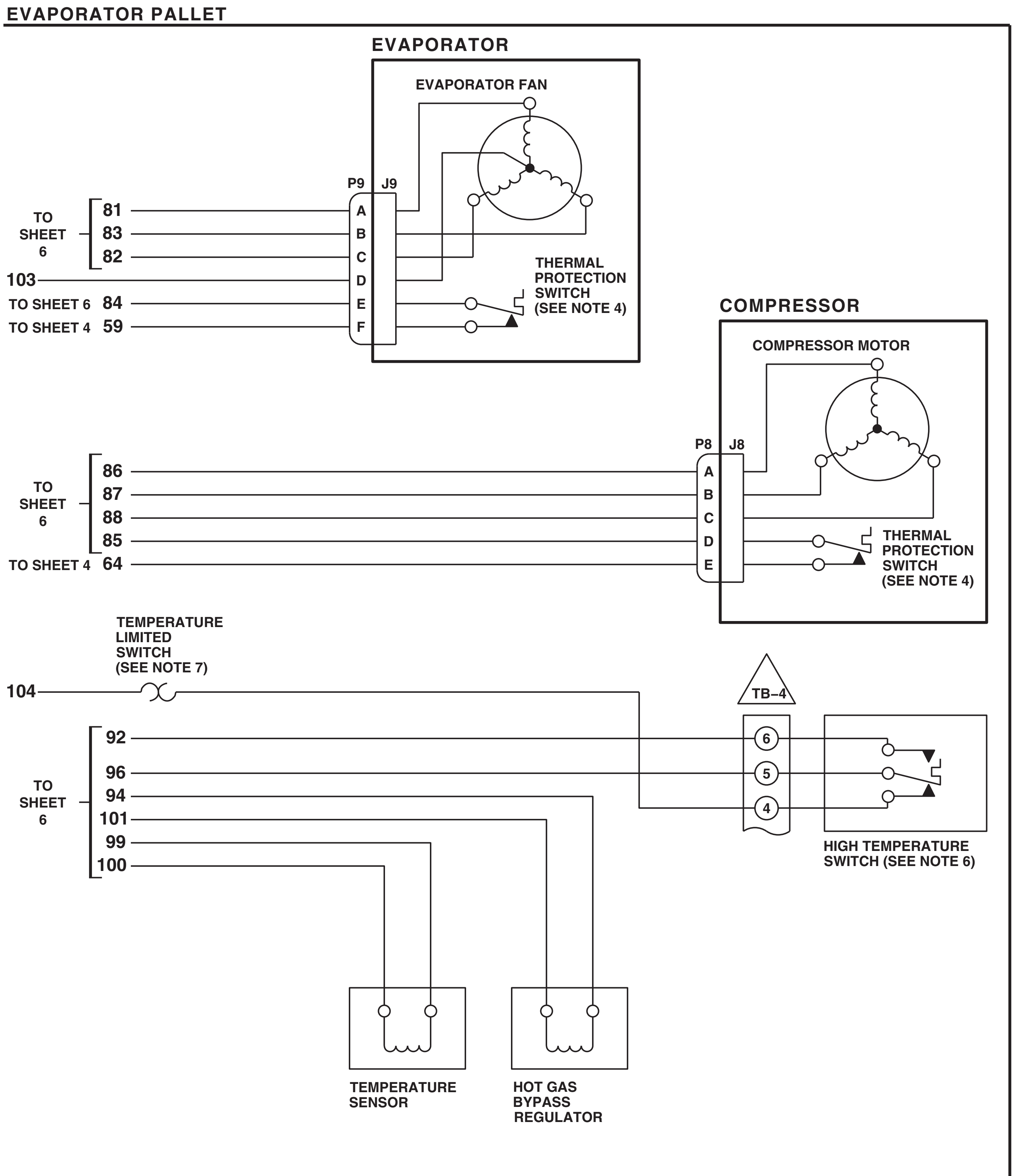

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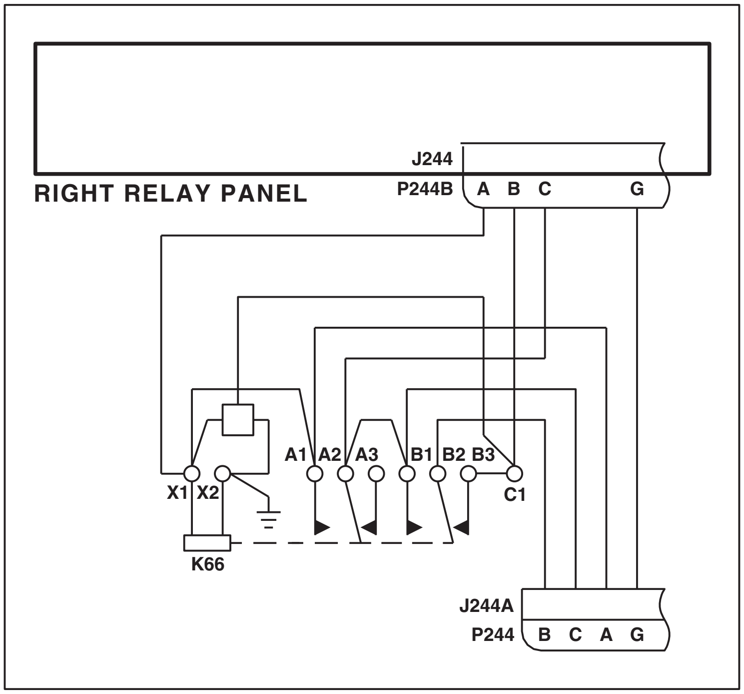
FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 8 OF 9)



DETAIL A
(SEE NOTE 9)

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FIGURE 13-2-2-1. ENVIRONMENTAL CONTROL SYSTEM SCHEMATIC DIAGRAM (SHEET 9 OF 9)

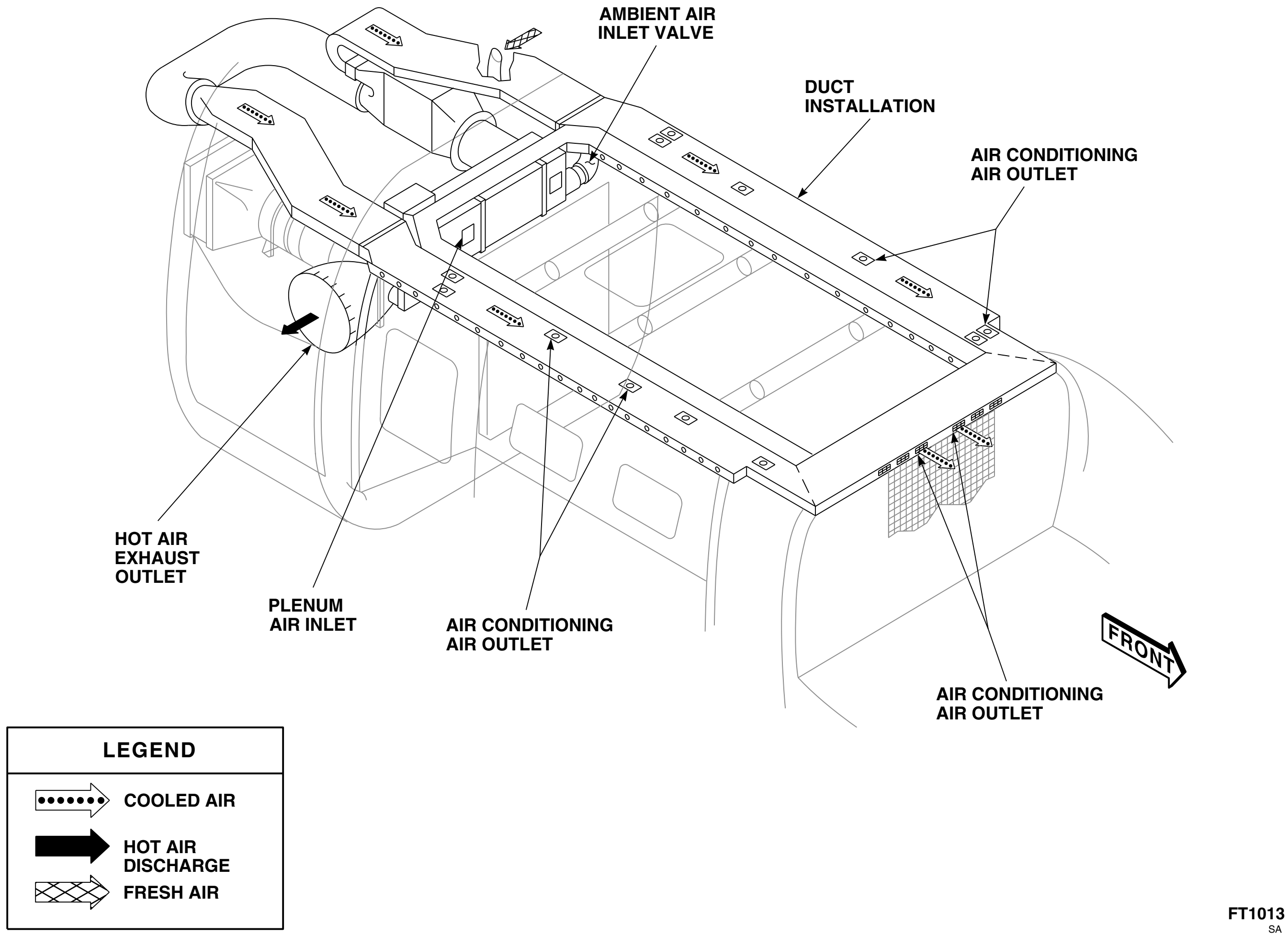


FIGURE 13-2-2-2. ENVIRONMENTAL CONTROL SYSTEM AIRFLOW DIAGRAM

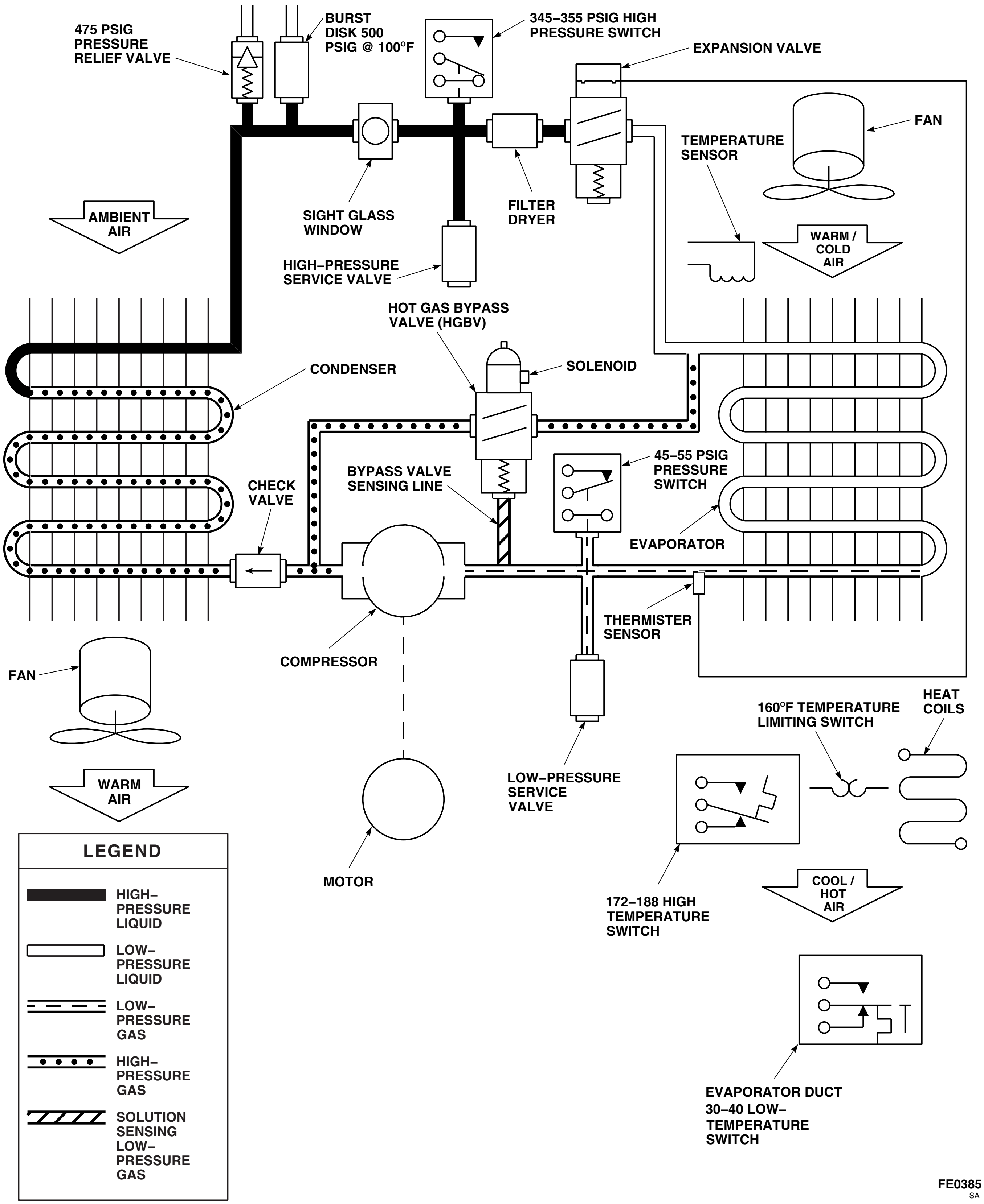


FIGURE 13-2-2-3. ENVIRONMENTAL CONTROL SYSTEM REFRIGERANT FLOW DIAGRAM

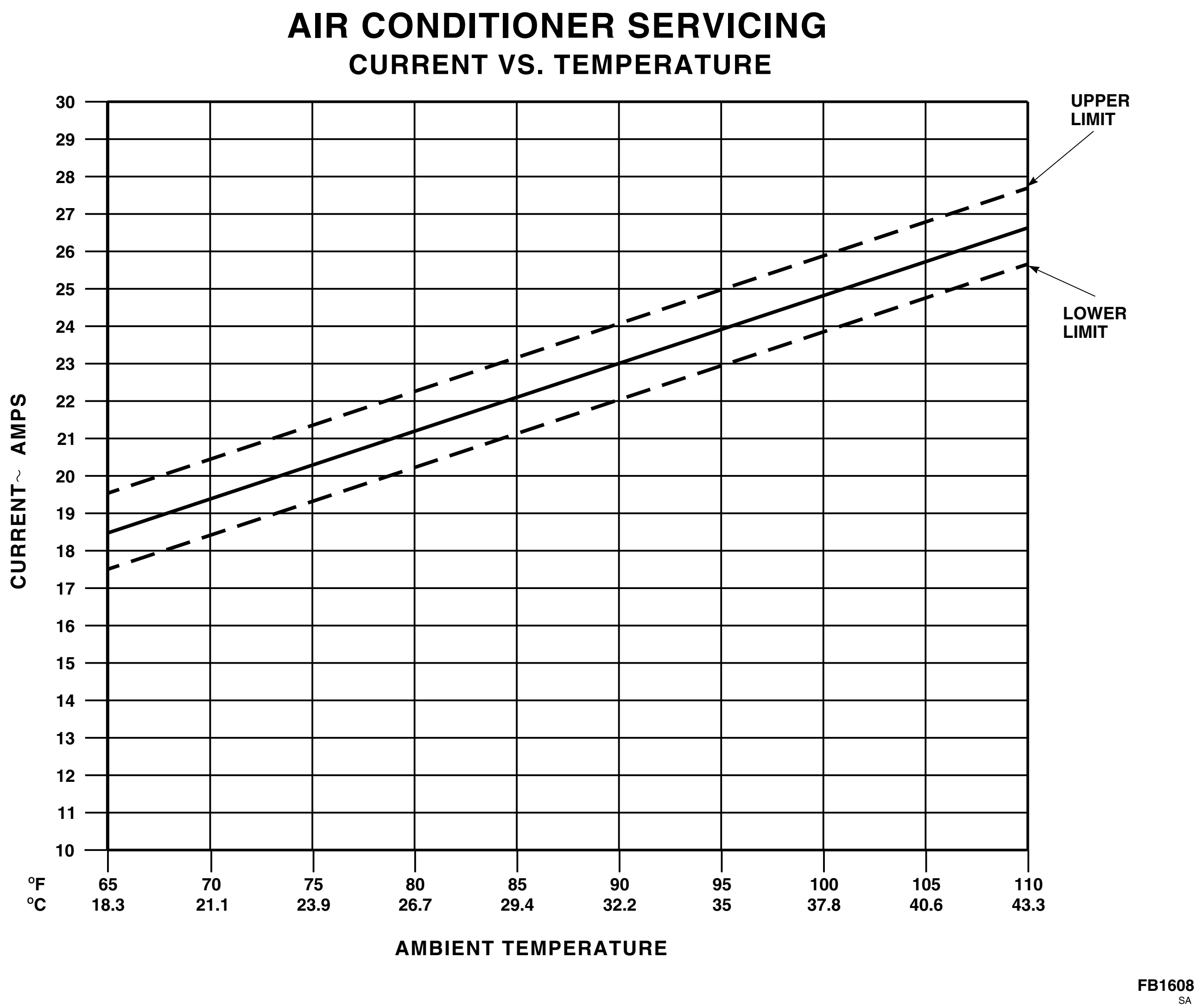


FIGURE 13-2-2-4. AIR CONDITIONER SERVICING CURRENT VS. TEMPERATURE

SECTION 13-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

13-3-1 HEATING AND VENTILATION SYSTEM INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 13-4-4

13-3-1.1 Bleed-Air Tube Inspection.

a. Inspect tubes for dents. **DENTS UP TO 25% OF TUBE DIAMETER ARE ACCEPTABLE.** If any dents exceed limit, replace tube(s) (PARA 13-4-4).

b. Inspect all welds on tubes for cracks. **NONE ALLOWED.** If any cracks at weld joints are found, replace tube(s) (PARA 13-4-4).

c. Inspect tube couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** If coupling is too stiff to slide over tube, or if chafed or torn, or show signs of other damage, replace coupling(s) (PARA 13-4-4).

d. Inspect brackets for bending, cracks, and security. If loose, tighten bracket(s). **NO BENDS OR CRACKS ALLOWED.** If bent or cracked, replace bracket(s).

e. Inspect bleed-air tubing insulation for breaks and tears. **NONE ALLOWED.** If breaks or tears are found, repair or replace insulation (PARA 13-4-4).

13-3-2 ENVIRONMENTAL CONTROL SYSTEM INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 2-4-69

13-3-2.1 Environmental Control System Duct Inspection.

a. Remove aft right bulkhead soundproofing panel (PARA 2-4-69).

b. Inspect condenser pallet intake transition duct, exhaust transition duct, condenser exhaust duct, evaporator pallet inlet duct, and outlet transition ducts for chaffing, damage, stiffness, and tears. **NONE ALLOWED.**

c. Inspect duct couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** Replace duct couplings if too stiff to slide over tube, or if chafed or torn.

d. Check brackets for cracks, bending, and security.

e. Install aft right bulkhead soundproofing panel (PARA 2-4-69).

SECTION 13-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

13-4-1 HEATING AND VENTILATION SYSTEM MIXING VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- File
- Grinding Wheel
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 13-2-1

13-4-1.1 Replace Heating and Ventilation System Mixing Valve.

13-4-1.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect electrical connector, P436, from mixing valve (Figure 13-4-1-1, Sheet 1, Detail B).

d. Install protective caps and plugs on electrical connectors.

e. Loosen clamps securing couplings to mixing valve inlet and outlet ports.

f. Disconnect upper temperature sensing tube from elbow on mixing valve (Figure 13-4-1-1, Sheet 2, Detail D). Install protective cap and plug on tube and elbow.

g. Remove mixing valve as follows:

(1) Loosen clamps and remove tube coupling to disconnect front bleed-air tube from mixing valve (Figure 13-4-1-1, Sheet 2, Detail C).

(2) Remove bolts, washers, and mixing valve from support bracket (Figure 13-4-1-1, Sheet 1, Detail B).

h. If mixing valve is being replaced, remove elbow, nut, and packing (Figure 13-4-1-1, Sheet 2, Detail D). Discard packing.

13-4-1.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If removed, install elbow, nut, and packing on mixing valve. **TORQUE NUT TO 85 - 105 INCH-POUNDS** (Figure 13-4-1-1, Sheet 2, Detail D).

c. Install mixing valve as follows:

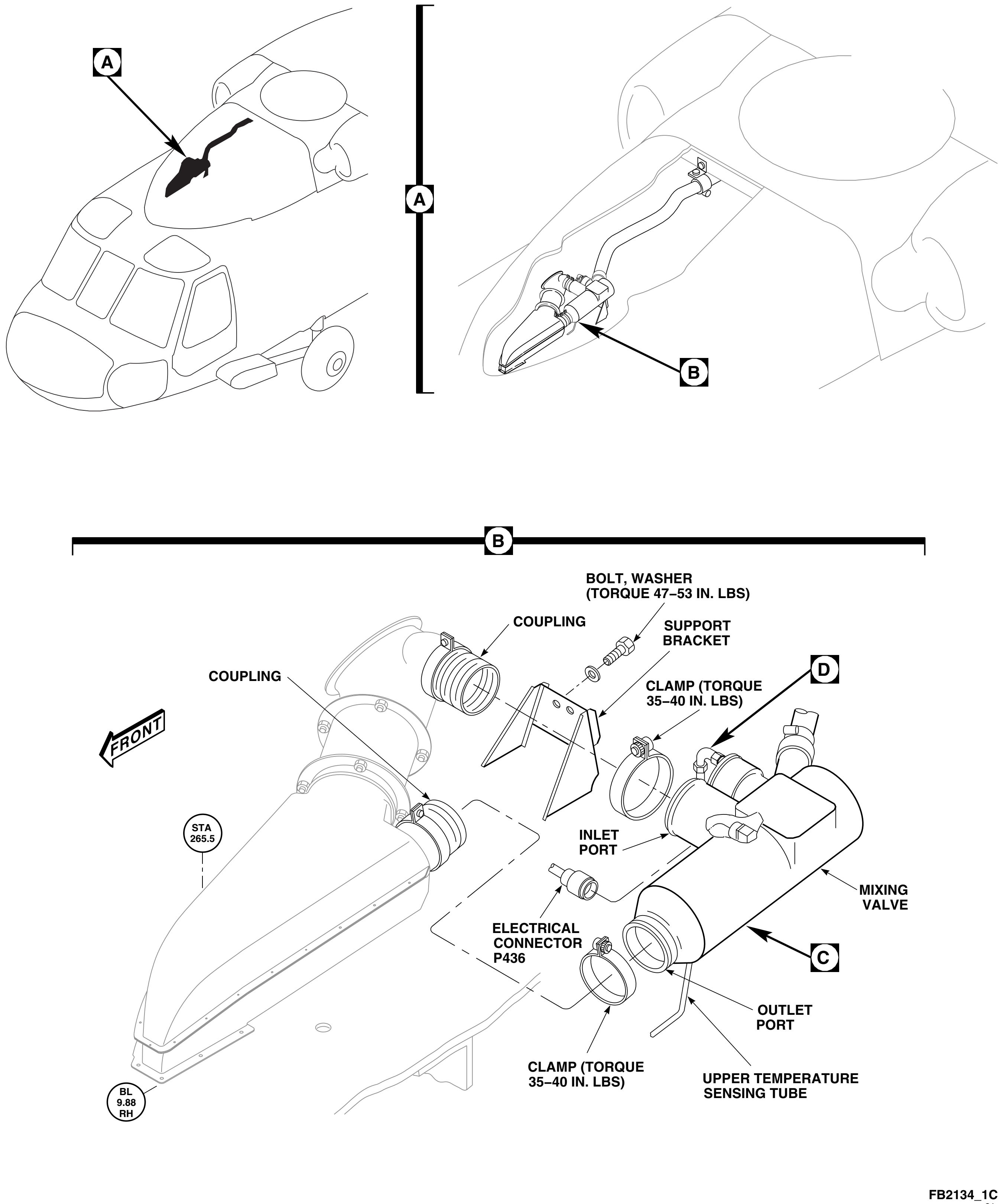


FIGURE 13-4-1-1. REPLACE HEATING AND VENTILATION SYSTEM MIXING VALVE (SHEET 1 OF 2)

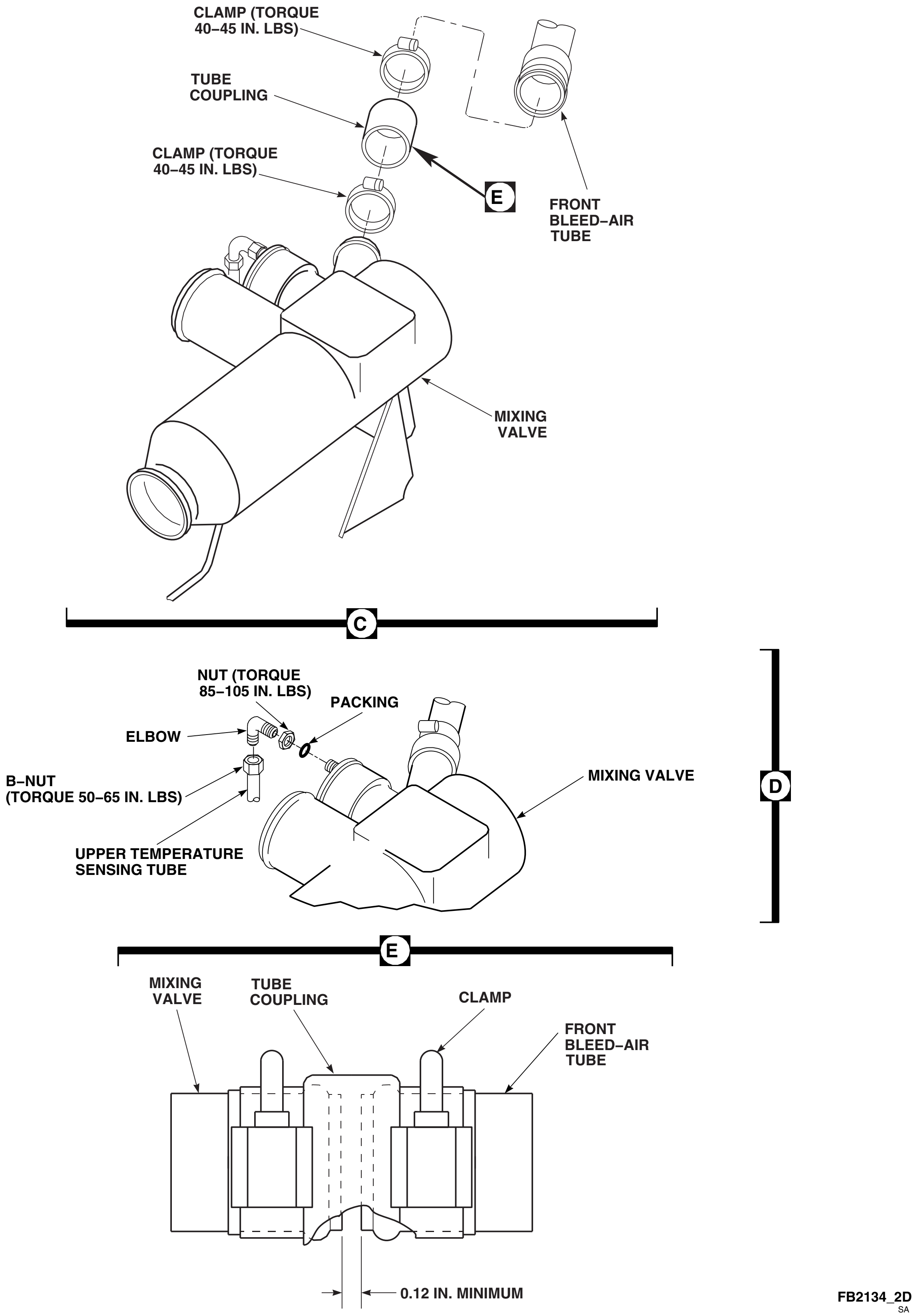


FIGURE 13-4-1-1. REPLACE HEATING AND VENTILATION SYSTEM MIXING VALVE (SHEET 2 OF 2)

(1) Install mixing valve on support bracket with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS** (Figure 13-4-1-1, Sheet 1, Detail B).

NOTE

Before connecting front bleed-air tube to mixing valve, distance between tube and valve must be measured for proper gap.

(2) Align front bleed-air tube with mixing valve. Measure gap between tubes. **GAP SHALL BE A MINIMUM OF 0.12-INCH** (Figure 13-4-1-1, Sheet 2, Detail E).

(3) If gap is less than 0.12-inch, perform the following:

(a) Remove mixing valve.

(b) Remove metal from connecting tube on mixing valve, using grinding wheel or suitable file. Do not remove any material from bead on end of tube.

(4) Repeat step c. (1) through step c. (3) until gap is within limits.

(5) Connect front bleed-air tube to mixing valve with tube coupling and clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS** (Figure 13-4-1-1, Sheet 2, Detail C).

d. Remove protective cap and plug and connect upper temperature sensing tube to elbow on mixing valve. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS** (Figure 13-4-1-1, Sheet 2, Detail D).

e. Make sure couplings are properly positioned over inlet and outlet ports of mixing valve. **TORQUE CLAMPS TO 35 - 40 INCH-POUNDS** (Figure 13-4-1-1, Sheet 1, Detail B).

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connector, P436, to mixing valve.

i. Make sure area is clean and free of foreign material.

j. Perform operational check of heating and ventilation system (PARA 13-2-1).

k. Close and latch main rotor pylon sliding cover.

13-4-2 HEATING AND VENTILATION SYSTEM UPPER TEMPERATURE SENSING TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 13-2-1
- PARA 13-4-3

13-4-2.1 Replace Heating and Ventilation System Upper Temperature Sensing Tube.

13-4-2.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect upper temperature sensing tube from elbow on mixing valve (Figure 13-4-2-1, Detail A). Install protective cap and plug on tube and elbow.

d. Disconnect upper temperature sensing tube from elbow on cabin skin. Install protective cap and plug on tube and elbow.

e. Remove screw, spacer, nut, and clamp securing upper temperature sensing tube to cabin skin. Remove upper temperature sensing tube (Figure 13-4-2-1, Detail B).

f. If required, remove elbow at cabin skin (PARA 13-4-3).

13-4-2.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If required, install elbow on cabin skin (PARA 13-4-3).

c. Remove protective caps and plugs and connect upper temperature sensing tube to elbow on cabin skin and elbow on mixing valve. **TORQUE B-NUTS TO 50 - 65 INCH-POUNDS** (Figure 13-4-2-1, Detail A).

d. Secure upper temperature sensing tube to cabin skin with clamp, screw, spacer, and nut (Figure 13-4-2-1, Detail B).

e. Perform operational check of heating and ventilation system (PARA 13-2-1).

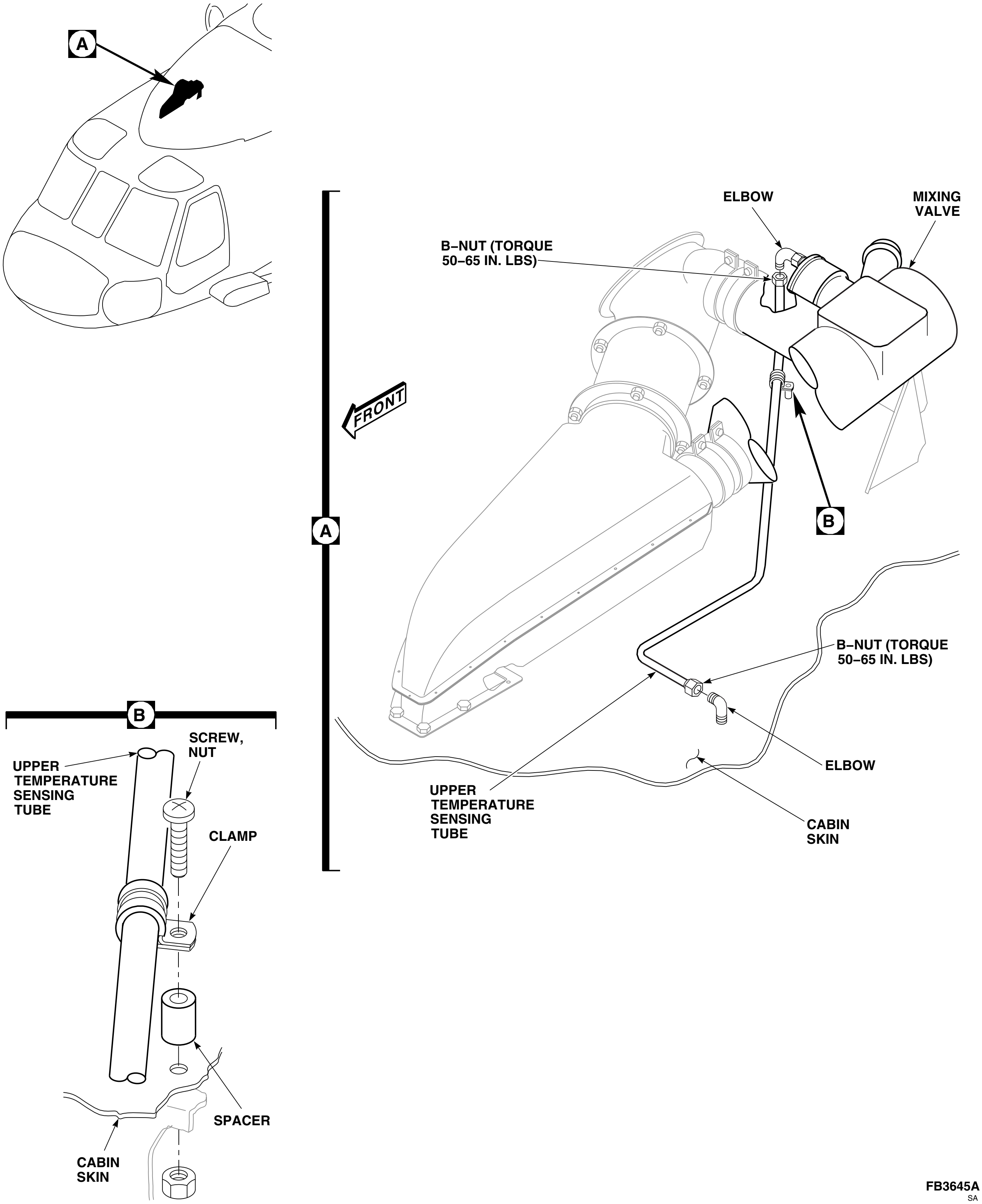


FIGURE 13-4-2-1. REPLACE HEATING AND VENTILATION SYSTEM UPPER TEMPERATURE SENSING TUBE

f. Make sure area is clean and free of foreign material.

g. Close and latch main rotor pylon sliding cover.

13-4-3 HEATING AND VENTILATION SYSTEM LOWER TEMPERATURE SENSING TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-42
- PARA 11-4-43
- PARA 11-4-46
- PARA 13-2-1

13-4-3.1 Replace Heating and Ventilation System Lower Temperature Sensing Tube.

13-4-3.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from STA 247.0 through STA 265.0 in cabin overhead (PARA 2-4-69).

CAUTION

Rigging of main and tail rotor controls will be affected if cabin control rod lengths are changed. Make sure rod lengths on cabin control rods are not changed when disconnected.

c. Remove yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

d. Disconnect lower temperature sensing tube from elbow in cabin skin and from union on mixture temperature sensor (Figure 13-4-3-1, Detail B). Install protective plugs in tube and caps on elbow and union.

e. If required, remove elbow from cabin skin as follows:

(1) Disconnect upper temperature sensing tube from elbow on cabin skin. Install protective plug in tube.

(2) Remove sealing compound with nonmetallic scraper.

(3) From inside cabin, remove nut and washer from elbow.

(4) On cabin skin, remove sealing washers and elbow. Install protective caps on elbow.

13-4-3.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If removed, install elbow on cabin skin as follows (Figure 13-4-3-1, Detail B):

(1) Wipe cabin skin area with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D. Wipe area dry with machinery towel, Item 211, Appendix D.

(2) On cabin skin, remove protective caps and install elbow and sealing washers.

(3) From inside cabin, install washer and nut and position elbow for mating with upper temperature sensing tube. **TORQUE NUT TO 85 - 105 INCH-POUNDS.**

(4) Remove protective plug and connect upper temperature sensing tube to elbow on cabin skin. **TORQUE B-NUT TO 50 - 65 INCH POUNDS.**

(5) Apply sealing compound, Item 292, Appendix D, around top of elbow where it passes through cabin skin. Wipe off excess sealing compound.

c. From inside cabin, remove protective caps and plugs and connect lower temperature sensing tube to elbow in overhead cabin and to union on mixture temperature sensor. **TORQUE B-NUTS TO 50 - 65 INCH-POUNDS.**

d. Install yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

e. Perform operational check of heating and ventilation system (PARA 13-2-1).

f. Make sure area is clean and free of foreign material.

g. Install soundproofing panels (PARA 2-4-69).

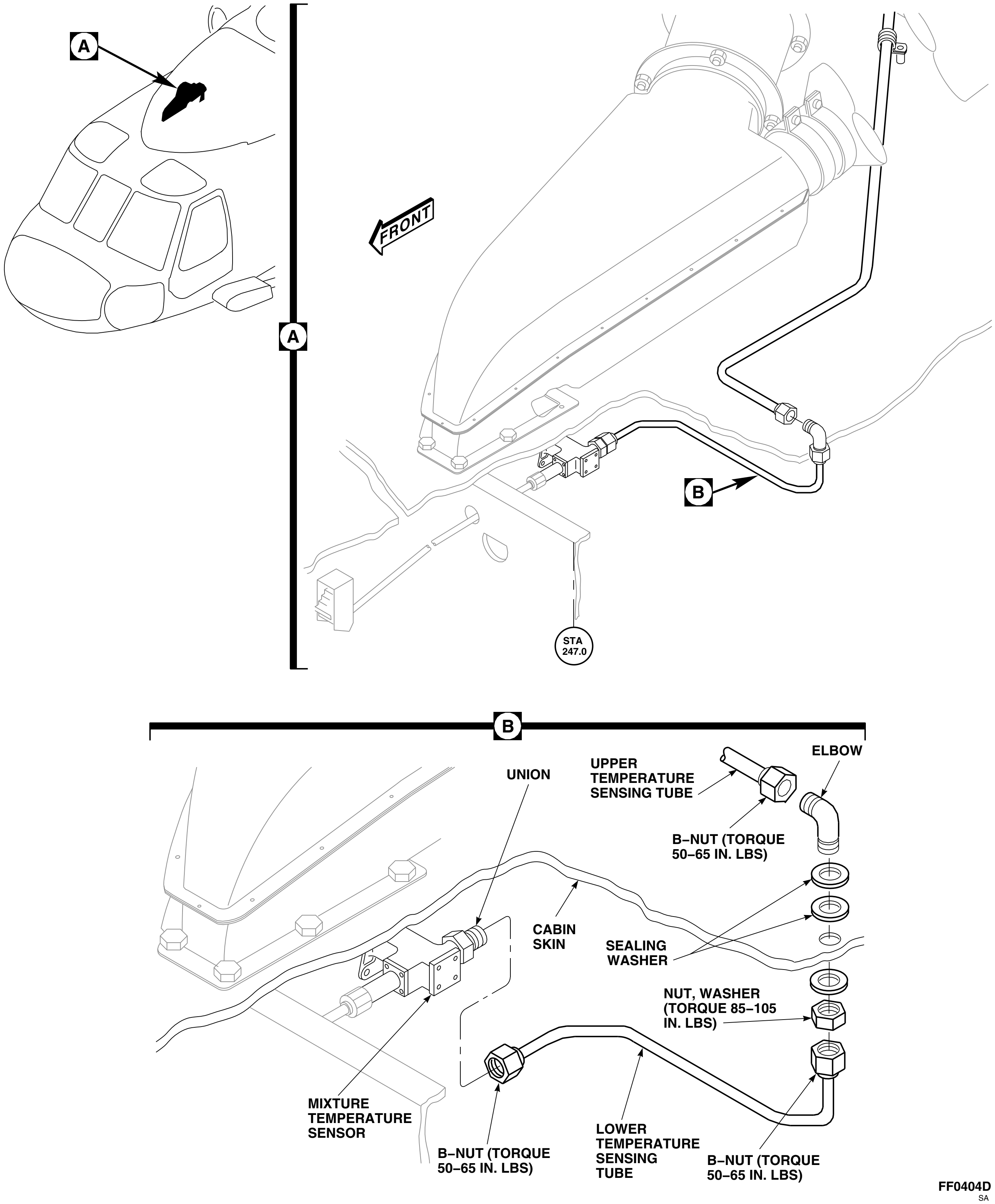


FIGURE 13-4-3-1. REPLACE HEATING AND VENTILATION SYSTEM LOWER TEMPERATURE SENSING TUBE

13-4-4 HEATING AND VENTILATION SYSTEM BLEED-AIR TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- File
- Grinding Wheel
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Tape, Insulation, Item 320, Appendix D
- Thermal Insulation, 70309-02108-101

References

- Appendix D
- PARA 1-6-16
- PARA 13-2-1

13-4-4.1 Replace Heating and Ventilation System Bleed-Air Tube.

13-4-4.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Open oil cooler compartment access doors.

NOTE

Cap or plug all tubes, ports, and fittings as parts are removed and disconnected.

d. Remove clamps and tube coupling to disconnect front bleed-air tube from mixing valve (Figure 13-4-4-1, Sheet 1, Detail B).

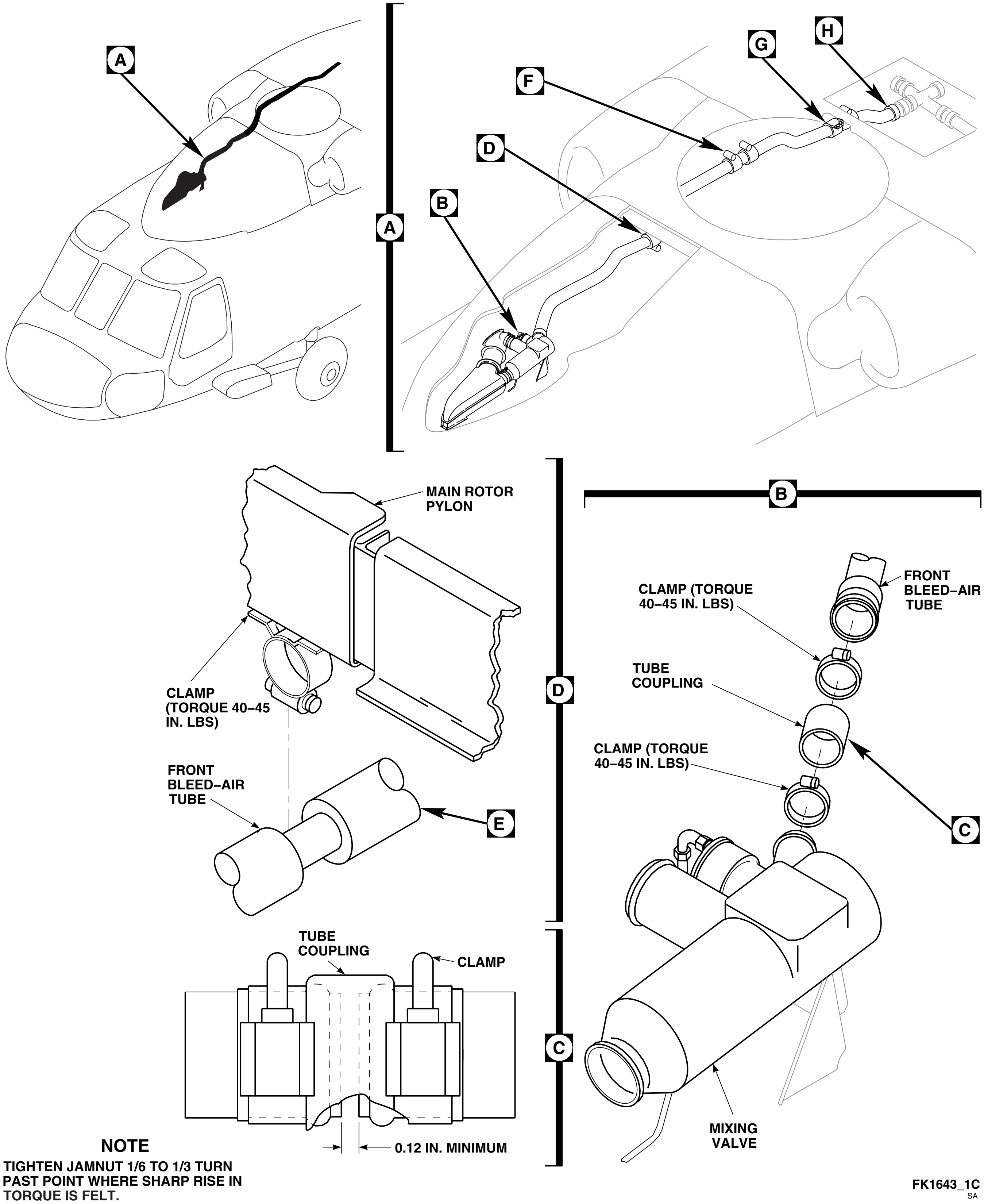
e. On main rotor pylon, remove front bleed-air tube from clamp (Figure 13-4-4-1, Sheet 1, Detail D).

f. 70588–SUBQ To disconnect front bleed-air tube, perform the following:

(1) Loosen clamps on left and right bleed-air lines (Figure 13-4-4-1, Sheet 2, Detail E).

(2) Disconnect bleed-air lines from tee on front bleed-air tube. Gently flex bleed-air lines over cone of tee.

(3) Remove tee from front bleed-air tube. Remove packing and jamnut from tee. Cap or plug tee. Discard packing.



NOTE

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

FIGURE 13-4-4-1. REPLACE HEATING AND VENTILATION SYSTEM BLEED-AIR TUBE (SHEET 1 OF 2)

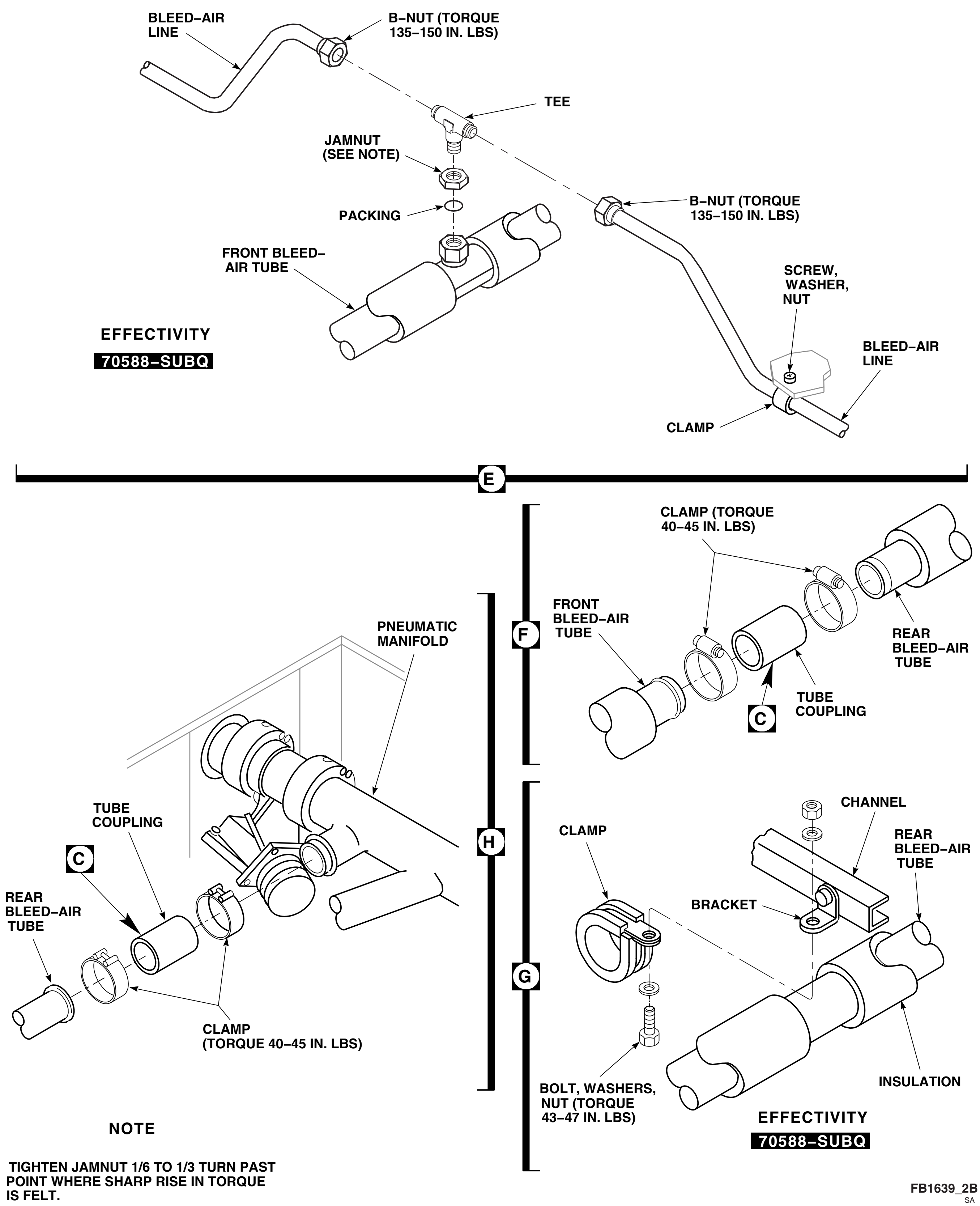


FIGURE 13-4-4-1. REPLACE HEATING AND VENTILATION SYSTEM BLEED-AIR TUBE (SHEET 2 OF 2)

(4) Remove bolt, washers, clamp, and nut securing rear bleed-air tube to bracket on channel (Figure 13-4-4-1, Sheet 2, Detail G).

g. On right side of transmission area, remove clamps securing tube coupling connecting front and rear bleed-air tubes (Figure 13-4-4-1, Sheet 2, Detail F). Remove tube coupling.

h. Slide front bleed-air tube toward front and remove from helicopter.

i. Remove clamps connecting tube coupling to rear bleed-air tube and pneumatic manifold. Remove rear bleed-air tube from helicopter (Figure 13-4-4-1, Sheet 2, Detail H).

13-4-4.1.2 Repair.

NOTE

Do not compress insulation when wrapping tube.

a. Using insulation tape, Item 320, Appendix D, repair minor breaks or tears in insulation.

b. To replace thermal insulation, perform the following:

(1) Remove all old thermal insulation from tube.

(2) Cut wedges out of new thermal insulation, as required, to fit insulation over tube bends.

(3) Tape all seams with insulation tape, Item 320, Appendix D, being careful not to compress insulation.

(4) Apply full wrap (about 1 ½ turns) of tape at each end of thermal insulation, and at 1-foot intervals.

13-4-4.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Remove caps and plugs from tubes, ports, and fittings as parts are removed.

b. Using filtered, dry compressed air, clean all tubes and mixing valve ports.

c. Position rear bleed-air tube (Figure 13-4-4-1, Sheet 2, Detail H).

d. Connect rear bleed-air tube to pneumatic manifold with tube coupling and clamps. Leave clamps loose.

e. Position front bleed-air tube between mixing valve and rear bleed-air tube (Figure 13-4-4-1, Sheet 1, Detail A).

f. 70588–SUBQ To install front bleed-air tube, perform the following:

(1) Using petrolatum, Item 234, Appendix D, lubricate packing. Install jamnut and packing on tee (Figure 13-4-4-1, Sheet 2, Detail E).

(2) Using antiseize compound, Item 74, Appendix D, coat threads of tee, and install tee in front bleed-air tube.

(3) Connect bleed-air lines to tee in front bleed-air tube. **DO NOT TORQUE NUTS NOW.**

(4) **TIGHTEN JAMNUT ON TEE ⅙ - ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(5) Install front bleed-air tube in clamp on main rotor pylon. **TORQUE CLAMP TO 40 - 45 INCH-POUNDS** (Figure 13-4-4-1, Sheet 1, Detail D).

(6) **TORQUE B-NUTS ON BLEED-AIR LINES TO 135 - 150 INCH-POUNDS** (Figure 13-4-4-1, Sheet 2, Detail E). **TIGHTEN CLAMPS ON BLEED AIR LINES.**

NOTE

Before connecting tube sections, distance between tubes must be measured for proper gap.

g. Align front and rear bleed-air tubes. Measure gap at rear bleed-air tube to pneumatic manifold, mixer valve connection to front bleed-air tube, and between front and rear bleed-air tubes (Figure 13-4-4-1, Sheet 1, Detail C). **GAP MUST BE A MINIMUM OF 0.12-INCH AT EACH CONNECTION.**

h. If possible, obtain 0.12-inch minimum gap by readjusting tubes. If gap is still less than 0.12-inch, perform the following:

(1) Remove front or rear bleed-air tube.

CAUTION

Bleed-air tube will be damaged if bead on end of tube is removed. Do not remove any material from bead on end of tube.

(2) Remove metal from end of tube using grinding wheel or suitable file.

(3) Install tube.

(4) Repeat step h. (1) through h. (3) until gap is within limit.

i. Connect front and rear tubes with tube couplings and clamps. **TORQUE CLAMPS AT FRONT AND REAR TUBE CONNECTION AND AT PNEUMATIC MANIFOLD TO 40 - 45 INCH-POUNDS** (Figure 13-4-4-1, Sheet 2, Detail F and Detail H).

j. 70588–SUBQ Clamp rear bleed-air tube to channel with clamp, bolt, washers, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 13-4-4-1, Sheet 2, Detail G).

k. Connect front bleed-air tube to mixing valve with tube coupling and clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS** (Figure 13-4-4-1, Sheet 1, Detail B).

l. Perform operational check of heating and ventilation system (PARA 13-2-1).

m. Make sure area is clean and free of foreign material.

n. Close main rotor pylon sliding cover.

o. Close oil cooler compartment access doors.

13-4-5 HEATING AND VENTILATION SYSTEM MIXTURE TEMPERATURE SENSOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-42
- PARA 11-4-43
- PARA 11-4-46
- PARA 13-2-1
- PARA 13-4-9

13-4-5.1 Replace Heating and Ventilation System Mixture Temperature Sensor.

13-4-5.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Remove soundproofing panels from cabin overhead from STA 247.0 through STA 265.0 (PARA 2-4-69).

CAUTION

Rigging of main and tail rotor controls will be affected if cabin control rod lengths are changed. Make sure rod length on cabin control rods are not changed when disconnected.

d. Remove yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

e. Remove heating and ventilation system heater muffler (PARA 13-4-9).

f. From inside cabin, disconnect heater control shaft from mixture temperature sensor (Figure 13-4-5-1, Detail B).

g. Disconnect lower temperature sensing tube from mixture temperature sensor. Install protective cap and plug in tube and port.

h. Remove bolts and washers securing mixture temperature sensor to interior cabin duct. Remove sensor.

13-4-5.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. From inside cabin, position replacement mixture temperature sensor in interior cabin duct (Figure 13-4-5-1, Detail B).

c. Install bolts and washers securing mixture temperature sensor to duct. **TORQUE BOLTS TO 48 - 53 INCH-POUNDS.**

d. Remove protective cap and plug and connect lower temperature sensing tube to mixture temperature sensor. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**

e. Connect heater control shaft to mixture temperature sensor and adjust control knob as follows:

(1) Install slot in end of heater control shaft, on key in mixture temperature sensor. **INSTALL AND TIGHTEN SERRATED NUT.**

(2) On upper console, turn heater control knob counterclockwise until it stops (Figure 13-4-5-1, Detail C). Do not force knob.

(3) Loosen setscrews of heater control knob and line up mark on knob with OFF position on information plate. **TIGHTEN SETSCREWS.**

f. Install yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

g. Inspect and install heating and ventilation system heater muffler (PARA 13-4-9).

h. Perform operational check of heating and ventilation system (PARA 13-2-1).

i. Make sure area is clean and free of foreign material.

j. Install soundproofing panels (PARA 2-4-69).

k. Close main rotor pylon sliding cover.

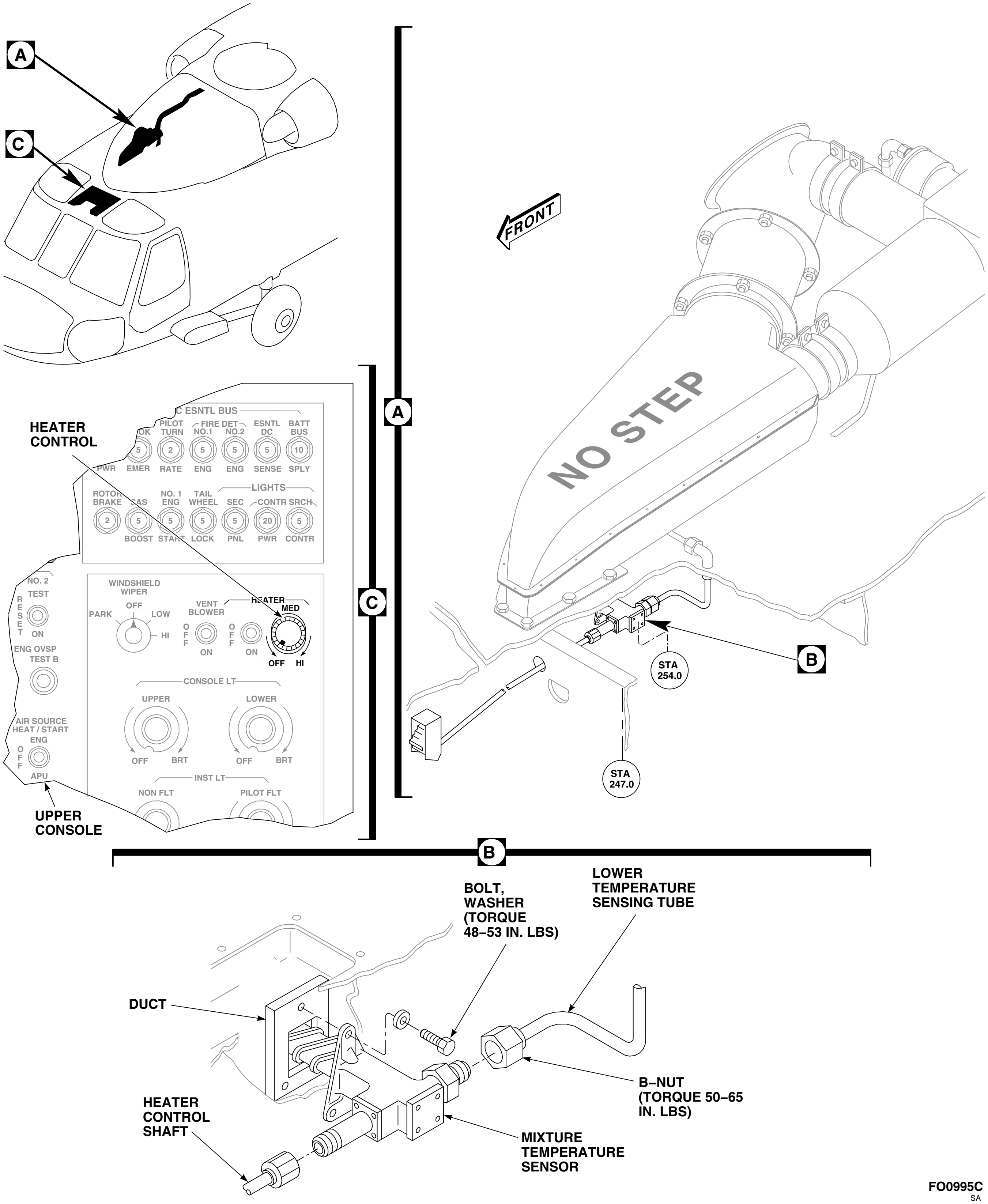


FIGURE 13-4-5-1. REPLACE HEATING AND VENTILATION SYSTEM MIXTURE TEMPERATURE SENSOR

13-4-6 HEATING AND VENTILATION SYSTEM AIR OUTLETS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit

References

- PARA 1-6-16

13-4-6.1 Replace Heating and Ventilation System Air Outlets.

NOTE

Heating and ventilating system has eight air outlets in different locations. Replacement is the same at each location, except some are installed with rivets, some with screws.

13-4-6.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws or rivets securing air outlets to heating/ventilating duct in cabin front section ceiling (Figure 13-4-6-1, Detail A or Detail B). Remove air outlet.

13-4-6.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install air outlet on heating/ventilating duct with screws or rivets (Figure 13-4-6-1, Detail A or Detail B).

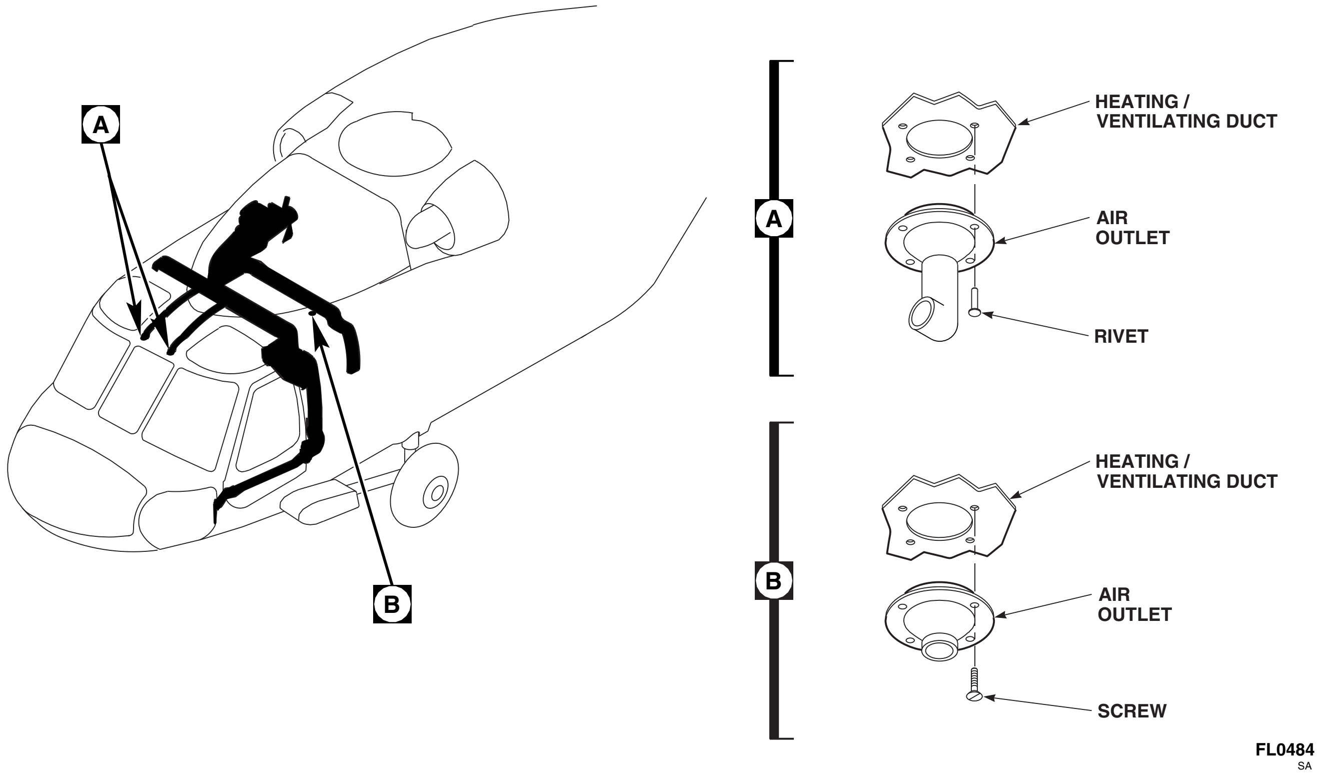


FIGURE 13-4-6-1. REPLACE HEATING AND VENTILATION SYSTEM AIR OUTLETS

13-4-7 HEATING AND VENTILATION SYSTEM HEATER CONTROL SHAFT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Spiral Wrap, Item 314, Appendix D
- Tiedown Strap, Item 328, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-42
- PARA 11-4-43
- PARA 11-4-46
- PARA 13-2-1

13-4-7.1 Replace Heating and Ventilation System Heater Control Shaft.

13-4-7.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from cabin overhead (PARA 2-4-69).

CAUTION

Rigging of main and tail rotor controls will be affected if cabin control rod lengths are changed. Make sure rod length on cabin control rods are not changed when disconnected.

c. Remove yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

d. Loosen fasteners securing upper console and carefully lower (Figure 13-4-7-1, Detail A).

e. Remove tiedown straps securing ends of heater control shaft.

f. Disconnect heater control shaft from adapter (Figure 13-4-7-1, Detail B).

g. In cabin, disconnect heater control shaft from mixture temperature sensor.

h. Slide heater control shaft out of grommet in frame (Figure 13-4-7-1, Detail A). Remove heater control shaft from helicopter.

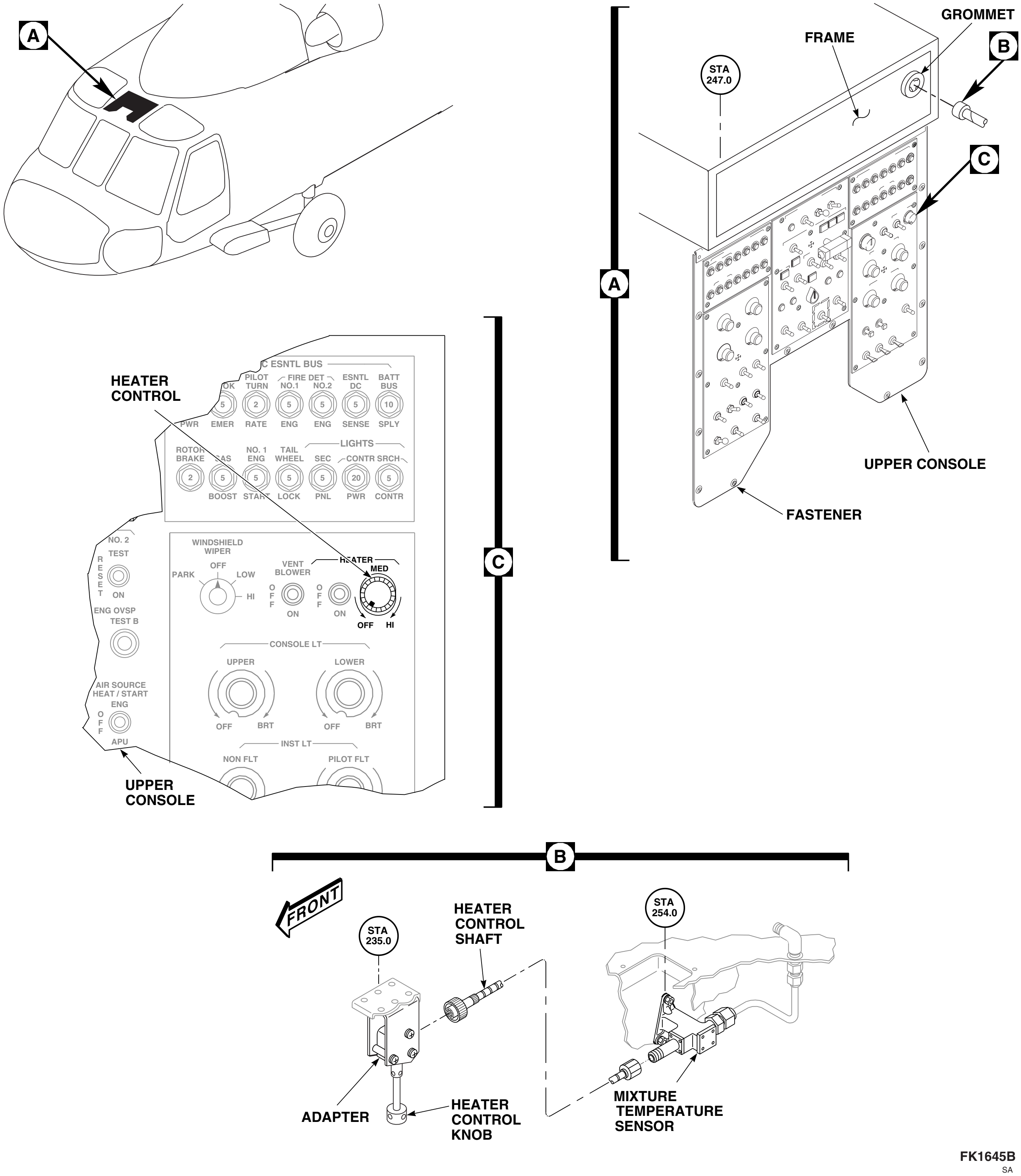


FIGURE 13-4-7-1. REPLACE HEATING AND VENTILATION SYSTEM HEATER CONTROL SHAFT

13-4-7.1.2 Inspect.

Inspect grommet in frame for damage or deterioration (Figure 13-4-7-1, Detail A). **NONE ALLOWED.** If damaged or deteriorated, replace grommet.

13-4-7.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install spiral wrap, Item 314, Appendix D, on heater control shaft before installing through grommet.

c. Insert heater control shaft through grommet in frame at rear of upper console (Figure 13-4-7-1, Detail A).

d. In cabin, connect heater control shaft to mixture temperature sensor (Figure 13-4-7-1, Detail B).

e. Connect opposite end of heater control shaft to adapter.

f. Install tiedown straps, Item 328, Appendix D, on each end of heater control rod shaft.

g. Make sure area is clean and free of foreign material.

h. Carefully raise upper console and secure with fasteners (Figure 13-4-7-1, Detail A).

i. Line up heater control with mixture temperature sensor as follows:

(1) On upper console, turn heater control knob counterclockwise until it stops (Figure 13-4-7-1, Detail C). Do not force knob.

(2) Loosen setscrews of heater control knob and line up mark on knob with OFF position on information plate.

(3) **TIGHTEN SETSCREWS.**

j. Install yaw control rod (PARA 11-4-46) and both cyclic control rods (PARA 11-4-42 and PARA 11-4-43).

k. Perform operational check of heating and ventilation system (PARA 13-2-1).

l. Make sure area is clean and free of foreign material.

m. Install soundproofing panels (PARA 2-4-69).

13-4-8 HEATING AND VENTILATION SYSTEM HEATER CONTROL ADAPTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16
- PARA 13-2-1

13-4-8.1 Replace Heating and Ventilation System Heater Control Adapter.

13-4-8.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Loosen fasteners securing upper console and carefully lower (Figure 13-4-8-1, Detail A).

c. Disconnect heater control shaft from adapter (Figure 13-4-8-1, Detail C).

d. Loosen setscrews and remove heater control knob and shaft from adapter.

e. Remove front screw, washers, nut, and spacer from attachment bracket.

f. Loosen remaining two screws.

g. Slide adapter out from between remaining spacers in attachment bracket and remove adapter from attachment bracket.

13-4-8.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Slide adapter in between spacers in attachment bracket (Figure 13-4-8-1, Detail C). Install spacer between bracket ears with front screw, washers, and nut.

c. **TIGHTEN REMAINING TWO SCREWS.**

d. Install heater control knob and shaft into adapter. **TIGHTEN SETSCREWS.**

e. Connect heater control shaft to adapter.

f. Carefully line up upper console to make sure heater control knob and shaft line up with hole in upper console (Figure 13-4-8-1, Detail A).

g. If heater control knob and shaft do not line up with hole in upper console, perform the following (Figure 13-4-8-1, Detail C):

(1) Loosen front screw in attachment bracket.

(2) Reposition adapter in bracket until they line up.

(3) **TIGHTEN SCREW.**

h. Make sure area is clean and free of foreign material.

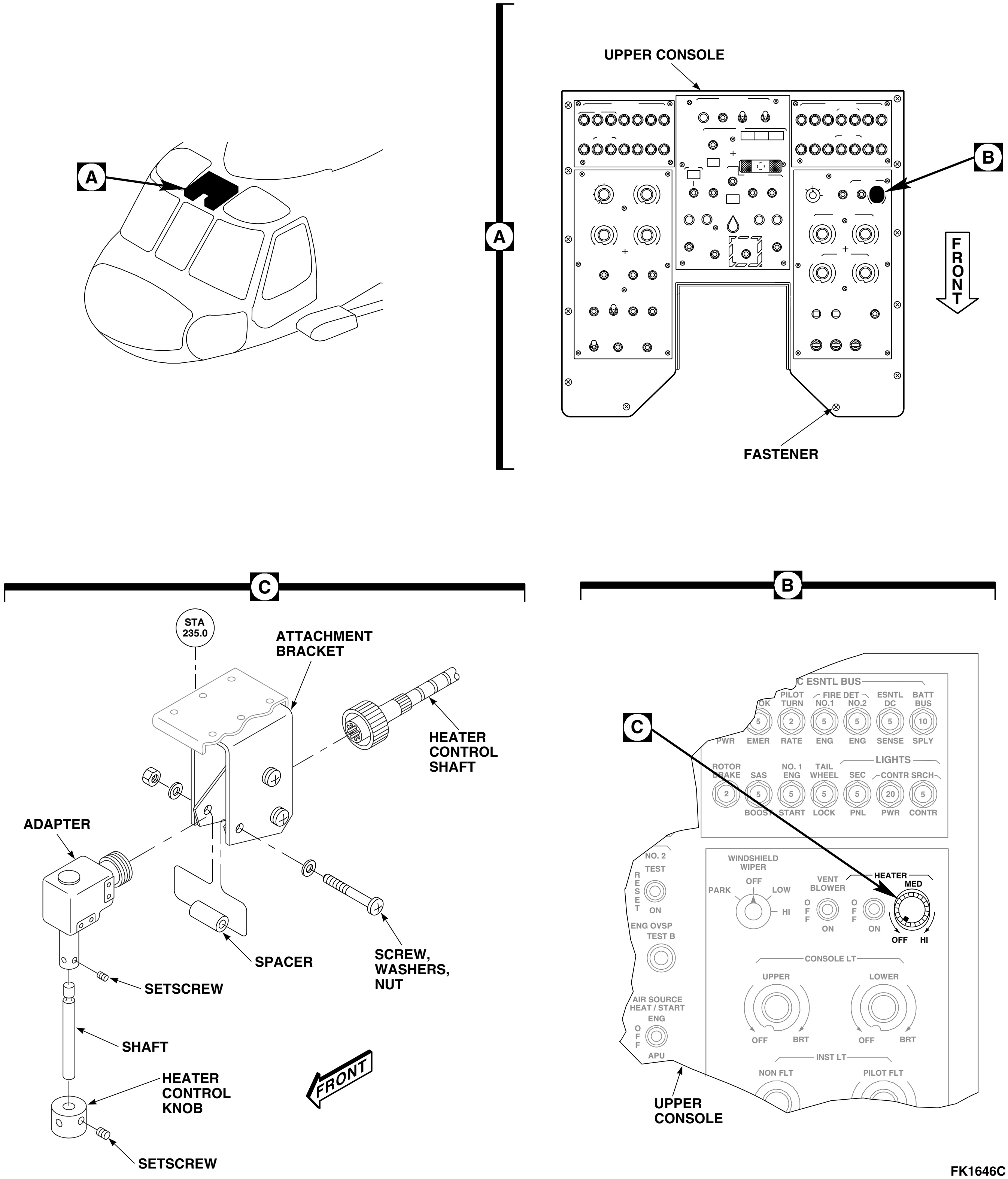


FIGURE 13-4-8-1. REPLACE HEATING AND VENTILATION SYSTEM HEATER CONTROL ADAPTER

i. Carefully raise upper console and secure with fasteners (Figure 13-4-8-1, Detail A).

j. Line up heater control with mixture temperature sensor as follows:

CAUTION

Heater control knob will be damaged if turned past stop. Do not attempt to turn knob past stop.

(1) On upper console, turn heater control knob counterclockwise until it stops (Figure 13-4-8-1, Detail B).

(2) Loosen setscrews of heater control knob and line up mark on knob with OFF position on information plate. **TIGHTEN SETSCREWS.**

k. Perform operational check of heating and ventilation system (PARA 13-2-1).

13-4-9 HEATING AND VENTILATION SYSTEM HEATER MUFFLER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 57, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 13-4-14

13-4-9.1 Replace Heating and Ventilation System Heater Muffler.

13-4-9.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Remove bolts and washers securing muffler to cabin skin and move wire and bracket to one side. Note location of wire and bracket for installation (Figure 13-4-9-1, Detail A).

d. Loosen clamps securing coupling to muffler. Slip coupling off muffler.

e. Remove bolts, washers, and nuts securing muffler to heater blower. Carefully remove muffler from helicopter.

f. Use cleaning cloth, Item 102, Appendix D, dampened in methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper to remove old adhesive from hardware, muffler, and cabin skin.

13-4-9.1.2 Inspect.

a. Check condition of rubber seals on cabin skin for damage. **NONE ALLOWED.** If damaged, replace seals. (PARA 13-4-14).

b. Check interior of duct for foreign material. **NONE ALLOWED.** If any evidence of obstruction is found, remove as necessary.

13-4-9.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

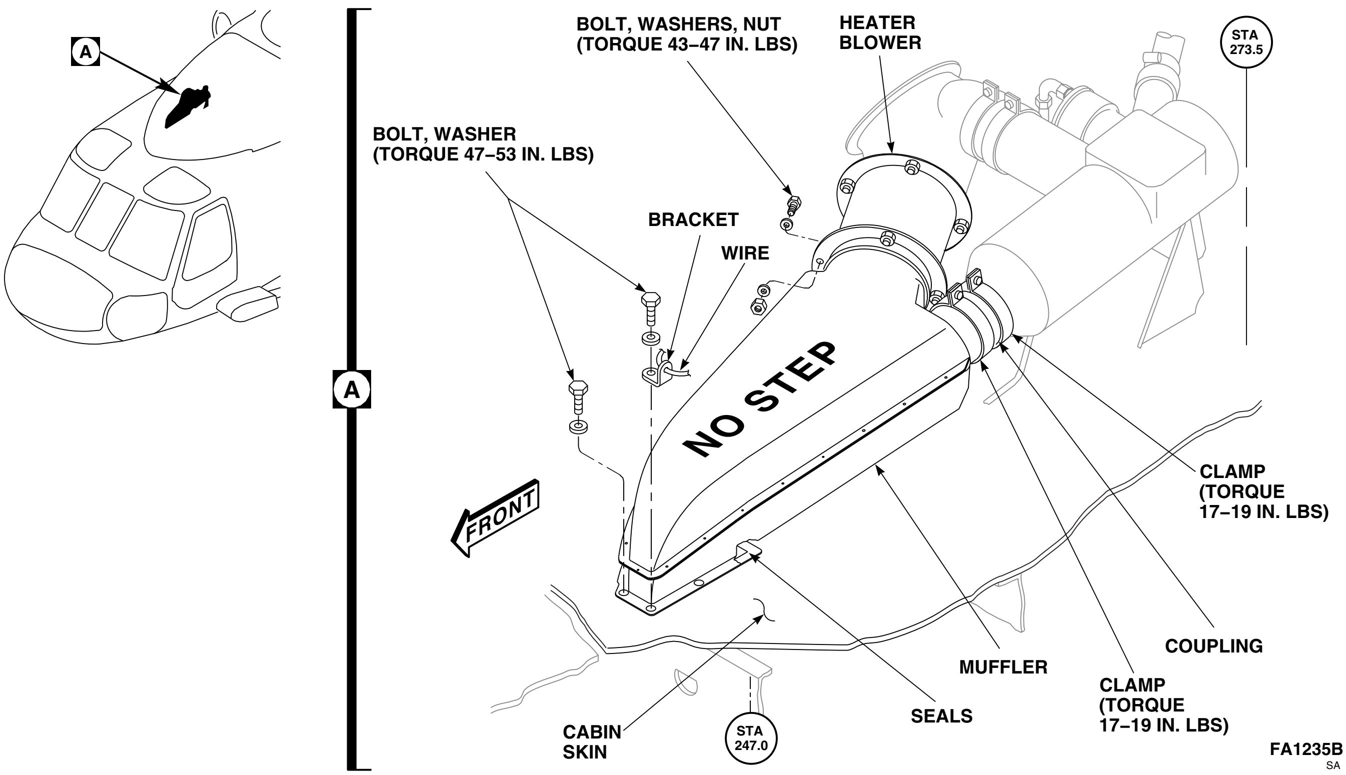


FIGURE 13-4-9-1. REPLACE HEATING AND VENTILATION SYSTEM HEATER MUFFLER

NOTE

To prevent gaps at heater muffler flanged joints, existing seal may be built up with adhesive.

b. If necessary, build up seal with adhesive, Item 57, Appendix D, to a maximum thickness of 0.060-inch.

c. Apply light bead of sealing compound, Item 292, Appendix D, around boltholes in muffler.

d. Install muffler on seal on cabin skin and position wire and bracket on muffler. Secure muffler to cabin skin with bracket, bolts, and washers (Figure 13-4-9-1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

e. Apply light bead of sealing compound, Item 292, Appendix D, around edge of muffler where it mates with cabin skin.

f. Connect muffler to heater blower with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

g. Slide coupling over muffler and install clamps. **TORQUE CLAMPS TO 17 - 19 INCH-POUNDS.**

h. Make sure area is clean and free of foreign material.

i. Close and latch main rotor pylon sliding cover.

13-4-10 HEATING AND VENTILATION SYSTEM COCKPIT DOOR DUCTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Rivet Gun

References

- PARA 1-6-16
- PARA 2-6-3

13-4-10.1 Replace Heating and Ventilation System Cockpit Door Ducts.

13-4-10.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open cockpit door.

c. Remove rivets securing air outlet to heating/ventilating duct (Figure 13-4-10-1, Detail B or Detail C).

d. To remove upper window duct, perform the following (Figure 13-4-10-1, Detail B):

(1) Remove screws and washers, and carefully separate heating/ventilating duct.

(2) Remove heating/ventilating duct.

e. To remove lower window duct, perform the following (Figure 13-4-10-1, Detail C):

(1) Drill out rivets securing lower window duct to door frame.

(2) Remove heating/ventilating duct from door by removing screws and washers.

(3) Push down on heating/ventilating duct to disengage hidden retaining latch.

(4) Remove heating/ventilating duct.

13-4-10.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. To install lower window duct, perform the following (Figure 13-4-10-1, Detail C):

(1) Install lower window duct on door frame with rivets using rivet gun.

(2) Install heating/ventilating duct on door by sliding duct forward from rear to engage hidden retaining latch.

(3) Install screws and washers securing heating/ventilating duct to door.

(4) Trim upper edge of flange as necessary to stop interference between edge of flange and door jettison handle (PARA 2-6-3).

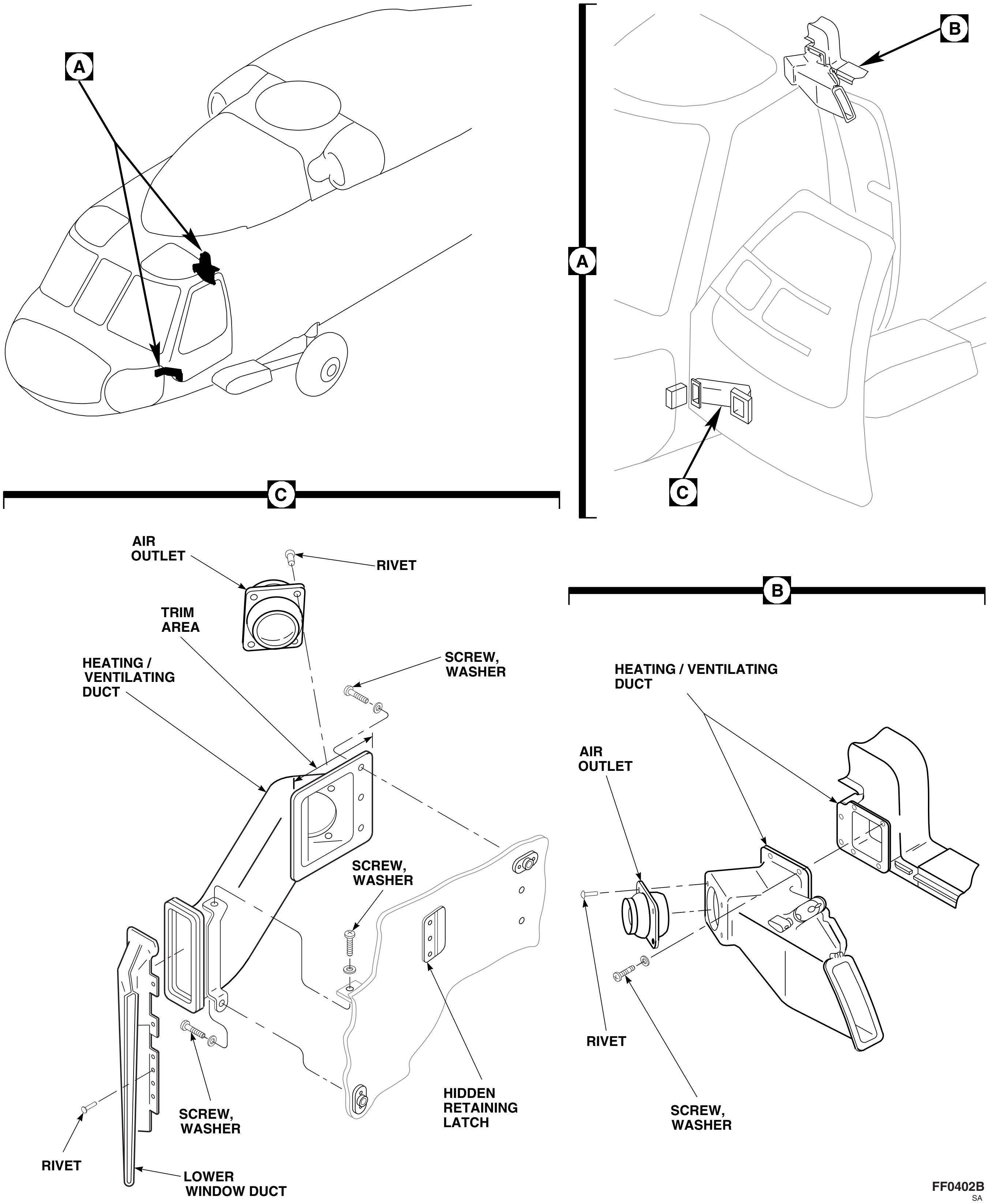


FIGURE 13-4-10-1. REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DOOR DUCTS

c. To install upper window duct, carefully position heating/ventilating duct and secure with screws and washers (Figure 13-4-10-1, Detail B).

d. If installing new duct, do this (Figure 13-4-10-1, Detail B or Detail C):

(1) Trial-fit duct.

(2) Mark holes that do not line up with nutplates.

(3) Fill marked holes (PARA 2-6-3).

(4) Mark and drill new holes using mounting duct as template.

e. Install air outlet in heating/ventilating duct with rivets using rivet gun.

f. Close cockpit door.

13-4-11 HEATING AND VENTILATION SYSTEM HEATER DUCT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16
- PARA 13-4-14

13-4-11.1 Replace Heating and Ventilation System Heater Duct.

13-4-11.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Loosen clamps securing coupling to heater duct and mixing valve. Slip coupling off heater duct (Figure 13-4-11-1, Detail A).

d. Remove bolts, washers, and nuts securing heater duct to heater blower. Carefully remove heater duct from helicopter.

13-4-11.1.2 Inspect.

a. Inspect condition of seals on heater duct flange for damage. **NONE ALLOWED.** If damaged, replace seals (PARA 13-4-14).

b. Check interior of heater duct for foreign material. **NONE ALLOWED.** If any evidence of obstruction is found, remove as necessary.

13-4-11.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position heater duct on heater blower and secure with bolts, washers, and nuts (Figure 13-4-11-1, Detail A). **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**

c. Slide coupling over heater duct and mixing valve, and install clamps. **TORQUE CLAMPS TO 17 - 19 INCH-POUNDS.**

d. Make sure area is clean and free of foreign material.

e. Close and latch main rotor pylon sliding cover.

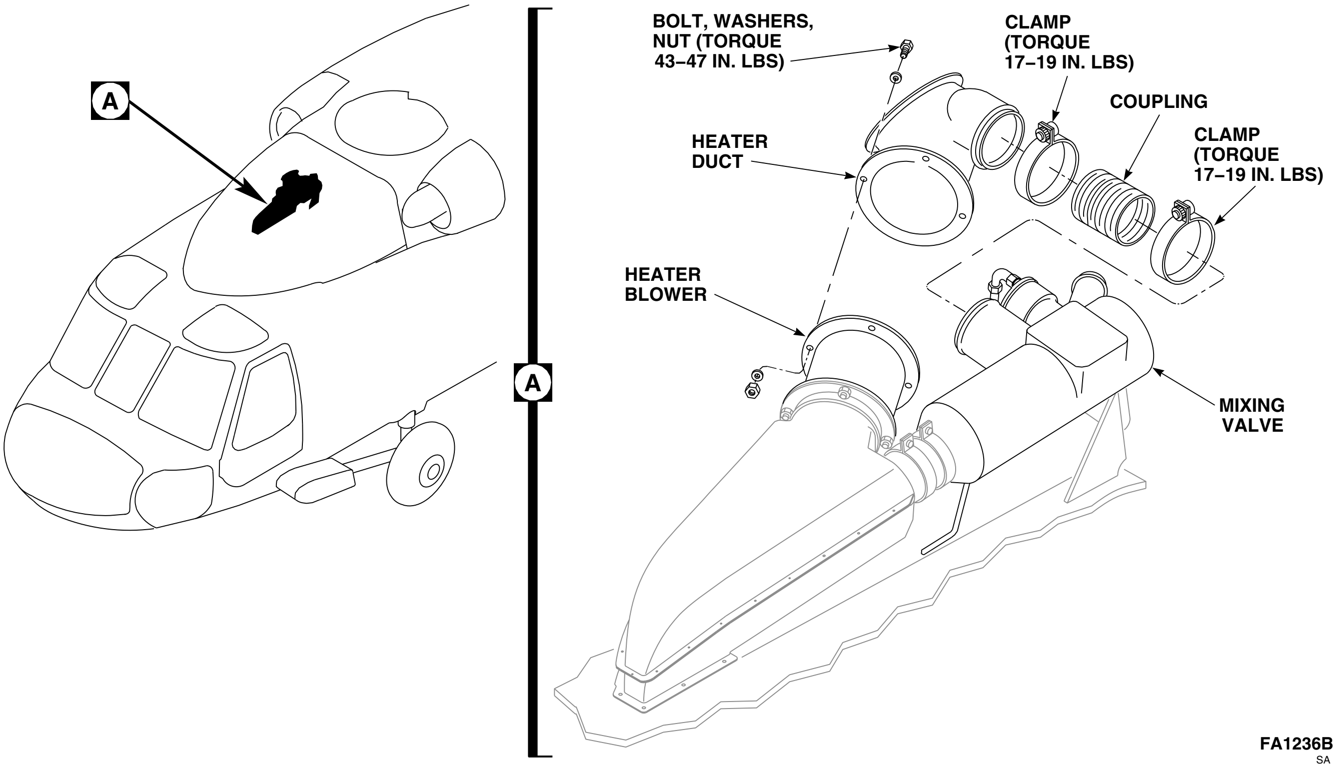


FIGURE 13-4-11-1. REPLACE HEATING AND VENTILATION SYSTEM HEATER DUCT

13-4-12 HEATING AND VENTILATION SYSTEM BLOWER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 57, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 13-2-1

13-4-12.1 Replace Heating and Ventilation System Blower.

13-4-12.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect electrical connector, J437 (Figure 13-4-12-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove bolt, washers, nut, clamp, and blower wiring from rear bracket.

f. Using nonmetallic scraper, remove adhesive from flange of blower.

g. Remove bolts, washers, and nuts securing blower to heater duct and muffler.

h. Remove blower from helicopter.

13-4-12.1.2 Inspect.

Inspect blower and make sure impeller spins freely. If impeller does not spin freely, replace blower.

13-4-12.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

NOTE

In the following step, install all hardware securing blower, except location on rear bracket where blower wiring is to be secured.

b. Place blower between heater duct and muffler so that air direction flow arrow points toward muffler (Figure 13-4-12-1, Detail A). Line up

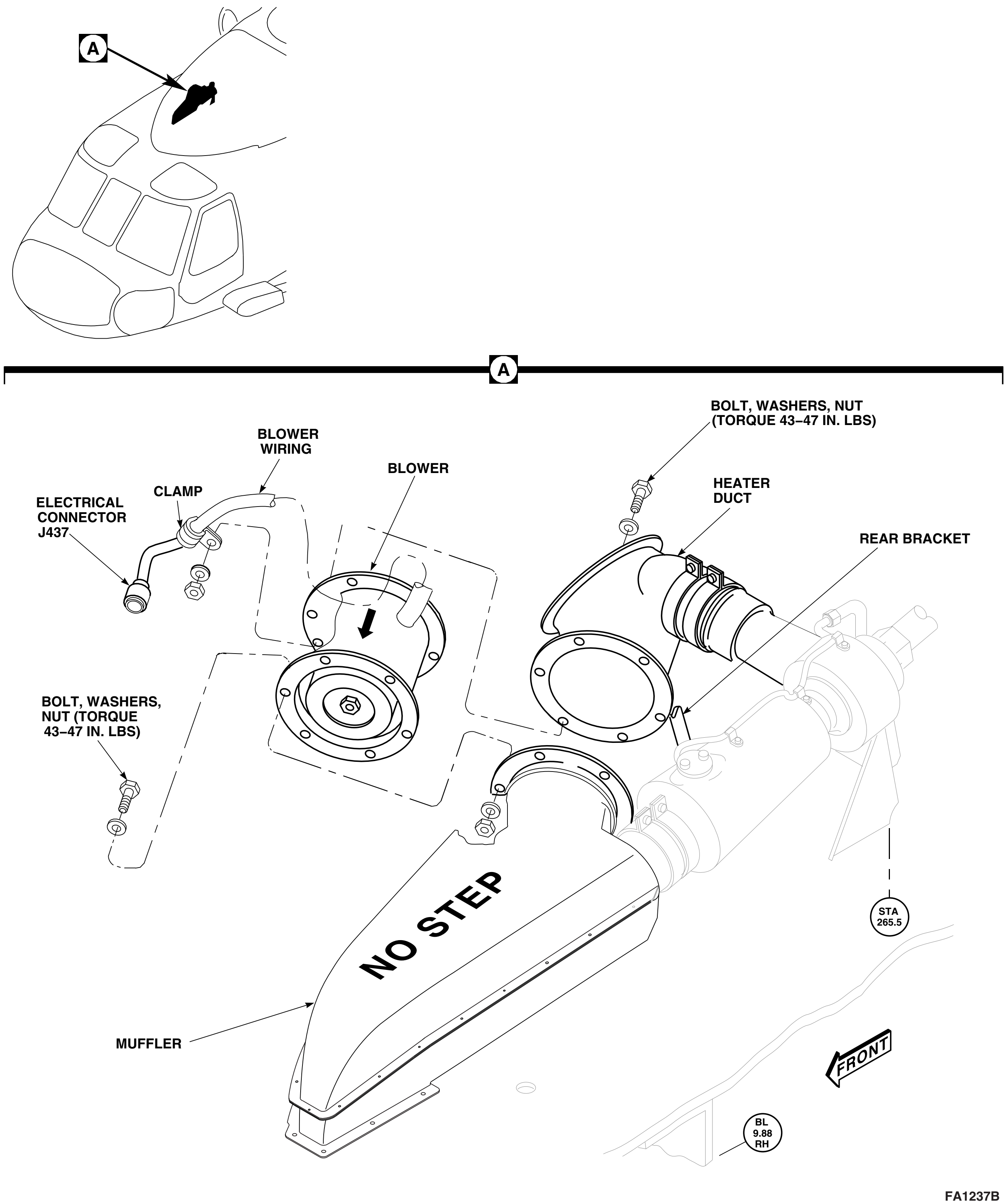


FIGURE 13-4-12-1. REPLACE HEATING AND VENTILATION SYSTEM BLOWER boltholes in blower with boltholes in heater duct flanges and muffler, and install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

c. Remove protective caps and plugs on electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, J437.

f. Install blower wiring on rear bracket with bolt, washers, clamp, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**

g. Apply bead of adhesive, Item 57, Appendix D, around front flange of blower.

h. Perform operational check of heating and ventilation system (PARA 13-2-1).

i. Make sure area is clean and free of foreign material.

j. Close main rotor pylon sliding cover.

13-4-13 HEATING AND VENTILATION SYSTEM AIR DUCT VALVE ASSEMBLIES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

13-4-13.1 Replace Heating and Ventilation System Air Duct Valve Assemblies.

NOTE

Replacement of both air duct valve assemblies is the same.

13-4-13.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from cabin overhead from STA 247.0 through STA 265.0 (PARA 2-4-69).

c. Remove screw and washer and separate air duct valve assembly from center duct section (Figure 13-4-13-1, Detail A).

d. Remove screw and washer and separate air duct valve assembly from outer duct assembly.

13-4-13.1.2 Inspect/Repair.

Inspect air duct valve assembly for damage and proper operation. **NO DAMAGE ALLOWED.** If required, disassemble air duct valve assembly as necessary (PARA 13-4-13.1.3), and replace any damaged components (PARA 13-4-13.1.4).

13-4-13.1.3 Disassemble.

a. Disconnect end of spring from pin (Figure 13-4-13-1, Detail B).

b. Disconnect other end of spring from bracket.

c. Remove cotter pin, washers, and pin securing rod end clevis (with rod) to lever.

d. Loosen nut securing rod to rod end clevis.

e. Remove rod from rod end clevis.

f. Remove nut from rod.

g. Remove rivets securing damper to lever. Remove damper (Figure 13-4-13-1, Detail C).

h. Remove lever and bushings from duct slide assembly (Figure 13-4-13-1, Detail B).

i. Remove screw and washer securing bracket to duct slide assembly.

13-4-13.1.4 Assemble.

a. Position bracket on duct slide assembly and secure with screw and washer (Figure 13-4-13-1, Detail B).

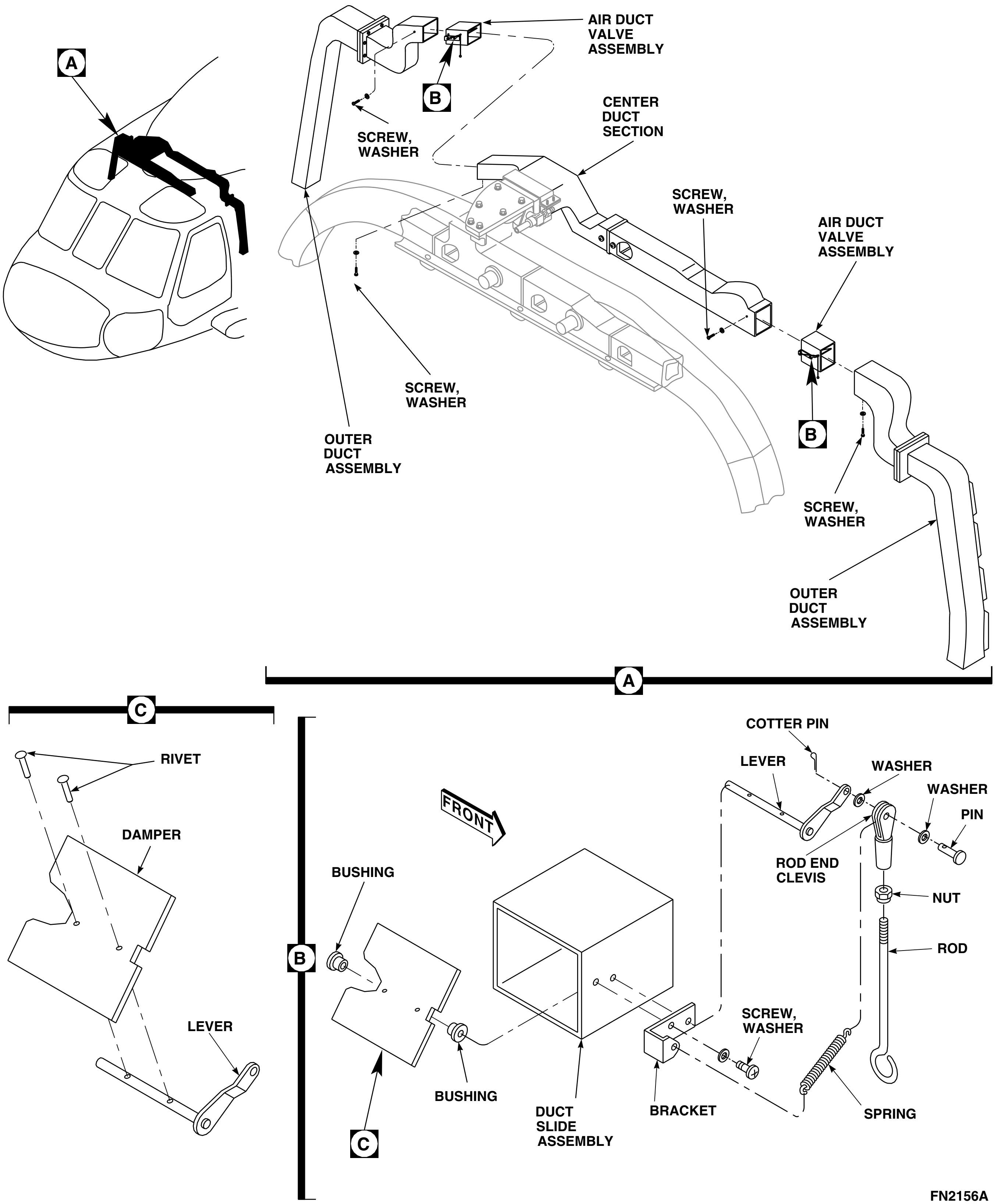


FIGURE 13-4-13-1. REPLACE HEATING AND VENTILATION SYSTEM AIR DUCT VALVE ASSEMBLIES

b. Partially insert lever in duct slide assembly.

NOTE

Bushings are installed as lever is installed.

c. Install bushings with flanges facing sides of duct assembly on lever. Fully insert lever.

d. Position damper on lever and secure with rivets (Figure 13-4-13-1, Detail C).

e. Install nut on rod (Figure 13-4-13-1, Detail B).

f. Install rod in rod end clevis and secure with nut. **TIGHTEN NUT.**

g. Position rod end clevis (with rod) and washer on lever and secure with pin, washer, and cotter pin.

h. Connect end of spring to bracket.

i. Connect other end of spring to pin.

13-4-13.1.5 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Slide air duct valve assembly into outer duct assembly and secure with screw and washer (Figure 13-4-13-1, Detail A). **TIGHTEN SCREW.**

c. Slide air duct valve assembly into center duct section and secure with screw and washer. **TIGHTEN SCREW.**

d. Install soundproofing panels (PARA 2-4-69).

13-4-14 HEATING AND VENTILATION SYSTEM REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Rivet Gun

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Adhesive, Item 30, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 35, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 56, Appendix D
- Adhesive, Item 57, Appendix D
- Adhesive, Item 66, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- MIL-R-46089
- MIL-R-6130
- MIL-R-6855
- MIL-S-22499
- MIL-S-6721
- MIL-S-6758
- MIL-STD-670
- PARA 1-6-16
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- PARA 13-4-9

- QQ-A-250

13-4-14.1 General Repairs.

NOTE

Heat and vent ducts are made from Kevlar/epoxy (reinforced plastic). Dampers, slide assemblies, brackets, and angles are aluminum alloy. Shafts for dampers are made from 4130 steel rod. Refer to Figure 13-4-14-1, Sheet 1, Sheet 2, and Sheet 3.

a. Evaluate and repair damage to Kevlar ducts (PARA 2-6-3).

b. Evaluate damage to aluminum parts (PARA 2-6-1), and if necessary, repair (PARA 2-6-2).

c. Rebond seals and gaskets (PARA 13-4-14.2, PARA 13-4-14.3, PARA 13-4-14.4, and PARA 13-4-14.5).

d. Rebond ducts using adhesive, Item 32, and Item 66, Appendix D. Apply adhesives per manufacturer's instructions.

e. Touch up paint as required (PARA 2-6-5).

13-4-14.2 Replace Cockpit Duct Seal.

13-4-14.2.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers securing duct assembly to door panel (Figure 13-4-14-2, Sheet 2, Detail F).

c. Remove old seal from duct assembly.

13-4-14.2.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean bond area of old adhesive using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

c. Roughen bonding surface of replacement seal with abrasive cloth, Item 1A, Appendix D.

d. Apply smooth even coat of adhesive, Item 37, Appendix D, to mating surfaces of seal and duct assembly. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.

e. Press seal firmly in place. Make sure all surfaces are in good contact (Figure 13-4-14-2, Sheet 2, Detail F). Allow seal to cure for 24 hours at 68°F (20°C).

f. Install duct assembly to door panel with screws and washers.

13-4-14.3 Replace Upper Cabin Duct Seals.

13-4-14.3.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers securing duct assembly (Figure 13-4-14-2, Sheet 2, Detail E).

c. Remove old seals from duct assembly.

13-4-14.3.2 Install.

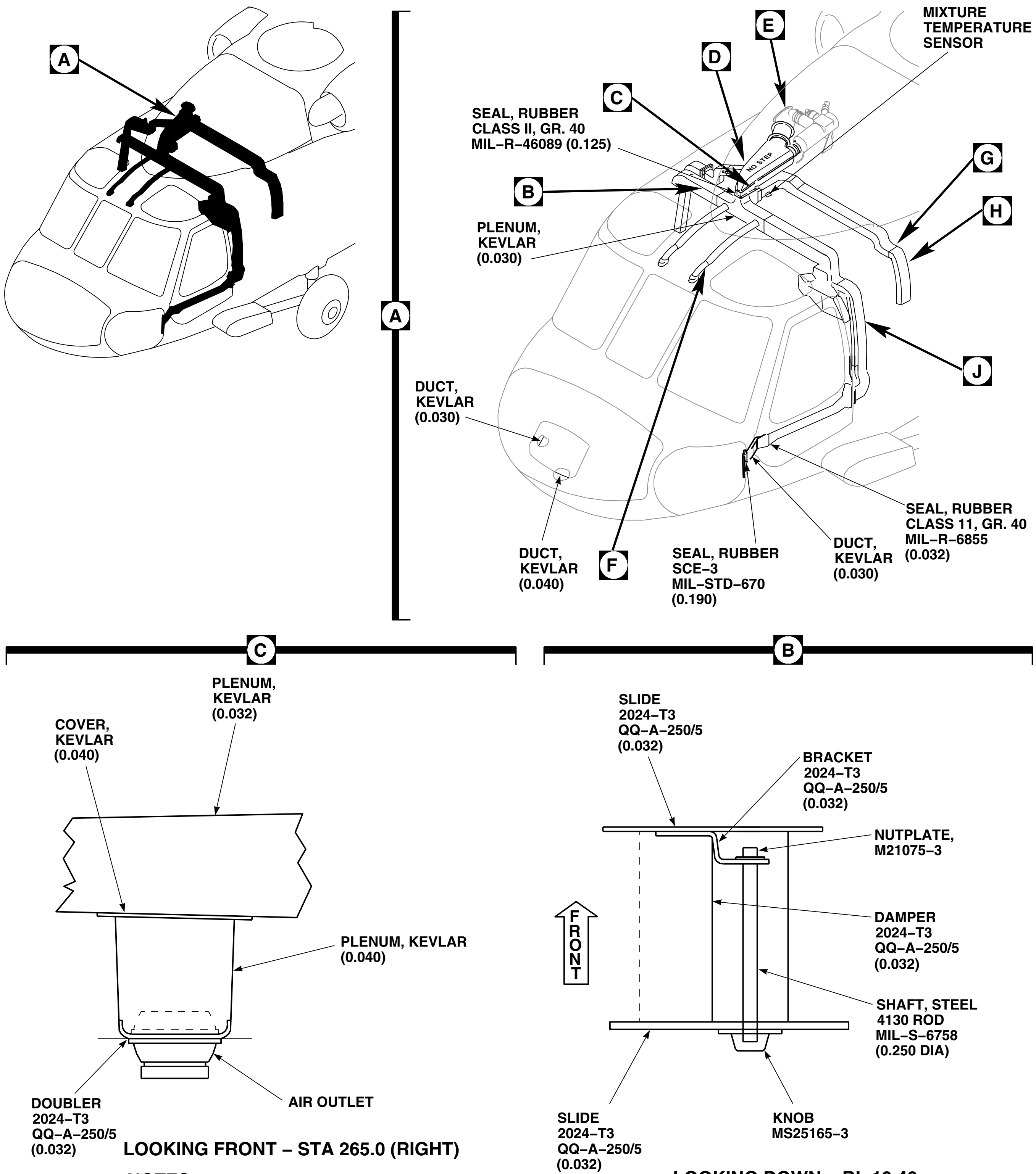
a. Turn off all electrical power (PARA 1-6-16).

b. Clean bond area of old adhesive using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

c. Roughen bonding surface of replacement seal with abrasive cloth, Item 1A, Appendix D.

d. Apply smooth even coat of adhesive, Item 37, Appendix D, to mating surfaces of seal and duct assembly. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.

e. Press seal firmly in place. Make sure all surfaces are in good contact (Figure 13-4-14-2, Sheet 2, Detail E). Allow seal to cure for 24 hours at 68°F (20°C).



NOTES

1. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.
2. POSITION SEAL JOINT ON CENTER OF FLANGE ON INBOARD SIDE.

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FIGURE 13-4-14-1. REPAIR HEATING AND VENTILATION SYSTEM (SHEET 1 OF 3)

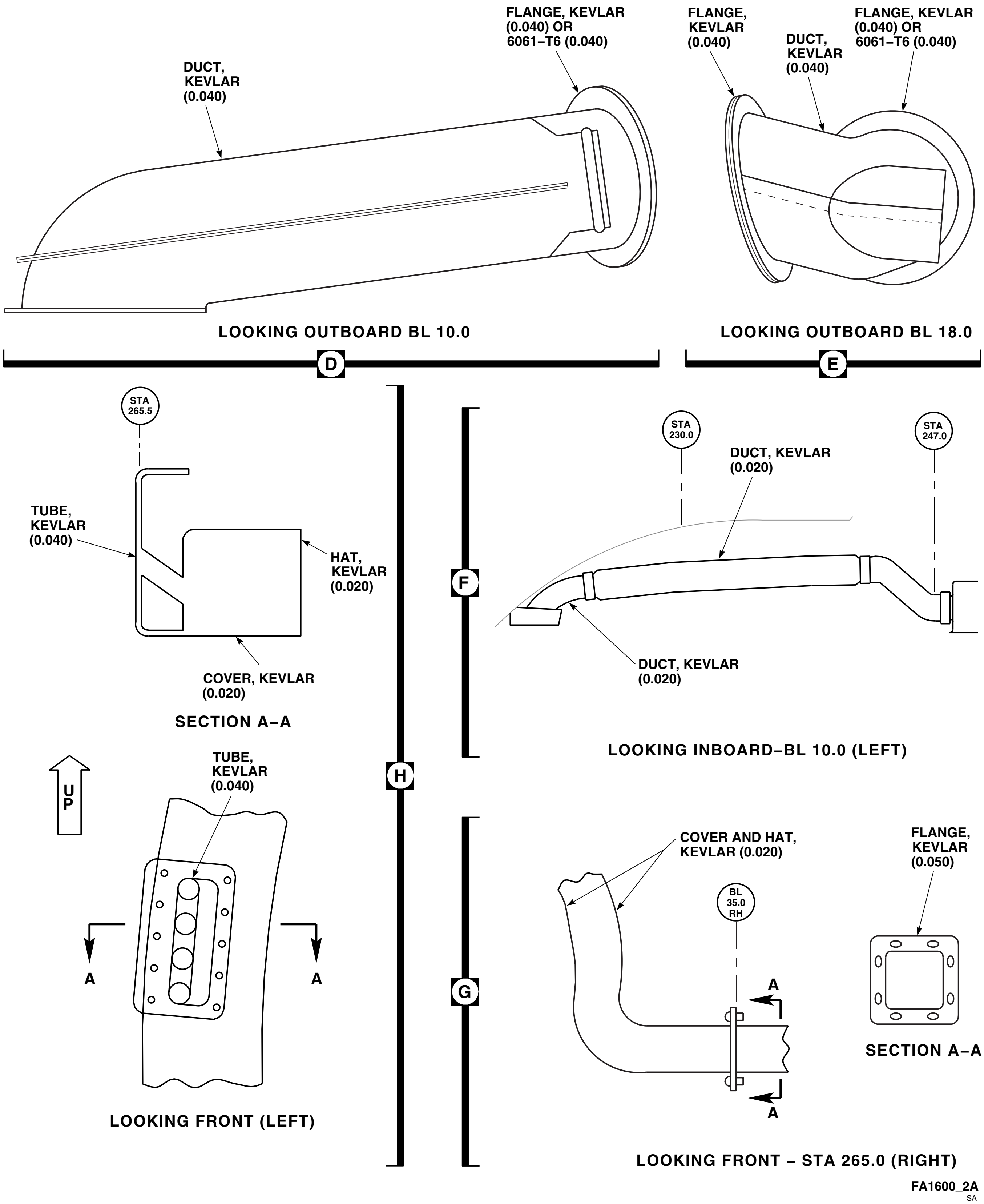


FIGURE 13-4-14-1. REPAIR HEATING AND VENTILATION SYSTEM (SHEET 2 OF 3)

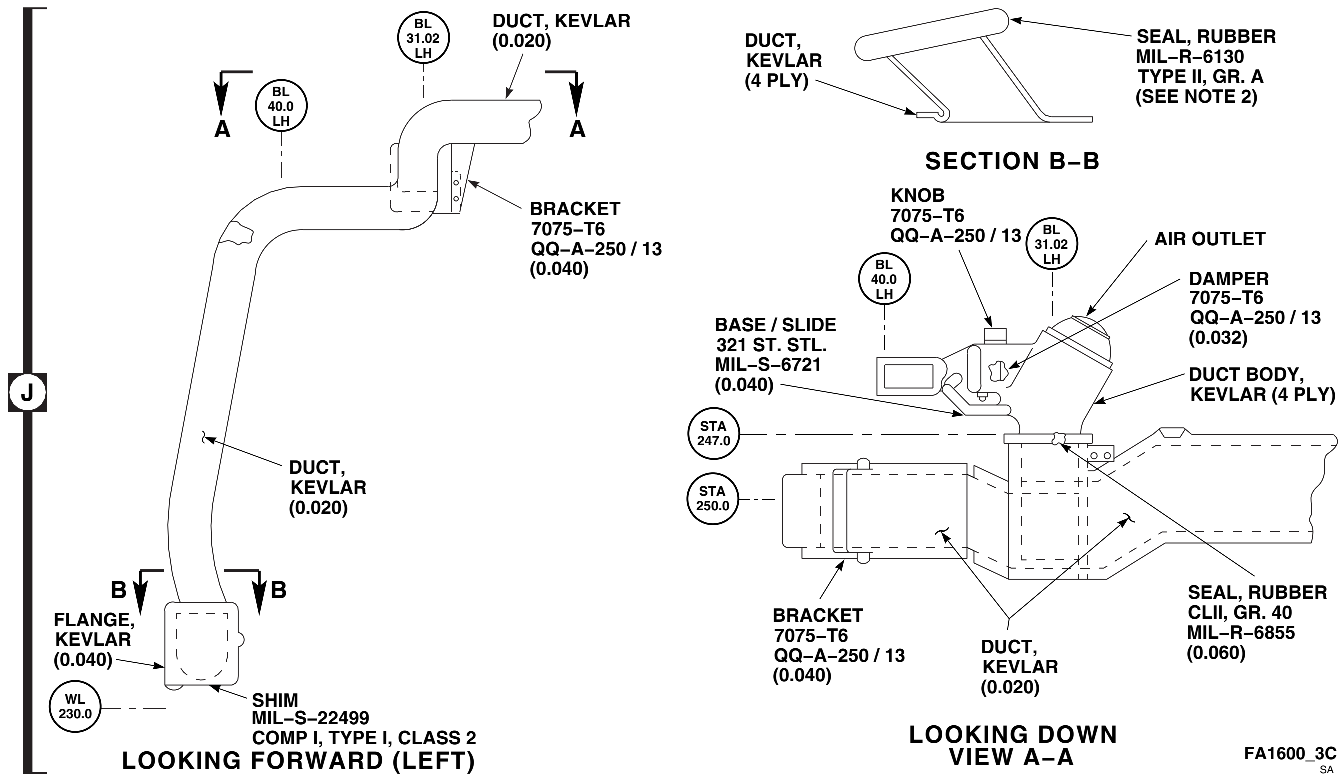


FIGURE 13-4-14-1. REPAIR HEATING AND VENTILATION SYSTEM (SHEET 3 OF 3)

f. Install duct assembly with screws and washers.

13-4-14.4 Replace Lower Cabin Duct Seal.

13-4-14.4.1 Remove.

Remove old seal from lower duct assembly (Figure 13-4-14-2, Sheet 2, Detail D).

13-4-14.4.2 Install.

a. Clean bond area of old adhesive using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

b. Roughen bonding surface of replacement seal with abrasive cloth, Item 1A, Appendix D.

c. Apply smooth even coat of adhesive, Item 37, Appendix D, to mating surfaces of seal and lower duct assembly. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.

NOTE

When bonding seal to lower duct assembly, locate split of seal at about midpoint of lower duct assembly on inboard side.

d. Press seal firmly in place. Make sure all surfaces are in good contact (Figure 13-4-14-2, Sheet 2, Detail D). Allow seal to cure for 24 hours at 68°F (20°C).

13-4-14.5 Replace Muffler/Plenum Seals and Gaskets.

13-4-14.5.1 Remove.

a. Remove heater muffler (PARA 13-4-9).

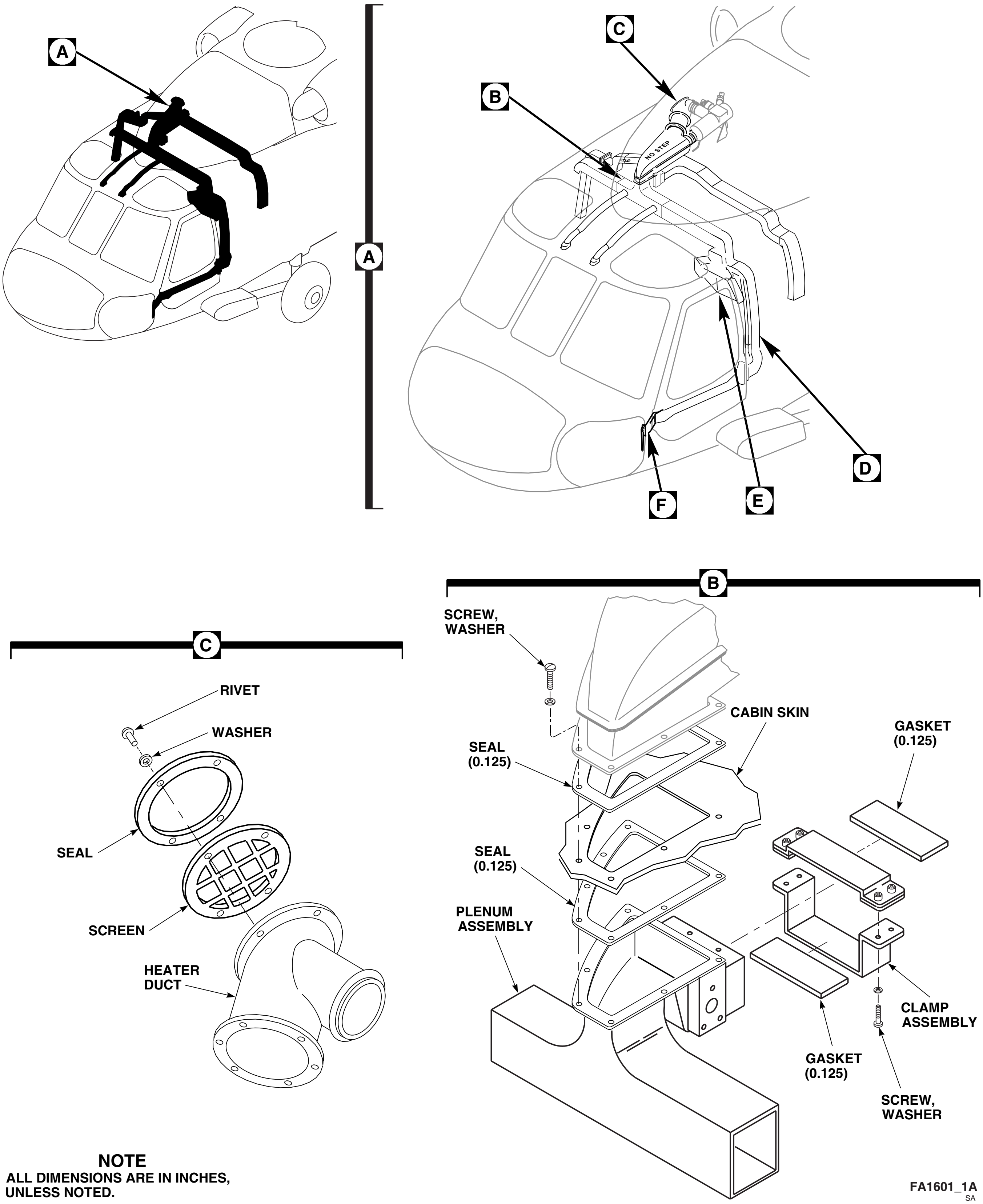
b. Remove screws and washers from clamp assembly (Figure 13-4-14-2, Sheet 1, Detail B).

c. Using nonmetallic scraper, remove plenum seal and gaskets from clamp assembly.

d. Using nonmetallic scraper, remove seal from muffler cabin skin area.

13-4-14.5.2 Install.

a. Turn off all electrical power (PARA 1-6-16).



NOTE
ALL DIMENSIONS ARE IN INCHES,
UNLESS NOTED.

FIGURE 13-4-14-2. REPLACE SEALS AND GASKETS (SHEET 1 OF 2)

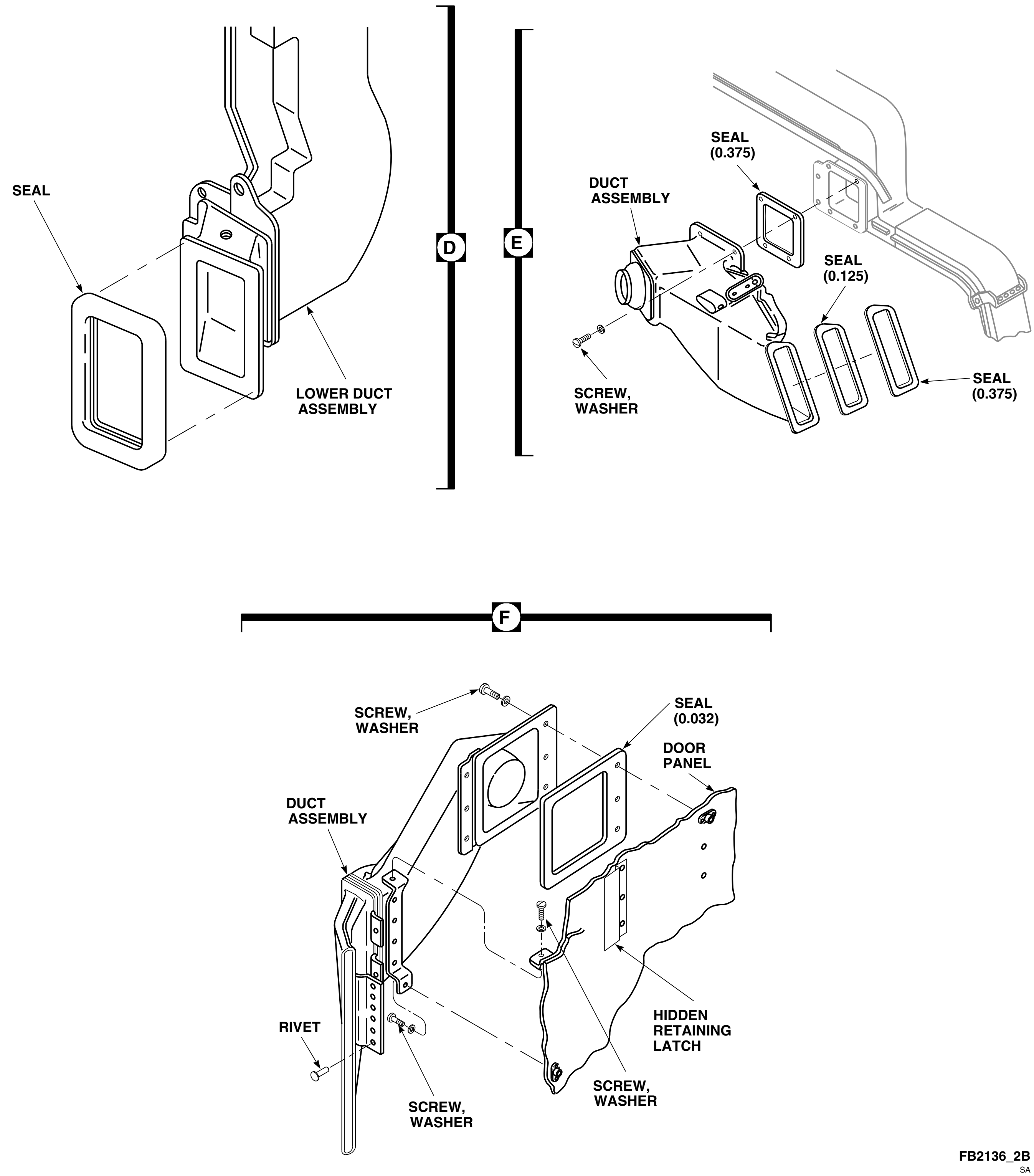


FIGURE 13-4-14-2. REPLACE SEALS AND GASKETS (SHEET 2 OF 2)

b. Clean bond area of old adhesive using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

c. Roughen bonding surface of replacement seal and gaskets with abrasive cloth, Item 1A, Appendix D. Wipe clean using cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

d. Apply smooth even coat of adhesive, Item 35, Appendix D, to mating surfaces. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.

e. Press plenum seal and gaskets firmly in place. Make sure all surfaces are in good contact (Figure 13-4-14-2, Sheet 1, Detail B). Allow adhesive to cure for 24 hours at 68°F (20°C).

f. Install clamp assembly with screws and washers.

g. Install heater muffler (PARA 13-4-9).

13-4-14.6 Replace Heater Duct Seal and Screen.

13-4-14.6.1 Remove.

a. Remove rivets and washers from screen and seal (Figure 13-4-14-2, Sheet 1, Detail C).

b. Remove screen and seal from heater duct.

c. Using nonmetallic scraper, remove seal from screen.

d. Clean bonding surfaces of screen and heater duct using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

13-4-14.6.2 Install.

a. Roughen bonding surfaces of replacement seal with abrasive cloth, Item 1A, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

b. Apply smooth even coat of adhesive, Item 32, Appendix D, to mating surface of heater duct.

c. Install screen on heater duct (Figure 13-4-14-2, Sheet 1, Detail C).

d. Apply smooth even coat of adhesive, Item 56 or Item 57, Appendix D, and adhesive, Item 30, Appendix D, to mating surfaces of seal.

e. Install seal on screen. Using rivet gun, secure with washers and rivets. Allow adhesive to cure for 24 hours at room temperature.

13-4-15 ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR BLOWER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Putty Knife
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 61, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Protective Caps and Plugs
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69

13-4-15.1 Replace Environmental Control System Evaporator Blower.

13-4-15.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove left soundproofing panel from rear cabin bulkhead (PARA 2-4-69).

c. Disconnect electrical connector, P9, from ECS evaporator blower (Figure 13-4-15-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. If left plenum is installed, perform the following:

(1) Remove screws and washers securing access door and rubber seal to left plenum.

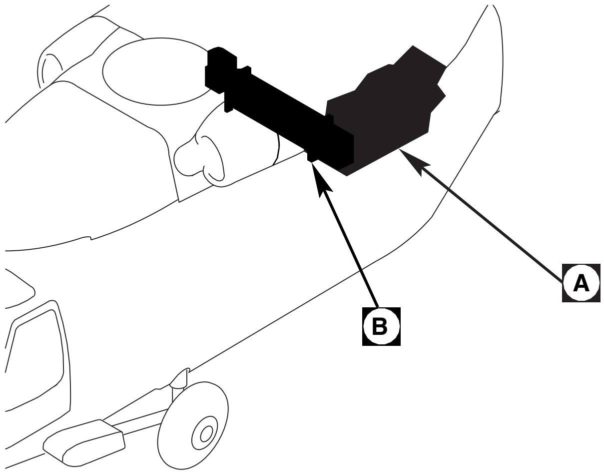
(2) Remove bolts and nuts securing left plenum to front support bracket, rubber seal, and ECS evaporator blower (Figure 13-4-15-1, Sheet 2, Detail B).

(3) Using putty knife, carefully separate left plenum from ECS evaporator blower. Remove rubber seal.

f. If left plenum is not installed, perform the following:

(1) Remove bolts and washers securing ECS evaporator blower to front support bracket.

(2) Remove bolts and washers securing ECS evaporator blower to rear support bracket and evaporator inlet transition duct (Figure 13-4-15-1, Sheet 1, Detail A).



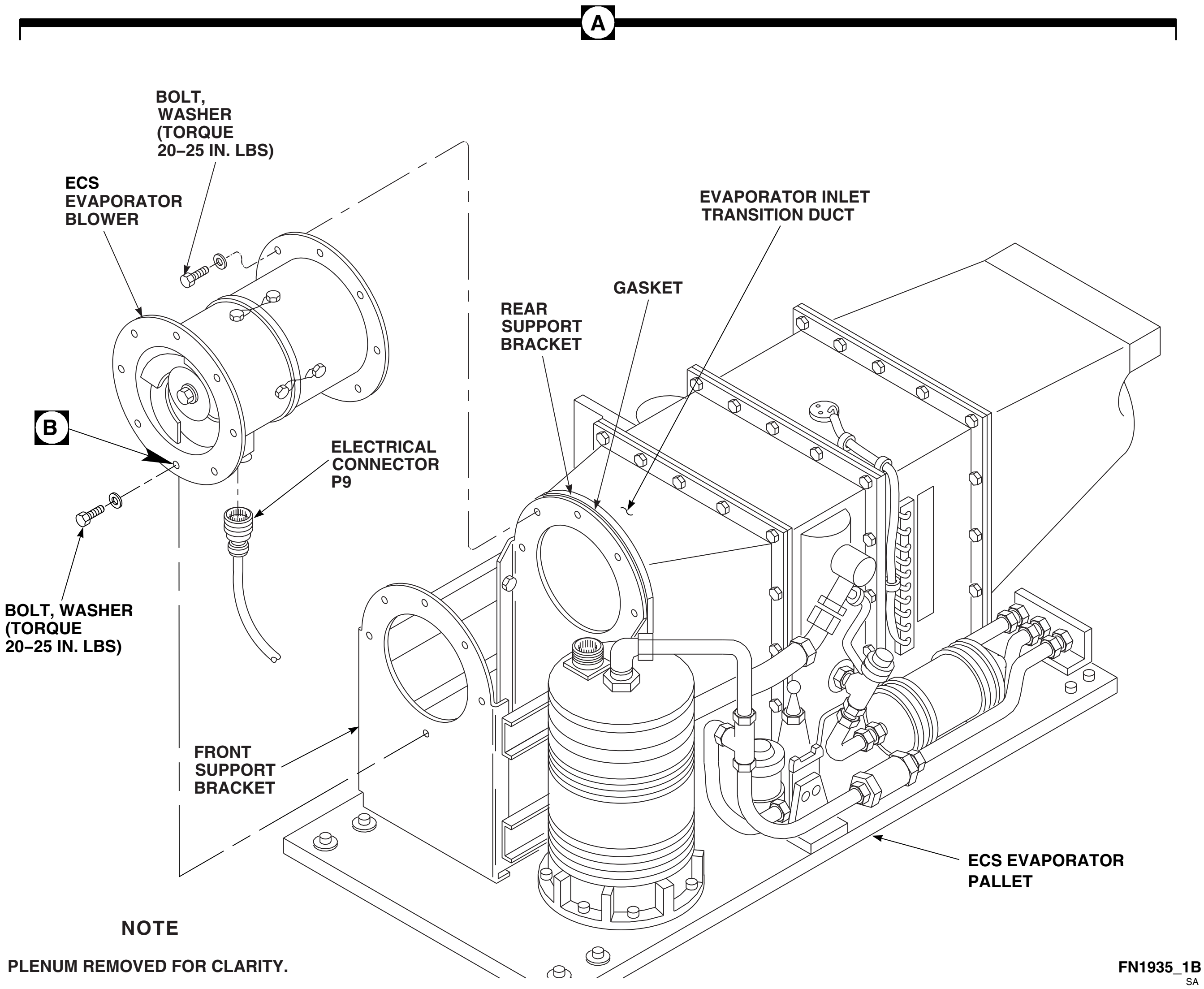


FIGURE 13-4-15-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR BLOWER (SHEET 1 OF 2)

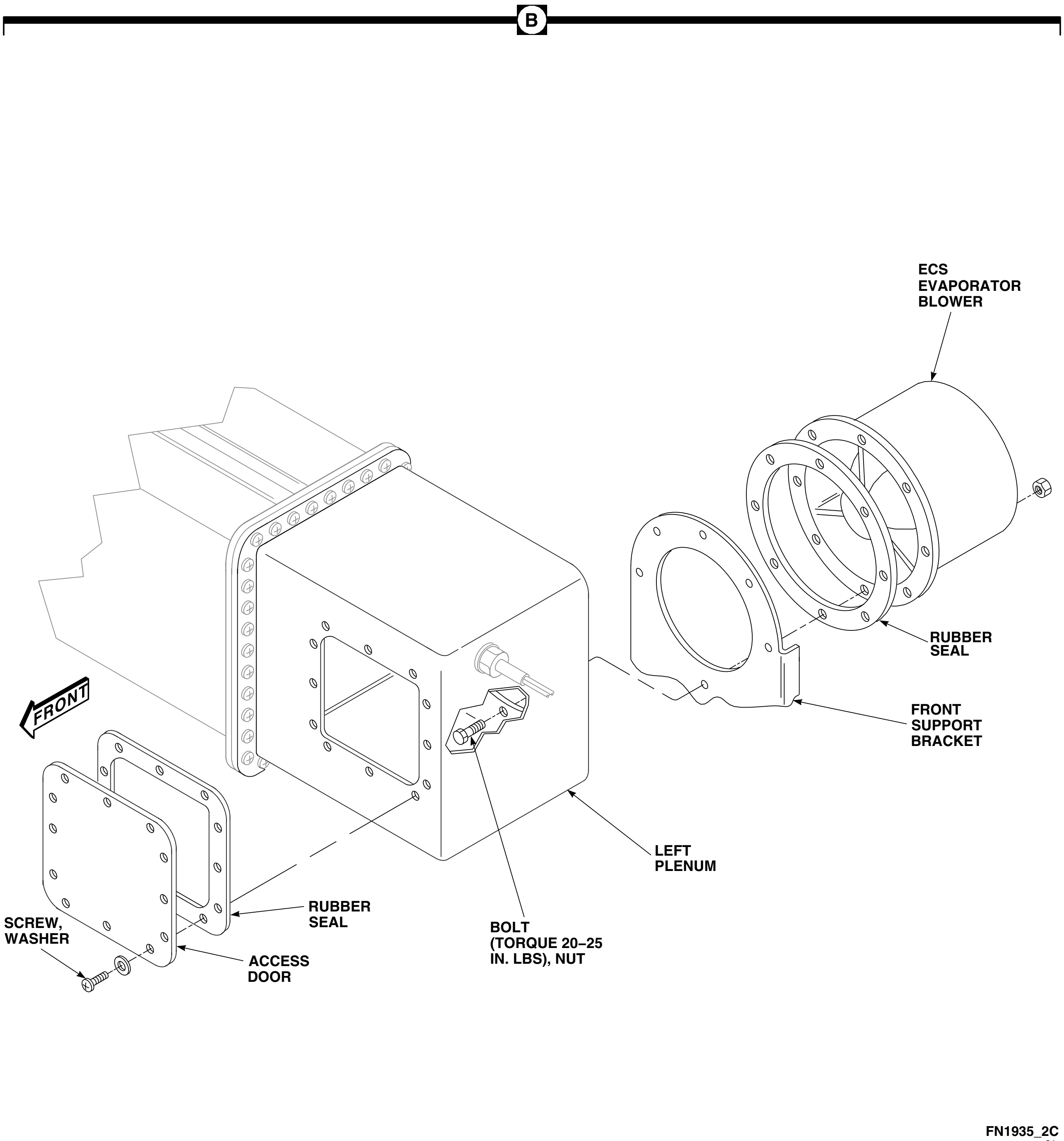


FIGURE 13-4-15-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR BLOWER (SHEET 2 OF 2)

(3) Remove ECS evaporator blower from ECS evaporator pallet.

13-4-15.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Make sure gasket is in place between rear support bracket and evaporator inlet transition duct.

c. Install ECS evaporator blower on rear support bracket and evaporator inlet transition duct and secure with bolts and washers (Figure 13-4-15-1, Sheet 1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

d. If left plenum was removed, perform the following:

(1) Install rubber seal and ECS evaporator blower on front support bracket and secure to left plenum with bolts and washers (Figure 13-4-15-1, Sheet 2, Detail B).

(2) **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

e. If left plenum is not installed, perform the following:

(1) Install ECS evaporator blower to ECS evaporator pallet.

(2) Install bolts and washers securing ECS evaporator blower to rear support bracket and evaporator inlet transition duct (Figure 13-4-15-1, Sheet 1, Detail A).

(3) Install bolts and washers securing ECS evaporator blower to front support bracket.

(4) **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connector, P9, to ECS evaporator blower.

i. If left plenum was installed, install access door as follows:

(1) Using cleaning cloth, Item 102, Appendix D, dampened with trichloroethane 1,1,1, Item 341, Appendix D, clean mating surface of access door.

(2) Using adhesive, Item 61, Appendix D, apply light coat to sealing surface of access door and both sides of rubber seal. Install rubber seal on access door.

(3) Install access door on left plenum and secure with screws and washers.

j. Install left soundproofing panel on rear cabin bulkhead (PARA 2-4-69).

13-4-16 ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR ELECTRICAL HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 13-5-8

13-4-16.1 Replace Environmental Control System Evaporator Electrical Harness.

13-4-16.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove left soundproofing panel from rear cabin bulkhead (PARA 2-4-69).

c. Remove ECS hot gas bypass (HGBP) valve (PARA 13-5-8).

d. Remove bolts, washers, and clamps securing ECS evaporator electrical harness to ECS evaporator pallet and ECS heater/demister (Figure 13-4-16-1, Detail A).

e. Disconnect electrical connector, P9, from ECS evaporator blower.

f. Install protective caps and plugs on electrical connectors.

g. Disconnect electrical connector, P8, from compressor.

h. Install protective caps and plugs on electrical connectors.

i. Remove chafe guard and spiral wraps and separate electrical leads for hot gas bypass valve, thermistor, expansion valve and heater/demister temperature switches.

j. Remove nuts, lockwashers, and all wires from terminal board, TB4, on heater/demister. Tag wires (Figure 13-4-16-1, Detail B).

k. Disconnect and tag electrical leads to heater elements.

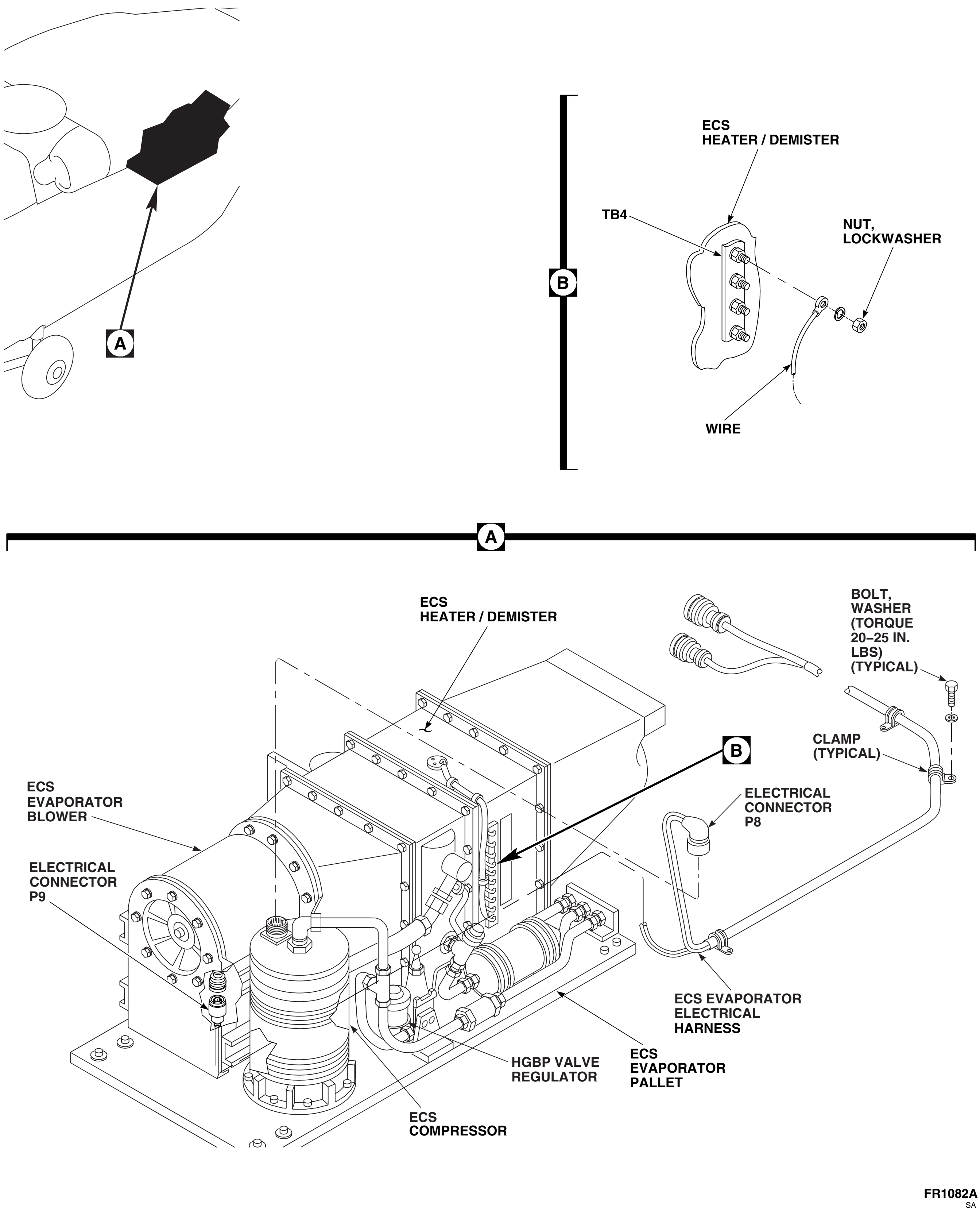


FIGURE 13-4-16-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR ELECTRICAL HARNESS

13-4-16.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Connect leads to heater elements. Remove tags.

c. Connect wires to terminal board, TB4, on heater/demister housing with lockwashers and nuts (Figure 13-4-16-1, Detail B). Remove tags.

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P9, to ECS evaporator blower (Figure 13-4-16-1, Detail A).

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connector (PARA 1-7-20).

i. Connect electrical connector, P8, to ECS compressor.

j. Add leads for hot gas bypass valve, thermistor, expansion valve and heater/demister temperature switches to ECS evaporator electrical harness. Install spiral wraps and chafe guards.

k. Install electrical harness on ECS evaporator pallet with bolts, washers, and clamps. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

l. Install ECS hot gas bypass (HGBP) valve (PARA 13-5-8).

m. Install left soundproofing panel on rear cabin bulkhead (PARA 2-4-69).

13-4-17 ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE LIMITING SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 55, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 13-2-2

13-4-17.1 Replace Environmental Control System Temperature Limiting Switch.

13-4-17.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Slit foam from around ECS temperature limiting switch.

c. Remove screws, washers, and temperature limiting switch from ECS heater/demister (Figure 13-4-17-1, Detail B).

d. Remove nuts, lockwashers, and wires from terminals 4 and 7, of terminal board, TB4. Tag wires.

13-4-17.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Connect wires to terminals 4 and 7, of terminal board, TB4, with lockwashers and nuts (Figure 13-4-17-1, Detail B). Remove tags.

c. Install ECS temperature limiting switch to ECS heater/demister with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

d. Using adhesive, Item 55, Appendix D, repair foam insulation. If necessary, replace foam insulation.

e. Perform operational check of ECS (PARA 13-2-2).

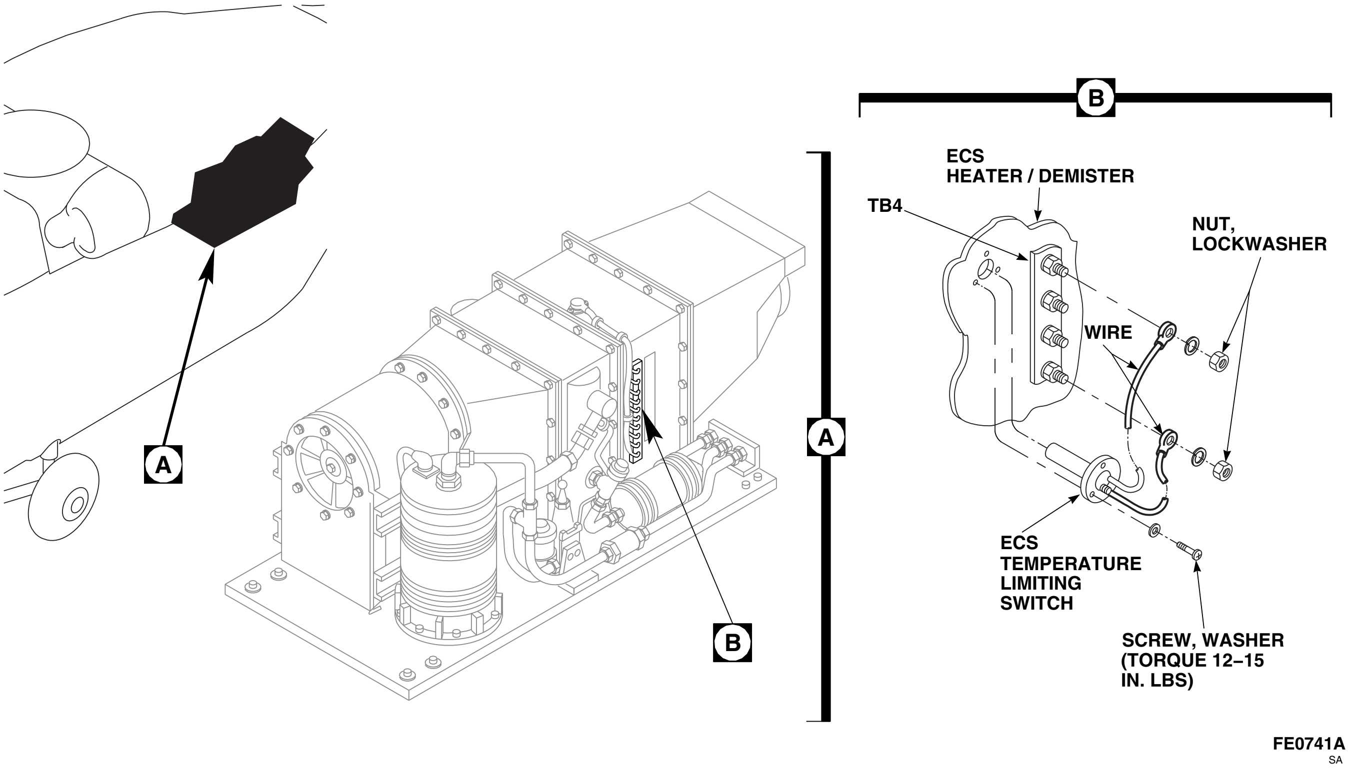


FIGURE 13-4-17-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE LIMITING SWITCH

13-4-18 ENVIRONMENTAL CONTROL SYSTEM LOW AND HIGH TEMPERATURE SWITCHES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 55, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 13-2-2

13-4-18.1 Replace Environmental Control System Low and High Temperature Switches.

13-4-18.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove ECS low temperature switch as follows (Figure 13-4-18-1, Detail B):

(1) Slit foam from around low temperature switch.

(2) Remove screws, washers, and low temperature switch from ECS heater/demister.

(3) Remove nuts, lockwashers, and switch wires from terminals, 8, 9, and 10, of terminal board, TB4. Tag wires.

c. Remove ECS high temperature switch as follows (Figure 13-4-18-1, Detail C):

(1) Remove screws, washers, clamps, nuts, and high temperature switch from brackets on ECS heater/demister.

(2) Remove nuts, lockwashers, and wires from terminals, 4, 5, and 6, of terminal board, TB4. Tag wires.

13-4-18.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install ECS low temperature switch as follows (Figure 13-4-18-1, Detail B):

(1) Connect wires to terminals, 8, 9, and 10, of terminal board, TB4. Remove tags.

(2) Install low temperature switch to ECS heater/demister with screws and washers.

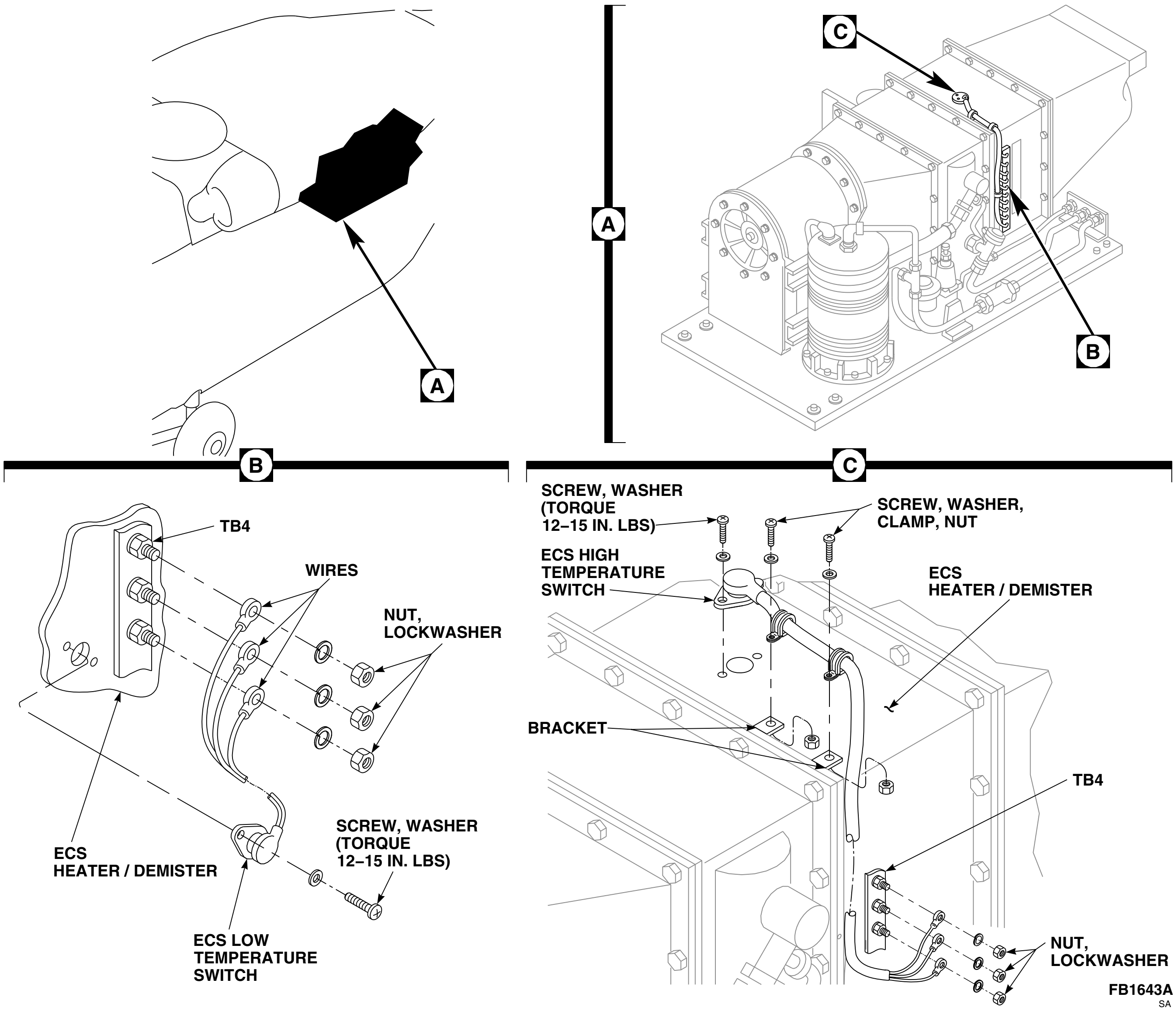


FIGURE 13-4-18-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM LOW AND HIGH TEMPERATURE SWITCHES

TORQUE SCREWS TO 12 - 15 INCH-POUNDS.

c. Install ECS high temperature switch as follows (Figure 13-4-18-1, Detail C):

(1) Connect wires to terminals, 4, 5, and 6, of terminal board, TB4. Remove tags.

(2) Install high temperature switch to ECS heater/demister with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

d. Clamp high temperature switch to brackets on ECS heater/demister with screws, washers, and nuts.

e. Using adhesive, Item 55, Appendix D, repair foam insulation. If necessary, replace foam insulation.

f. Perform operational check of ECS (PARA 13-2-2).

13-4-19 ENVIRONMENTAL CONTROL SYSTEM CONDENSER BLOWER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 13-4-23

13-4-19.1 Replace Environmental Control System Condenser Blower.

13-4-19.1.1 Remove.

a. Remove ECS condenser exhaust duct (PARA 13-4-23).

b. Disconnect electrical connector, P10 from J10 (Figure 13-4-19-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove clamp securing front end of ECS condenser blower and duct sleeve to cradle bracket.

e. Remove bolts and washers securing ECS condenser blower to ECS condenser transition duct and support bracket. Remove condenser blower from ECS condenser pallet.

f. Remove duct sleeve and install on new ECS condenser blower.

13-4-19.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position ECS condenser blower on ECS condenser pallet (Figure 13-4-19-1, Detail A).

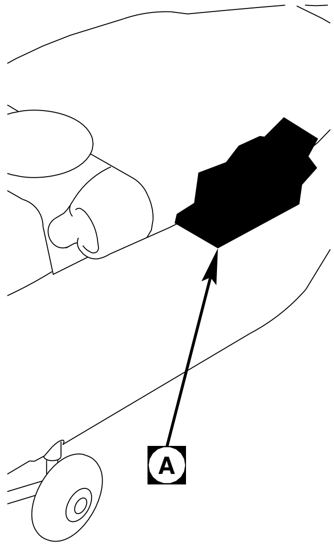
c. Make sure gasket is in place between rear support bracket and ECS condenser transition duct.

d. Install condenser blower on condenser transition duct and support bracket and secure with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

e. Install duct sleeve and front end of condenser blower on cradle bracket with clamp.

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).



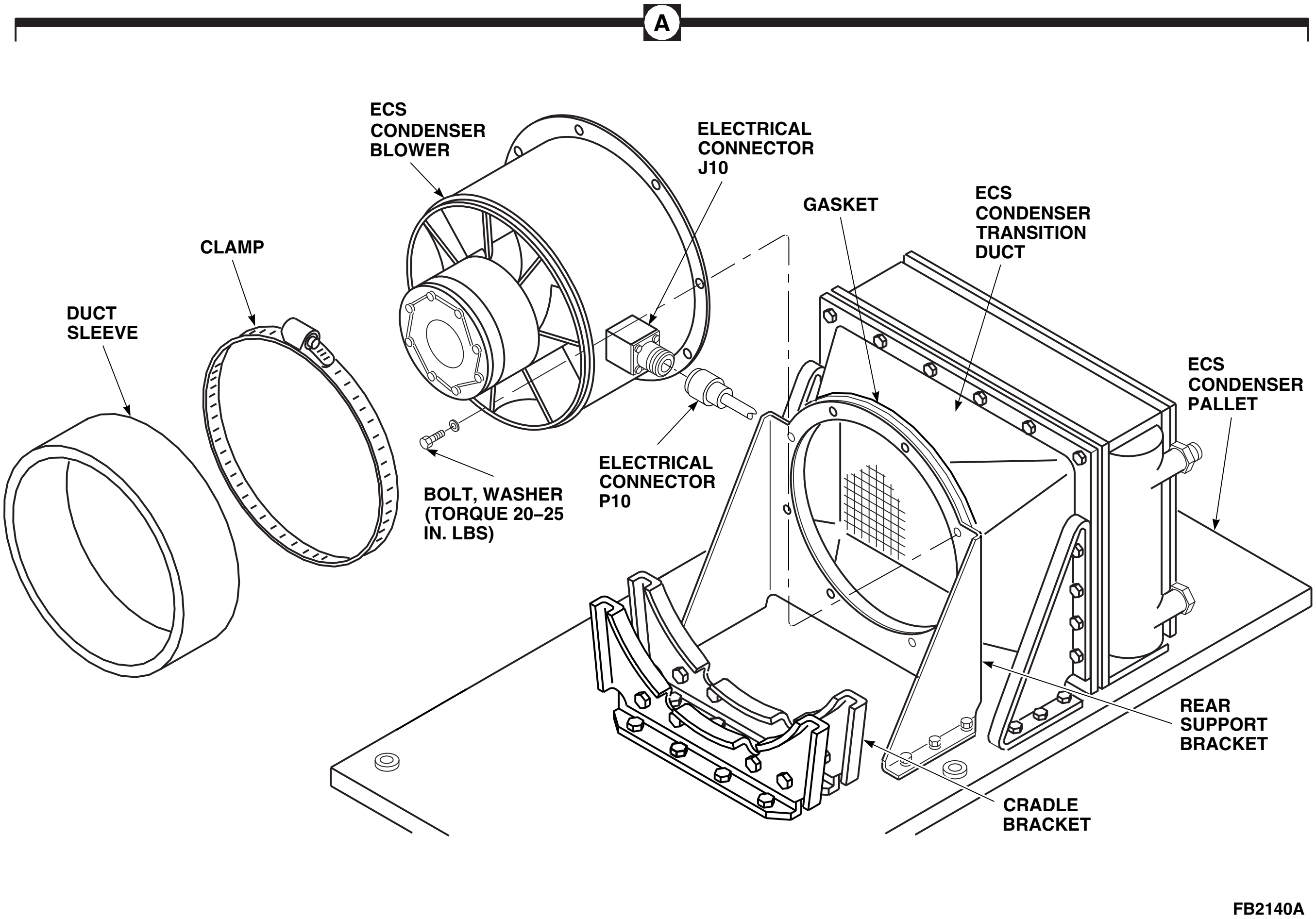


FIGURE 13-4-19-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER BLOWER

h. Connect electrical connector, P10 to J10.

i. Install ECS condenser exhaust duct (PARA 13-4-23).

13-4-20 ENVIRONMENTAL CONTROL SYSTEM CONDENSER ELECTRICAL HARNESS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 13-4-23

13-4-20.1 Replace Environmental Control System Condenser Electrical Harness.

13-4-20.1.1 Remove.

a. Remove ECS condenser exhaust duct (PARA 13-4-23).

b. Disconnect electrical connector, P10 from J10, on ECS condenser blower (Figure 13-4-20-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove screws, bolt, washers, and clamps securing ECS condenser electrical harness to ECS condenser pallet.

e. Disconnect electrical connector, P4 from J4, on electrical pallet.

f. Install protective caps and plugs on electrical connectors.

g. Remove electrical harness from ECS condenser pallet.

13-4-20.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove protective caps and plugs from electrical connectors.

c. Inspect and treat electrical connector (PARA 1-7-20).

d. Connect electrical connector, P4 to J4, on electrical pallet (Figure 13-4-20-1, Detail A).

e. Install electrical harness on ECS condenser pallet with screws, bolt, washers, and clamps. **TORQUE SCREWS AND BOLT TO 20 - 25 INCH-POUNDS.**

f. Remove protective caps and plugs from electrical connectors.

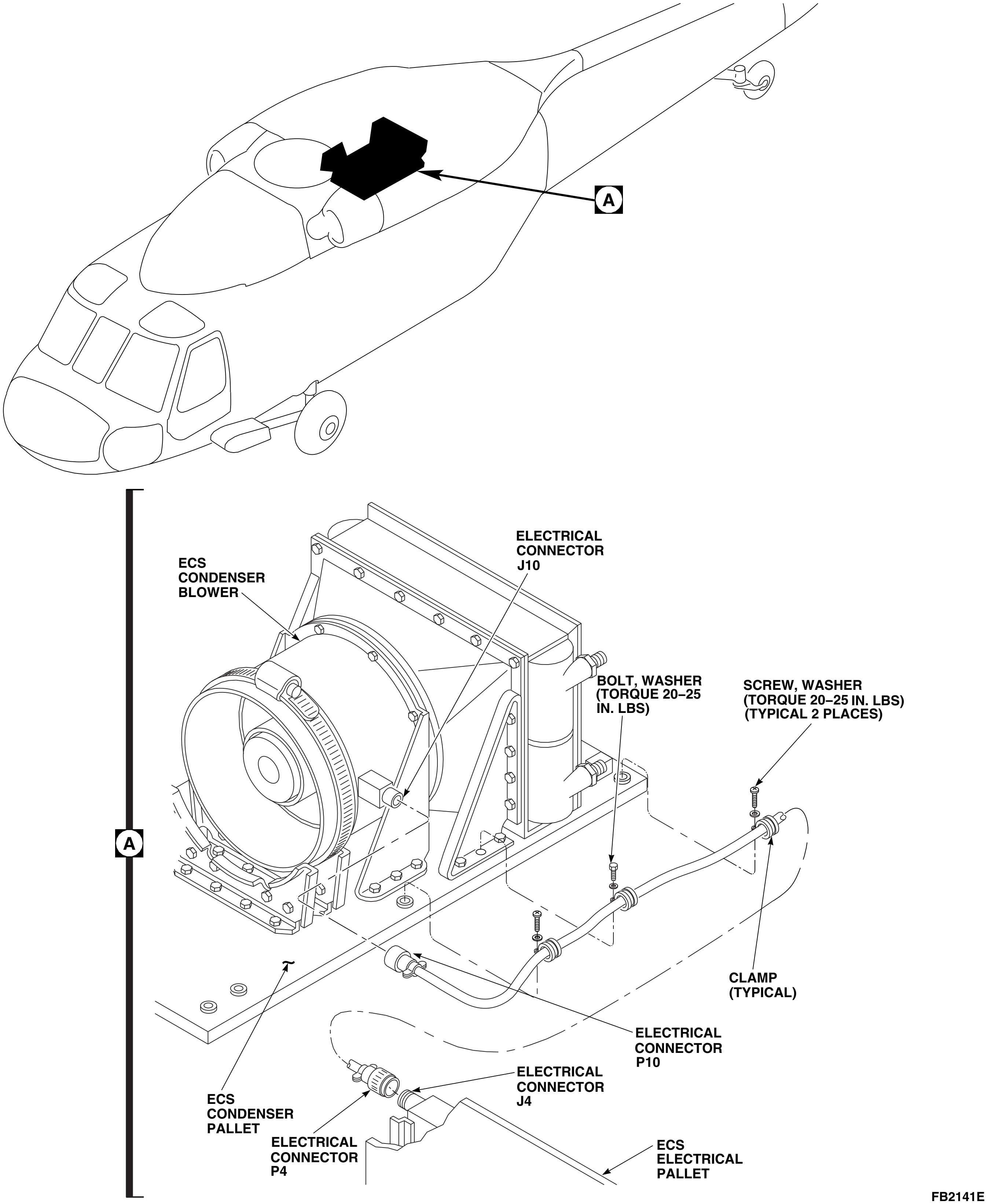


FIGURE 13-4-20-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER ELECTRICAL HARNESS

g. Inspect and treat electrical connector (PARA 1-7-20).

h. Connect electrical connector, P10 to J10, on ECS condenser blower.

i. Install ECS condenser exhaust duct (PARA 13-4-23).

13-4-21 ENVIRONMENTAL CONTROL SYSTEM PLENUM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Putty Knife
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 61, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Tetrachloroethylene, Item 324, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

13-4-21.1 Replace Environmental Control System Plenum.

13-4-21.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from rear cabin bulkhead and rear section of cabin ceiling (PARA 2-4-69).

c. Remove screws and washers securing access door and rubber seal to left plenum (Figure 13-4-21-1, Sheet 1, Detail A).

d. To remove left plenum, perform the following:

(1) Remove nut and washers securing ECS temperature sensor to left plenum. Remove temperature sensor.

(2) Support left plenum and remove screws and washers securing left plenum and rubber seal to center plenum.

(3) Remove bolts and nuts securing left plenum to front support bracket, rubber seal, and ECS evaporator blower.

(4) Using putty knife, carefully separate left plenum from front support bracket. Remove left plenum and rubber seal.

e. To remove right plenum, perform the following:

(1) Support right plenum and remove screws and washers securing right plenum and rubber seal to center plenum and duct.

(2) Using putty knife, carefully separate right plenum from center plenum. Remove right plenum and rubber seal.

f. To remove center plenum, perform the following:

NOTE

To remove center plenum from left side, the ECS evaporator pallet must be removed. To remove center plenum from right side, the ECS condenser pallet must be removed.

(1) Remove ECS condenser pallet (PARA 13-5-3) or ECS evaporator pallet (PARA 13-5-6).

(2) Unlock turnlock fasteners securing panel to airframe.

(3) While supporting center plenum, remove bolts securing center plenum to panel and remove panel.

(4) Remove center plenum.

13-4-21.1.2 Install.

a. If same plenum is being installed, clean all mating surfaces with clean cloth, Item 102, Appendix D, dampened with tetrachloroethylene, Item 324, Appendix D.

b. To install center plenum, perform the following (Figure 13-4-21-1, Sheet 2, Detail B):

(1) Install panel and lock turnlock fasteners while holding center plenum.

(2) Install one bolt on left side and one bolt on right side to hold center plenum in place.

(3) Install ECS condenser pallet (PARA 13-5-3) or ECS evaporator pallet (PARA 13-5-6).

(4) Apply light coat of adhesive, Item 61, Appendix D, to sealing surface and both sides of rubber seal. Place rubber seal on each end of center plenum. Place center plenum in rear cabin.

c. To install right plenum, perform the following:

(1) Apply light coat of adhesive, Item 61, Appendix D, to sealing surface on right plenum and both sides of rubber seal. Place rubber seal on right plenum.

(2) Remove bolt from center plenum.

(3) Install right plenum to center plenum and duct and secure with screws and washers.

d. To install left plenum, perform the following:

(1) Apply light coat of adhesive, Item 61, Appendix D, to sealing surface on left plenum and both sides of rubber seal. Place rubber seal on left plenum.

(2) Remove bolt from center plenum.

(3) Install left plenum to center plenum and secure with screws and washers.

(4) Apply light coat of adhesive, Item 61, Appendix D, to sealing surface of ECS evaporator blower, left plenum, and both sides of rubber seal. Place rubber seal on ECS evaporator blower.

(5) Install ECS evaporator blower on front support bracket and secure to bracket and left plenum with bolts and nuts. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

(6) Install ECS temperature sensor and washer in left plenum and secure with washer and nut (Figure 13-4-21-1, Sheet 1, Detail A). **TIGHTEN NUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

e. Install access door as follows:

(1) Using cleaning cloth, Item 102, Appendix D, dampened with trichloroethane 1,1,1, Item 341, Appendix D, clean mating surface of access door.

(2) Using adhesive, Item 61, Appendix D, apply light coat to sealing surface of access door and both sides of rubber seal. Install rubber seal on access door.

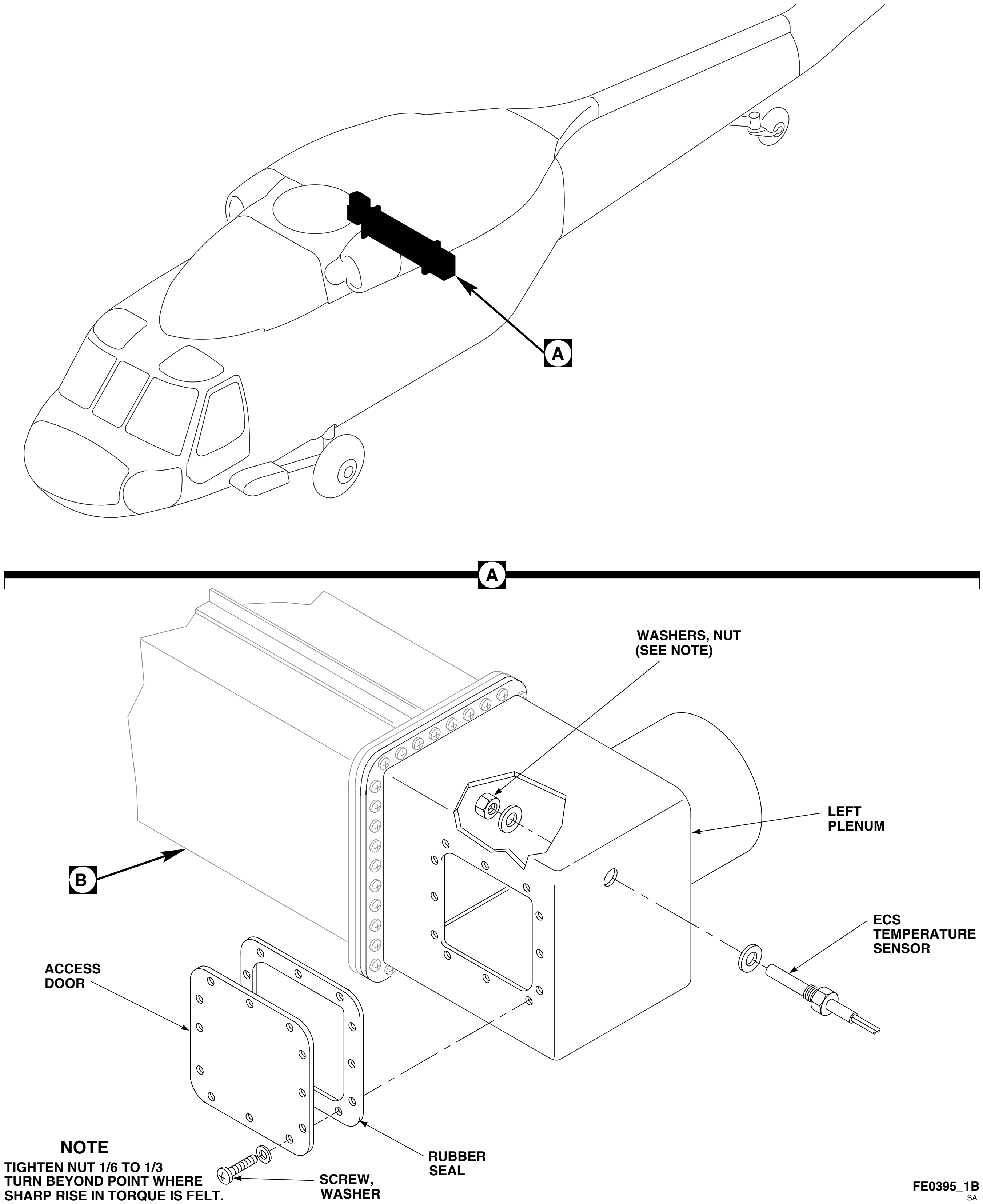


FIGURE 13-4-21-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM PLENUM
(SHEET 1 OF 2)

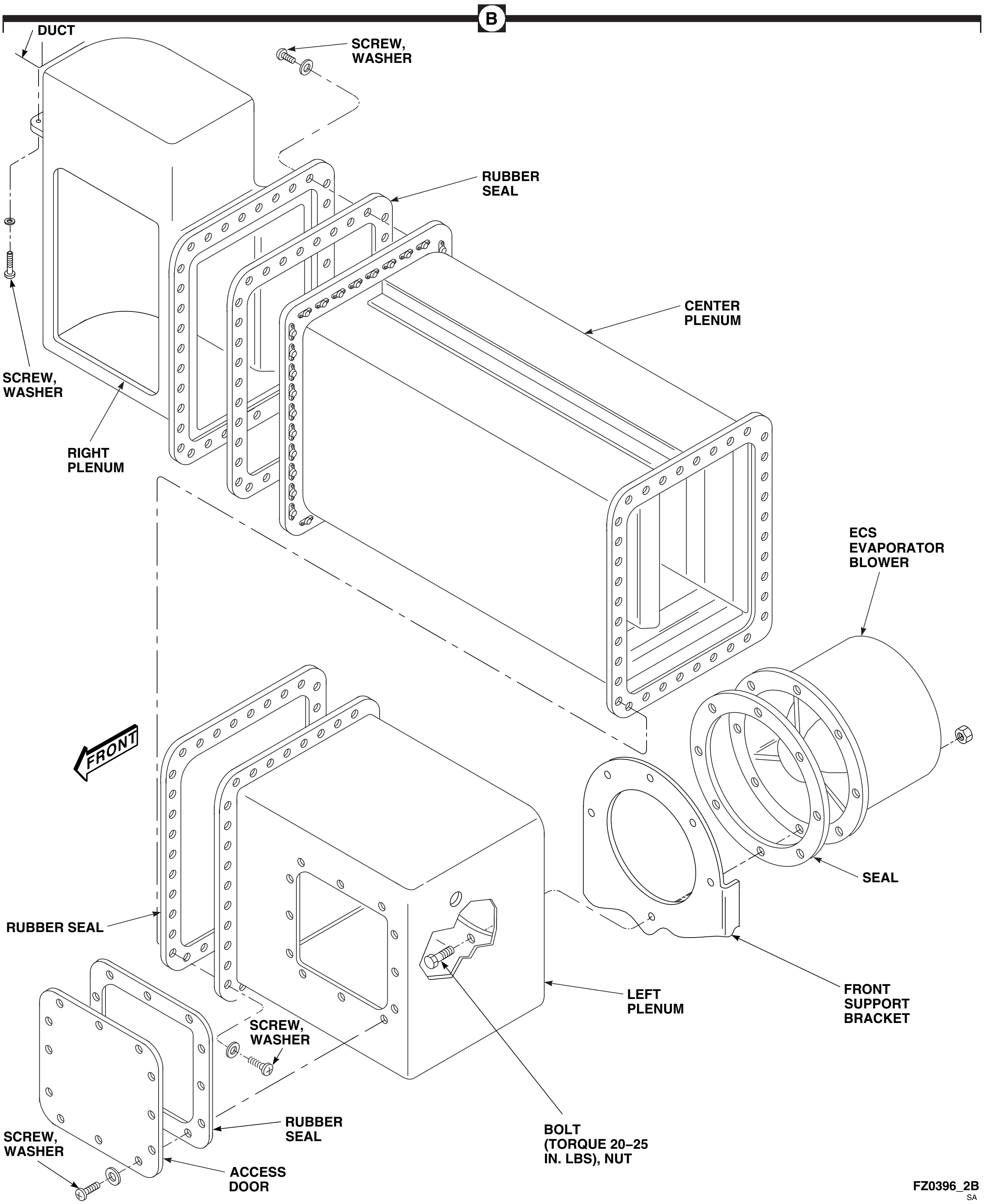


FIGURE 13-4-21-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM PLENUM (SHEET 2 OF 2)

(3) Install access door on left plenum and secure with screws and washers.

f. Install soundproofing panels on rear cabin bulkhead and rear section of cabin ceiling (PARA 2-4-69).

13-4-22 ENVIRONMENTAL CONTROL SYSTEM SUPPLY DUCTING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Adhesive, Item 37, Appendix D
- Adhesive, Item 57, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Pressure Sensitive Tape, Item 251B, Appendix D
- Tetrachloroethylene, Item 324, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69

13-4-22.1 Replace Environmental Control System Supply Ducting.

NOTE

Replacement of left and right ducts is same, except as noted.

13-4-22.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open avionics compartment access door.

c. Remove right and left soundproofing panels from rear cabin bulkhead (PARA 2-4-69).

d. Remove screws, washers, and strap securing cabin ducting to supply ducting (Figure 13-4-22-1, Detail A).

e. In transition section, remove two clamps securing right flex duct to airframe, lower evaporator outlet duct, and right supply duct.

f. Remove flex duct from helicopter.

g. Remove screws and strap securing left supply duct to upper evaporator outlet duct.

h. Remove screws, washers, and nuts securing straps and hangers to airframe and angle. Remove supply ducts from helicopter.

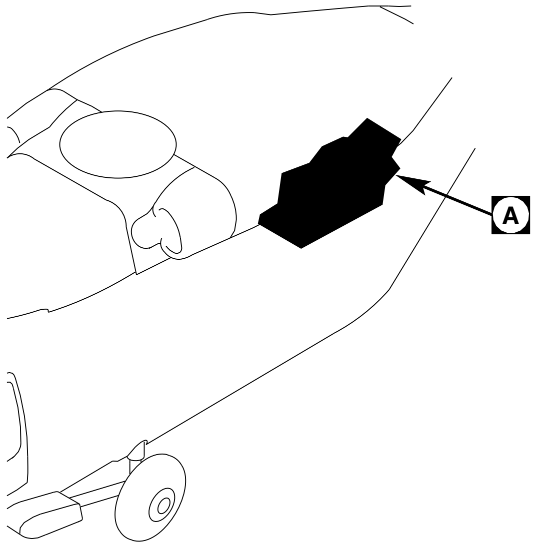
13-4-22.1.2. Repair/Replace Insulation.

a. Inspect insulation for damage. If damaged, repair or replace insulation. Proceed to step b. or step c. as necessary.

b. To repair damaged insulation, perform the following:

(1) Wrap insulation and patch with reinforced plastic film.

(2) Apply a thin coat of adhesive, Item 37, Appendix D, to surfaces of duct and insulation.



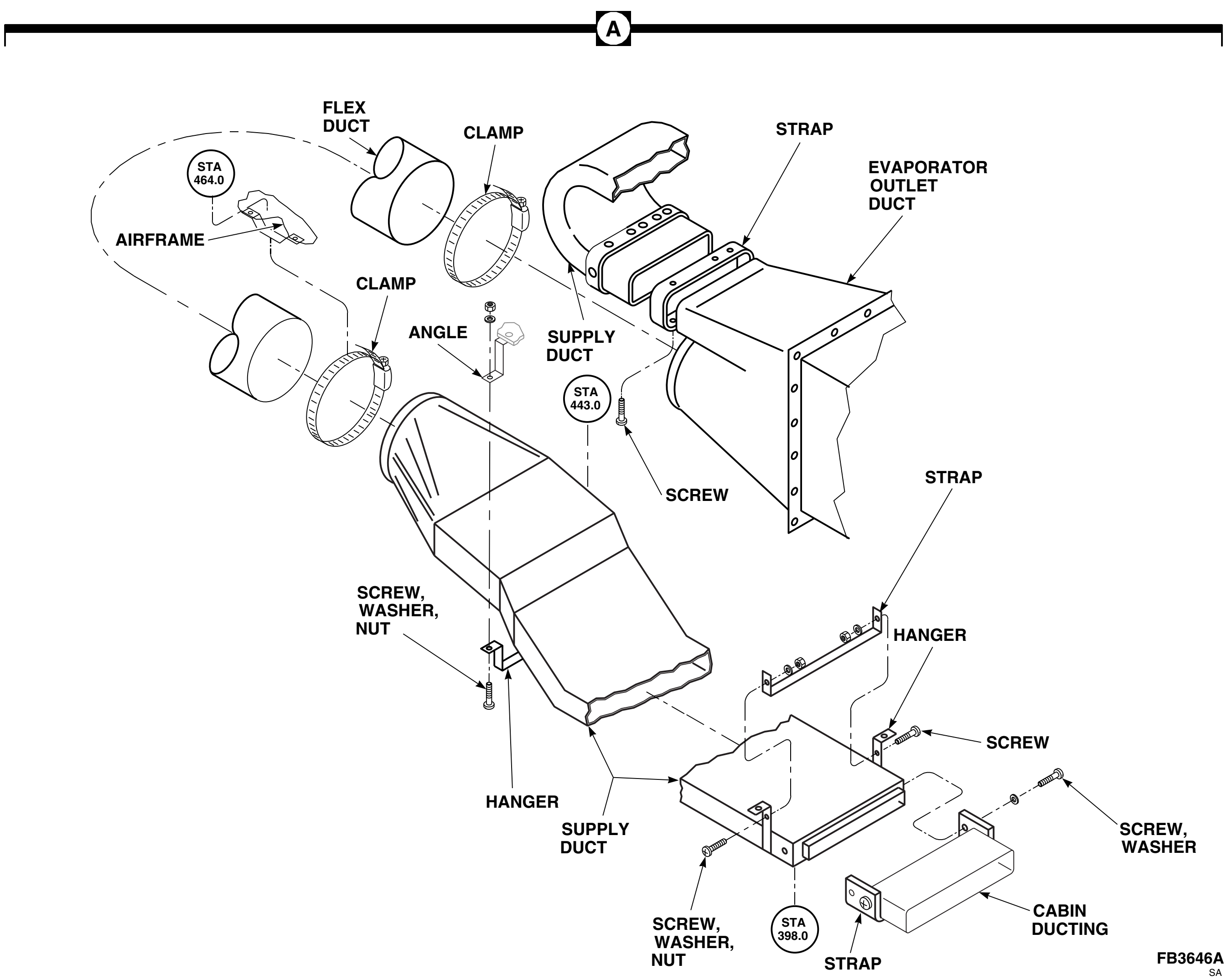


FIGURE 13-4-22-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM SUPPLY DUCTING

(3) Allow to air-dry for 10 minutes or until tacky.

(4) Press film against insulation.

(5) Tape joints and splices with pressure sensitive tape, Item 251B, Appendix D.

c. To replace insulation, perform the following:

(1) Remove damaged insulation from duct.

(2) Clean surface of duct using cleaning cloth, Item 102, Appendix D, dampened with tetrachloroethylene, Item 324, Appendix D. Wipe dry with clean cloth.

(3) Seal all edges of new insulation with adhesive, Item 57, Appendix D.

(4) Apply a thin coat of adhesive, Item 37, Appendix D, to surfaces of duct and insulation.

(5) Allow to air-dry for 10 minutes or until tacky.

(6) Line up insulation and press into place.

(7) Tape all joints with pressure sensitive tape, Item 251B, Appendix D.

13-4-22.1.3. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install supply ducts on helicopter airframe and angle with straps, hangers, screws, washers, and nuts (Figure 13-4-22-1, Detail A).

c. Install left supply duct on upper evaporator outlet duct with strap and screws.

d. Install flex duct on right supply duct, lower evaporator outlet duct, and airframe with clamps.

e. Connect supply ducting and cabin ducting. Install strap and secure with screws and washers.

f. Install right and left soundproofing panels on rear cabin bulkhead (PARA 2-4-69).

g. Close and latch avionics compartment access door.

13-4-23 ENVIRONMENTAL CONTROL SYSTEM CONDENSER EXHAUST DUCT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 13-4-21

13-4-23.1 Replace Environmental Control System Condenser Exhaust Duct.

13-4-23.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove right soundproofing panel from rear cabin bulkhead (PARA 2-4-69).

c. Remove right plenum (PARA 13-4-21).

d. Open transition section avionics door.

e. Loosen clamps securing flex duct to ECS condenser intake duct and condenser intake transition duct. Remove flex duct.

f. Disconnect drain tube from bottom of condenser exhaust duct (Figure 13-4-23-1, Detail A).

g. Remove screws and washers securing outlet end of condenser exhaust duct to outlet screen flange.

h. Loosen clamp securing condenser exhaust duct to condenser duct sleeve.

i. Remove bolts and washers securing ECS condenser pallet to support channels (Figure 13-4-23-1, Detail B).

j. Slide ECS condenser pallet to rear position.

k. Remove condenser exhaust duct from helicopter.

13-4-23.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Apply bead of sealing compound, Item 292, Appendix D, to outboard side of duct.

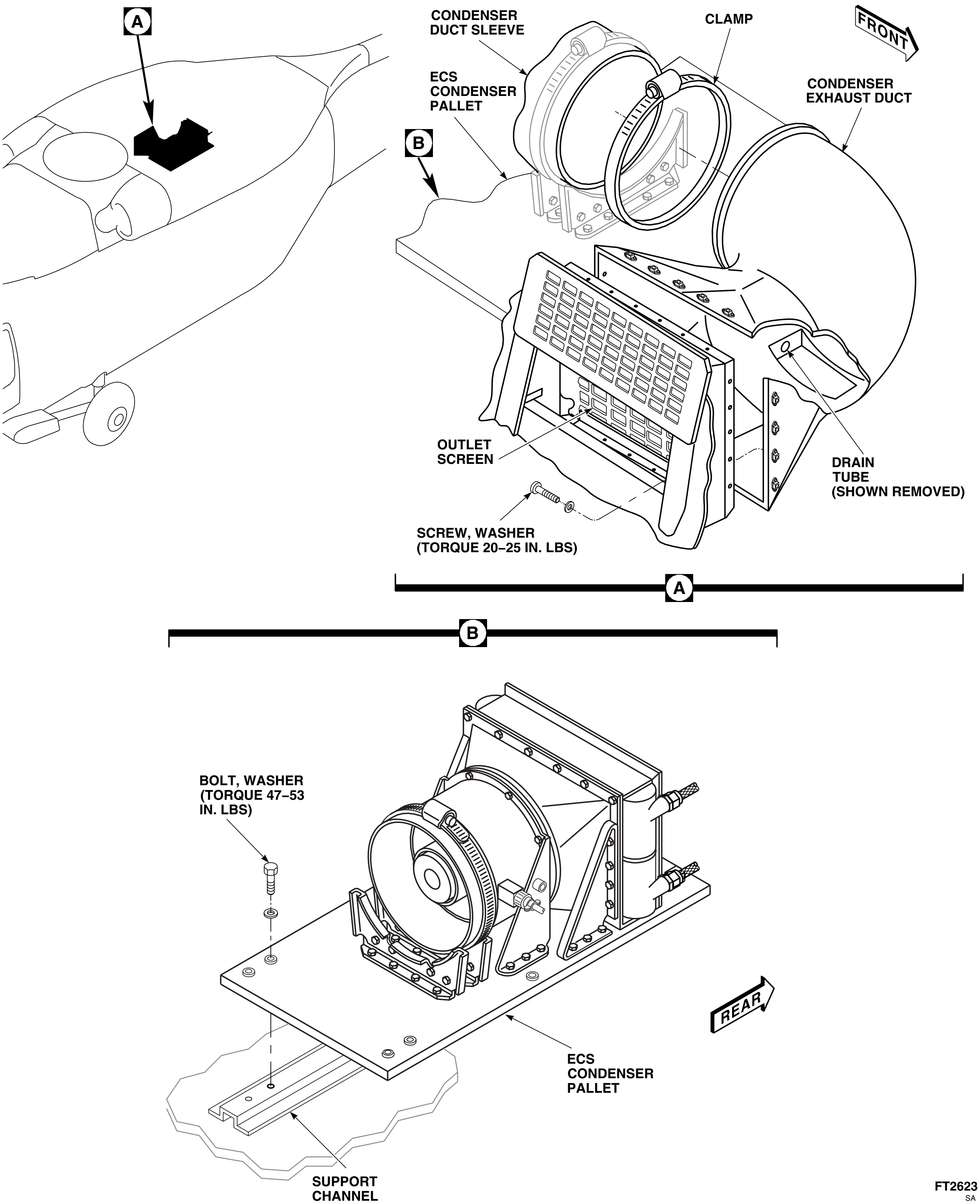


FIGURE 13-4-23-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER EXHAUST DUCT

c. Position condenser exhaust duct in helicopter. Loosely install screws and washers in condenser exhaust duct and outlet screen (Figure 13-4-23-1, Detail A).

d. Connect drain tube to bottom of condenser exhaust duct.

e. Slide ECS condenser pallet forward until boltholes in pallet line up with boltholes in support channels.

f. Connect condenser duct sleeve to condenser exhaust duct with clamp.

g. Install bolts and washers securing ECS condenser pallet to support channels. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS** (Figure 13-4-23-1, Detail B).

h. **TORQUE SCREWS AT OUTLET SCREEN TO 20 - 25 INCH-POUNDS.**

i. Install flex duct between ECS condenser intake duct and condenser intake transition duct.

NOTE

Clamps installed on flex duct are two hose clamps connected.

j. Install clamps on both ends of flex duct, at condenser intake transition duct and ECS condenser intake duct. **TIGHTEN CLAMPS.**

k. Install right plenum (PARA 13-4-21).

l. Install right soundproofing panel on rear cabin bulkhead (PARA 2-4-69).

m. Close and latch transition section avionics door.

13-4-24 ENVIRONMENTAL CONTROL SYSTEM CONDENSER INTAKE DUCT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 37, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69

13-4-24.1 Replace Environmental Control System Condenser Intake Duct.

13-4-24.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove rear cabin bulkhead panel (PARA 2-4-69).

c. Open avionics compartment access door.

d. Loosen clamps securing flex duct to ECS condenser intake duct and condenser intake transition duct. Remove flex duct (Figure 13-4-24-1, Detail A).

e. Position assistant inside aft transition to support ECS condenser intake duct.

f. Remove screws and washers securing ECS condenser intake duct and intake screen to doubler.

g. Remove ECS condenser intake duct from helicopter.

13-4-24.1.2. Install.

a. If necessary, replace worn or damaged gasket as follows:

(1) Remove damaged gasket using nonmetallic scraper (Figure 13-4-24-1, Detail A).

(2) Clean mating surfaces of new gasket and ECS condenser intake duct with cleaning cloth, Item 102, Appendix D, dampened with trichloroethane 1,1,1, Item 341, Appendix D.

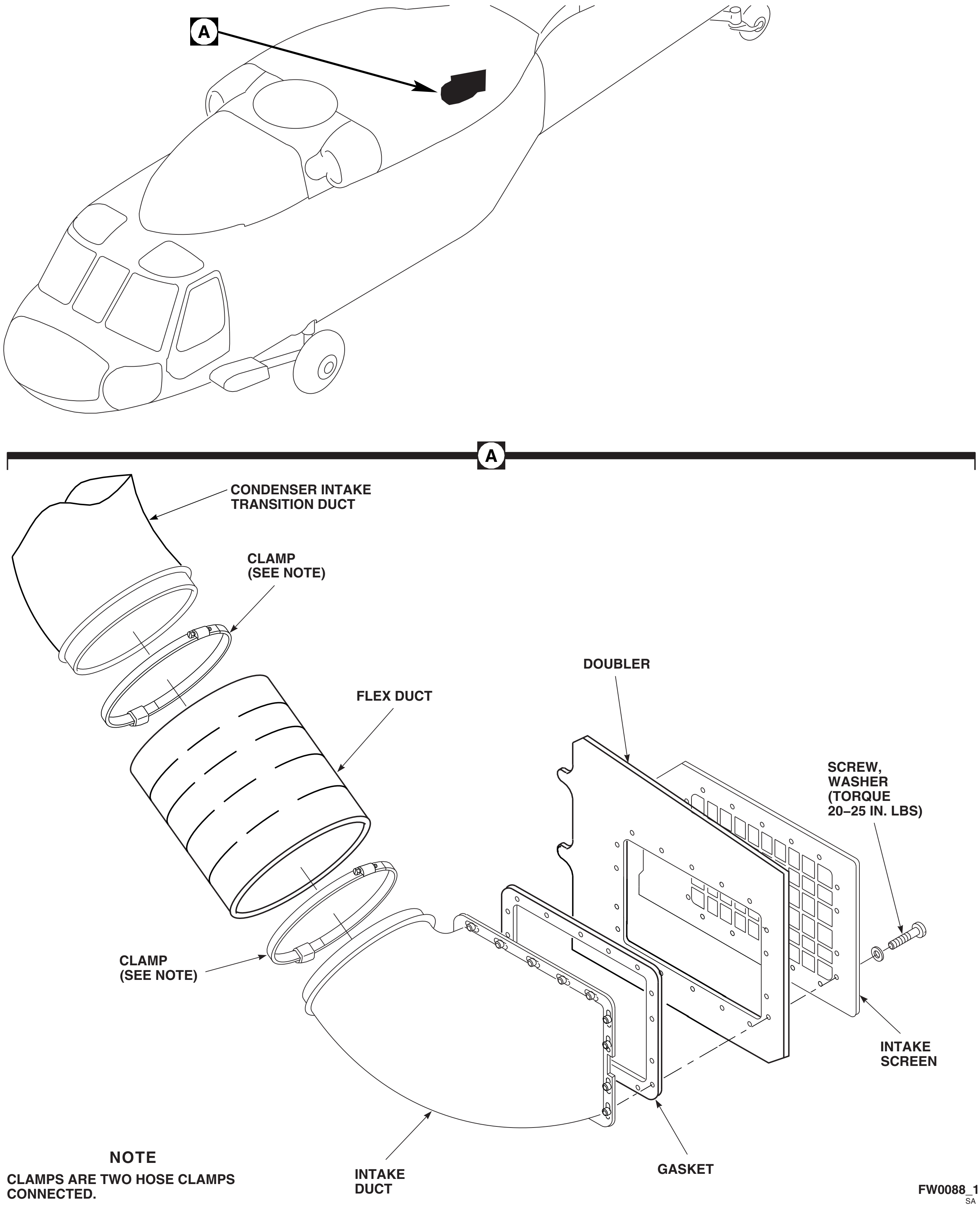


FIGURE 13-4-24-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER INTAKE DUCT

(3) Apply a thin coat of adhesive, Item 37, Appendix D, to surface of ECS condenser intake duct and gasket.

(4) Allow to air dry 10 minutes or until tacky.

(5) Line up holes in gasket and ECS condenser intake duct. Press gasket onto ECS condenser intake duct.

b. Have assistant position ECS condenser intake duct to doubler.

c. Position intake screen on doubler lining up screwholes. Install screws and washers securing intake screen and ECS condenser intake duct to doubler. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS.**

d. Install flex duct between ECS condenser intake duct and condenser intake transition duct.

NOTE

Clamps installed on flex duct are two hose clamps connected.

e. Install clamps on both ends of flex duct, at condenser intake transition duct and ECS condenser intake duct. **TIGHTEN CLAMPS.**

f. Install rear cabin bulkhead panel (PARA 2-4-69).

g. Close and latch avionics compartment access door.

13-4-25 ENVIRONMENTAL CONTROL SYSTEM AMBIENT AIR DUCT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

13-4-25.1 Replace Environmental Control System Ambient Air Duct.

13-4-25.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from rear cabin bulkhead (PARA 2-4-69).

c. Open avionics compartment access door.

d. Remove clamps, screws, washers, and nuts holding flexible ducting to ambient air valve, airframe, and coupling assembly (Figure 13-4-25-1, Detail A).

e. Remove flexible ducting from ambient air valve and coupling assembly. Slide flexible ducting through airframe and remove.

13-4-25.1.2. Install.

a. Slide flexible ducting through airframe and install on ambient air valve and coupling assembly using hose clamps (Figure 13-4-25-1, Detail A).

b. Install ducting to airframe using clamps, screws, washers, and nuts.

c. Install soundproofing panels (PARA 2-4-69).

d. Close and latch avionics compartment access door.

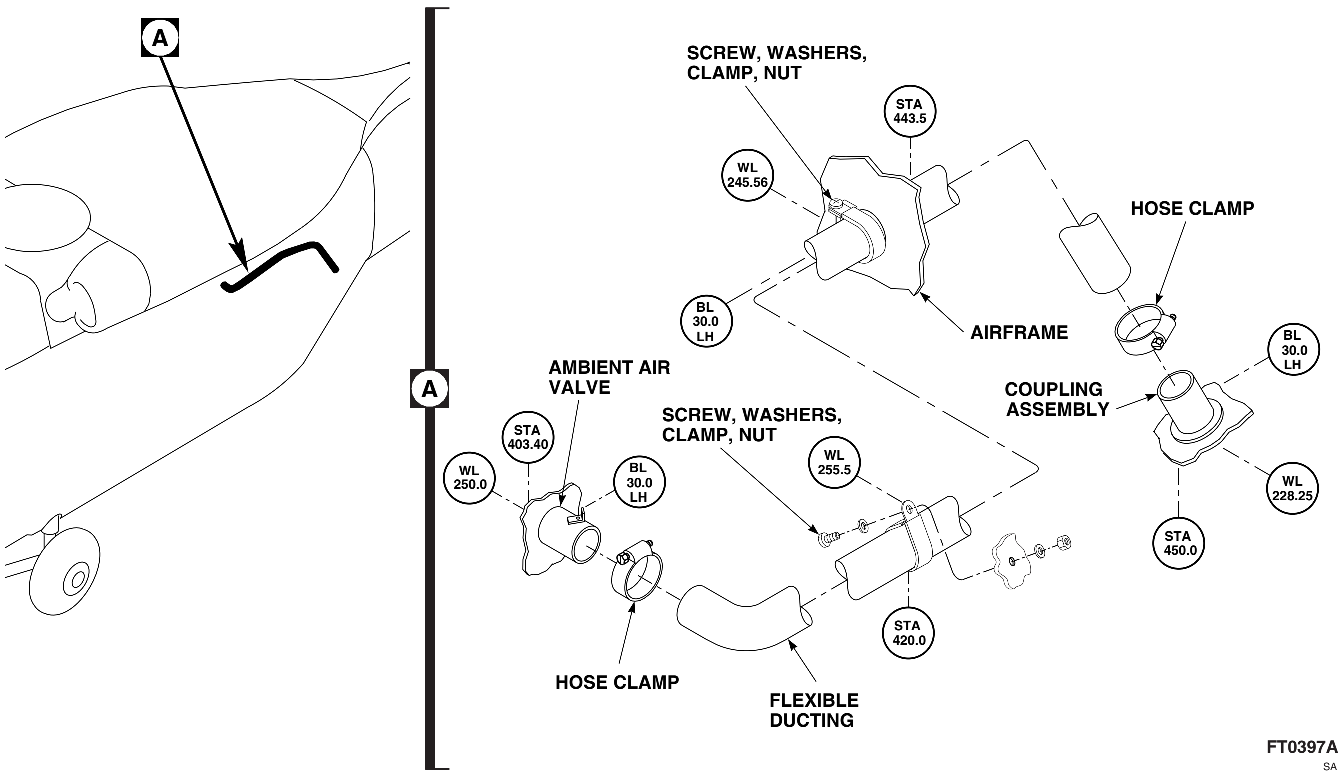


FIGURE 13-4-25-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM AMBIENT AIR DUCT

13-4-26 ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE SENSOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Adhesive, Item 61, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Tags
- Tiedown Straps, Item 329, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 13-2-2
- PARA 13-4-16

13-4-26.1 Replace Environmental Control System Temperature Sensor.

13-4-26.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels from rear cabin bulkhead and rear section of cabin ceiling (PARA 2-4-69).

c. Remove screws and washers securing access door and rubber seal to left plenum (Figure 13-4-26-1, Detail A).

d. Remove nut and washers securing ECS temperature sensor to left plenum.

e. Remove ECS evaporator electrical harness (PARA 13-4-16).

f. Remove tiedown straps from electrical harness.

g. Tag and remove wires from pins, N and P, of electrical connector, P3.

h. Remove temperature sensor and its wires from electrical harness.

13-4-26.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Inspect and treat electrical connector (PARA 1-7-20).

c. Position sensor wires in ECS evaporator electrical harness and install in pins, N and P, of electrical connector, P3. Remove tags.

d. Install new tiedown straps, Item 329, Appendix D.

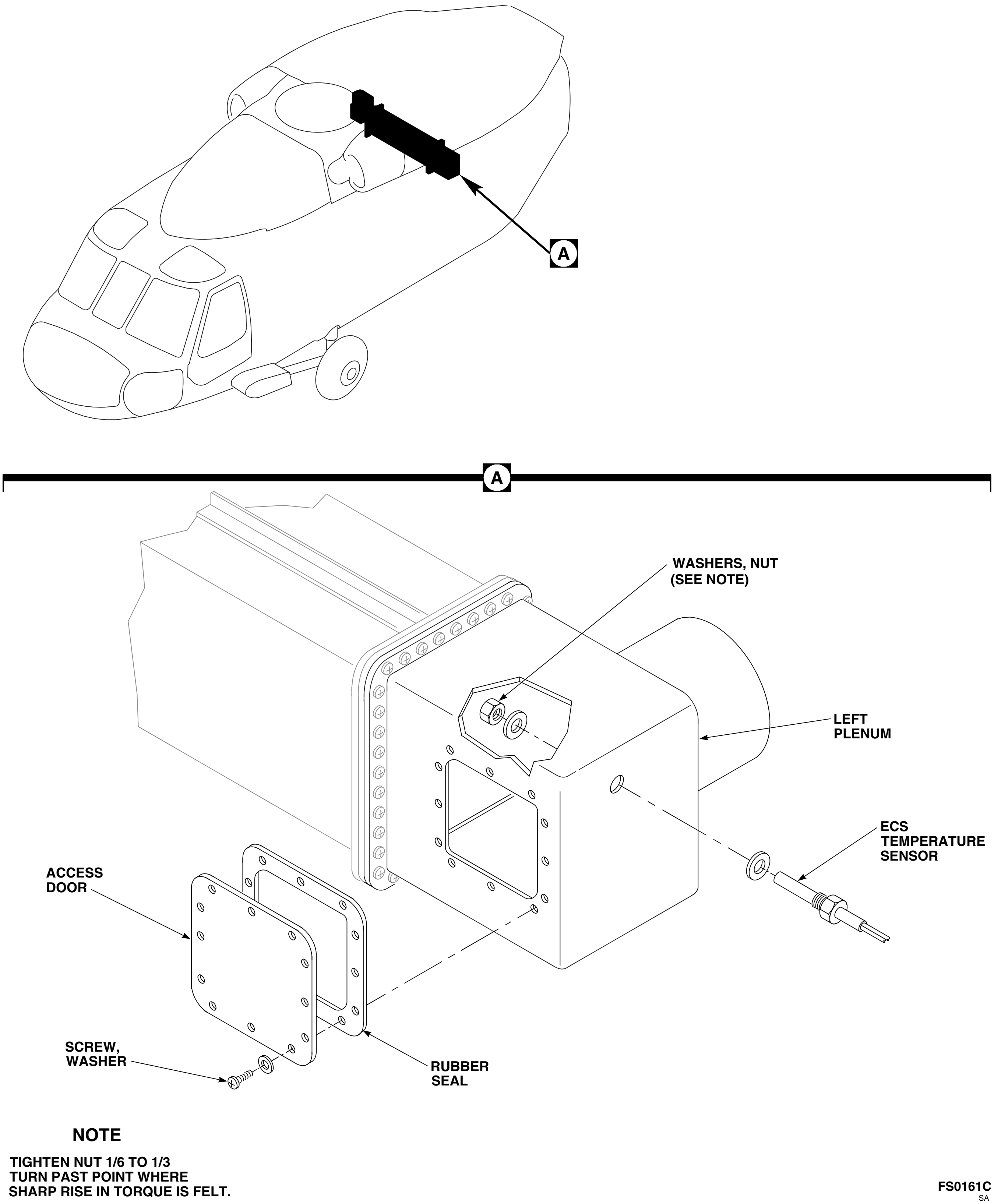


FIGURE 13-4-26-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE SENSOR

e. Install ECS evaporator electrical harness (PARA 13-4-16).

f. Install ECS temperature sensor and washer in left plenum and secure with washer and nut. **TIGHTEN NUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 13-4-26-1, Detail A).

g. Install access door as follows:

(1) Using cleaning cloth, Item 102, Appendix D, dampened with trichloroethane 1,1,1, Item 341, Appendix D, clean mating surface of access door.

(2) Using adhesive, Item 61, Appendix D, apply light coat to sealing surface of access door and both sides of rubber seal. Install rubber seal on access door.

(3) Install access door on left plenum with screws and washers.

h. Perform operational check of environmental control system (PARA 13-2-2).

i. Install soundproofing panels on rear cabin bulkhead and rear section of cabin ceiling (PARA 2-4-69).

13-4-27 ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL CONTROL UNIT (ECU).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 13-2-2

13-4-27.1 Replace Environmental Control System Electrical Control Unit (ECU).

13-4-27.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connectors from ECS electrical control unit as follows (Figure 13-4-27-1, Detail A):

(1) P456 from J1.

(2) P457 from J2.

(3) P3 from J3.

(4) P4 from J4.

(5) P5 from J5.

(6) P6 from J6.

(7) P7 from J7.

c. Install protective caps and plugs on electrical connectors.

CAUTION

To prevent damage to ECS, do not use lines, fittings, or assemblies as a handhold.

d. Support ECS electrical control unit and remove bolts and washers securing electrical control unit to electrical pallet.

e. Remove ECS electrical control unit from electrical pallet.

13-4-27.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position ECS electrical control unit on electrical pallet and secure with bolts and washers

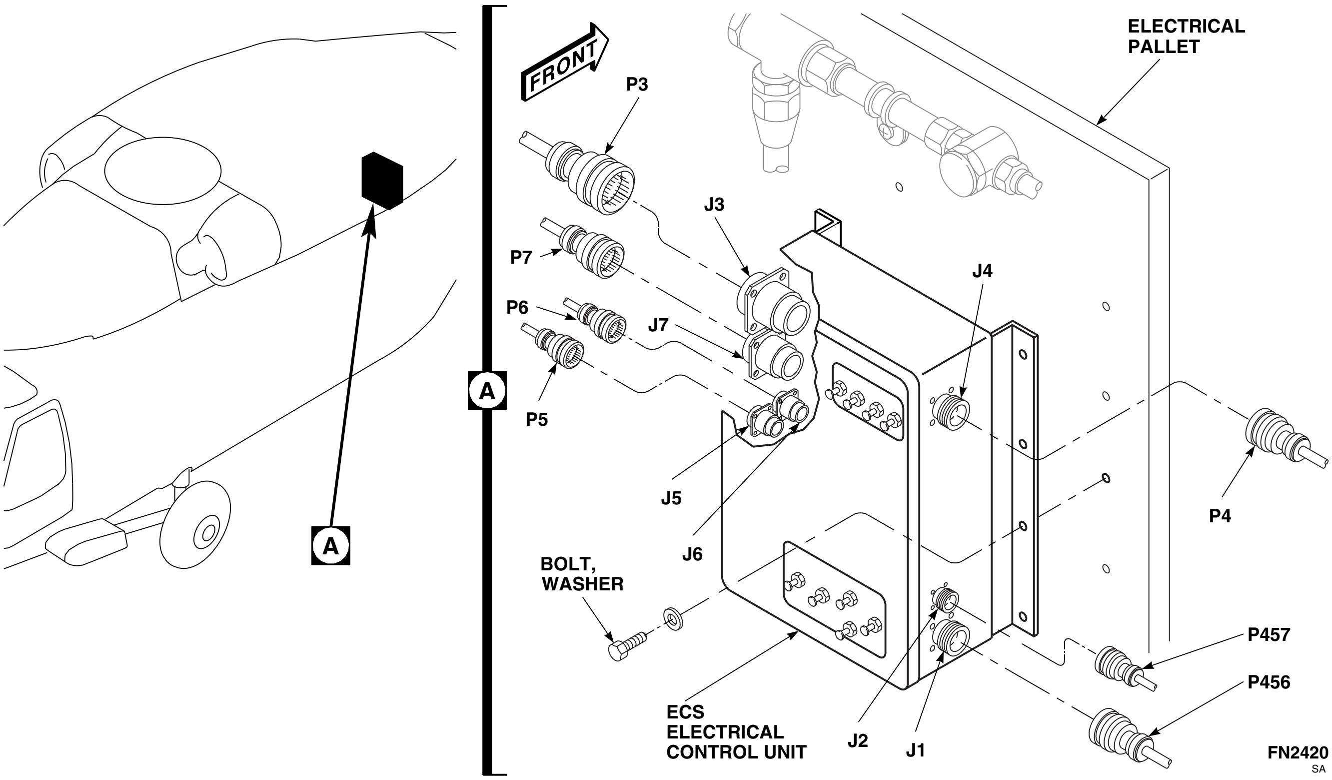


FIGURE 13-4-27-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL CONTROL UNIT (ECU)

(Figure 13-4-27-1, Detail A). **TIGHTEN BOLTS.**

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connectors to ECS electrical control unit as follows:

(1) P456 to J1.

(2) P457 to J2.

(3) P3 to J3.

(4) P4 to J4.

(5) P5 to J5.

(6) P6 to J6.

(7) P7 to J7.

f. Make sure area is clean and free of foreign material.

g. Perform operational check of ECS (PARA 13-2-2).

13-4-28 ENVIRONMENTAL CONTROL SYSTEM DUCTING REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lacquer, Item 185, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- AMS-QQ-A-200
- AMS4049
- Appendix D
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3

13-4-28.1 Repair Environmental Control System Ducting.

NOTE

The environmental control system ducting is composed of fiberglass duct sections with insulation covering the left and right supply ducts. The supply ducts are mounted between STA 398.0 and STA 460.0 and are suspended from aluminum alloy angles, hangers, and straps. All ducting is secondary structure (Figure 13-4-28-1, Sheet 1, Detail A and Sheet 2, Detail B). The condenser exhaust duct is composed of fiberglass (Figure 13-4-28-1, Sheet 2, Detail C).

a. Evaluate damage to ducting (PARA 2-6-1).

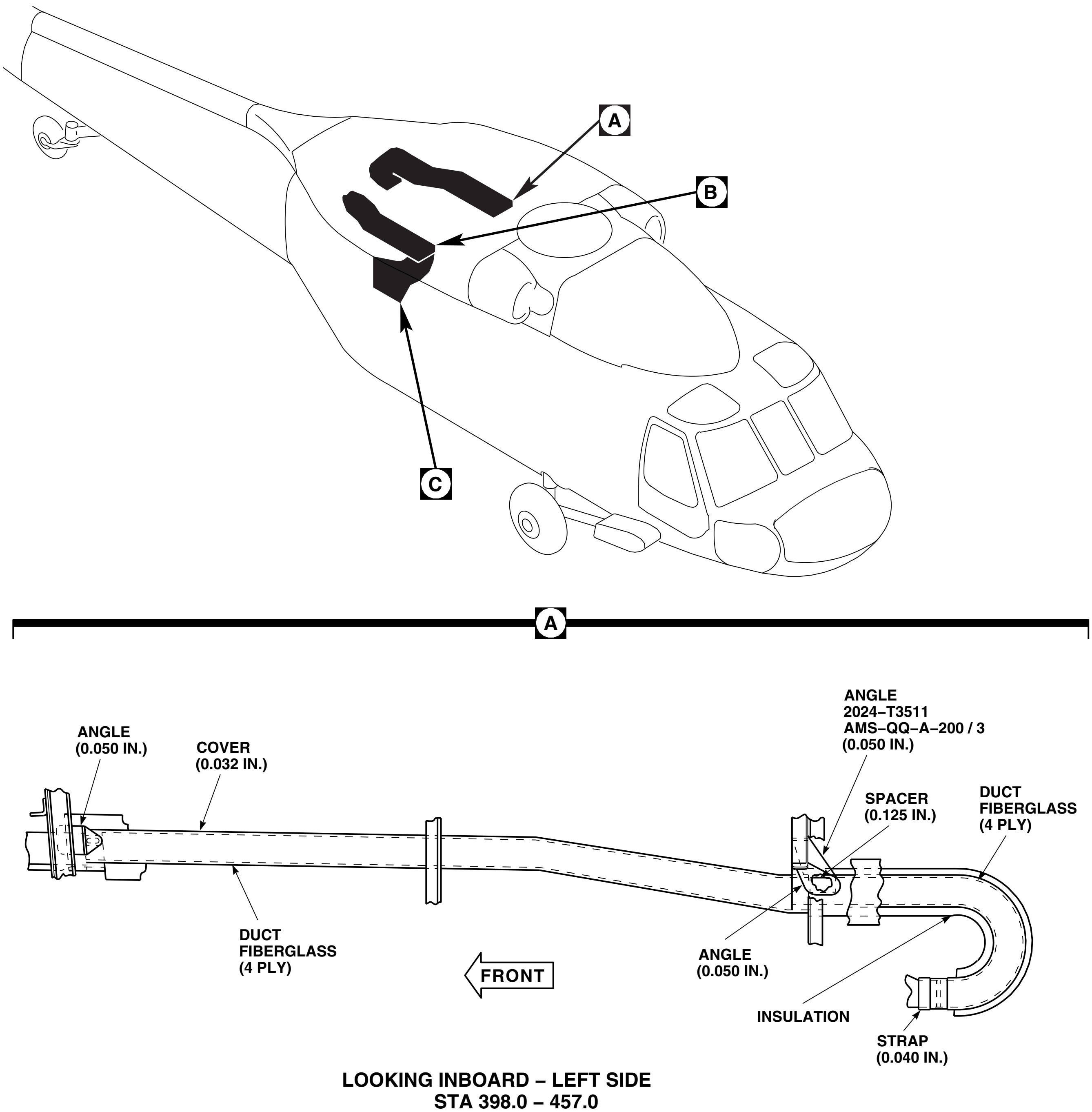
b. If necessary, repair aluminum (PARA 2-6-2).

c. Evaluate damage, and if necessary, repair fiberglass parts (PARA 2-6-3).

d. When making riveted patch repairs, using sealing compound, Item 292, Appendix D, seal for water integrity corrosion protection (PARA 2-6-1).

e. Using epoxy primer coating kit, Item 135, Appendix D, touch up repair areas, and paint with lacquer, Item 185, Appendix D.

f. Make sure area is clean and free of foreign material.



NOTE

UNLESS OTHERWISE NOTED ON THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075–T6, AMS4049, AND ALL EXTRUSION PARTS ARE ALUMINUM ALLOY, 7075–T6511, AMS–QQ–A–200 / 3.

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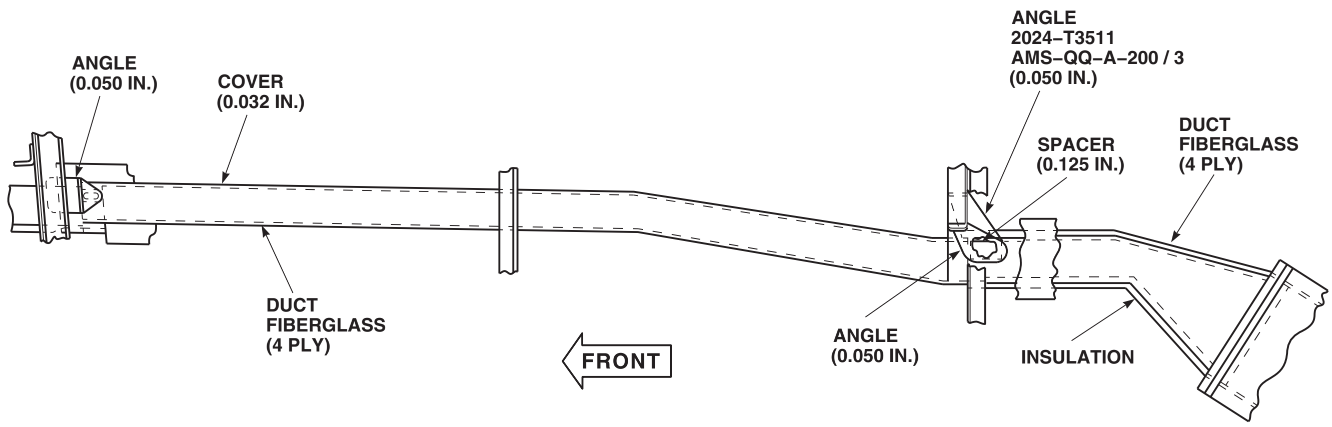
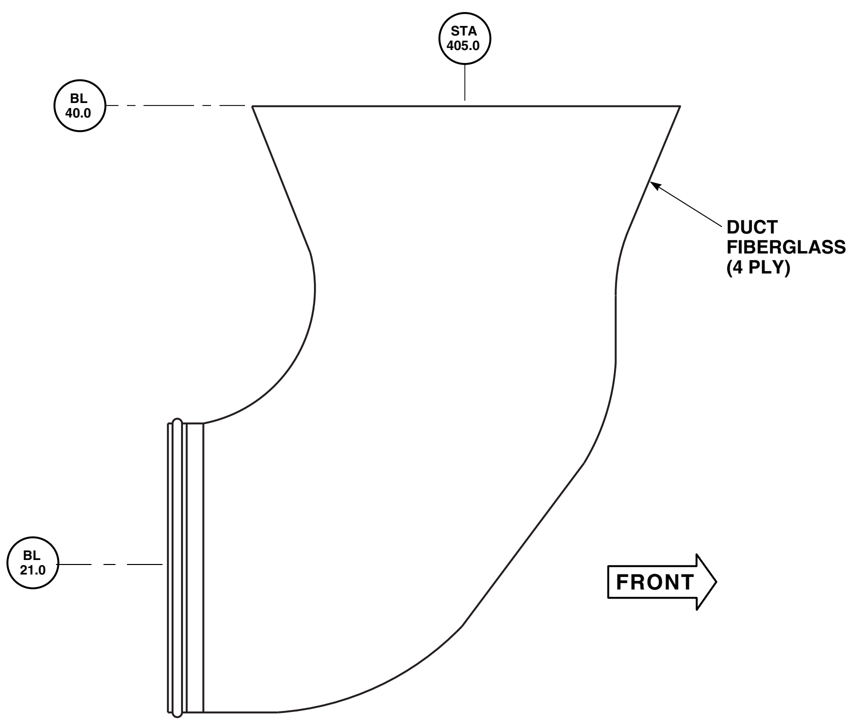
FIGURE 13-4-28-1. REPAIR ENVIRONMENTAL CONTROL SYSTEM DUCTING (SHEET 1 OF 2)



LOOKING OUTBOARD – RIGHT SIDE
STA 398.0 – 457.0

B

C



CONDENSER EXHAUST DUCT – LOOKING UP

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FIGURE 13-4-28-1. REPAIR ENVIRONMENTAL CONTROL SYSTEM DUCTING (SHEET 2 OF 2)

13-4-29 ENVIRONMENTAL CONTROL SYSTEM FILTER/DRYER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-3-27
- PARA 13-2-2

13-4-29.1 Replace Environmental Control System Filter/Dryer.

NOTE

The following procedure applies to filter/dryer mounted on or near the electrical pallet in the transition section.

13-4-29.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Deservice ECS (PARA 1-3-27).

CAUTION

- To prevent moisture from entering the air-conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.
- Filter/dryer is extremely sensitive to moisture and humidity. Replace filter/dryer if inlet and/or outlet connections are left uncapped for more than 10 minutes.

c. Disconnect lines from filter/dryer (Figure 13-4-29-1, Detail A).

d. Install caps and plugs on lines and filter/dryer.

e. Remove bolts, washers, clamps, and brackets securing filter/dryer to bracket assembly.

13-4-29.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

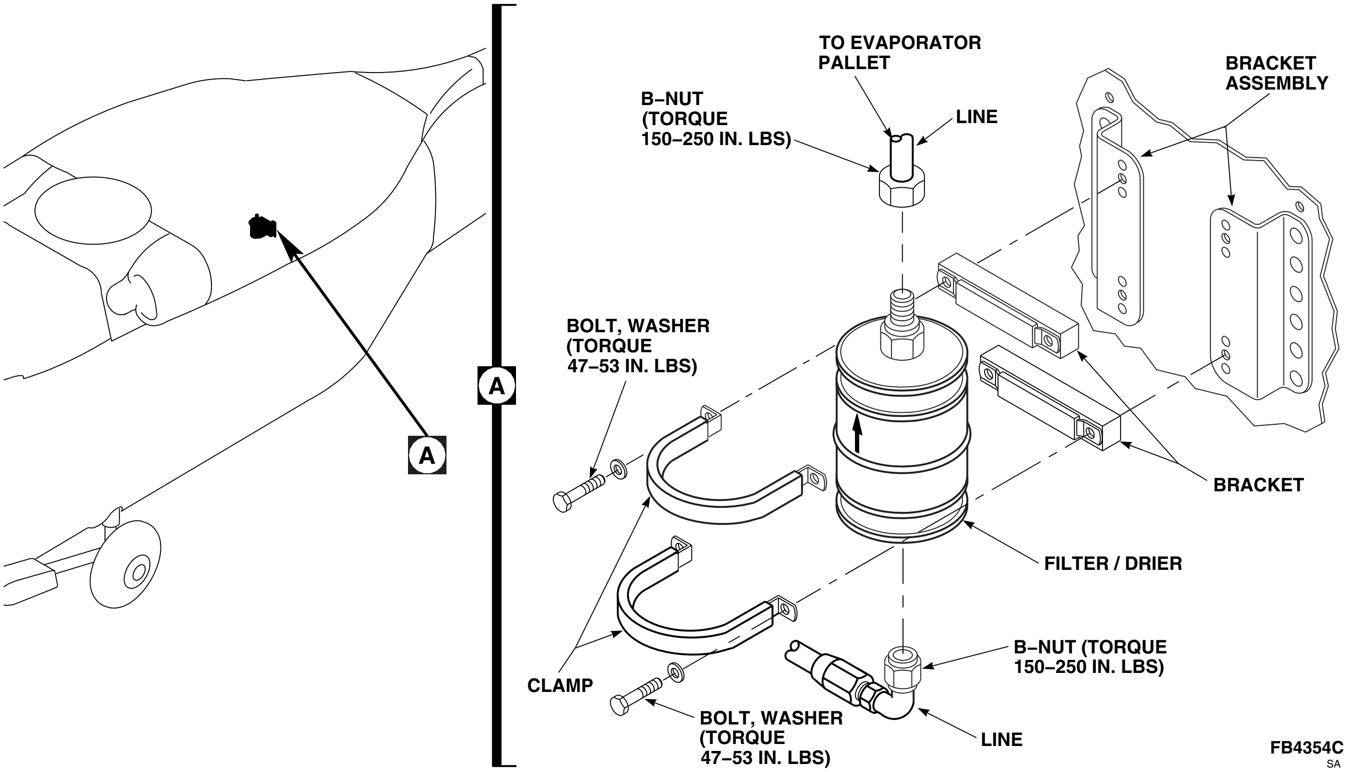


FIGURE 13-4-29-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM FILTER/DRYER

CAUTION

Filter/dryer is extremely sensitive to moisture and humidity. If inlet and/or outlet connections are left uncapped for more than 10 minutes, replace filter/dryer.

b. Install filter/dryer on bracket assembly and secure with bolts, washers, clamps, and brackets (Figure 13-4-29-1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

c. Remove caps and plugs from lines and filter/dryer.

d. Lubricate threads of line B-nuts with refrigerant compressor lubricating oil, Item 209A, Appendix D.

e. Connect lines to filter/dryer. **TORQUE B-NUTS TO 150 - 250 INCH-POUNDS.**

f. Service ECS (PARA 1-3-27).

g. Perform operational check of ECS (PARA 13-2-2).

SECTION 13-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

13-5-1 MIXTURE TEMPERATURE SENSOR PRESSURE BELLOWS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

13-5-1.1 Repair Mixture Temperature Sensor Pressure Bellows (BENCH).

13-5-1.1.1. Remove.

a. Remove screws holding pressure bellows to mixture temperature sensor (Figure 13-5-1-1).

b. Remove pressure bellows and actuating pin.

13-5-1.1.2. Install.

a. Insert actuating pin in mixture temperature sensor (Figure 13-5-1-1).

b. Install pressure bellows over actuating pin on mixture temperature sensor with screws.

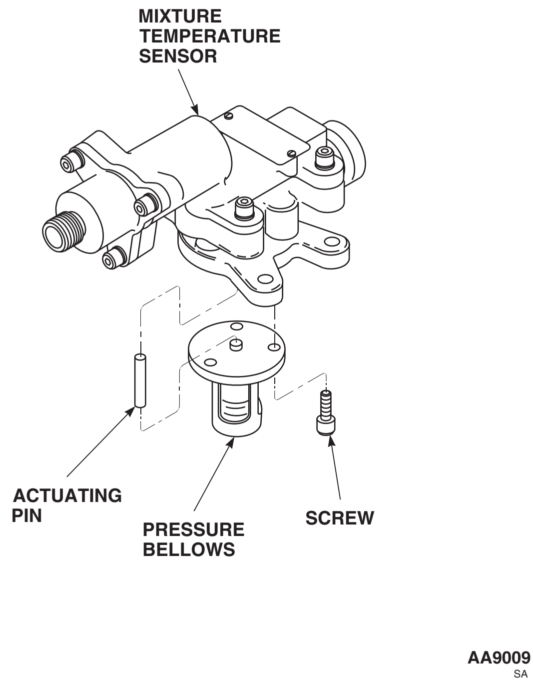


FIGURE 13-5-1-1. REPAIR MIXTURE TEMPERATURE SENSOR PRESSURE BELLOWS (BENCH)

13-5-2 BLOWER UNIT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Insert/Extraction Tool
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix D

13-5-2.1 Repair Blower Unit (BENCH).

13-5-2.1.1 Disassemble.

a. Remove nut, washer, and rotor from motor shaft (Figure 13-5-2-1).

b. Remove screw, washer, and retainer securing valve plate to stator.

c. Remove Woodruff key from motor shaft.

d. Disassemble electrical connector as follows:

(1) Remove backshell, ferrule, and coupling nut.

(2) Using insert/extraction tool, remove contact pins.

(3) Slide backshell, ferrule, and coupling nut off wiring harness.

e. Remove screws and washers securing motor in stator.

f. Remove motor from stator.

g. Remove wiring harness through grommet.

13-5-2.1.2 Inspect.

a. Inspect the following. If necessary, replace any unserviceable parts:

(1) Rotor for bent, cracked, or worn blades.

(2) Wiring harness for signs of overheating.

(3) Grommet for crack(s).

(4) Stator for damage or crack(s).

(5) Valve plate for broken spring(s) or dented plate.

(6) Motor for signs of overheating. Make sure motor shaft turns freely.

13-5-2.1.3 Assemble.

a. Install wiring harness through grommet in stator (Figure 13-5-2-1).

b. Position motor in stator and secure with screws and washers. Using lockwire, Item 196, Appendix D, lockwire screws.

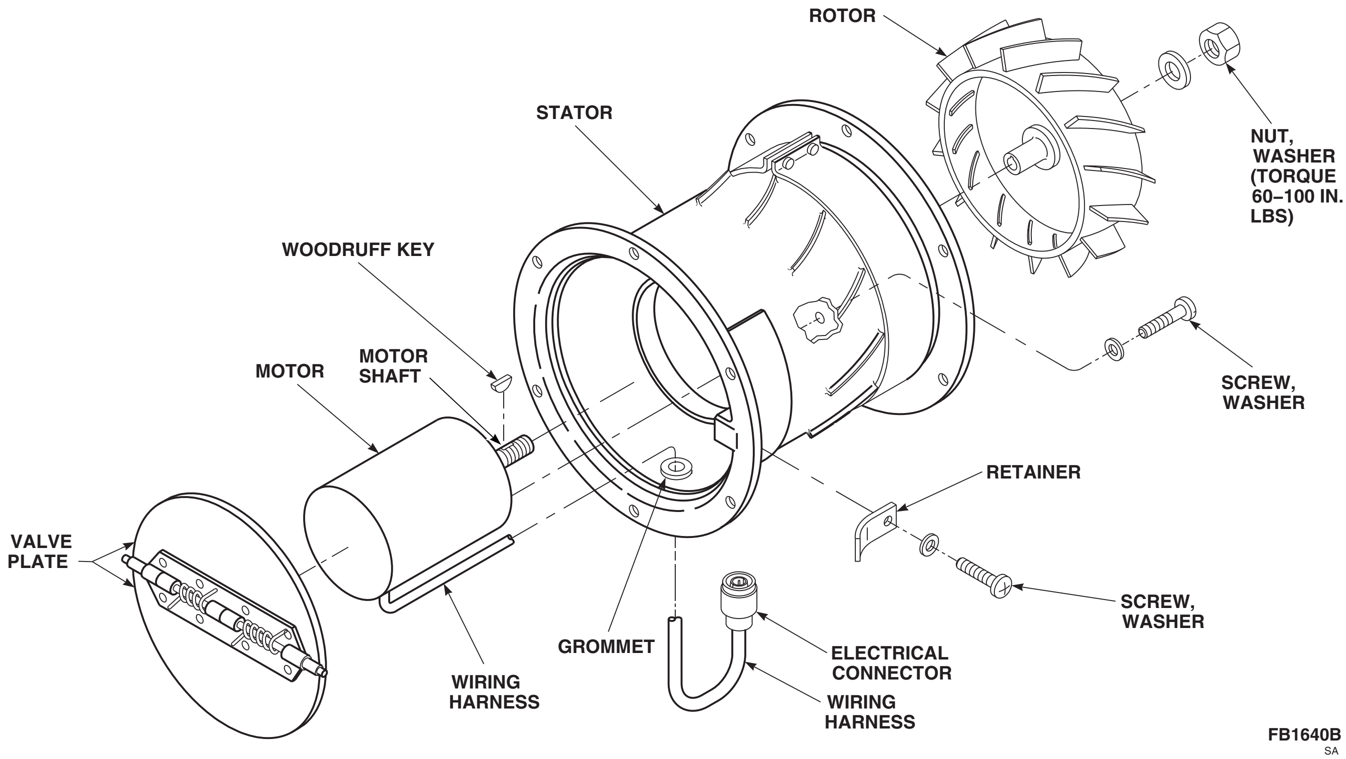


FIGURE 13-5-2-1. REPAIR BLOWER UNIT (BENCH)

c. Assemble electrical connector as follows:

(1) Slide coupling nut, ferrule, and backshell onto wiring harness.

(2) Using insert/extraction tool, insert contact pins into electrical connector.

(3) Install backshell, ferrule, and coupling nut.

d. Install Woodruff key in motor shaft.

e. Position rotor on motor shaft, aligning Woodruff key with notch in rotor.

f. Secure rotor to motor shaft with washer and nut. **TORQUE NUT TO 60 - 100 INCH-POUNDS.**

g. Position valve plate on stator and secure with retainer, screw, and washer.

13-5-3 ENVIRONMENTAL CONTROL SYSTEM CONDENSER PALLET (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Container, 5-Gallon
- Halogen Leak Detector
- Protection, Eye and Skin
- Refrigeration Unit Service Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cap, NAS817-8
- Cap, NAS817-10
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Nitrogen, Item 221, Appendix D
- Plug, NAS818-8
- Plug, NAS818-10
- Protective Cap and Plug

References

- Appendix D
- PARA 1-3-27
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 13-2-2
- PARA 13-4-23
- PARA 13-4-24

13-5-3.1 Replace Environmental Control System Condenser Pallet (BENCH).

13-5-3.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open transition section avionics door.

NOTE

Make sure the refrigerant recovery and recycling station, 17500B, is used during any deservicing of the air conditioning system.

c. Deservice ECS (PARA 1-3-27).

d. Remove right soundproofing panel from rear cabin bulkhead (PARA 2-4-69).

e. Remove ECS condenser exhaust duct (PARA 13-4-23).

f. If installed, remove ECS condenser intake duct (PARA 13-4-24).

g. Disconnect electrical connector, P4, from right side of ECS electrical pallet (Figure 13-5-3-1, Detail A).

h. Install protective cap and plug on electrical connectors.

CAUTION

To prevent moisture from entering air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting.

i. Disconnect inlet and outlet hoses from inlet and outlet ports on ECS condenser. Plug, NAS818-8 and NAS818-10, hoses and cap, NAS817-8 and NAS817-10, fittings.

j. Remove bolts and washers securing ECS condenser pallet to support channel.

CAUTION

To prevent damage to ECS, do not use lines, fittings, or other components as a handhold. Handle condenser pallet by base only.

k. Carefully remove condenser pallet from helicopter.

13-5-3.1.2 Pressure Test.

WARNING

- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with the eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of the pressurized cooling system without first carefully releasing the pressure.
- Freon-500 will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- The use of heat on any part of the pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.

a. Make sure servicing manifold valves are closed (Figure 13-5-3-1, Detail B).

b. Connect high pressure hose (red) to high side valve (REF) of servicing manifold.

c. Uncap inlet port on ECS condensor and connect other end of high pressure hose (red) to inlet port.

d. Install cap on outlet port on condenser.

e. Connect low pressure hose (yellow) to Freon-500 bottle and low side valve (VAC) of servicing manifold.

NOTE

Make sure Freon-500 bottle is kept upright so that only vapors will transfer.

f. Open valve on Freon-500 bottle and record bottle pressure.

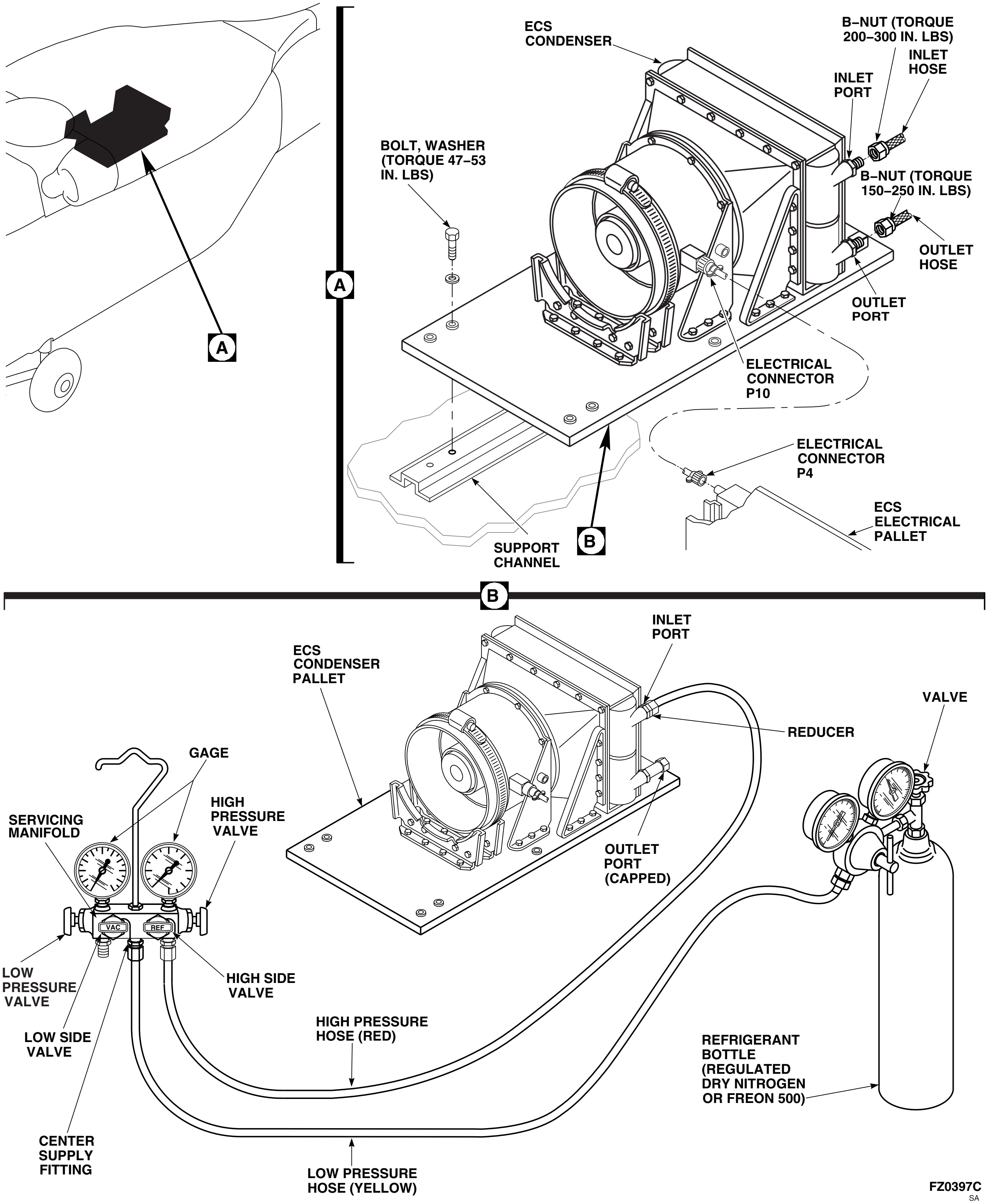


FIGURE 13-5-3-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER PALLET (BENCH)

g. Close valve on Freon-500 bottle and disconnect low pressure hose (yellow) from VAC side of servicing manifold.

h. Connect low pressure hose (yellow) to center supply fitting of servicing manifold.

i. Open Freon-500 bottle.

j. Purge low pressure hose (yellow) of air by loosening fitting at servicing manifold until a little Freon-500 vapor comes out. **IMMEDIATELY TIGHTEN FITTING.**

k. Open high side valve (REF) on servicing manifold until high side pressure gage equals Freon-500 bottle.

l. Close high side valve (REF) and Freon-500 bottle valve. Disconnect low pressure hose (yellow) from Freon-500 bottle.

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close manifold valves prior to opening supply valve on nitrogen bottle.

m. Unscrew adjustment on regulated dry nitrogen, Item 221, Appendix D, source, until it is loose. Connect low pressure hose (yellow) to regulator.

n. Open valve on nitrogen bottle and slowly increase regulator pressure until it equals Freon-500 system pressure.

o. On servicing manifold, open REF valve.

p. Slowly increase nitrogen pressure until system pressure indicated on high pressure gage of servicing manifold reaches 300 psi.

q. Using Halogen leak detector calibrated to detect leaks of 2 oz. per year or less, check all fittings, connections, hoses, and components for leaks. Repair as necessary.

r. Repeat step q. until no leaks are found.

s. On servicing manifold, close high side valve (REF).

t. Close valve on nitrogen bottle and slowly crack connection fitting.

u. Disconnect low pressure hose (yellow) from nitrogen regulator fitting and place loose end into a clean 5-gallon container, ¼-full of water.

v. Slowly open high side valve until a small stream of bubbles appears in water.

w. Continue to drain system until no bubbles appear in container. Pressure gage should read 0 psig.

x. Close high side valve and disconnect low pressure hose (yellow).

y. Disconnect high pressure hose (red) from condenser inlet.

z. **IMMEDIATELY CAP INLET FITTING.**

13-5-3.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

CAUTION

To prevent damage to ECS, do not use lines, fittings, or other components as a handhold. Handle ECS condenser pallet by base only.

b. Install condenser pallet over fuel cells and secure to support channels with bolts and washers (Figure 13-5-3-1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

c. Remove caps and plugs from inlet and outlet hoses and fittings.

d. Lubricate threads of inlet and outlet hose B-nuts with refrigerant lubricating oil compressor, Item 209A, Appendix D.

e. Connect inlet hose to inlet port (UPPER). **TORQUE B-NUT TO 200 - 300 INCH-POUNDS.**

f. Connect outlet hose to outlet port (LOWER). **TORQUE B-NUT TO 150 - 250 INCH-POUNDS.**

g. Remove protective cap and plug from electrical connectors.

h. Inspect and treat electrical connector (PARA 1-7-20).

i. Connect electrical connector, P4, to right side of ECS electrical pallet.

j. Install ECS condenser exhaust duct (PARA 13-4-23).

k. If removed, install ECS condenser intake duct (PARA 13-4-24).

l. Service ECS (PARA 1-3-27).

m. Perform operational check of ECS (PARA 13-2-2).

n. Install right soundproofing panel on rear cabin bulkhead (PARA 2-4-69).

o. Close transition section avionics door.

13-5-4 ENVIRONMENTAL CONTROL SYSTEM CONDENSER LINES, BURST DISC, AND RELIEF VALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cap, NAS817-6
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-6
- Sealing Compound, Item 293, Appendix D

References

- Appendix D
- PARA 1-3-27
- PARA 13-4-23
- PARA 13-5-5

13-5-4.1 Replace Environmental Control System Condenser Lines, Burst Disc, and Relief Valve (BENCH).

13-5-4.1.1 Remove.

a. Deservice ECS (PARA 1-3-27).

b. If installed, remove intake transition duct (PARA 13-5-5).

c. Remove ECS condenser exhaust duct (PARA 13-4-23).

d. Remove bolts, washers, clamps, and spacers securing overpressure line assembly to ECS condenser pallet (Figure 13-5-4-1, Detail B).

e. Disconnect overpressure line assembly from outlet tee.

f. Slide outboard end of overpressure line assembly from grommet in ECS transition duct.

g. Remove overpressure line assembly, burst disc, and relief valve, as an assembly, from condenser pallet.

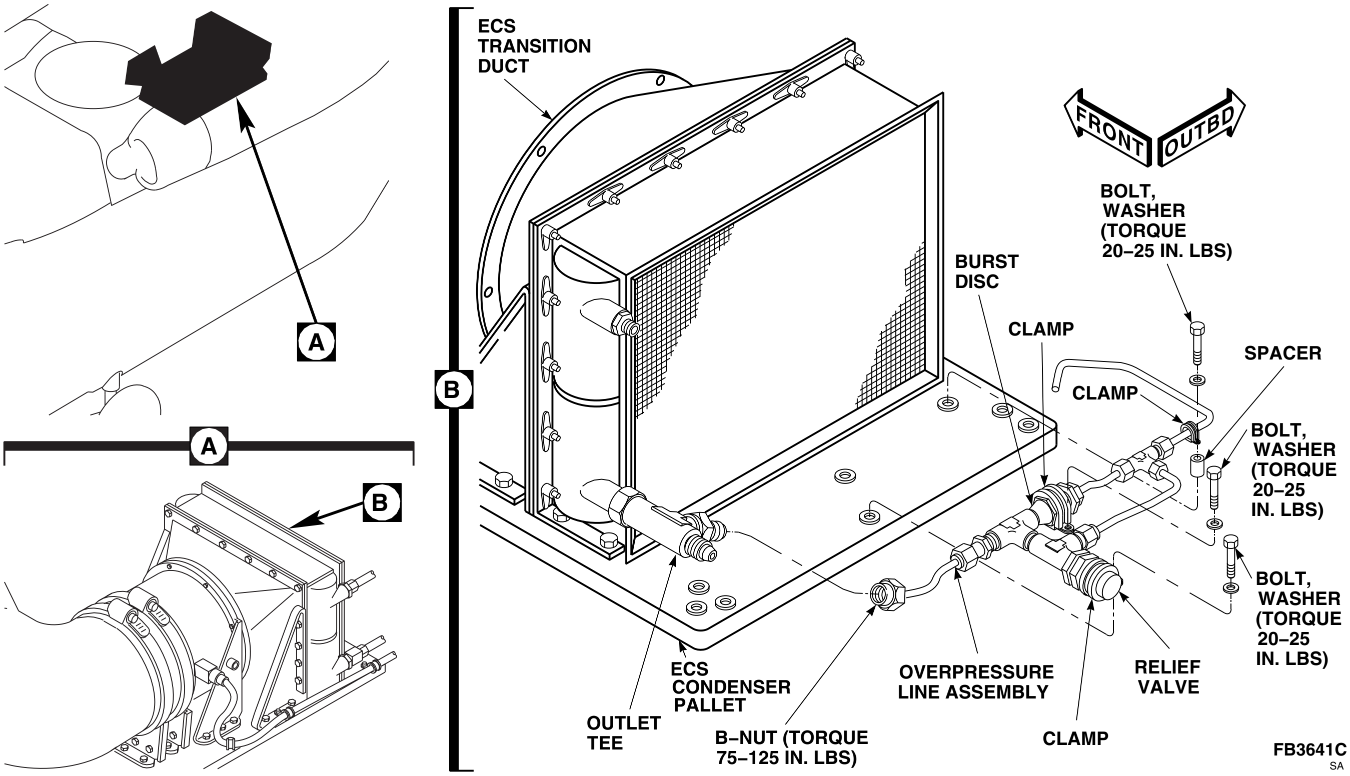


FIGURE 13-5-4-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER LINES, BURST DISC, AND RELIEF VALVE (BENCH)

CAUTION

To prevent moisture from entering air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting.

h. **IMMEDIATELY INSTALL PLUG, NAS818-6, ON LINE AND PLACE CAP, NAS817-6, ON OUTLET TEE.**

13-5-4.1.2 Install.

a. Apply sealing compound, Item 293, Appendix D, to burst disc inlet and outlet fittings and relief valve inlet fitting.

b. Lubricate threads of overpressure line B-nuts with refrigerant compressor lubricating oil, Item 209A, Appendix D, before connecting.

c. Loosely position overpressure line assembly, burst disc, and relief valve, as an assembly, on ECS condenser pallet (Figure 13-5-4-1, Detail B).

d. Slide outboard end of overpressure line assembly into grommet in exhaust transition duct.

e. Connect overpressure line assembly, burst disc, and relief valve to condenser pallet with bolts, washers, clamps and spacers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

f. Lubricate threads of remaining B-nut with refrigerant compressor lubricating oil, Item 209A, Appendix D, and loosely connect to outlet tee.

g. **TORQUE ALL OVERPRESSURE LINE ASSEMBLY B-NUTS TO 75 - 125 INCH-POUNDS.**

h. If removed, install intake transition duct (PARA 13-5-5).

i. Install ECS condenser exhaust duct (PARA 13-4-23).

j. Service ECS (PARA 1-3-27).

13-5-5 ENVIRONMENTAL CONTROL SYSTEM CONDENSER AND TRANSITION DUCTS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Nonmetallic Scraper, Locally-Made
- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cap, NAS817-6
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-6
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 13-5-3

13-5-5.1 Replace Environmental Control System Condenser and Transition Ducts (BENCH).

13-5-5.1.1 Remove.

a. Remove ECS condenser pallet (PARA 13-5-3).

CAUTION

To prevent moisture from entering the air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

b. Disconnect overpressure line assembly from outlet tee (Figure 13-5-5-1, Detail A).

c. **IMMEDIATELY INSTALL PLUG, NAS818-6, ON LINE AND PLACE CAP, NAS817-6, ON OUTLET TEE.**

d. Remove bolt, washer, spacer, and clamp securing overpressure line assembly at outboard side of condenser pallet (Figure 13-5-5-1, Detail B).

e. Disconnect overpressure line assembly from tee.

f. Slide outboard end of overpressure line assembly from grommet in ECS transition duct.

g. Remove bolts, washers, and brackets from ECS condenser pallet and transition duct (Figure 13-5-5-1, Detail A).

h. Remove bolts, washers, and nuts and break seal between ECS condenser and condenser pallet.

i. Remove ECS condenser and transition ducts, as an assembly, from condenser pallet.

j. Remove screws, washers, ECS transition duct, and, if installed, intake transition duct from condenser.

k. Using nonmetallic scraper, remove sealing compound from ECS transition duct mating flange and condenser pallet.

13-5-5.1.2 Install.

a. Using sealing compound, Item 292, Appendix D, seal interface of ECS condenser and transition duct (Figure 13-5-5-1, Detail A).

b. Install intake ECS transition ducts, if equipped, and ECS transition ducts on condenser and secure with screws and washers. **HANDTIGHTEN SCREWS SECURING TRANSITION DUCT TO CONDENSER. TORQUE SCREWS SECURING TRANSITION DUCT TO CONDENSER 20 - 25 INCH-POUNDS.**

c. Install ECS condenser and transition ducts, as an assembly, on condenser pallet with bolts, washers, and nuts. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

d. Using sealing compound, Item 292, Appendix D, seal interface of ECS condenser and condenser pallet.

e. Install brackets on ECS condenser pallet and secure with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

f. Secure brackets to ECS transition duct and condenser with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

g. Lubricate threads of overpressure line B-nut using refrigerant compressor lubricating oil, Item 209A, Appendix D. Slide overpressure line assembly into grommet in ECS transition duct.

h. Connect overpressure line assembly to tee on ECS condenser pallet. **TORQUE B-NUT TO 75 - 125 INCH-POUNDS** (Figure 13-5-5-1, Detail B).

i. Install overpressure line assembly on ECS condenser pallet with bolt, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

k. Pressure test ECS condenser pallet (PARA 13-5-3).

l. Install ECS condenser pallet (PARA 13-5-3).

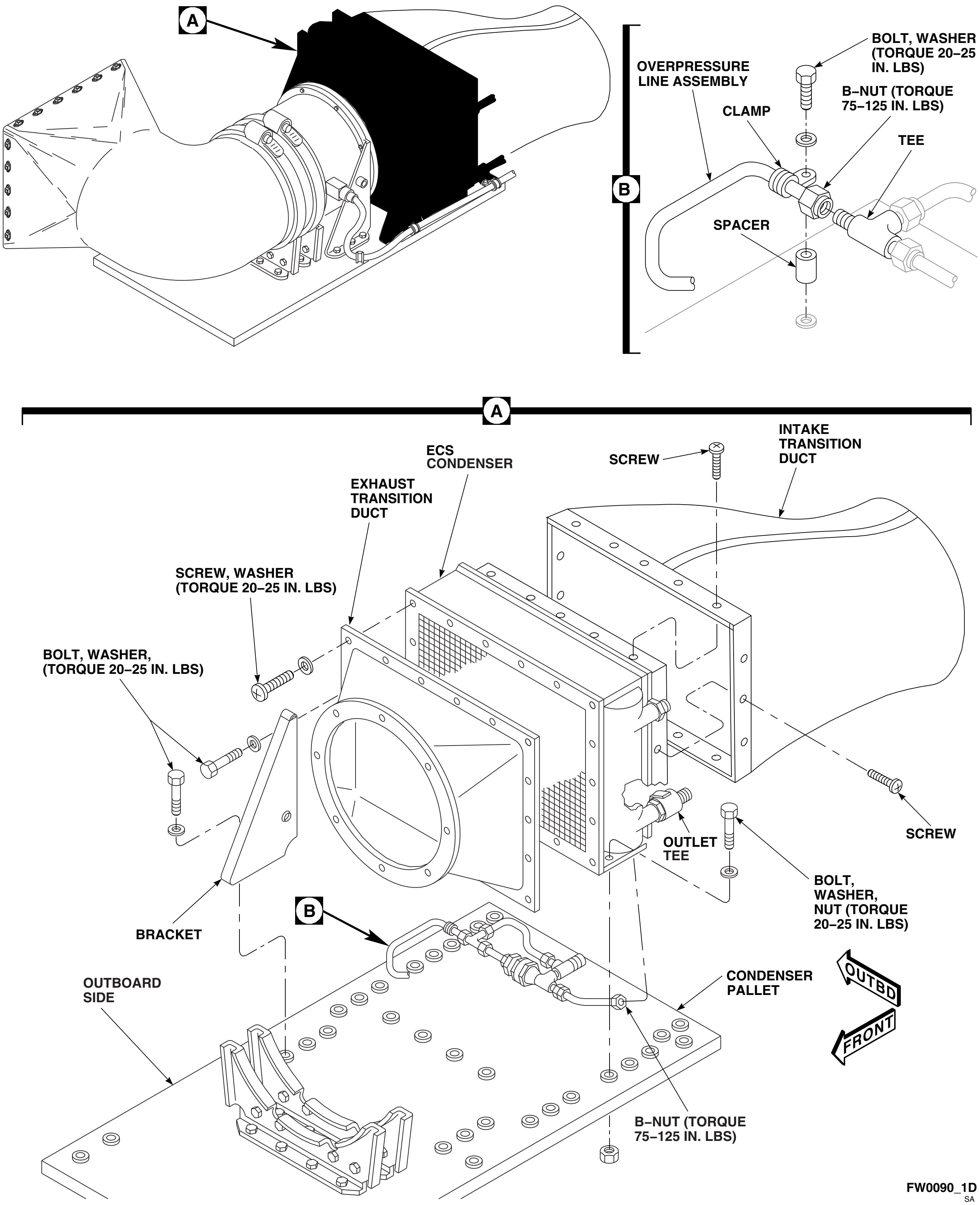


FIGURE 13-5-5-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM CONDENSER AND TRANSITION DUCTS (BENCH)

13-5-6 ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR PALLET (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Container, 5-Gallon
- Halogen Leak Detector
- Protection, Eye and Skin
- Refrigeration Unit Service Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 7500 in. lbs

Materials/Parts

- Cap, NAS817-4
- Cap, NAS817-8
- Cap, NAS817-10
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Nitrogen, Item 221, Appendix D
- Plug, NAS818-4
- Plug, NAS818-8
- Plug, NAS818-10
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-27
- PARA 1-7-20
- PARA 2-4-69
- PARA 13-2-2
- PARA 13-4-21

13-5-6.1 Replace Environmental Control System Evaporator Pallet (BENCH).

13-5-6.1.1 Remove.

a. Deservice ECS (PARA 1-3-27).

b. Remove left soundproofing panel from rear cabin bulkhead (PARA 2-4-69).

c. Remove left plenum (PARA 13-4-21).

d. Disconnect and remove electrical connectors, P3 and P7, from left side of electrical pallet (Figure 13-5-6-1, Sheet 1, Detail C).

e. Install protective caps and plugs on electrical connectors.

f. Loosen clamps securing lower supply duct to outlet transition duct and right supply duct (Figure 13-5-6-1, Sheet 1, Detail B). Remove lower supply duct.

g. Remove screw and strap securing upper supply duct to outlet transition duct. Slide upper supply duct off outlet transition duct.

h. Remove bolts and washers, and remove outlet transition duct.

CAUTION

To prevent moisture from entering the air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

i. Disconnect outlet line, inlet line, and equalization line at rear panel on evaporator pallet (Figure 13-5-6-1, Sheet 1, Detail B and Detail D). Cap, NAS817-4, NAS817-8, and NAS817-10, fittings and plug, NAS818-4, NAS818-8, and NAS818-10, hoses.

j. Remove bolts and washers securing evaporator pallet to support channels.

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as handhold. Handle evaporator pallet by base only.

k. Remove evaporator pallet from helicopter.

13-5-6.1.2 Pressure Test Evaporator Pallet.

WARNING

- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with the eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of the pressurized cooling system without first carefully releasing the pressure.
- Freon-500 will displace air and can cause suffocation, even though it is non toxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- The use of heat on any part of the pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.

a. Connect low pressure hose (yellow) to VAC side of servicing manifold and high pressure hose (red) to REF side of servicing manifold. Make sure servicing manifold valves are closed (Figure 13-5-6-1, Sheet 2).

b. Uncap center fitting at rear of ECS evaporator pallet and connect high pressure hose (red). Install plug in hose outlet.

c. Connect low pressure hose (yellow) to Freon-500 bottle.

NOTE

Make sure Freon-500 bottle is positioned upright so only vapors will transfer.

d. Open valve on Freon-500 bottle and record pressure.

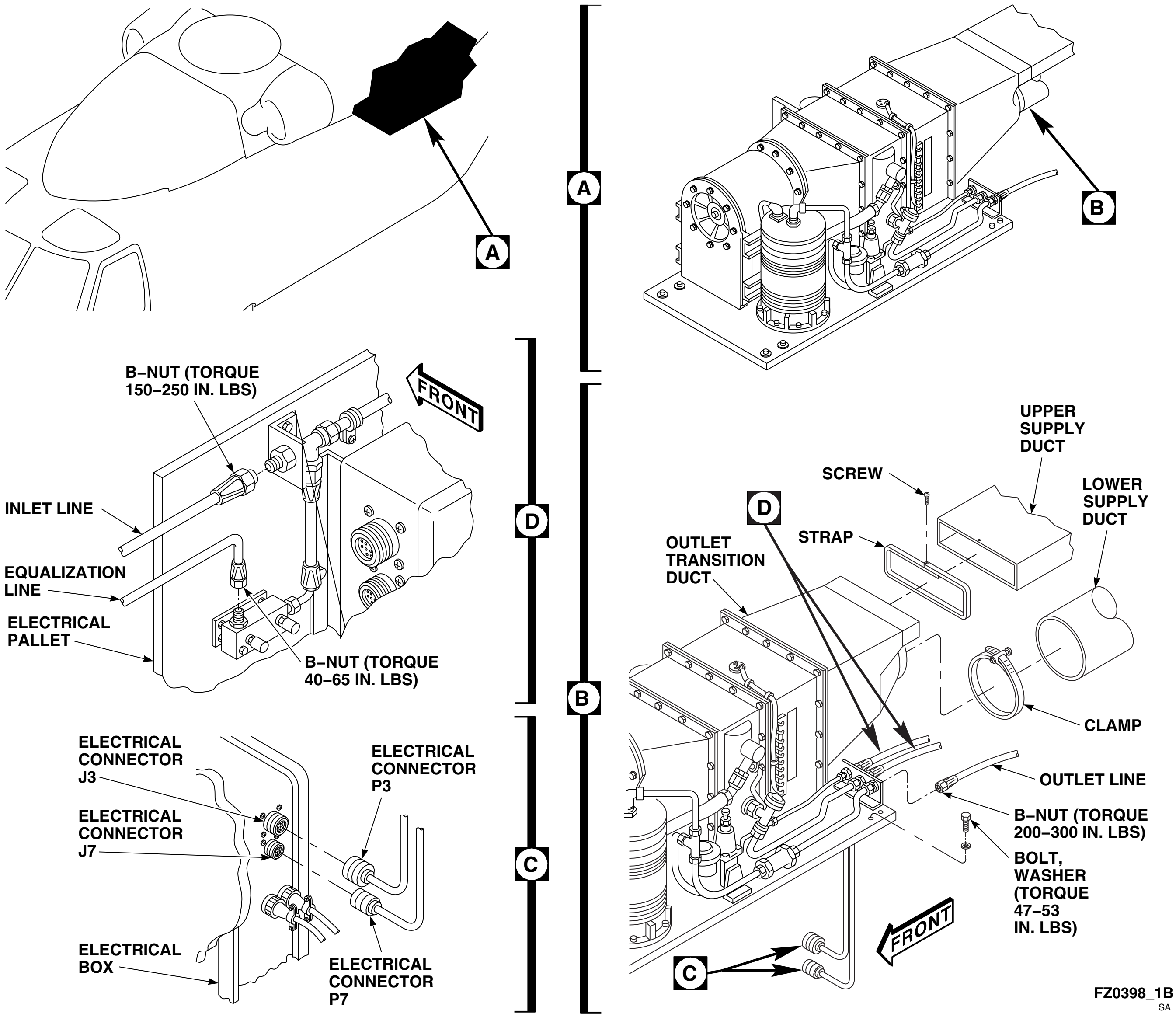


FIGURE 13-5-6-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR PALLET (BENCH) (SHEET 1 OF 2)

e. Close valve on Freon-500 bottle and disconnect low pressure hose (yellow) from VAC side of servicing manifold.

f. Connect end of low pressure hose (yellow) to center supply fitting of servicing manifold.

g. Open valve on Freon-500 bottle.

h. Purge low pressure hose (yellow) of air by loosening fitting at servicing manifold until a little Freon-500 vapor comes out. **IMMEDIATELY TIGHTEN FITTING.**

i. Open high side valve on servicing manifold until high side pressure gage equals Freon-500 pressure.

j. Close high side valve and Freon-500 bottle valve. Disconnect low pressure hose (yellow) from Freon-500 bottle.

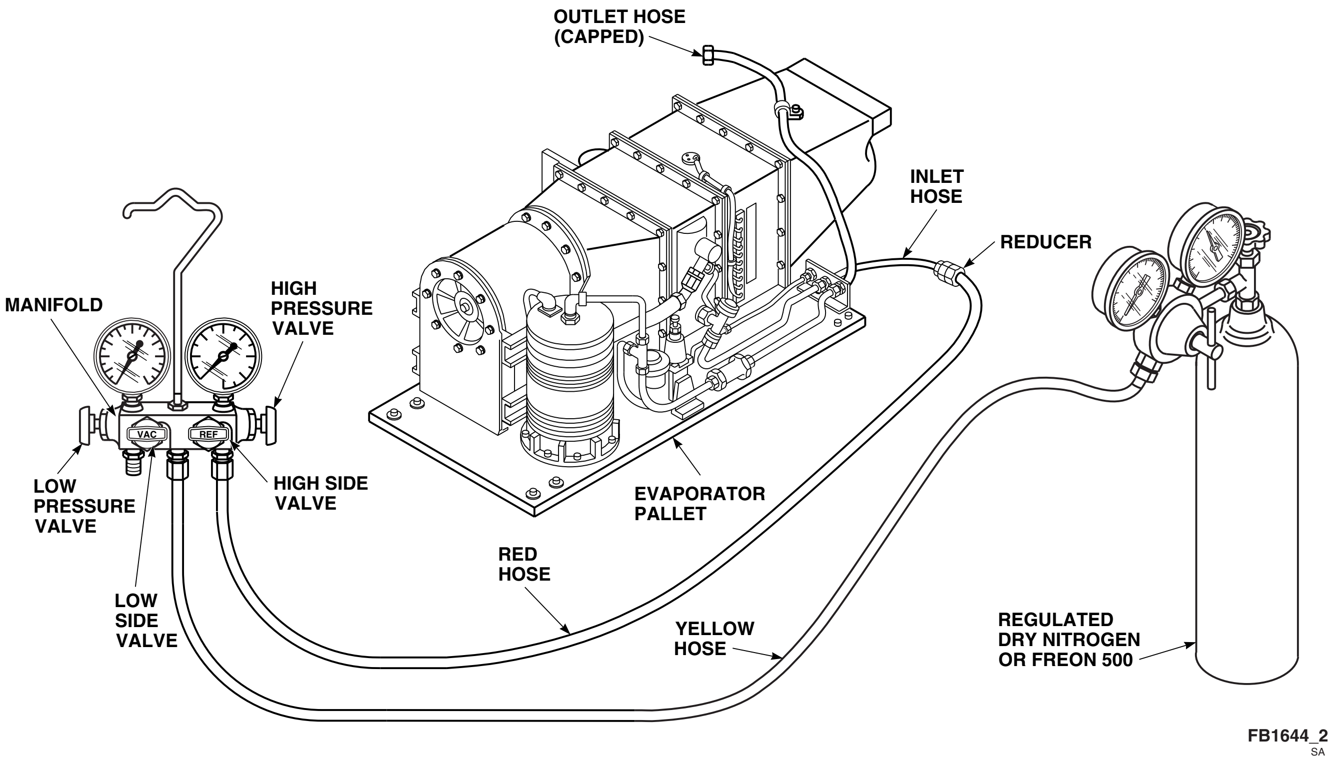


FIGURE 13-5-6-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR PALLET (BENCH) (SHEET 2 OF 2)

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close servicing manifold valves prior to opening supply valve on nitrogen bottle.

k. Unscrew adjustment on regulated dry nitrogen, Item 221, Appendix D, source until it is loose. Connect low pressure (yellow) to regulator.

l. Open valve on nitrogen bottle and slowly increase regulator pressure until it equals Freon-500 system pressure.

m. Open REF valve on servicing manifold.

n. Slowly increase nitrogen pressure until system pressure indicated on servicing manifold high pressure gage reaches 300 psi.

o. Using halogen leak detector calibrated to detect leaks of 2 oz. per year or less, check all fittings, connections, hoses, and components for leaks. Repair as necessary.

p. Repeat step o. until no leaks are found.

q. Close high side valve on servicing manifold.

r. Close valve on nitrogen bottle and slowly crack connection fitting.

s. Disconnect low pressure hose (yellow) from nitrogen regulator fitting and place hose end into a clean 5-gallon container, ¼-full of water.

t. Slowly open high side valve until a small stream of bubbles appears in water.

u. Continue to drain system until no bubbles appear in container. Pressure gages should read 0 psig.

v. Close high side valve and disconnect low pressure hose (yellow).

w. Disconnect high pressure hose (red) from rear of evaporator pallet.

x. **IMMEDIATELY CAP FITTING.**

13-5-6.1.3 Install.

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as handhold. Handle evaporator pallet by base only.

a. Position evaporator pallet over fuel cells and secure to support channels with bolts and washers (Figure 13-5-6-1, Sheet 1, Detail B). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

b. Before connecting, lubricate threads of line B-nuts using refrigerant compressor lubricating oil, Item 209A, Appendix D.

c. Connect outlet line to outboard fitting on rear panel of evaporator pallet (Figure 13-5-6-1, Sheet 1, Detail B). **TORQUE B-NUT TO 200 - 300 INCH-POUNDS.**

d. Connect inlet line to middle fitting on rear panel of evaporator pallet (Figure 13-5-6-1, Sheet 1, Detail D). **TORQUE B-NUT TO 150 - 250 INCH-POUNDS.**

e. Connect equalization line to inboard fitting on rear panel of evaporator pallet (Figure 13-5-6-1, Sheet 1, Detail D). **TORQUE B-NUT TO 40 - 65 INCH-POUNDS.**

f. Remove protective caps and plugs on electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connectors, P3 to J3 and P7 to J7, to left side of electrical pallet (Figure 13-5-6-1, Sheet 1, Detail C).

i. Install outlet transition duct on rear panel of evaporator pallet with bolts and washers.

j. Install upper supply duct on outlet transition duct with strap and screw (Figure 13-5-6-1, Sheet 1, Detail B).

k. Install lower supply duct on outlet transition duct and right supply duct with clamps.

l. Install left plenum (PARA 13-4-21).

m. Install left soundproofing panel on rear cabin bulkhead (PARA 2-4-69).

n. Service ECS (PARA 1-3-27).

o. Perform operational check of ECS (PARA 13-2-2).

13-5-7 ENVIRONMENTAL CONTROL SYSTEM COMPRESSOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Container, 2-Quart
- Funnel
- Refrigeration Unit Service Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Cap, NAS817-10
- Cap, NAS817-12
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Nitrogen, Item 221, Appendix D
- Plug, NAS818-10
- Plug, NAS818-12
- Protective Cap and Plug

References

- Appendix D
- PARA 1-3-27
- PARA 1-6-16
- PARA 1-7-20
- PARA 13-4-29
- PARA 13-5-3
- PARA 13-5-4
- PARA 13-5-6
- PARA 13-5-8
- PARA 13-5-10
- PARA 13-5-11

13-5-7.1 Replace Environmental Control System Compressor (BENCH).

13-5-7.1.1 Remove.

a. Remove ECS evaporator pallet (PARA 13-5-6).

b. Disconnect electrical connector, P8, from ECS compressor (Figure 13-5-7-1, Detail B).

c. Install protective cap and plug on electrical connectors.

CAUTION

To prevent moisture from entering air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting.

d. Remove discharge line from top of compressor and tee. Plug, NAS818-10, line and cap, NAS817-10, fitting.

e. Disconnect suction line from side of compressor. Plug, NAS818-12, line and cap, NAS817-12, fitting. Remove and discard packing.

f. Remove bolts, washers, and nuts, noting location of clamp holding plenum temperature sensor wire. Remove compressor from pallet.

13-5-7.1.2 Disassemble.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove nut and elbow from top of ECS compressor. Remove and discard packing.

c. Using funnel and 2-quart container, remove sight glass from compressor, catching fluid as it exits compressor. Remove and discard packing.

13-5-7.1.3 Clean.

CAUTION

Damage to ECS will result if system is not completely flushed during replacement of compressor. Failure of a compressor may release particulate matter into refrigerant/compressor oil passages throughout entire system. Make sure all parts, lines, and hoses which contain refrigerant/compressor oil are flushed to remove any possible contaminates.

a. Remove ECS condenser pallet (PARA 13-5-3).

b. Disconnect hoses from ECS electrical pallet (PARA 13-5-11).

c. Remove and discard ECS filter/dryer (PARA 13-5-10 or 13-4-29).

d. Remove hot gas bypass valve and associated lines (PARA 13-5-8).

e. Remove lines, burst disc, and relief valve (PARA 13-5-4).

f. Using denatured alcohol, Item 70, Appendix D, flush condenser and evaporator heat exchangers. Rotate heat exchangers several times on front and rear axis. Dump out alcohol and repeat until alcohol runs clear. Using dry nitrogen, Item 221, Appendix D, blow dry heat exchangers.

g. Using denatured alcohol, Item 70, Appendix D, flush all plumbing lines and hoses. Dump out alcohol and repeat until alcohol runs clear. Using dry nitrogen, Item 221, Appendix D, blow dry all plumbing.

h. Using denatured alcohol, Item 70, Appendix D, flush expansion valve and check valve. Dump out alcohol and repeat until alcohol runs clear. Using dry nitrogen, Item 221, Appendix D, blow dry valves.

i. Install lines, burst disc, and relief valve (PARA 13-5-4).

j. Install hot gas bypass valve and associated lines (PARA 13-5-8).

k. Install ECS filter/dryer (PARA 13-5-10 or 13-4-29).

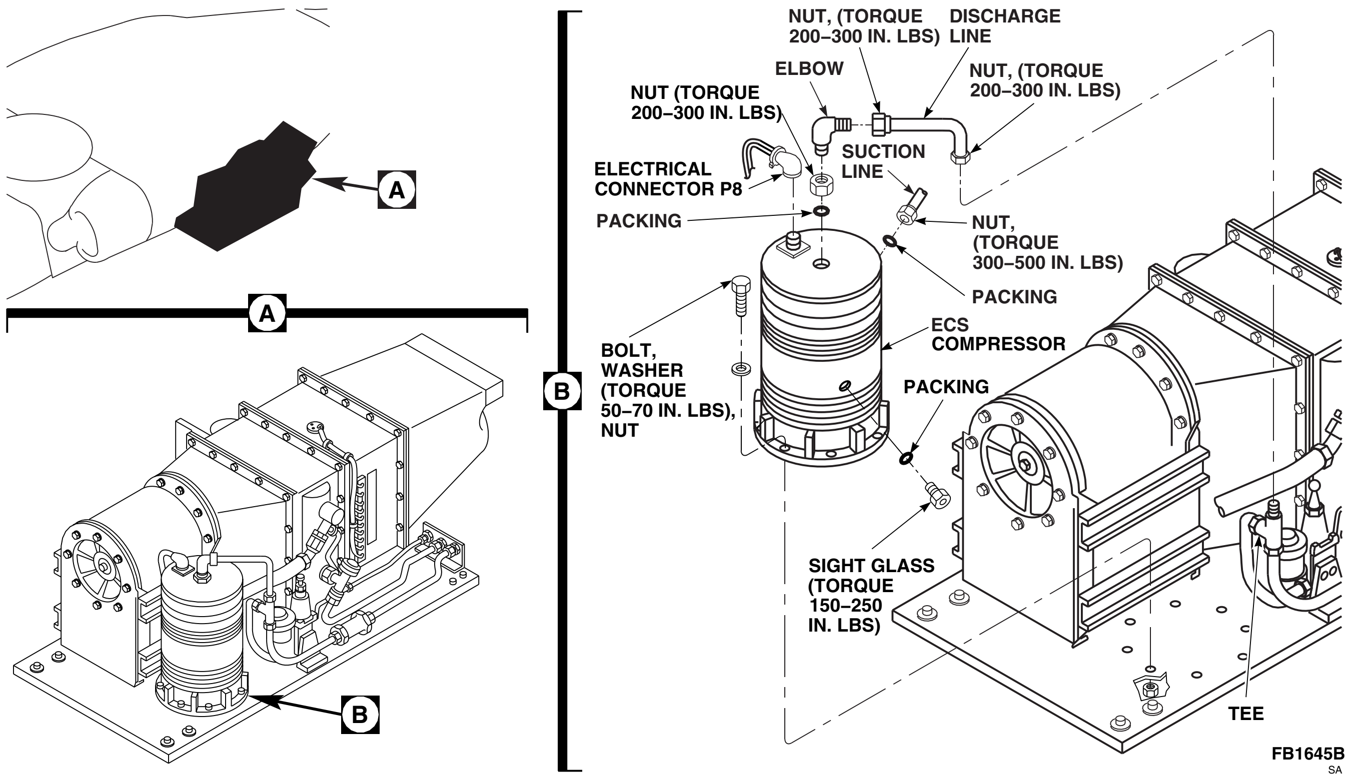


FIGURE 13-5-7-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM COMPRESSOR (BENCH)

l. Install ECS condenser pallet (PARA 13-5-3).

m. Connect hoses on ECS electrical pallet (PARA 13-5-11).

13-5-7.1.4 Assemble.

a. Turn off all electrical power (PARA 1-6-16).

b. Install sight glass on ECS compressor using packing lubricated with refrigerant lubricating oil compressor, Item 209A, Appendix D. Do not torque sight glass (Figure 13-5-7-1, Detail B).

c. Service compressor using compressor oil as follows:

(1) Check compressor oil level shown in sight glass. If completely full, overfilled condition is indicated, and compressor oil must be drained until level appears in sight glass.

(2) Proper compressor oil level is no more than ½ full. If required, add refrigerant lubricating oil compressor, Item 209A, Appendix D, in 4-ounce increments with 2 minutes between for setting, through discharge line.

(3) **TORQUE SIGHT GLASS TO 150 - 250 INCH-POUNDS.**

d. Lubricate packing using refrigerant lubricating oil compressor, Item 209A, Appendix D.

e. Install elbow with packing and nut on compressor. **TORQUE NUT TO 200 - 300 INCH-POUNDS.**

13-5-7.1.5 Install.

a. Install ECS compressor on pallet with bolts, washers, and nuts (Figure 13-5-7-1, Detail B). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**

b. Before connecting, lubricate threads of suction line B-nut using refrigerant lubricating oil compressor, Item 209A, Appendix D.

c. Remove cap and connect suction line on side of ECS compressor with packing. **TORQUE B-NUT TO 300 - 500 INCH-POUNDS.**

d. Before connecting, lubricate threads of discharge line B-nuts using refrigerant lubricating oil compressor, Item 209A, Appendix D.

e. Remove cap and connect discharge line between elbow and tee. **TORQUE B-NUTS TO 200 - 300 INCH-POUNDS.**

f. Remove protective cap and plug from electrical connectors.

g. Inspect and treat electrical connector (PARA 1-7-20).

h. Connect electrical connector, P8, on ECS compressor.

i. Pressure test ECS evaporator pallet (PARA 13-5-6).

j. Evacuate system for a minimum of two hours (PARA 1-3-27).

k. Service ECS using refrigerant (PARA 1-3-27).

13-5-8 ENVIRONMENTAL CONTROL SYSTEM HOT GAS BYPASS (HGBP) VALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cap, NAS817-4
- Cap, NAS817-6
- Cap, NAS817-8
- Cap, NAS817-10
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-4
- Plug, NAS818-6
- Plug, NAS818-8
- Plug, NAS818-10

References

- Appendix D
- PARA 13-5-6
- PARA 13-5-10

13-5-8.1 Replace Environmental Control System Hot Gas Bypass (HGBP) Valve (BENCH).

13-5-8.1.1 Remove.

a. Remove check valve as follows:

(1) Remove bolt, washer, spacer, and clamp, securing check valve to evaporator pallet (Figure 13-5-8-1, Sheet 1, Detail A).

(2) Disconnect lines from fitting bracket and discharge line tee.

CAUTION

To prevent moisture from entering the air-conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

(3) Remove lines, with check valve attached. Plug, NAS818-10, lines and cap, NAS817-10, fittings.

b. If filter/drier is installed on pallet, remove filter/drier (PARA 13-5-10). If expansion valve return line is installed, remove return line as follows:

(1) Disconnect lines from fitting bracket and expansion valve (Figure 13-5-8-1, Sheet 1, Detail B).

(2) Remove line. Plug, NAS818-8, lines and cap, NAS817-8, fitting.

c. Remove equalization line as follows:

CAUTION

To prevent moisture from entering the air-conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

(1) Loosen B-nuts securing equalization lines on HGBP valve, evaporator outlet fitting, and fitting bracket at rear of pallet (Figure 13-5-8-1, Sheet 2, Detail C).

(2) Remove line assembly from pallet. Plug, NAS818-4, lines and cap, NAS817-4, fittings.

d. Remove HGBP valve as follows:

CAUTION

To prevent moisture from entering the air-conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

(1) Loosen B-nuts securing line on outlet of HGBP valve and evaporator inlet (Figure 13-5-8-1, Sheet 2, Detail D). Remove and plug, NAS818-6 and NAS818-10, line. Cap, NAS817-6, and NAS817-10, fittings.

(2) Loosen B-nut securing line on inlet of HGBP valve. Plug, NAS818-10, lines and cap, NAS817-10, fittings.

(3) Loosen clamps securing electrical harness to pallet. Free harness enough to remove chafe guards and spiral wrap.

(4) Disconnect and free the electrical leads of HGBP valve regulator from terminals, 11 and 12, of terminal board, TB4.

(5) Remove bolts washers, HGBP valve, and bracket from pallet.

13-5-8.1.2 Install.

a. Install HGBP valve as follows:

(1) Install HGBP valve on bracket with bolts and washers (Figure 13-5-8-1, Sheet 2, Detail D). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.** Lockwire, Item 197, Appendix D, boltheads.

(2) Connect electrical leads of HGBP valve regulator to terminals, 11 and 12, of terminal board, TB4.

(3) Install spiral wrap and chafing strip around harness. **TORQUE CLAMP BOLTS TO 20 - 25 INCH-POUNDS.**

(4) Lubricate threads of compressor discharge line tee and HGBP valve inlet with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(5) Connect line between compressor discharge line tee and HGBP valve inlet. **TORQUE B-NUTS TO 200 - 300 INCH-POUNDS.**

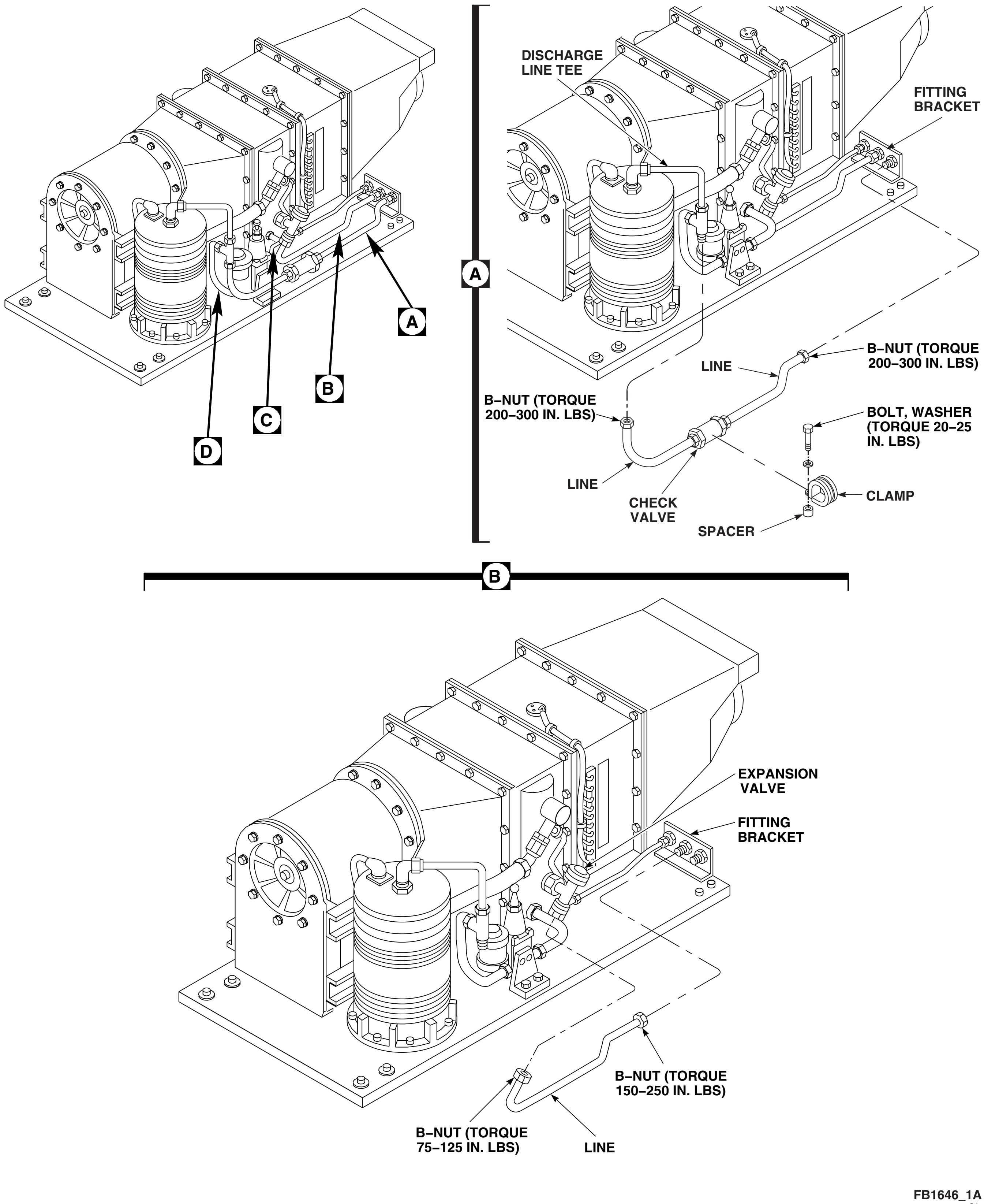


FIGURE 13-5-8-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM HOT GAS BYPASS (HGBP) VALVE (BENCH) (SHEET 1 OF 2)

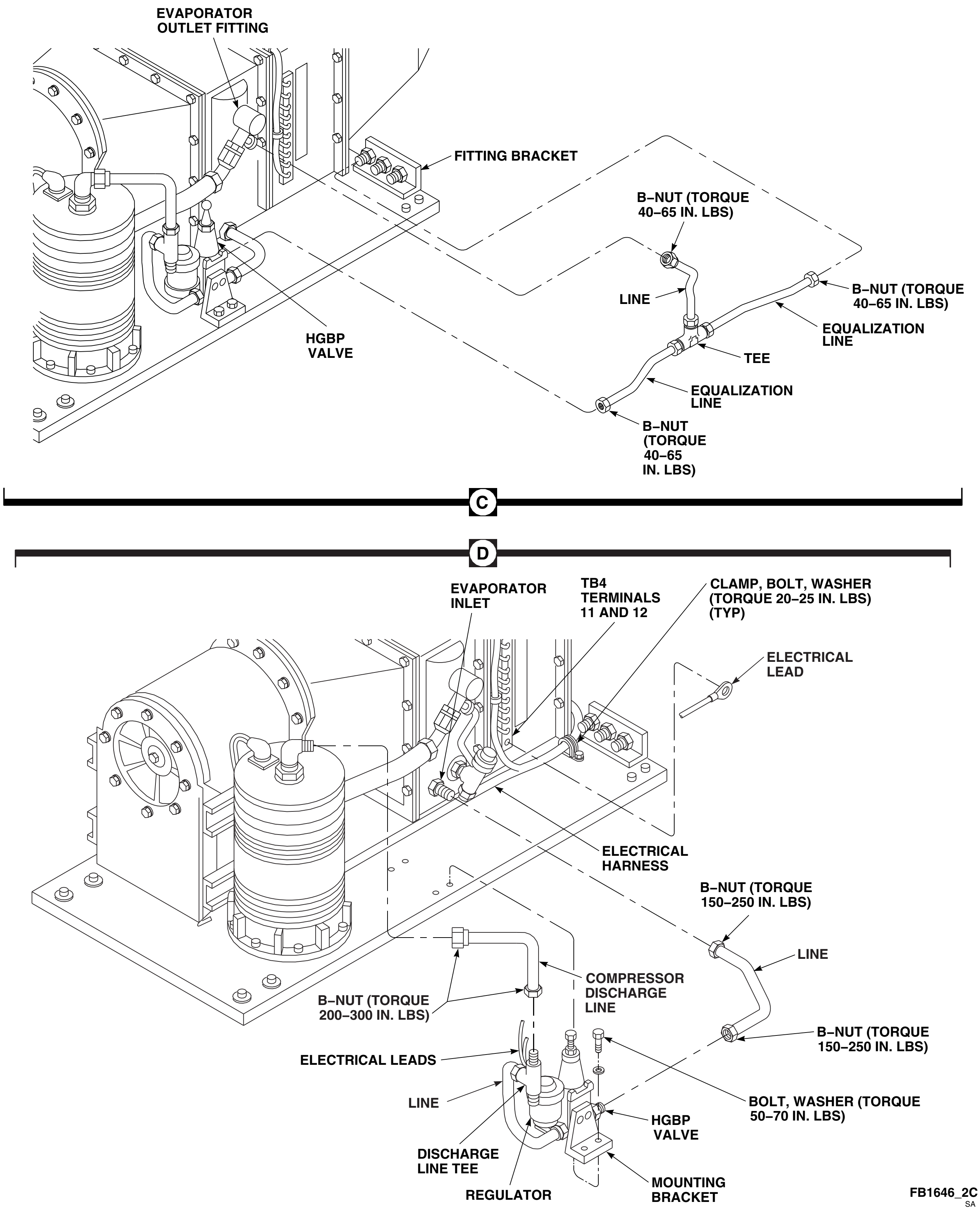


FIGURE 13-5-8-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM HOT GAS BYPASS (HGBP) VALVE (BENCH) (SHEET 2 OF 2)

(6) Lubricate threads of evaporator line B-nuts and HGBP outlet with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(7) Connect line between evaporator inlet and HGBP valve outlet. **TORQUE B-NUTS TO 150 - 250 INCH-POUNDS.**

b. Install equalization line as follows:

(1) Lubricate threads of equalization line B-nuts with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(2) Install equalization line on HGBP valve, evaporator outlet fitting, and fitting bracket at rear of pallet. **TORQUE ALL B-NUTS TO 40 - 65 INCH-POUNDS** (Figure 13-5-8-1, Sheet 2, Detail C).

c. If filter/drier was removed from pallet, install filter/drier (PARA 13-5-10).

d. If expansion valve return line was removed, install expansion valve return line as follows:

(1) Remove caps and plugs.

(2) Lubricate threads of expansion valve return line B-nuts with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(3) Connect expansion valve return line on expansion valve and fitting bracket at rear of pallet (Figure 13-5-8-1, Sheet 1, Detail B). **TORQUE NUT ON EXPANSION VALVE TO 75 - 125 INCH-POUNDS. TORQUE NUT ON FITTING BRACKET TO 150 - 250 INCH-POUNDS.**

e. Install check valve as follows:

(1) Lubricate threads of check valve line B-nuts with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(2) Position check valve line assembly on pallet. Install on fitting bracket at rear of pallet and compressor discharge line tee (Figure 13-5-8-1, Sheet 1, Detail A). **TORQUE B-NUTS TO 200 - 300 INCH-POUNDS.**

(3) Install bolt, washer, spacers, and clamp on check valve. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

(4) Pressure test evaporator pallet (PARA 13-5-6).

13-5-9 ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, AND HEATER/DEMISTER (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Adhesive, Item 57, Appendix D
- Cap, NAS817-8
- Cap, NAS817-12
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Loctite, Item 235, Appendix D
- Plug, NAS818-4
- Plug, NAS818-8
- Plug, NAS818-12
- Protective Cap and Plug
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-7-20
- PARA 13-4-15
- PARA 13-5-6
- PARA 13-5-8

13-5-9.1 Replace Environmental Control System Ducts, Evaporator, and Heater/ Demister (BENCH).

13-5-9.1.1. Remove.

NOTE

The inlet transition duct, evaporator heat exchanger, and the heater/demister are removed as an assembly from the pallet and disassembled thereafter.

a. Remove bolts and washers holding evaporator outlet transition duct to heater/demister (Figure 13-5-9-1, Sheet 4, Detail E).

b. Remove evaporator blower (PARA 13-4-15).

CAUTION

To prevent moisture from entering the air conditioning system and damaging parts, make sure to cap all fittings and plug all lines immediately after disconnecting them.

c. Loosen B-nut holding compressor suction line to evaporator outlet (Figure 13-5-9-1, Sheet 4, Detail G). Plug, NAS818-12, line and cap, NAS817-12, fitting.

d. Remove hot gas bypass valve (PARA 13-5-8).

e. Remove thermistor from evaporator outlet. Plug, NAS818-4, fitting.

f. Disconnect electrical connector from expansion valve.

g. Install protective cap and plug on electrical connectors.

h. Slit foam from around low temperature switch. Remove screws and washers securing switch to heater/demister. Tag and remove switch (Figure 13-5-9-1, Sheet 1, Detail B).

i. Slit foam from around high temperature switch. Remove screws and washers securing switch to heater/demister. Tag and remove switch.

j. Slit foam from around limiting switch. Remove screws and washers securing switch to heater/ demister. Tag and remove switch.

k. Remove nuts and lockwashers securing electrical leads to heater/demister terminal board, TB4. Tag and remove electrical leads (Figure 13-5-9-1, Sheet 1, Detail A).

l. Disconnect and tag heater element leads on heater/demister (Figure 13-5-9-1, Sheet 2, Detail C).

m. Remove screws, wahers, nuts, and clamps securing electrical harness to clips (Figure 13-5-9-1, Sheet 4, Detail F).

n. Loosen clamp and disconnect heater/demister drain tube (Figure 13-5-9-1, Sheet 2, Detail C).

o. Remove bolts, washers, and nuts holding heater/demister assembly to pallet (Figure 13-5-9-1, Sheet 1, Detail B).

p. Remove bolts and washers holding heater/ demister to evaporator.

q. Remove sealing compound from evaporator, heater/demister, and pallet with nonmetallic scraper.

13-5-9.1.2. Disassemble.

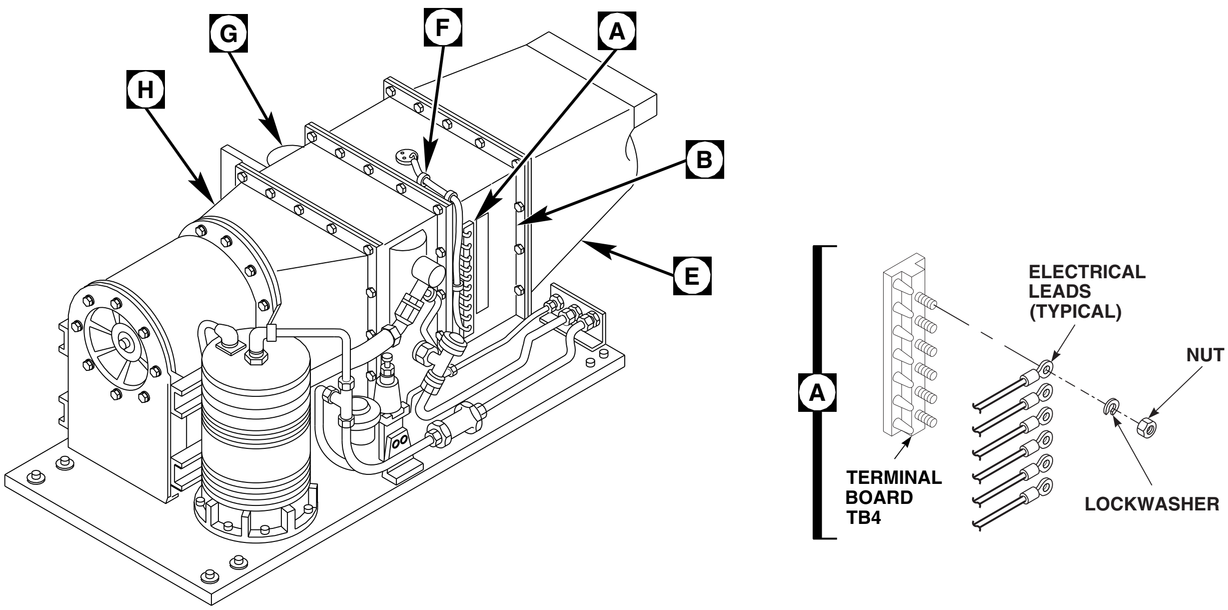
a. To disassemble heater/demister, do this:

NOTE

Each heat element is different. Do not attempt to interchange them.

(1) Remove nuts, washers, lockwashers, and fiber washers (inside housing) holding heater elements to housing. Remove heater elements. Discard nuts (Figure 13-5-9-1, Sheet 2, Detail C).

(2) Slit foam from around heater/demister and remove screws and nuts holding retaining brackets to heater/demister housing (Figure 13-5-9-1, Sheet 3, Detail D).



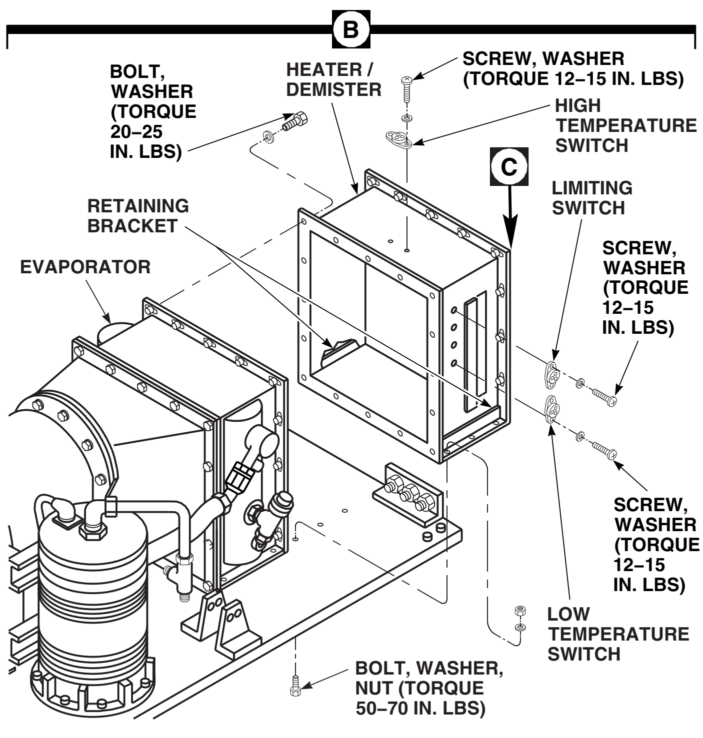


FIGURE 13-5-9-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, AND HEATER/DEMISTER (BENCH) (SHEET 1 OF 4)

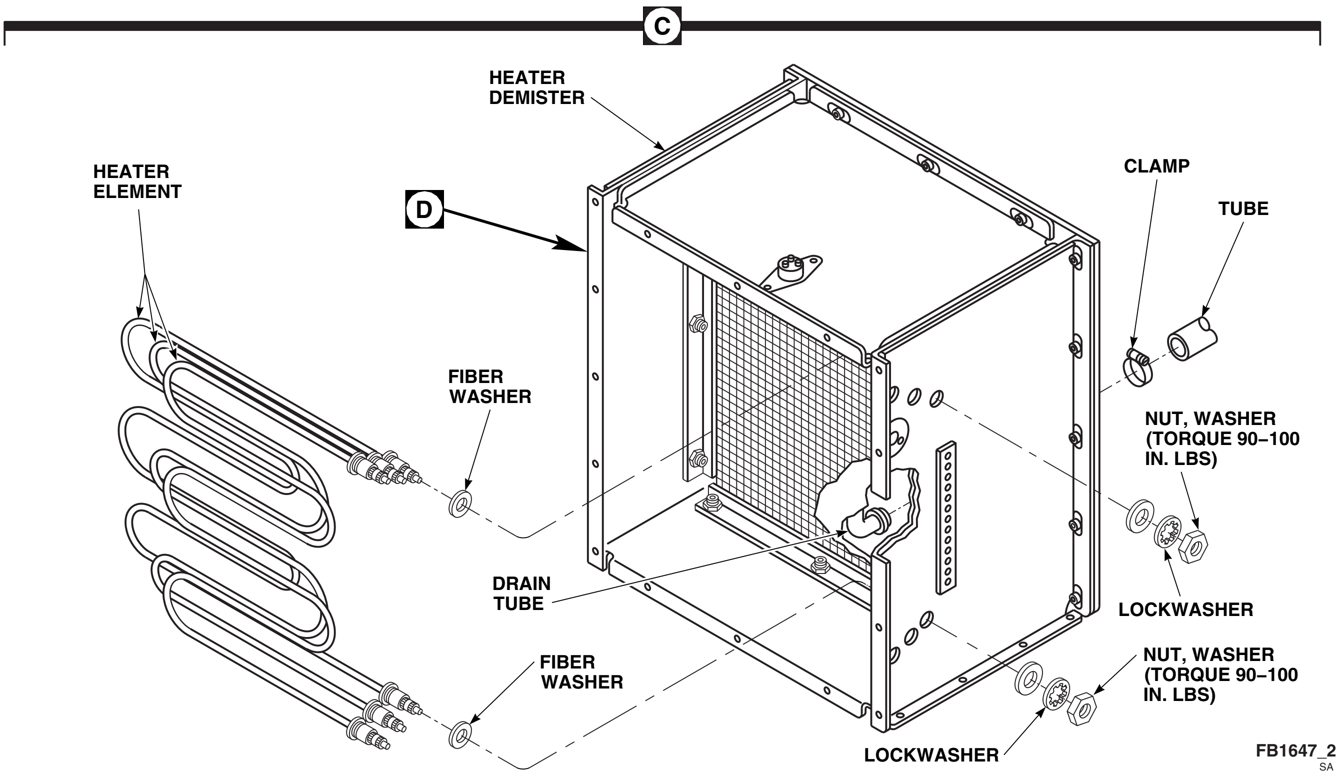


FIGURE 13-5-9-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, AND HEATER/DEMISTER (BENCH) (SHEET 2 OF 4)

(3) Remove demister pad and screen.

b. Remove bolts and washers holding evaporator to evaporator inlet transition duct (Figure 13-5-9-1, Sheet 4, Detail G).

c. To disassemble evaporator, do this:

CAUTION

To prevent moisture from entering the air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

(1) Remove union from outlet fitting. Plug, NAS818-4, fitting. Discard packing.

CAUTION

To avoid cracking fitting, be careful when loosening nut holding expansion valve to evaporator inlet.

(2) Loosen nut holding expansion valve to evaporator. Plug, NAS818-8, fitting and cap, NAS817-8, expansion valve.

d. Remove bolts, washers, and evaporator inlet transition duct from bracket (Figure 13-5-9-1, Sheet 4, Detail H).

e. Remove bolts, washers, and baffles from evaporator inlet transition duct.

13-5-9.1.3. Assemble.

a. Install baffles to evaporator inlet transition duct with bolts and washers (Figure 13-5-9-1, Sheet 4, Detail H). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

b. Install evaporator inlet transition duct to bracket with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

c. To assemble evaporator, do this:

(1) Remove cap and plug from expansion valve and fitting.

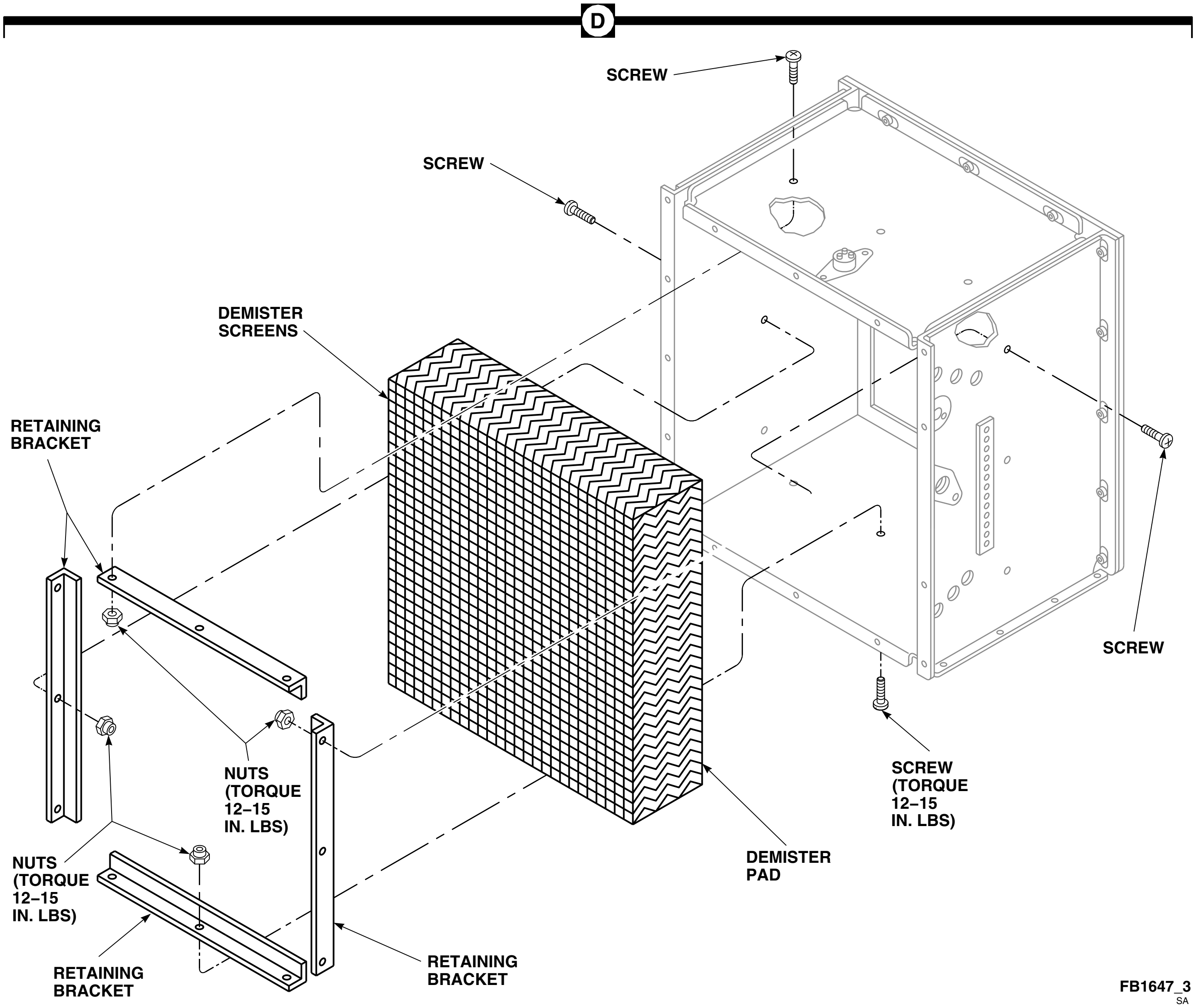


FIGURE 13-5-9-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, AND HEATER/DEMISTER (BENCH) (SHEET 3 OF 4)

CAUTION

To avoid cracking fitting, be careful when tightening nut holding expansion valve to evaporator inlet.

(2) Lubricate nut on evaporator fittings with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

(3) Connect expansion valve to evaporator fitting (Figure 13-5-9-1, Sheet 4, Detail G). **TORQUE NUT TO 150 - 250 INCH-POUNDS.**

(4) Lubricate threads on union with refrigerant lubricating oil compressor , Item 209A, Appendix D, before connecting.

(5) Remove plug and connect union to evaporator outlet fitting using packing. **TORQUE UNION TO 40 - 65 INCH-POUNDS.**

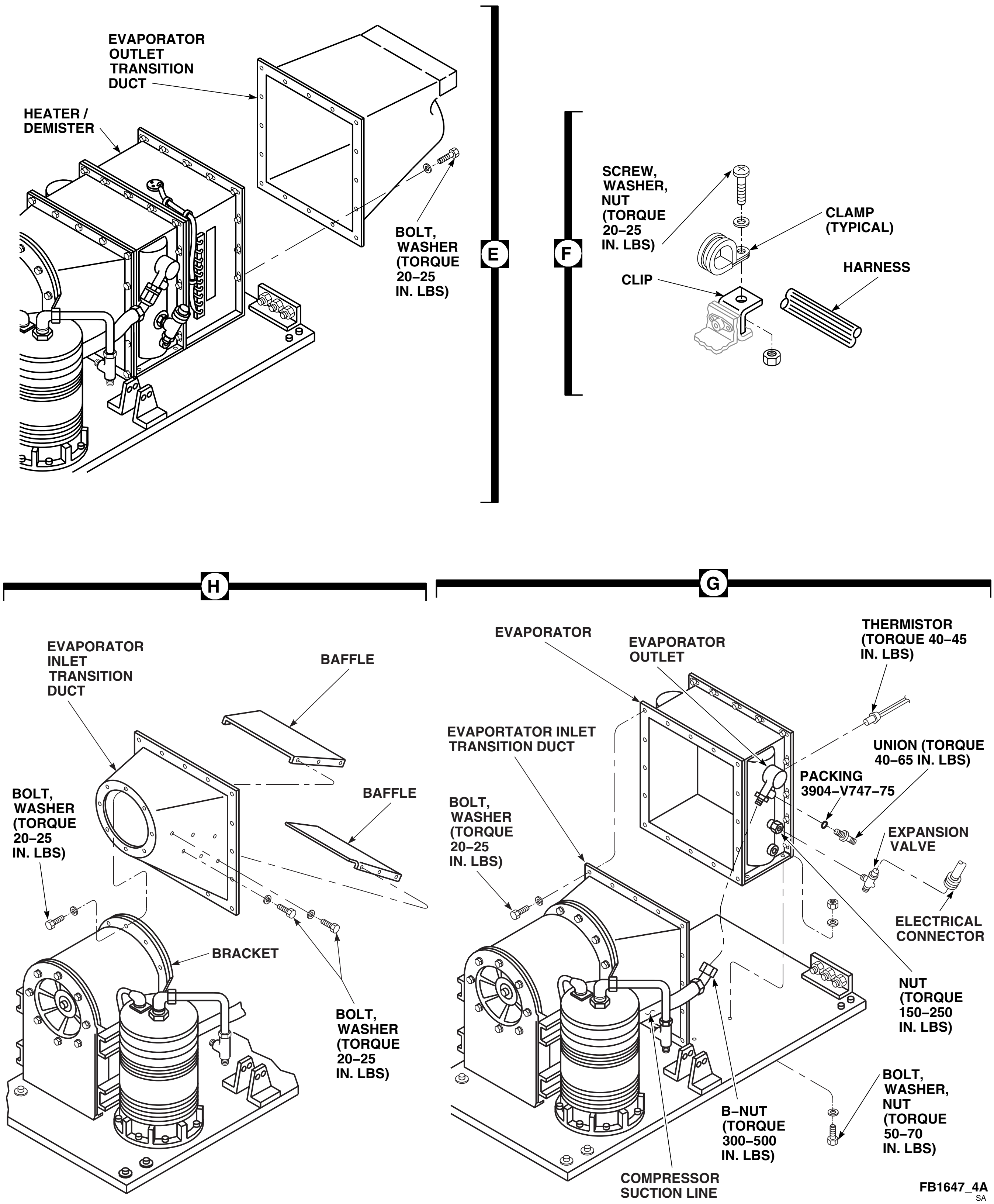


FIGURE 13-5-9-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, AND HEATER/DEMISTER (BENCH) (SHEET 4 OF 4)

d. Install evaporator on evaporator inlet transition duct with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

e. To assemble heater/demister, do this:

(1) Position demister pad and screen in housing. Install with retaining brackets, screws, and nuts (Figure 13-5-9-1, Sheet 3, Detail D). **TORQUE NUTS TO 12 - 15 INCH-POUNDS.**

NOTE

Each heater element is different. Do not attempt to interchange them.

(2) Install heater elements in housing with new nuts, washers, lockwashers, and fiber washers (inside housing) (Figure 13-5-9-1, Sheet 2, Detail C). **TORQUE NUTS TO 90 - 100 INCH-POUNDS.**

f. Install heater/demister on evaporator with bolts and washers (Figure 13-5-9-1, Sheet 1, Detail B). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

13-5-9.1.4. Install.

a. Install and seal evaporator and heater/demister to pallet with sealing compound, Item 292, Appendix D, positioning gasket between transition duct and evaporator blower rear support bracket. Install evaporator mount bolts, washers and nuts (Figure 13-5-9-1, Sheet 1, Detail B). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**

b. Connect heater/demister drain tube and secure with clamp (Figure 13-5-9-1, Sheet 2, Detail C).

c. Remove plug and cap.

d. Lubricate threads of compressor line with refrigerant lubricating oil compressor, Item 209A, Appendix D, before connecting.

e. Connect compressor suction line to evaporator outlet (Figure 13-5-9-1, Sheet 4, Detail G). **TORQUE B-NUT TO 300 - 500 INCH-POUNDS.**

f. Connect electrical harness to clips with clamps, screws, washers, and nuts (Figure 13-5-9-1, Sheet 4, Detail F). **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**

g. Install limiting switch on housing with screws and washers (Figure 13-5-9-1, Sheet 1, Detail B). **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

h. Install high temperature switch on housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

i. Install low temperature switch on housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

j. Repair foam insulation with adhesive, Item 57, Appendix D. Replace if required.

k. Remove plug and install thermistor into evaporator heat exchanger outlet fitting (Figure 13-5-9-1, Sheet 4, Detail G). Coat threads with Loctite, Item 235, Appendix D. **TORQUE TO 40 - 45 INCH-POUNDS.**

l. Secure all electrical leads to terminal board, TB4, on heater/demister with nuts and lockwashers. Remove tags from electrical leads. (Figure 13-5-9-1, Sheet 1, Detail A).

m. Connect heater element leads to heater/demister (Figure 13-5-9-1, Sheet 2, Detail C).

n. Remove cap and plug from electrical connectors.

o. Inspect and treat electrical connector (PARA 1-7-20).

p. Install electrical connector to expansion valve (Figure 13-5-9-1, Sheet 4, Detail G).

q. Install hot gas bypass valve (PARA 13-5-8).

r. Install evaporator blower (PARA 13-4-15).

s. Install evaporator outlet transition duct on heater/demister with bolts and washers (Figure 13-5-9-1, Sheet 4, Detail E). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

t. Pressure test evaporator pallet (PARA 13-5-6).

13-5-10 ENVIRONMENTAL CONTROL SYSTEM FILTER/DRIER (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cap, NAS817-8
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-8

References

- Appendix D
- PARA 1-3-27
- PARA 1-6-16

13-5-10.1 Replace Environmental Control System Filter/Drier (BENCH).

NOTE

The following procedure applies to filter/drier mounted on the evaporator pallet.

13-5-10.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Deservice ECS (PARA 1-3-27).

c. Remove bolts, washers, spacers, and saddle clamps from filter/drier (Figure 13-5-10-1, Detail A).

CAUTION

Filter/drier is extremely sensitive to moisture and humidity. Replace filter/drier if inlet and/or outlet connections are left uncapped for more than 10 minutes.

d. Loosen B-nuts that hold filter/drier lines on expansion valve and fitting bracket on rear of pallet.

e. Remove filter/drier from pallet. Plug, NAS818-8, lines and cap, NAS817-8, fittings.

13-5-10.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

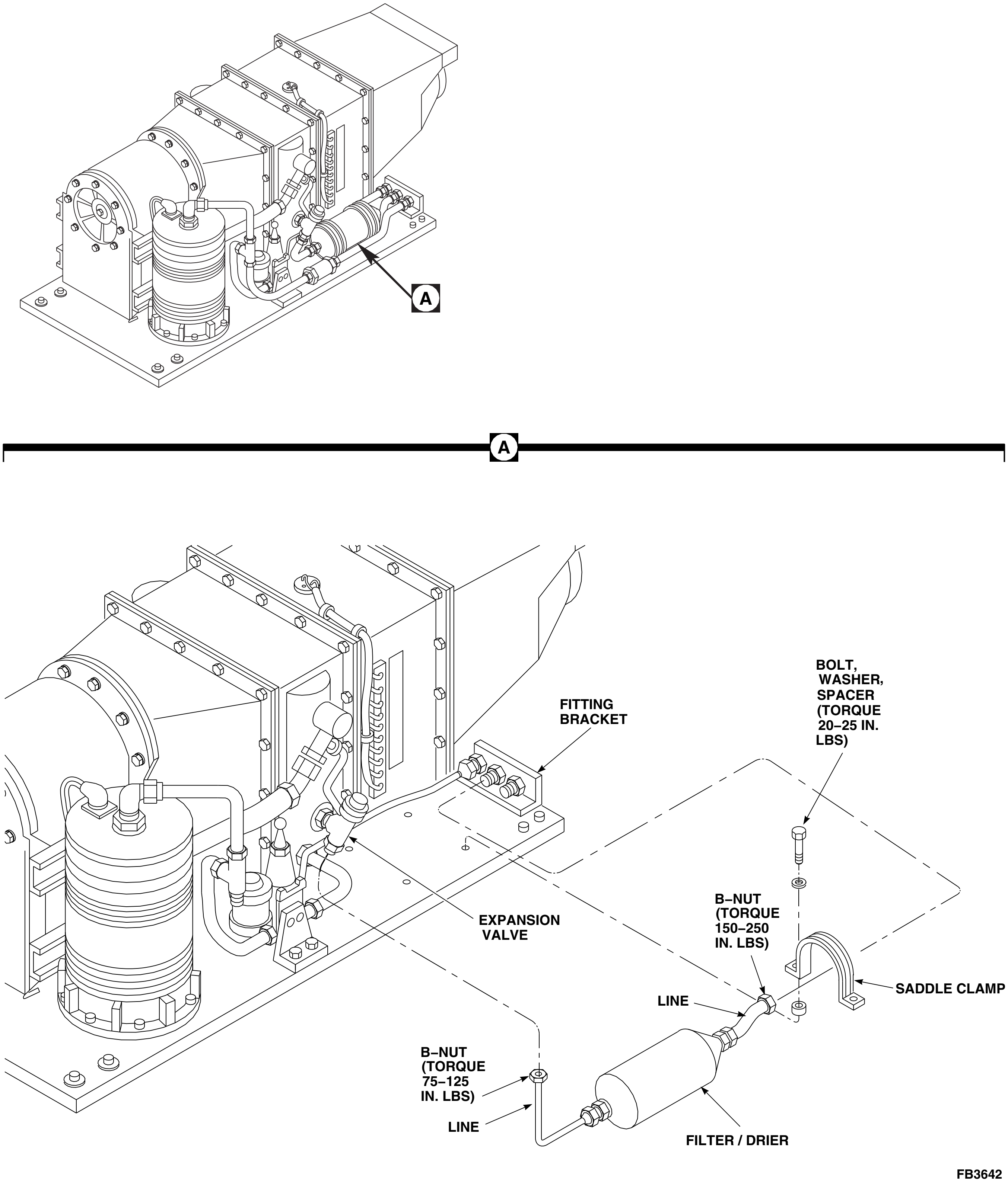


FIGURE 13-5-10-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM FILTER/DRIER

CAUTION

Filter/drier is extremely sensitive to moisture and humidity. Replace filter drier if inlet and outlet connections are left uncapped for more than 10 minutes.

b. Remove caps and plugs.

c. Lubricate threads with refrigerant lubricating oil compressor, Item 209A, Appendix D.

d. Install filter/drier on pallet with bolts, washers, spacers, and saddle clamps (Figure 13-5-10-1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

e. Install filter/drier lines on expansion valve and fitting bracket at rear of pallet. **TORQUE B-NUT ON EXPANSION VALVE TO 75 - 125 INCH-POUNDS. TORQUE B-NUT ON FITTING BRACKET TO 150 - 250 INCH-POUNDS.**

f. Service ECS (PARA 1-3-27).

13-5-11 ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL PALLET (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cap, NAS817-4
- Cap, NAS817-8
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-4
- Plug, NAS818-8
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-27
- PARA 1-7-20
- PARA 13-2-2

13-5-11.1 Replace Environmental Control System Electrical Pallet (BENCH).

13-5-11.1.1 Remove.

a. Deservice ECS (PARA 1-3-27).

b. Disconnect electrical connectors, P3, P4, P7, P456, and P457, from electrical box (Figure 13-5-11-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove bolt, washer, and spacer from clamp securing inlet hose to ECS electrical pallet.

CAUTION

To prevent moisture from entering air conditioning system and damaging parts, make sure to plug all lines and cap all fittings after disconnecting.

e. Disconnect inlet lines from electrical pallet and sight glass. Plug, NAS818-8, lines and cap, NAS817-8, fittings.

f. Disconnect hose from ECS servicing manifold. Plug, NAS818-4, line and cap, NAS817-4, fitting.

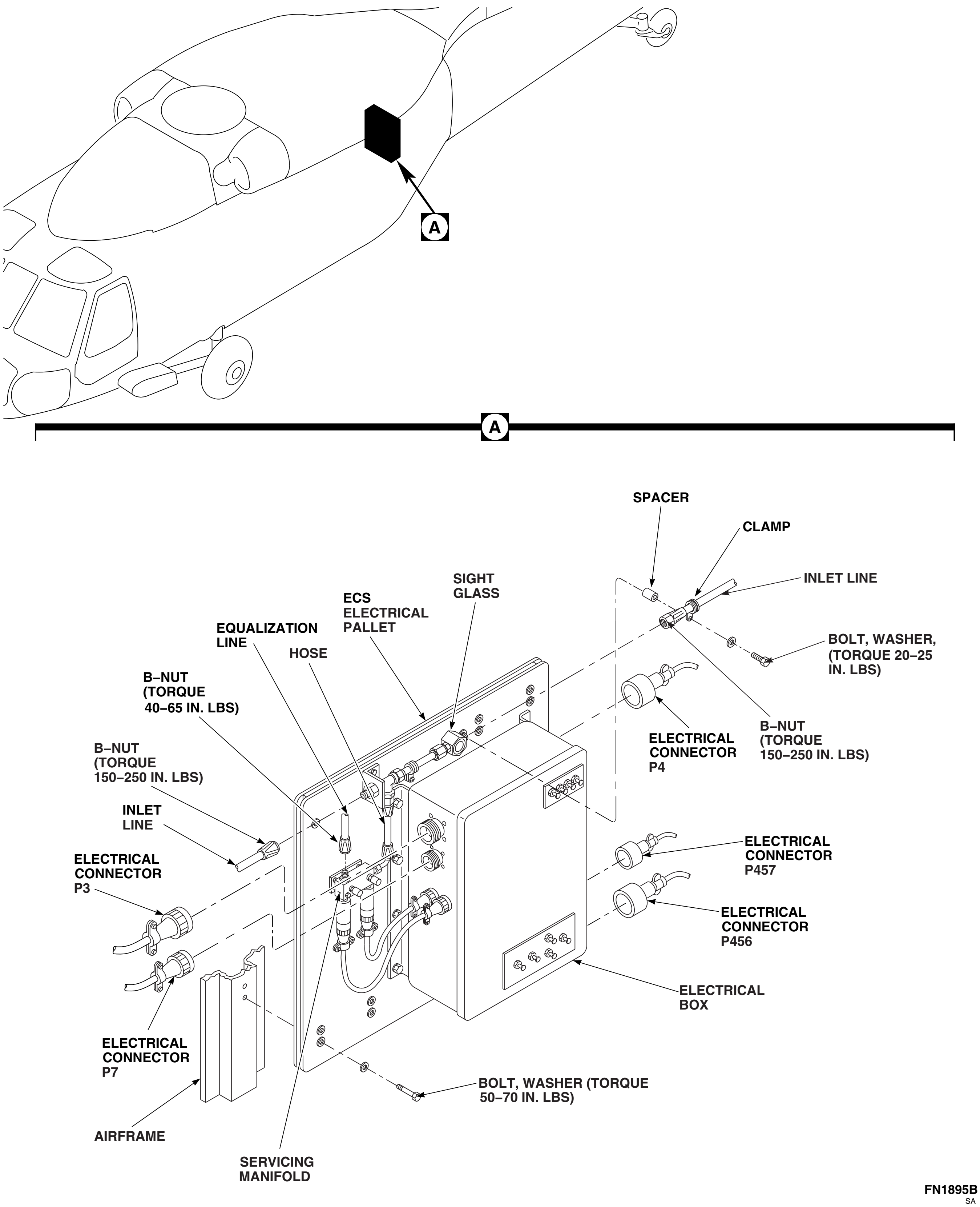


FIGURE 13-5-11-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL PALLET (BENCH)

g. Remove bolts and washers securing electrical pallet to airframe.

CAUTION

To prevent damage to ECS, do not use lines, fittings, and other components as handhold. Handle pallet by base only.

h. Remove electrical pallet from helicopter.

13-5-11.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install ECS electrical pallet in helicopter with bolts and washers (Figure 13-5-11-1, Detail A). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.**

c. Before connecting, lubricate threads of hose and line B-nuts using refrigerant lubricating oil compressor, Item 209A, Appendix D.

d. Remove plug and cap and connect hose to ECS servicing manifold. **TORQUE B-NUT TO 40 - 65 INCH-POUNDS.**

e. Before connecting, lubricate threads of hose B-nuts using refrigerant lubricating oil compressor, Item 209A, Appendix D.

f. Remove plugs and caps and connect inlet lines to electrical pallet and sight glass. **TORQUE B-NUTS TO 150 - 250 INCH-POUNDS.**

g. Install inlet line on electrical pallet with bolt, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

h. Remove protective caps and plugs from electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. Connect electrical connectors, P3, P4, P7, P456, and P457, to electrical box .

k. Service ECS (PARA 1-3-27).

l. Perform operational check of ECS (PARA 13-2-2).

13-5-12 ENVIRONMENTAL CONTROL SYSTEM SERVICING MANIFOLD (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cap, NAS817-4
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-4

References

- Appendix D
- PARA 1-6-16
- PARA 13-5-14

13-5-12.1 Replace Environmental Control System Servicing Manifold (BENCH).

13-5-12.1.1 Remove.

a. Remove ECS high and low pressure switches (PARA 13-5-14).

CAUTION

To prevent moisture from entering the air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting them.

b. Disconnect high and low pressure hoses from ECS servicing manifold. Plug, NAS818-4, hoses and cap, NAS817-4, fittings (Figure 13-5-12-1, Detail B).

c. Remove bolts, washers, and spacers securing servicing manifold to ECS electrical pallet.

d. Remove servicing manifold from ECS electrical pallet.

13-5-12.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove any caps and plugs prior to lubricating and installing hoses.

c. Before connecting, lubricate threads of high and low pressure hose assembly B-nuts using refrigerant lubricating oil compressor, Item 209A, Appendix D.

d. Install ECS servicing manifold on ECS electrical pallet with bolts, washers, and spacers (Figure 13-5-12-1, Detail B). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

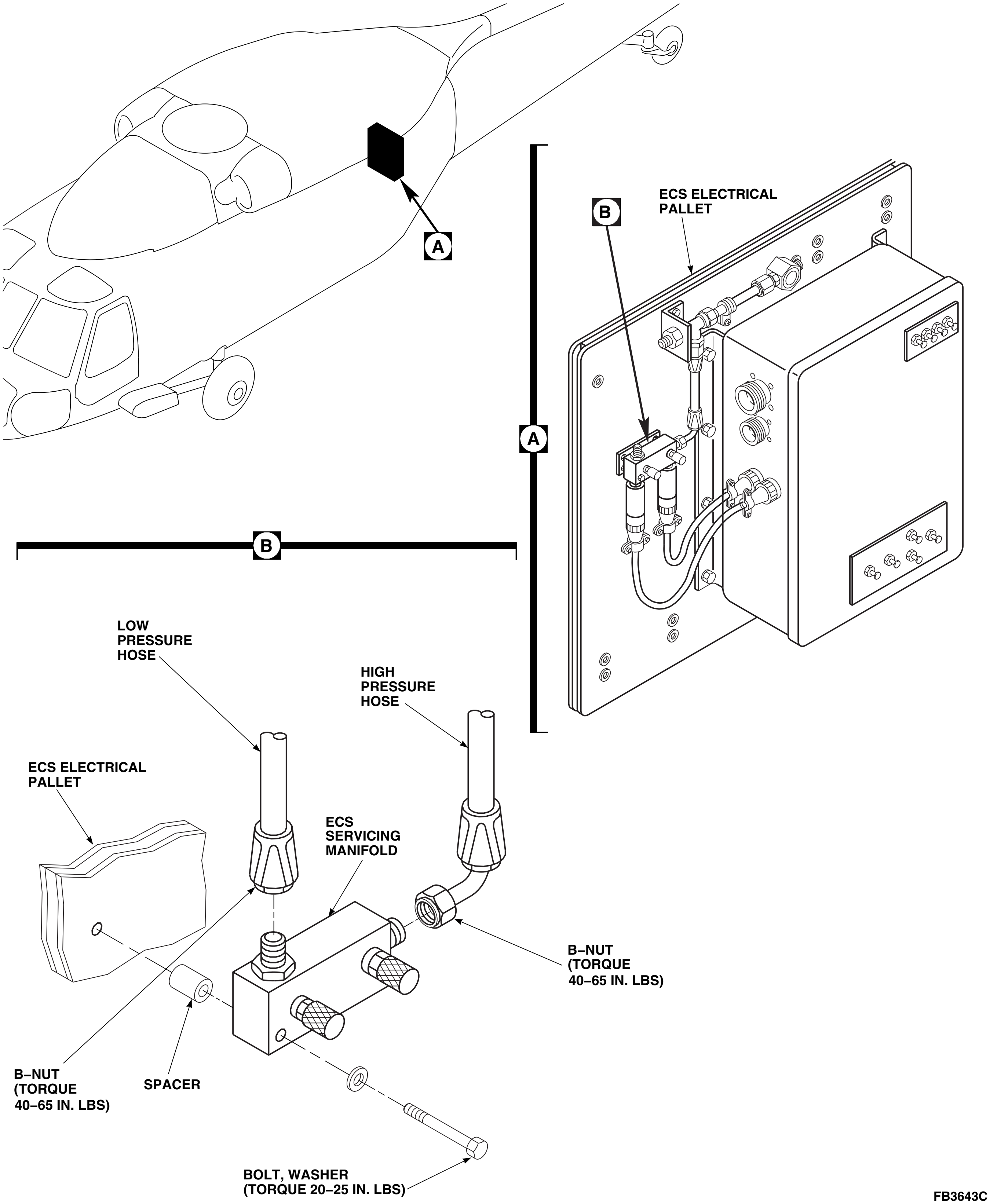


FIGURE 13-5-12-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM SERVICING MANIFOLD (BENCH)

e. Connect high and low pressure hoses to servicing manifold. **TORQUE B-NUTS TO 40 - 65 INCH-POUNDS.**

f. Install ECS high and low pressure switches (PARA 13-5-14).

13-5-13 ENVIRONMENTAL CONTROL SYSTEM SIGHT GLASS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cap, NAS817-8 (2)
- Lubricating Oil, Refrigerant Compressor, Item 209A, Appendix D
- Plug, NAS818-8 (2)

References

- Appendix D
- PARA 1-3-27

13-5-13.1 Replace Environmental Control System Sight Glass (BENCH).

13-5-13.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Deservice ECS (PARA 1-3-27).

CAUTION

To prevent moisture from entering air conditioning system and damaging parts, make sure to cap all lines immediately after disconnecting.

c. Disconnect lines from ends of sight glass. Plug lines and cap fittings (Figure 13-5-13-1, Detail B).

d. Remove bolt, washer, and spacer from clamp securing inlet line to ECS electrical pallet.

e. Remove sight glass.

13-5-13.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position sight glass between ends of lines (Figure 13-5-13-1, Detail B).

c. Remove plugs from lines and caps from fittings.

d. Lubricate threads of line B-nuts with refrigerant compressor lubricating oil, Item 209A, Appendix D, before connecting.

e. Loosely connect lines to sight glass.

f. Install inlet line on ECS electrical pallet with bolt, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

g. **TORQUE B-NUTS ON LINES TO 150 - 250 INCH-POUNDS.**

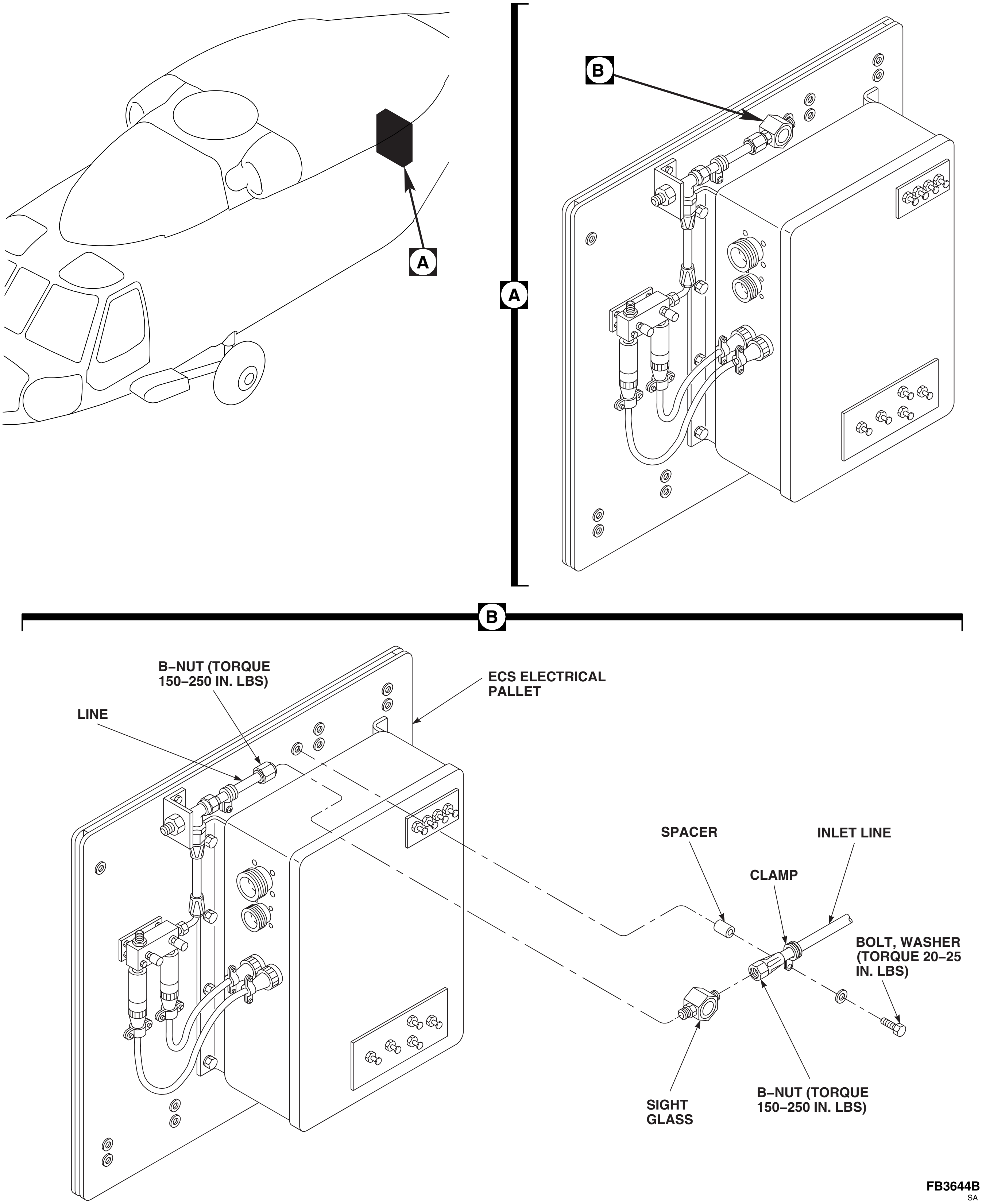


FIGURE 13-5-13-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM SIGHT GLASS (BENCH)

h. Service ECS (PARA 1-3-27).

13-5-14 ENVIRONMENTAL CONTROL SYSTEM HIGH AND LOW PRESSURE SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Refrigeration Unit Service Toolkit

Materials/Parts

- Protective Cap and Plug

References

- PARA 1-3-27
- PARA 1-6-16
- PARA 1-7-20
- PARA 13-2-2

13-5-14.1 Replace Environmental Control System High and Low Pressure Switches (BENCH).

13-5-14.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Deservice ECS (PARA 1-3-27).

c. Disconnect electrical connector, P11, from high pressure switch (Figure 13-5-14-1, Detail A).

d. Disconnect electrical connector, P12, from low pressure switch.

e. Install protective caps and plugs on electrical connectors.

NOTE

One or both fittings may come out with switches. The fitting must be removed from switch for use with replacement switch.

f. Remove high and low pressure switches from fittings on servicing manifold. If fitting(s) came out, retain fitting(s). Discard packing(s).

13-5-14.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If either fitting was removed from servicing manifold, reinstall fittings, with new packings, in servicing manifold.

c. Install high and low pressure switches in fittings (Figure 13-5-14-1, Detail A).

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

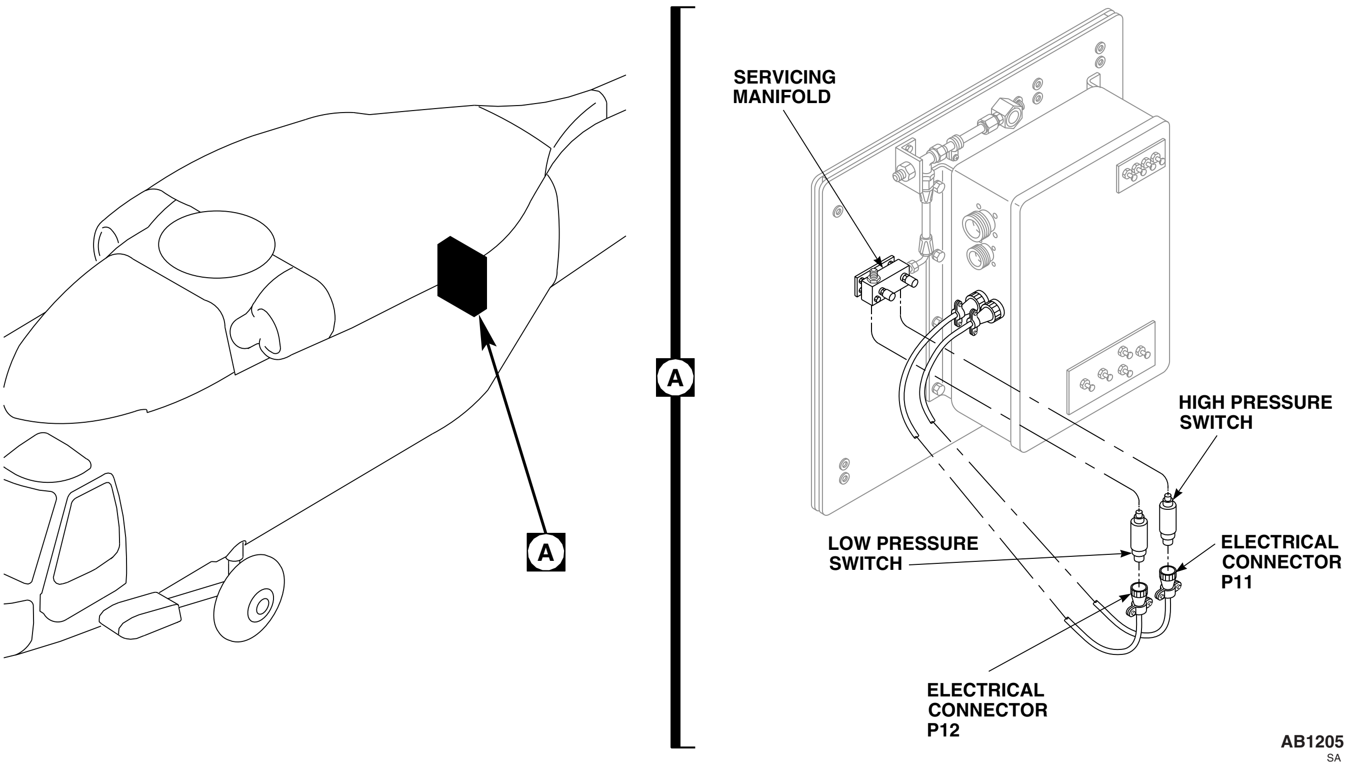


FIGURE 13-5-14-1. REPLACE ENVIRONMENTAL CONTROL SYSTEM HIGH AND LOW PRESSURE SWITCHES (BENCH)

f. Connect electrical connector, P11, to high pressure switch.

g. Connect electrical connector, P12, to low pressure switch.

h. Service ECS (PARA 1-3-27).

i. Perform operational check of ECS (PARA 13-2-2).